

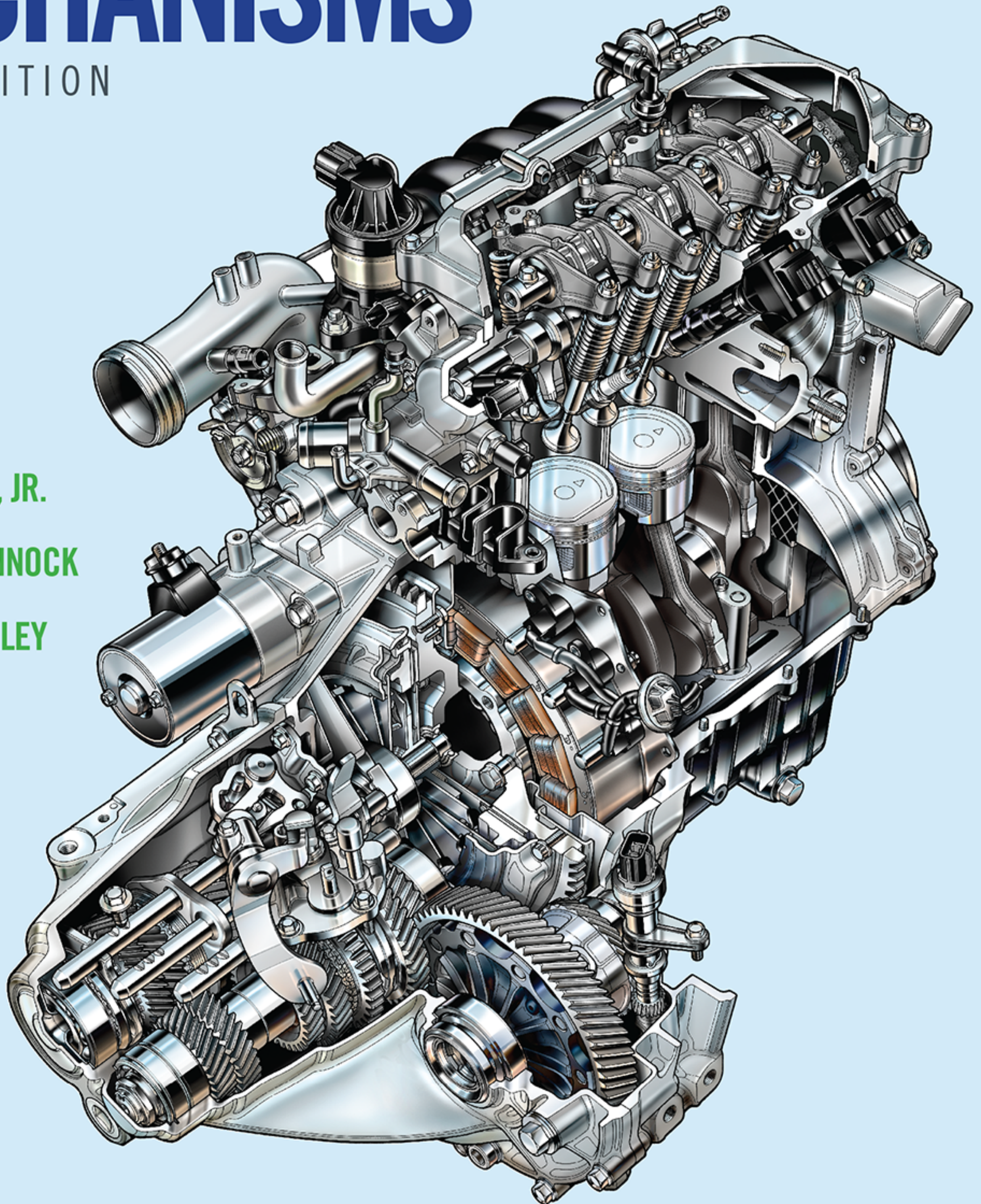
THEORY OF MACHINES AND MECHANISMS

FIFTH EDITION

JOHN J. VICKER, JR.

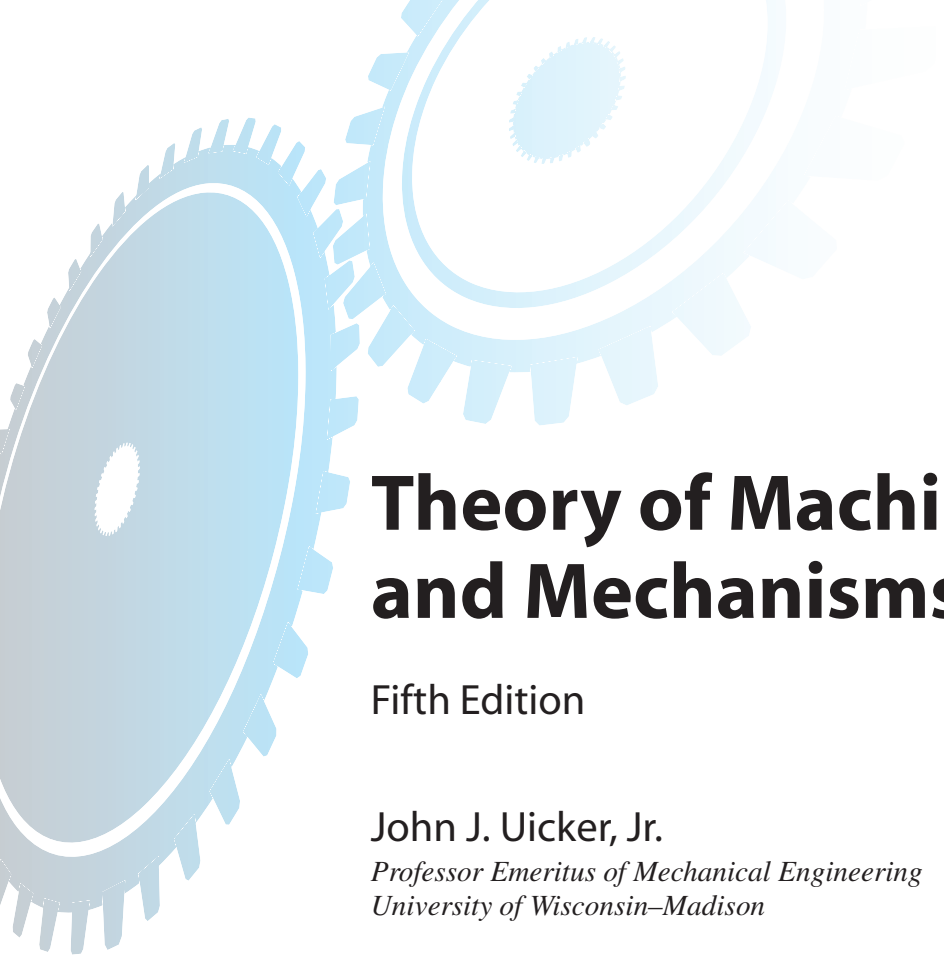
GORDON R. PENNOCK

JOSEPH E. SHIGLEY



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Theory of Machines and Mechanisms



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Fifth Edition

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University of Wisconsin–Madison*

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This textbook is dedicated to the memory of my parents, John J. Uicker, Emeritus Dean of Engineering, University of Detroit, Elizabeth F. Uicker, and to my six children, Theresa A. Zenchenko, John J. Uicker III, Joseph M. Uicker, Dorothy J. Winger, Barbara A. Peterson, and Joan E. Horne.

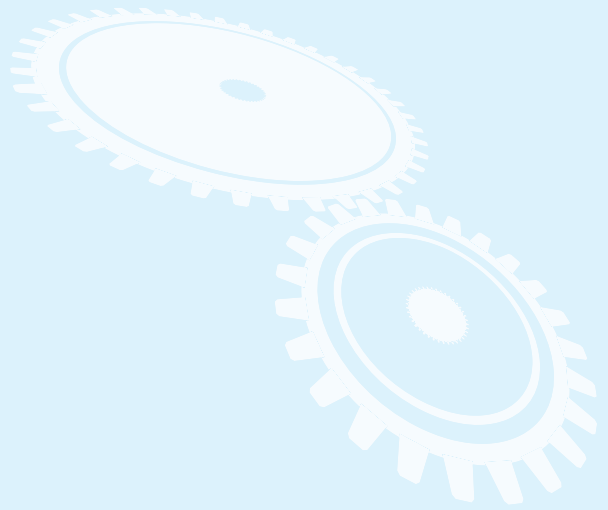
—**John J. Uicker, Jr.**

This work is also dedicated first and foremost to my wife, Mollie B., and my son, Callum R. Pennock. The work is also dedicated to my friend and mentor, the late Dr. An (Andy) Tzu Yang, and my colleagues in the School of Mechanical Engineering, Purdue University, West Lafayette, Indiana.

—**Gordon R. Pennock**

Finally, this text is dedicated to the memory of the late **Joseph E. Shigley**, Professor Emeritus, Mechanical Engineering Department, University of Michigan, Ann Arbor. Although this fifth edition contains significant changes from earlier editions, the text remains consistent with his previous writings.

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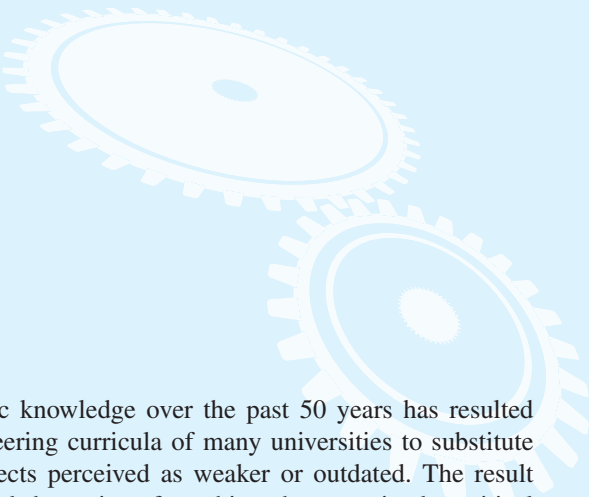
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Preface

The tremendous growth of scientific knowledge over the past 50 years has resulted in an intense pressure on the engineering curricula of many universities to substitute “modern” subjects in place of subjects perceived as weaker or outdated. The result is that, for some, the kinematics and dynamics of machines has remained a critical component of the curriculum and a requirement for all mechanical engineering students, while at others, a course on these subjects is only made available as an elective topic for specialized study by a small number of engineering students. Some schools, depending largely on the faculty, require a greater emphasis on mechanical design at the expense of depth of knowledge in analytical techniques. Rapid advances in technology, however, have produced a need for a textbook that satisfies the requirement of new and changing course structures.

Much of the new knowledge in the theory of machines and mechanisms currently exists in a large variety of technical journals and manuscripts, each couched in its own singular language and nomenclature and each requiring additional background for clear comprehension. It is possible that the individual published contributions could be used to strengthen engineering courses if the necessary foundation was provided and a common notation and nomenclature was established. These new developments could then be integrated into existing courses to provide a logical, modern, and comprehensive whole. The purpose of this book is to provide the background that will allow such an integration.

This book is intended to cover that field of engineering theory, analysis, design, and practice that is generally described as mechanisms or as kinematics and dynamics of machines. Although this text is written primarily for students of mechanical engineering, the content can also be of considerable value to practicing engineers throughout their professional careers.

To develop a broad and basic comprehension, the text presents numerous methods of analysis and synthesis that are common to the literature of the field. The authors have included graphic methods of analysis and synthesis extensively throughout the book, because they are firmly of the opinion that graphic methods provide visual feedback that enhances the student’s understanding of the basic nature of, and interplay between, the underlying equations. Therefore, graphic methods are presented as one possible solution technique, but are always accompanied by vector equations defined by the fundamental laws of mechanics, rather than as graphic “tricks” to be learned by rote and applied blindly. In addition, although graphic techniques, performed by hand, may lack accuracy, they can be performed quickly, and even inaccurate sketches can often provide reasonable estimates of a solution and can be used to check the results of analytic or numeric solution techniques.

The authors also use conventional methods of vector analysis throughout the book, both in deriving and presenting the governing equations and in their solution. Raven's methods using complex algebra for the solution of two-dimensional vector equations are included because of their compactness, because of the ease of taking derivatives, because they are employed so frequently in the literature, and because they are so easy to program for computer evaluation. In the chapter dealing with three-dimensional kinematics and robotics, the authors present a brief introduction to Denavit and Hartenberg's methods using transformation matrices.

Another feature of this text is its focus on the method of kinematic coefficients, which are derivatives of motion variables with respect to the input position variable(s) rather than with respect to time. The authors believe that this analytic technique provides several important advantages, namely: (1) Kinematic coefficients clarify for the student those parts of a motion problem that are kinematic (geometric) in their nature, and clearly separate these from the parts that are dynamic or speed dependent. (2) Kinematic coefficients help to integrate the analysis of different types of mechanical systems, such as gears, cams, and linkages, which might not otherwise seem similar.

One dilemma that all writers on the subject of this book have faced is how to distinguish between the motions of different points of the same moving body and the motions of coincident points of different moving bodies. In other texts, it has been customary to describe both of these as "relative motion"; however, because they are two distinctly different situations and are described by different equations, this causes the student confusion in distinguishing between them. We believe that we have greatly relieved this problem by the introduction of the terms *motion difference* and *apparent motion* and by using different terminology and different notation for the two cases. Thus, for example, this book uses the two terms *velocity difference* and *apparent velocity*, instead of the term "relative velocity," which will not be found when speaking rigorously. This approach is introduced beginning with position and displacement, used extensively in the chapter on velocity, and brought to fulfillment in the chapter on accelerations, where the Coriolis component *always* arises in, and *only* arises in, the apparent acceleration equation.

Access to personal computers, programmable calculators, and laptop computers is commonplace and is of considerable importance to the material of this book. Yet engineering educators have told us very forcibly that they do not want computer programs included in the text. They prefer to write their own programs, and they expect their students to do so as well. Having programmed almost all the material in the book many times, we also understand that the book should not include such programs and thus become obsolete with changes in computers or programming languages.

The authors have endeavored to use US Customary units and SI units in about equal proportions throughout the book. However, there are certain exceptions. For example, in Chapter 14 (Dynamics of Reciprocating Engines), only SI units are presented, because engines are designed for an international marketplace, even by US companies. Therefore, they are always rated in kilowatts rather than horsepower, they have displacements in liters rather than cubic inches, and their cylinder pressures are measured in kilopascals rather than pounds per square inch.

Part 1 of this book deals mostly with theory, nomenclature, notation, and methods of analysis. Serving as an introduction, Chapter 1 tells what a mechanism is, what

a mechanism can do, how mechanisms can be classified, and what some of their limitations are. Chapters 2, 3, and 4 are concerned totally with analysis, specifically with kinematic analysis, because they cover position, velocity, and acceleration analyses, respectively, of single-degree-of-freedom planar mechanisms. Chapter 5 expands this background to include multi-degree-of-freedom planar mechanisms.

Part 2 of the book goes on to demonstrate engineering applications involving the selection, the specification, the design, and the sizing of mechanisms to accomplish specific motion objectives. This part includes chapters on cam systems, gears, gear trains, synthesis of linkages, spatial mechanisms, and an introduction to robotics. Chapter 6 is a study of the geometry, kinematics, proper design of high-speed cam systems, and now includes material on the dynamics of elastic cam systems. Chapter 7 studies the geometry and kinematics of spur gears, particularly of involute tooth profiles, their manufacture, and proper tooth meshing, and then studies gear trains, with an emphasis on epicyclic and differential gear trains. Chapter 8 expands this background to include helical gears, bevel gears, worms, and worm gears. Chapter 9 is an introduction to the kinematic synthesis of planar linkages. Chapter 10 is a brief introduction to the kinematic analysis of spatial mechanisms and robotics, including the forward and inverse kinematics problems.

Part 3 of the book adds the dynamics of machines. In a sense, this part is concerned with the consequences of the mechanism design specifications. In other words, having designed a machine by selecting, specifying, and sizing the various components, what happens during the operation of the machine? What forces are produced? Are there any unexpected operating results? Will the proposed design be satisfactory in all respects? Chapter 11 presents the static force analysis of machines. This chapter also includes sections focusing on the buckling of two-force members subjected to axial loads. Chapter 12 studies the planar and spatial aspects of the dynamic force analysis of machines. Chapter 13 then presents the vibration analysis of mechanical systems. Chapter 14 is a more detailed study of one particular type of mechanical system, namely the dynamics of both single- and multi-cylinder reciprocating engines. Chapter 15 next addresses the static and dynamic balancing of rotating and reciprocating systems. Finally, Chapter 16 is on the study of the dynamics of flywheels, governors, and gyroscopes.

As with all texts, the subject matter of this book also has limitations. Probably the clearest boundary on the coverage in this text is that it is limited to the study of rigid-body mechanical systems. It does study planar multibody systems with movable connections or constraints between them. However, all motion effects are assumed to come within the connections; the shapes of the individual bodies are assumed constant, except for the dynamics of elastic cam systems. This assumption is necessary to allow the separate study of kinematic effects from those of dynamics. Because each individual body is assumed rigid, it can have no strain; therefore, except for buckling of axially loaded members, the study of stress is also outside the scope of this text. It is hoped, however, that courses using this text can provide background for the later study of stress, strength, fatigue life, modes of failure, lubrication, and other aspects important to the proper design of mechanical systems.

Despite the limitations on the scope of this book, it is still clear that it is not reasonable to expect that all of the material presented here can be covered in a single-semester

course. As stated above, a variety of methods and applications have been included to allow the instructor to choose those topics that best fit the course objectives and to still provide a reference for follow-on courses and help build the student's library. Yet, many instructors have asked for suggestions regarding a choice of topics that might fit a 3-hour per week, 15-week course. Two such outlines follow, as used by two of the authors to teach such courses at their institutions. It is hoped that these might be used as helpful guidelines to assist others in making their own parallel choices.

Tentative Schedule I

Kinematics and Dynamics of Machine Systems

Week	Topics	Sections
1	Introduction to Mechanisms	1.1–1.10
	Kutzbach and Grashof Criteria	1.6, 1.9
	Advance-to-Return Time Ratio	1.7
	Overlay Method of Synthesis	9.8
2	Vector Loop-Closure Equation	2.6, 2.7
	Velocity Difference Equation	3.1–3.3
	Velocity Polygons; Velocity Images	3.4
3	Apparent Velocity Equation	3.5, 3.6, 3.8
	Direct and Rolling Contact Velocity	3.7
4	Instantaneous Centers of Velocity	3.12
	Aronhold–Kennedy Theorem of Three Centers	3.13, 3.14
	Use of Instant Centers to Find Velocities	3.15, 3.16
5	Exam #1	
	Acceleration Difference Equation	4.1–4.3
	Acceleration Polygons; Acceleration Images	4.4
6	Apparent Acceleration Equation	4.5, 4.6
	Coriolis Component of Acceleration	
7	Direct and Rolling Contact Acceleration	4.7, 4.8
	Review of Velocity and Acceleration Analyses	
8	Raven's Method of Kinematic Analysis	2.10, 3.10, 4.10
	Kinematic Coefficients	3.11, 4.11
	Computer Methods in Kinematics	10.9

9	Exam #2	
	Static Forces	11.1–11.6
	Two-, Three-, and Four-Force Members	11.7, 11.8
	Force Polygons	
10	Coulomb Friction Forces in Machines	11.9, 11.10
11	D'Alembert's Principle	12.1–12.4
	Dynamic Forces in Machine Members	12.4, 12.5
12	Introduction to Cam Design	6.1–6.4
	Choice of Cam Profiles; Matching Displacement Curves	6.5–6.8
13	First-Order Kinematic Coefficients; Face Width; Pressure Angle	6.9
	Second-Order Kinematic Coefficients; Pointing and Undercutting	6.10
14	Exam #3	
	Introduction to Gearing	7.1–7.6
	Involute Tooth Geometry; Contact Ratio; Undercutting	7.7–7.9, 7.11
15	Epicyclic and Differential Gear Trains	7.15–7.17
	Review	
	Final Exam	

Tentative Schedule II

Machine Design I

Week	Topics	Sections
1	The World of Mechanisms	1.1–1.6
	Measures of Performance (Indices of Merit)	1.10, 3.19
	Quick Return Mechanisms	1.7
2	Position Analysis. Vector Loops	2.1–2.7
	Newton–Raphson Technique	2.8, 2.11
3	Velocity Analysis	3.1–3.9
	First-Order Kinematic Coefficients	3.11
	Instant Centers of Zero Velocity	3.12–3.17

4	Rolling Contact, Rack and Pinion, Two Gears	3.10
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Supplement packages for this fifth edition have been designed to support both the student and the instructor in the kinematics and dynamics course. The Companion Website (<http://www.oup.com/us/uicker>) will include a list of any errors discovered in the text and their corrections. This website also includes over 100 animations of key figures from the text; these are marked with a  symbol in the text. These animations, created by Zhong Hu of South Dakota State University, are presented in both Working Model and .avi file formats, and are meant to help students visualize and comprehend the movement of important mechanisms.

An Ancillary Resource Center site is available for instructors only (registration is required). A complete solutions manual for all problems is available on that site. Solutions are also available on that site for 100 problems in the text worked out using MatLab software, for instructors wishing to incorporate MatLab code into their courses. Problems marked with a † signify that there is a MatLab-based solution available on that site; thank you to Bob Williams at Ohio University for his help with those solutions.

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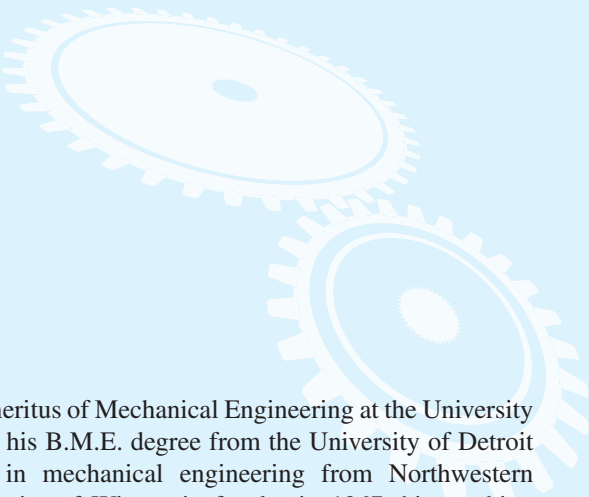
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*John J. Uicker, Jr.
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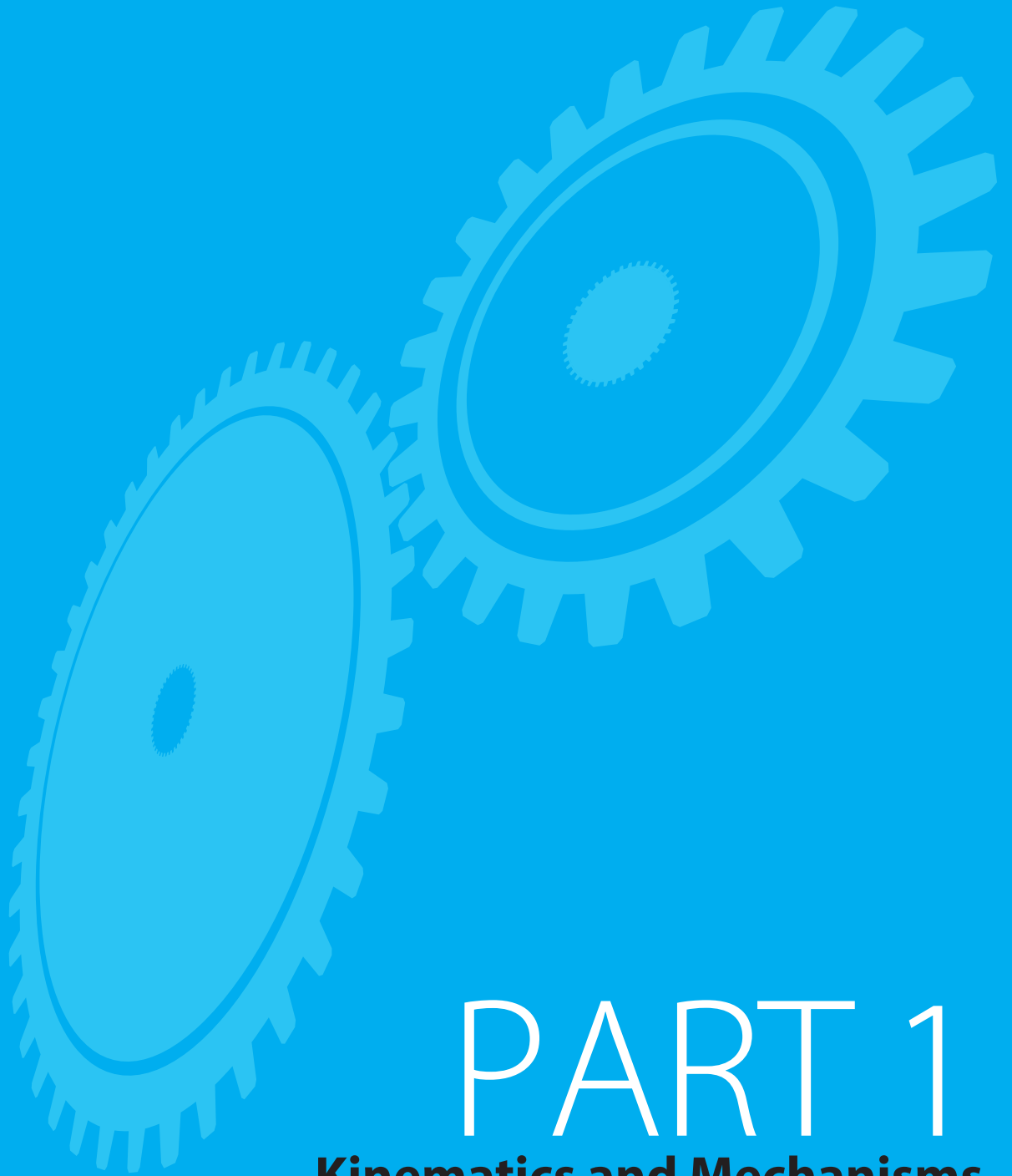
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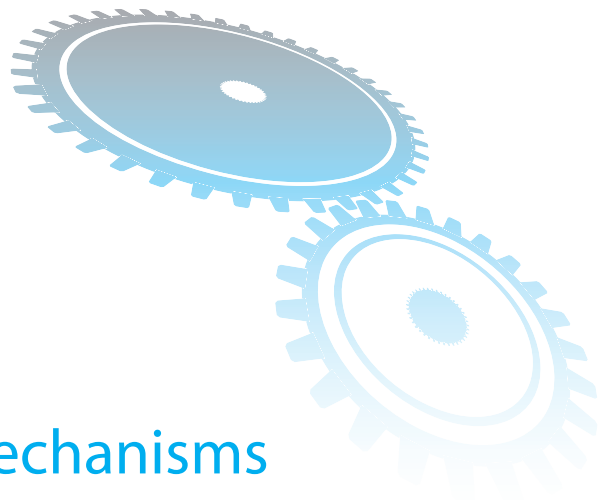
Joseph E. Shigley (deceased May 1994) was Professor Emeritus of Mechanical Engineering at the University of Michigan and a fellow in the ASME. He received the Mechanisms Committee Award in 1974, the Worcester Reed Warner medal in 1977, and the Machine Design Award in 1985. He was author of eight books, including *Mechanical Engineering Design* (with Charles R. Mischke) and *Applied Mechanics of Materials*. He was coeditor-in-chief of the *Standard Handbook of Machine Design*. He first wrote *Kinematic Analysis of Mechanisms* in 1958 and then wrote *Dynamic Analysis of Machines* in 1961, and these were published in a single volume titled *Theory of Machines* in 1961; they have evolved over the years to become the current text, *Theory of Machines and Mechanisms*, now in its fifth edition.

He was awarded the B.S.M.E. and B.S.E.E. degrees of Purdue University and received his M.S. at the University of Michigan. After several years in industry, he devoted his career to teaching, writing, and service to his profession, first at Clemson University and later at the University of Michigan. His textbooks have been widely used throughout the United States and internationally.



PART 1

Kinematics and Mechanisms



1

The World of Mechanisms

1.1 INTRODUCTION

The theory of machines and mechanisms is an applied science that is used to understand the relationships between the geometry and motions of the parts of a machine, or mechanism, and the forces that produce these motions. The subject, and therefore this book, divides itself naturally into three parts. Part 1, which includes Chaps. 1 through 5, is concerned with mechanisms and the kinematics of mechanisms, which is the analysis of their motions. Part 1 lays the groundwork for Part 2, comprising Chaps. 6 through 10, in which we study methods of designing mechanisms. Finally, in Part 3, which includes Chaps. 11 through 16, we take up the study of kinetics, the time-varying forces in machines and the resulting dynamic phenomena that must be considered in their design.

The design of a modern machine is often very complex. In the design of a new engine, for example, the automotive engineer must deal with many interrelated questions. What is the relationship between the motion of the piston and the motion of the crankshaft? What are the sliding velocities and the loads at the lubricated surfaces, and what lubricants are available for this purpose? How much heat is generated, and how is the engine cooled? What are the synchronization and control requirements, and how are they satisfied? What is the cost to the consumer, both for initial purchase and for continued operation and maintenance? What materials and manufacturing methods are used? What are the fuel economy, noise, and exhaust emissions; do they meet legal requirements? Although all these and many other important questions must be answered before the design is completed, obviously not all can be addressed in a book of this size. Just as people with diverse skills must be brought together to produce an adequate design, so too must many branches of science be brought together. This book assembles material that falls into the science of mechanics as it relates to the design of mechanisms and machines.

1.2 ANALYSIS AND SYNTHESIS

There are two completely different aspects of the study of mechanical systems: *design* and *analysis*. The concept embodied in the word “design” is more properly termed *synthesis*, the process of contriving a scheme or a method of accomplishing a given purpose. Design is the process of prescribing the sizes, shapes, material compositions, and arrangements of parts so that the resulting machine will perform the prescribed task.

Although there are many phases in the design process that can be approached in a well-ordered, scientific manner, the overall process is by its very nature as much an art as a science. It calls for imagination, intuition, creativity, judgment, and experience. The role of science in the design process is merely to provide tools to be used by designers as they practice their art.

In the process of evaluating the various interacting alternatives, designers find a need for a large collection of mathematical and scientific tools. These tools, when applied properly, provide more accurate and more reliable information for judging a design than one achieves through intuition or estimation. Thus, the tools are of tremendous help in deciding among alternatives. However, scientific tools cannot make decisions for designers; designers have every right to exert their imagination and creative abilities, even to the extent of overruling the mathematical recommendations.

Probably the largest collection of scientific methods at the designer’s disposal fall into the category called *analysis*. These are techniques that allow the designer to critically examine an already existing, or proposed, design to judge its suitability for the task. Thus, analysis in itself is not a creative science but one of evaluation and rating things already conceived.

We should bear in mind that, although most of our effort may be spent on analysis, the real goal is synthesis: the design of a machine or system. Analysis is simply a tool; however, it is a vital tool and will inevitably be used as one step in the design process.

1.3 SCIENCE OF MECHANICS

The branch of scientific analysis that deals with motions, time, and forces is called *mechanics* and is made up of two parts: statics and dynamics. *Statics* deals with the analysis of stationary systems—that is, those in which time is not a factor—and *dynamics* deals with systems that change with time.

As shown in Fig. 1.1, dynamics is also made up of two major disciplines, first recognized as separate entities by Euler* in 1765 [2]:†

The investigation of the motion of a rigid body may be conveniently separated into two parts, the one, geometrical, and the other mechanical. In the first part, the transference of the body from a given position to any other position must be investigated without respect to the causes of the motion, and must be represented by analytical formulae, which will define the position of each point of the body. This

* Leonhard Euler (1707–1783).

† Numbers in square brackets refer to references at the end of each chapter.

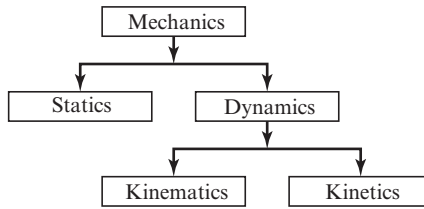


Figure 1.1

investigation will therefore be referable solely to geometry, or rather to stereotomy [the art of stonecutting, now referred to as descriptive geometry].

It is clear that by the separation of this part of the question from the other, which belongs properly to Mechanics, the determination of the motion from dynamical principles will be made much easier than if the two parts were undertaken conjointly.

These two aspects of dynamics were later recognized as the distinct sciences of *kinematics* (*cinématique* was a term coined by Ampère* and derived from the Greek word *kinema*, meaning motion) and *kinetics* and deal with motion and the forces producing the motion, respectively.

The initial problem in the design of a mechanical system, therefore, is understanding the kinematics. *Kinematics* is the study of motion, quite apart from the forces that produce the motion. In particular, kinematics is the study of position, displacement, rotation, speed, velocity, acceleration, and jerk. The study, say, of planetary or orbital motion is also a problem in kinematics, but in this book we shall concentrate our attention on kinematic problems that arise in the design and operation of mechanical systems. Thus, the kinematics of machines and mechanisms is the focus of the next several chapters of this book. In addition, statics and kinetics are also vital parts of a complete design analysis, and they are also covered in later chapters.

It should be carefully noted in the previous quotation that Euler based his separation of dynamics into kinematics and kinetics on the assumption that they deal with *rigid* bodies. It is this very important assumption that allows the two to be treated separately. For flexible bodies, the shapes of the bodies themselves, and therefore their motions, depend on the forces exerted on them. In this situation, the study of force and motion must take place simultaneously, thus significantly increasing the complexity of the analysis.

Fortunately, although all real machine parts are flexible to some degree, machines are usually designed from relatively rigid materials, keeping part deflections to a minimum. Therefore, it is common practice to assume that deflections are negligible and parts are rigid while analyzing a machine's kinematic performance and then, during dynamic analysis when loads are sought, to design the parts so that the assumption is justified. A more detailed discussion of a rigid body compared to a deformable, or flexible, body is presented in the introduction to static force analysis in Sec. 11.1.

* André-Marie Ampère (1775–1836).

1.4 TERMINOLOGY, DEFINITIONS, AND ASSUMPTIONS

Reuleaux* defines a *machine*[†] as a “combination of resistant bodies so arranged that by their means the mechanical forces of nature can be compelled to do work accompanied by certain determinate motions.” He also defines a *mechanism* as an “assemblage of resistant bodies, connected by movable joints, to form a closed kinematic chain with one link fixed and having the purpose of transforming motion.”

Some light can be shed on these definitions by contrasting them with the term *structure*. A structure is also a combination of resistant (rigid) bodies connected by joints, but the purpose of a structure (such as a truss) is not to do work or to transform motion, but to be rigid. A truss can perhaps be moved from place to place and is movable in this sense of the word; however, it has no *internal* mobility. A structure has no *relative motions* between its various links, whereas both machines and mechanisms do. Indeed, the whole purpose of a machine or mechanism is to utilize these relative internal motions in transmitting power or transforming motion.

A machine is an arrangement of parts for doing work, a device for applying power or changing the direction of motion. It differs from a mechanism in its purpose. In a machine, terms such as force, torque, work, and power describe the predominant concepts. In a mechanism, though it may transmit power or force, the predominant idea in the mind of the designer is one of achieving a desired motion. There is a direct analogy between the terms structure, mechanism, and machine and the branches of mechanics illustrated in Fig. 1.1. The term “structure” is to statics as the term “mechanism” is to kinematics and as the term “machine” is to kinetics.

We use the word *link* to designate a machine part or a component of a mechanism. As discussed in the previous section, a link is assumed to be completely rigid. Machine components that do not fit this assumption of rigidity, such as springs, usually have no effect on the kinematics of a device but do play a role in supplying forces. Such parts or components are not called links; they are usually ignored during kinematic analysis, and their force effects are introduced during force analysis (see the analysis of buckling in Secs. 11.14–11.18). Sometimes, as with a belt or chain, a machine part may possess one-way rigidity; such a body can be considered a link when in tension but not under compression.

The links of a mechanism must be connected in some manner in order to transmit motion from the *driver*, or input, to the *driven*, or *follower*, or output. The connections, the joints between the links, are called *kinematic pairs* (or simply *pairs*), because each joint consists of a pair of mating surfaces, two elements, one mating surface or element being a part of each of the joined links. Thus, we can also define a link as *the rigid connection between two or more joint elements*.

Stated explicitly, the assumption of rigidity is that there can be no relative motion (no change in distance) between two arbitrarily chosen points on the same link. In particular,

* Much of the material of this section is based on definitions originally set down by Franz Reuleaux (1829–1905), a German kinematician whose work marked the beginning of a systematic treatment of kinematics [7].

† There appears to be no agreement at all on the proper definition of a machine. In a footnote Reuleaux gives 17 definitions, and his translator gives 7 more and discusses the whole problem in detail [7].

the relative positions of joint elements on any given link do not change no matter what loads are applied. In other words, the purpose of a link is to hold a constant spatial relationship between its joint elements.

As a result of the assumption of rigidity, many of the intricate details of the actual part shapes are unimportant when studying the kinematics of a machine or mechanism. For this reason, it is common practice to draw highly simplified schematic diagrams that contain important features of the shape of each link, such as the relative locations of joint elements, but that completely subdue the real geometry of the manufactured part. The slider-crank linkage of the internal combustion engine, for example, can be simplified for purposes of analysis to the schematic diagram illustrated later in Fig. 1.3*b*. Such simplified schematics are a great help since they eliminate confusing factors that do not affect the analysis; such diagrams are used extensively throughout this text. However, these schematics also have the drawback of bearing little resemblance to physical hardware. As a result they may give the impression that they represent only academic constructs rather than real machinery. We should continually bear in mind that these simplified diagrams are intended to carry only the minimum necessary information so as not to confuse the issue with unimportant detail (for kinematic purposes) or complexity of the true machine parts.

When several links are connected together by joints, they are said to form a *kinematic chain*. Links containing only two joint elements are called *binary* links, those having three joint elements are called *ternary* links, those having four joint elements are called *quaternary* links, and so on. If every link in a chain is connected to at least two other links, the chain forms one or more closed loops and is called a *closed* kinematic chain; if not, the chain is referred to as *open*. If a chain consists entirely of binary links, it is a *simple-closed* chain. *Compound-closed* chains, however, include other than binary links and thus form more than a single closed loop.

Recalling Reuleaux's definition of a mechanism, we see that it is necessary to have a closed kinematic chain *with one link fixed*. When we say that one link is fixed, we mean that it is chosen as the frame of reference for all other links; that is, the motions of all points on the links of the mechanism are measured with respect to the fixed link. This link, in a practical machine, usually takes the form of a stationary platform or base (or a housing rigidly attached to such a base) and is commonly referred to as the *ground*, *frame*, or *base link*.^{*} The question of whether this reference frame is truly stationary (in the sense of being an inertial reference frame) is immaterial in the study of kinematics, but becomes important in the investigation of kinetics, where forces are considered. In either case, once a frame link is designated (and other conditions are met), the kinematic chain becomes a mechanism and, as the driver is moved through various positions, all other links have well-defined motions with respect to the chosen frame of reference. We use the term *kinematic chain* to specify a particular arrangement of links and joints when it is not clear which link is to be treated as the frame. When the frame link is specified, the kinematic chain is called a *mechanism*.

For a mechanism to be useful, the motions between links cannot be completely arbitrary; they too must be constrained to produce the proper relative motions—those chosen by the designer for the particular task to be performed. These desired relative

^{*} In this text, the ground, frame, or base of the mechanism is commonly numbered 1.

motions are achieved by proper choice of the number of links and the kinds of joints used to connect them. Thus we are led to the concept that, in addition to the distances between successive joints, the nature of the joints themselves and the relative motions they permit are essential in determining the kinematics of a mechanism. For this reason, it is important to look more closely at the nature of joints in general terms, and in particular at several of the more common types.

The controlling factors that determine the relative motions allowed by a given joint are the shapes of the mating surfaces or elements. Each type of joint has its own characteristic shapes for the elements, and each allows a given type of motion, which is determined by the possible ways in which these elemental surfaces can move with respect to each other. For example, the pin joint in Fig. 1.2*a*, has cylindric elements, and, assuming that the links cannot slide axially, these surfaces permit only relative rotational motion. Thus a pin joint allows the two connected links to experience relative rotation about the pin center. So, too, other joints each have their own characteristic element shapes and relative motions. These shapes restrict the totally arbitrary motion of two unconnected links to some prescribed type of relative motion and form constraining conditions (constraints) on the mechanism's motion.

It should be pointed out that the element shapes may often be subtly disguised and difficult to recognize. For example, a pin joint might include a needle bearing, so that two mating surfaces, as such, are not distinguishable. Nevertheless, if the motions of the individual rollers are not of interest, the motions allowed by the joints are equivalent, and the joints are of the same generic type. Thus the criterion for distinguishing different joint types is the relative motions they permit and not necessarily the shapes of the elements, though these may provide vital clues. The diameter of the pin used (or other dimensional data) is also of no more importance than the exact sizes and shapes of the connected links. As stated previously, the kinematic function of a link is to hold a fixed geometric relationship between the joint elements. Similarly, the only kinematic function of a joint, or pair, is to determine the relative motion between the connected links. All other features are determined for other reasons and are unimportant in the study of kinematics.

When a kinematic problem is formulated, it is necessary to recognize the type of relative motion permitted in each of the joints and to assign to it some variable parameter(s) for measuring or calculating the motion. There will be as many of these parameters as there are degrees of freedom of the joint in question, and they are referred to as *joint variables*. Thus, the joint variable of a pinned joint will be a single angle measured between reference lines fixed in the adjacent links, while a spheric joint will have three joint variables (all angles) to specify its three-dimensional rotation.

Reuleaux separated kinematic pairs into two categories: namely, *higher pairs* and *lower pairs*, with the latter category consisting of the six prescribed types to be discussed next. He distinguished between the categories by noting that lower pairs, such as the pin joint, have surface contact between the joint elements, while higher pairs, such as the connection between a cam and its follower, have line or point contact between the elemental surfaces. This criterion, however, can be misleading (as noted in the case of a needle bearing). We should rather look for distinguishing features in the relative motion(s) that the joint allows between the connected links.

Lower pairs consist of the six prescribed types shown in Fig. 1.2.

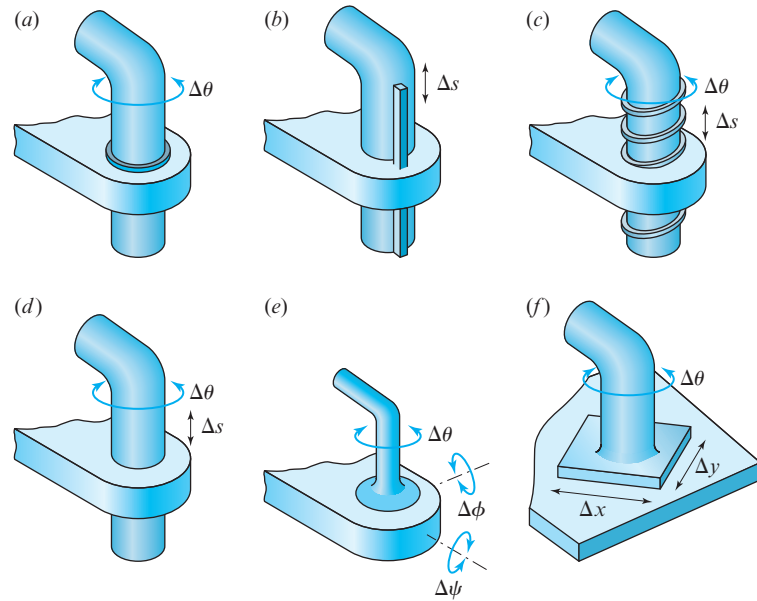


Figure 1.2 (a) Revolute; (b) prism; (c) screw; (d) cylinder; (e) sphere; (f) flat pairs.

The names and the symbols (Hartenberg and Denavit [4]) that are commonly employed for the six lower pairs are presented in Table 1.1. The table also includes the number of degrees of freedom and the joint variables that are associated with each lower pair.

The *revolute* or *turning pair*, R (Fig. 1.2a), permits only relative rotation and is often referred to as a pin joint. This joint has one degree of freedom.

The *prism* or *prismatic pair*, P (Fig. 1.2b), permits only relative sliding motion and therefore is often called a sliding joint. This joint also has one degree of freedom.

The *screw* or *helical pair*, H (Fig. 1.2c), permits both rotation and sliding motion. However, it only has one degree of freedom, since the rotation and sliding motions are related by the helix angle of the thread. Thus, the joint variable may be chosen as either Δs or $\Delta \theta$, but not both. Note that the helical pair reduces to a revolute if the helix angle is made zero, and to a prism if the helix angle is made 90° .

Table 1.1 Lower Pairs

Pair	Symbol	Pair Variable	Degrees of Freedom	Relative Motion
Revolute	R	$\Delta \theta$	1	Circular
Prism	P	Δs	1	Rectilinear
Screw	H	$\Delta \theta$ or Δs	1	Helical
Cylinder	C	$\Delta \theta$ and Δs	2	Cylindric
Sphere	S	$\Delta \theta$, $\Delta \phi$, $\Delta \psi$	3	Spheric
Flat	F	Δx , Δy , $\Delta \theta$	3	Planar

The *cylinder* or *cylindric pair*, C (Fig. 1.2d), permits both rotation and an independent sliding motion. Thus, the cylindric pair has two degrees of freedom.

The *sphere* or *globular pair*, S (Fig. 1.2e), is a ball-and-socket joint. It has three degrees of freedom, sometimes taken as rotations about each of the coordinate axes.

The *flat* or *planar pair*, sometimes called an *ebene pair* (German), F (Fig. 1.2f), is seldom found in mechanisms in its undisguised form, except at a support point. It has three degrees of freedom, that is, two translations and a rotation.

All other joint types are called higher pairs. Examples include mating gear teeth, a wheel rolling and/or sliding on a rail, a ball rolling on a flat surface, and a cam contacting its follower. Since an unlimited variety of higher pairs exist, a systematic accounting of them is not a realistic objective. We shall treat each separately as it arises.

Among the higher pairs is a subcategory known as *wrapping pairs*. Examples are the connections between a belt and a pulley, a chain and a sprocket, or a rope and a drum. In each case, one of the links has only one-way rigidity.

The treatment of various joint types, whether lower or higher pairs, includes another important limiting assumption. Throughout the book, we assume that the actual joint, as manufactured, can be reasonably represented by a mathematical abstraction having perfect geometry. That is, when a real machine joint is assumed to be a spheric joint, for example, it is also assumed that there is no “play” or clearance between the joint elements and that any deviation from spheric geometry of the elements is negligible. When a pin joint is treated as a revolute, it is assumed that no axial motion takes place; if it is necessary to study the small axial motions resulting from clearances between real elements, the joint must be treated as cylindric, thus allowing the axial motion.

The term “mechanism,” as defined earlier, can refer to a wide variety of devices, including both higher and lower pairs. A more limited term, however, refers to those mechanisms having only lower pairs; such a mechanism is commonly called a *linkage*. A linkage, then, is connected only by the lower pairs shown in Fig. 1.2.

1.5 PLANAR, SPHERIC, AND SPATIAL MECHANISMS

Mechanisms may be categorized in several different ways to emphasize their similarities and differences. One such grouping divides mechanisms into planar, spheric, and spatial categories. All three groups have many things in common; the criterion that distinguishes the groups, however, is to be found in the characteristics of the motions of the links.

A *planar mechanism* is one in which all particles describe planar curves in space, and all these curves lie in parallel planes; that is, the loci of all points are planar curves parallel to a single common plane. This characteristic makes it possible to represent the locus of any chosen point of a planar mechanism in its true size and shape in a single drawing or figure. The motion transformation of any such mechanism is called *coplanar*. The planar four-bar linkage, the slider-crank linkage, the plate cam-and-follower mechanism, and meshing gears are familiar examples of planar mechanisms.

Planar mechanisms utilizing only lower pairs are called *planar linkages*; they include only revolute and prismatic joints. Although the planar pair might theoretically be included in a planar linkage, this would impose no constraint on the motion. Planar motion also

requires that all revolute axes be normal to the plane of motion, and that all prismatic joint axes be parallel to the plane.

As already pointed out, it is possible to observe the motions of all particles of a planar mechanism in true size and shape from a single direction. In other words, all motions can be represented graphically in a single view. Thus, graphic techniques are well suited to their analysis, and this background is beneficial to the student once mastered. Since spheric and spatial mechanisms do not have this special geometry, visualization becomes more difficult and more powerful techniques must be used for their study.

A *spheric mechanism* is one in which each moving link has a point that remains stationary as the mechanism moves. Also, arbitrary points fixed in each moving link travel on spheric surfaces; the spheric surfaces must all be *concentric*. Therefore, the motions of all these points can be completely described by their radial projections (or shadows) on the surface of a sphere with a properly chosen center. Note that the only lower pairs (Table 1.1) that allow spheric motion are the revolute pair and the spheric pair. In a spheric linkage, the axes of all revolute pairs must intersect at a single point. In addition, a spheric pair center must be concentric with this point, and, then, it would not produce any constraint on the motions of the other links. Therefore, a *spheric linkage* must consist of only revolute pairs, and the axes of all such pairs must intersect at a single point. A familiar example of a spheric mechanism is the Hooke universal joint (also referred to as the Cardan joint) shown in Fig. 1.21*b*.

Spatial mechanisms include *no* restrictions on the relative motions of the links. For example, a mechanism that contains a screw joint (Fig. 1.2*c*) must be a spatial mechanism, since the relative motion within a screw joint is helical. An example of a spatial mechanism is the differential screw shown in Fig. 1.11. Because of the more complex motion characteristics of spatial mechanisms, and since these motions can not be analyzed graphically from a single viewing direction, more powerful techniques are required for their analysis. Such techniques are introduced in Chap. 10 for a detailed study of spatial mechanisms and robotics.

Since the majority of mechanisms in modern machinery are planar, one might question the need to study these complex mathematical techniques. However, even though the simpler graphic techniques for planar mechanisms may have been mastered, an understanding of the more complex techniques is of value for the following reasons:

1. They provide new, alternative methods that can solve problems in a different way. Thus, they provide a means for checking results. Certain problems by their nature may also be more amenable to one method than another.
2. Methods that are analytic in nature are better suited to solution by a calculator or a digital computer than by graphic techniques.
3. One reason why planar mechanisms are so common is that good methods for the analysis of spatial mechanisms have not been available until relatively recently. Without these methods, the design and application of spatial mechanisms has been hindered, even though they may be inherently better suited to certain applications.
4. We will discover that spatial mechanisms are, in fact, much more common in practice than their formal description indicates.

Consider the planar four-bar linkage (Fig. 1.3c), which has four links connected by four revolute pairs whose axes are parallel. This “parallelism” is a mathematical hypothesis; it is not a reality. The axes, as produced in a machine shop—in any machine shop, no matter how precise the machining—are only approximately parallel. If the axes are far out of parallel, there is binding in no uncertain terms, and the linkage moves only because the “rigid” links flex and twist, producing loads in the bearings. If the axes are nearly parallel, the linkage operates because of looseness of the running fits of the bearings or flexibility of the links. A common way of compensating for small nonparallelism is to connect the links with self-aligning bearings, actually spheric joints allowing three-dimensional rotation. Such a “planar” linkage is thus a low-grade spatial linkage.

Thus, the overwhelmingly large category of planar mechanisms and the category of spheric mechanisms are special cases, or subsets, of the all-inclusive category of spatial mechanisms. They occur as a consequence of the special orientations of their joint axes.

1.6 MOBILITY

One of the first concerns in either the design or the analysis of a mechanism is the number of degrees of freedom, also called the *mobility* of the device. The mobility* of a mechanism is the number of input parameters (usually joint variables) that must be controlled independently to bring the device into a particular posture. Ignoring, for the moment, certain exceptions to be mentioned later, it is possible to determine the mobility of a mechanism directly from a count of the number of links and the number and types of joints comprising the mechanism.

To develop this relationship, consider that—before they are connected together—each link of a planar mechanism has three degrees of freedom when moving with planar motion relative to the fixed link. Not counting the fixed link, therefore, an n -link planar mechanism has $3(n - 1)$ degrees of freedom before any of the joints are connected. Connecting two of the links by a joint that has one degree of freedom, such as a revolute, has the effect of providing two constraints between the connected links. If the two links are connected by a two-degree-of-freedom joint, it provides one constraint. When the constraints for all joints are subtracted from the total degrees of freedom of the unconnected links, we find the resulting mobility of the assembled mechanism.

If we denote the number of single-degree-of-freedom joints as j_1 and the number of two-degree-of-freedom joints as j_2 , then the resulting mobility, m , of a planar n -link mechanism is given by

$$m = 3(n - 1) - 2j_1 - j_2. \quad (1.1)$$

Written in this form, Eq. (1.1) is called the *Kutzbach criterion* for the mobility of a planar mechanism [8]. Its application is illustrated for several simple examples in Fig. 1.3.

* The German literature distinguishes between *movability* and *mobility*. Movability includes the six degrees of freedom of the device as a whole, as though the ground link were not fixed, and thus applies to a kinematic chain. Mobility neglects these degrees of freedom and considers only the internal relative motions, thus applying to a mechanism. The English literature seldom recognizes this distinction, and the terms are used somewhat interchangeably.

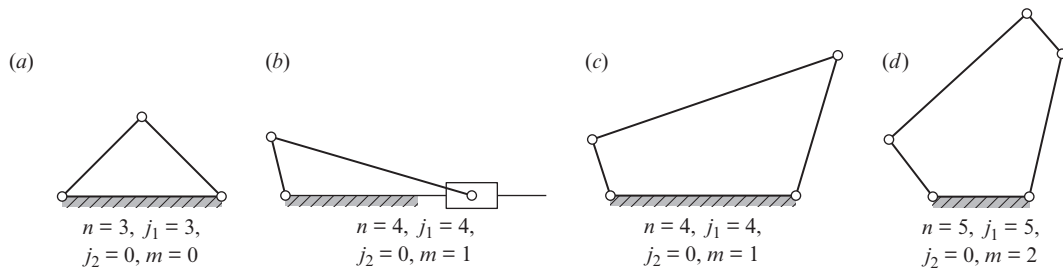


Figure 1.3 Applications of the Kutzbach criterion.

If the Kutzbach criterion yields $m > 0$, the mechanism has m degrees of freedom. If $m = 1$, the mechanism can be driven by a single input motion to produce constrained (uniquely defined) motion. Two examples are the slider-crank linkage and the four-bar linkage, shown in Figs. 1.3b and 1.3c, respectively. If $m = 2$, then two separate input motions are necessary to produce constrained motion for the mechanism; such a case is the five-bar linkage shown in Fig. 1.3d.

If the Kutzbach criterion yields $m = 0$, as in Figs. 1.3a and 1.4a, motion is impossible and the mechanism forms a structure.

If the criterion yields $m < 0$, then there are redundant constraints in the chain and it forms a statically indeterminate structure. An example is illustrated in Fig. 1.4b. Note in the examples of Fig. 1.4 that when three links are joined by a single pin, such a connection is treated as two separate but concentric joints; two j_1 joints must be counted.

Figure 1.5 shows two examples of the Kutzbach criterion applied to mechanisms with two-degree-of-freedom joints—that is, j_2 joints. Particular attention should be paid to the contact (joint) between the wheel and the fixed link in Fig. 1.5b. Here it is assumed that slipping is possible between the two links. If this contact prevents slipping, the joint would be counted as a one-degree-of-freedom joint—that is, a j_1 joint—because only one relative motion would then be possible between the links. Recall that, in this case, the mechanism is generally referred to as a “linkage.”

It is important to realize that the Kutzbach criterion can give an incorrect result. For example, note that Fig. 1.6a represents a structure and that the criterion properly predicts $m = 0$. However, if link 5 is arranged as in Fig. 1.6b, the result is a double-parallelgram linkage with a mobility of $m = 1$, even though Eq. (1.1) indicates that it is a structure. The actual mobility of $m = 1$ results only if the parallelogram geometry is achieved. In the development of the Kutzbach criterion, no consideration was given to the lengths of the

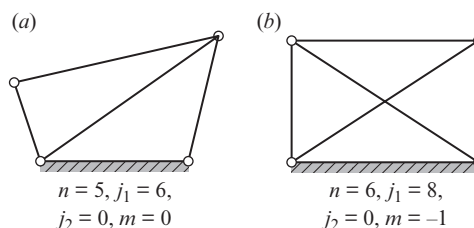


Figure 1.4 Applications of the Kutzbach criterion to structures.

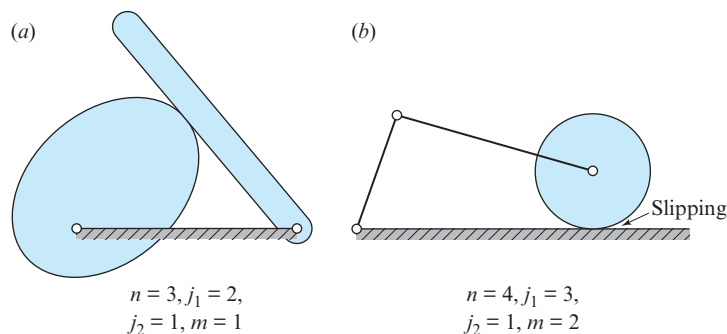


Figure 1.5

links or other dimensional properties. Therefore, it should not be surprising that exceptions to the criterion are found for particular cases with equal link lengths, parallel links, or other special geometric features.

Although there are exceptions, the Kutzbach criterion remains useful, because it is so easily applied during mechanism design. To avoid exceptions, it would be necessary to include all the dimensional properties of the mechanism. The resulting criterion would be very complex and would be useless at the early stages of design when dimensions may not be known.

An earlier mobility criterion, named after Grübler [3], applies to a planar linkage where the overall mobility is $m = 1$. Substituting $j_2 = 0$ and $m = 1$ into Eq. (1.1) and rearranging, we find that Grübler's criterion for planar linkages can be written as

$$3n - 2j_1 - 4 = 0. \quad (1.2)$$

Rearranging this equation, the number of links is

$$n = \frac{2j_1 + 4}{3}. \quad (1.3)$$

From this equation, we see that a planar linkage with a mobility of $m = 1$ cannot have an odd number of links. Also, the simplest possible linkage with all binary links has $n = j_1 = 4$. This explains one reason why the slider-crank linkage (Fig. 1.3b) and the four-bar linkage (Fig. 1.3c) appear so commonly in machines.

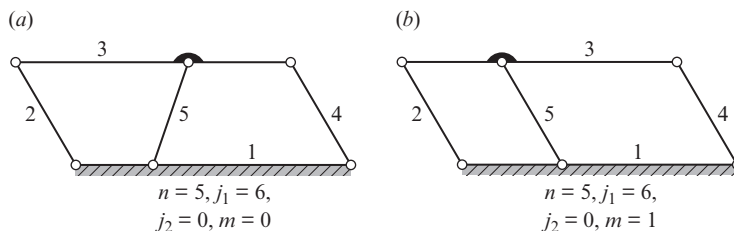


Figure 1.6

The Kutzbach criterion, Eq. (1.1), and the Grübler criterion, Eq. (1.2), were derived for the case of planar mechanisms and linkages, respectively. If similar criteria are developed for spatial mechanisms and linkages, which is the subject of Chap. 10, we must recall that each unconnected link has six degrees of freedom and each single-degree-of-freedom joint provides five constraints, each two-degree-of-freedom joint provides four constraints, and so on. Similar arguments then lead to the Kutzbach criterion for spatial mechanisms,

$$m = 6(n - 1) - 5j_1 - 4j_2 - 3j_3 - 2j_4 - j_5,$$

and the Grübler criterion for spatial linkages,

$$6n - 5j_1 - 7 = 0. \quad (1.4)$$

Therefore, the simplest form of a spatial linkage* with a mobility of $m = 1$ is $n = j_1 = 7$.

EXAMPLE 1.1

Determine the mobility of the planar mechanism shown in Fig. 1.7a.

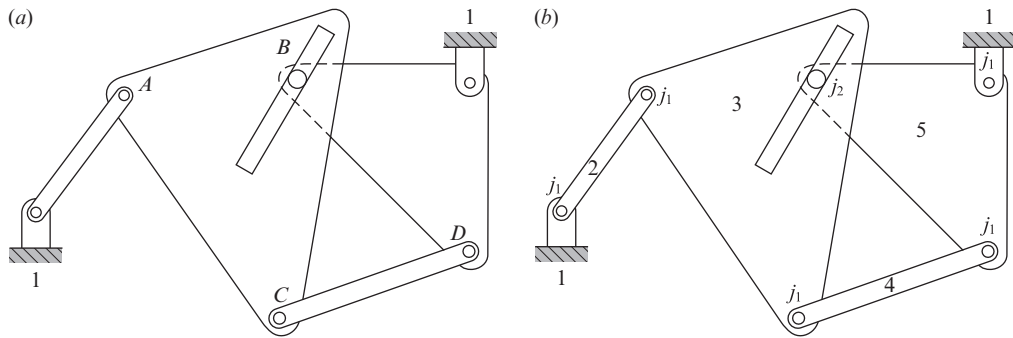


Figure 1.7 Planar mechanism.

SOLUTION

The link numbers and the joint types for the mechanism are shown in Fig. 1.7b. The number of links is $n = 5$, the number of lower pairs is $j_1 = 5$, and the number of higher pairs is $j_2 = 1$. Substituting these values into the Kutzbach criterion, Eq. (1.1), the mobility of the mechanism is

$$m = 3(5 - 1) - 2(5) - 1(1) = 1. \quad \text{Ans.}$$

Note that the Kutzbach criterion gives the correct answer for the mobility of this mechanism; that is, a single input motion is required to give a unique output motion.

* Note that all planar linkages are exceptions to the spatial mobility criterion. They have the special geometric characteristics that all revolute axes are parallel and perpendicular to the plane of motion and that all prismatic axes lie in the plane of motion.

For example, rotation of link 2 could be used as the input and rotation of link 5 could be used as the output.

EXAMPLE 1.2

For the mechanism shown in Fig. 1.8a, determine: (a) the number of lower pairs (j_1 joints) and the number of higher pairs (j_2 joints); and (b) the mobility of the mechanism using the Kutzbach criterion. Treating rolling contact to mean rolling with no slipping. Does this criterion provide the correct answer for the mobility of this mechanism? Briefly explain why or why not.

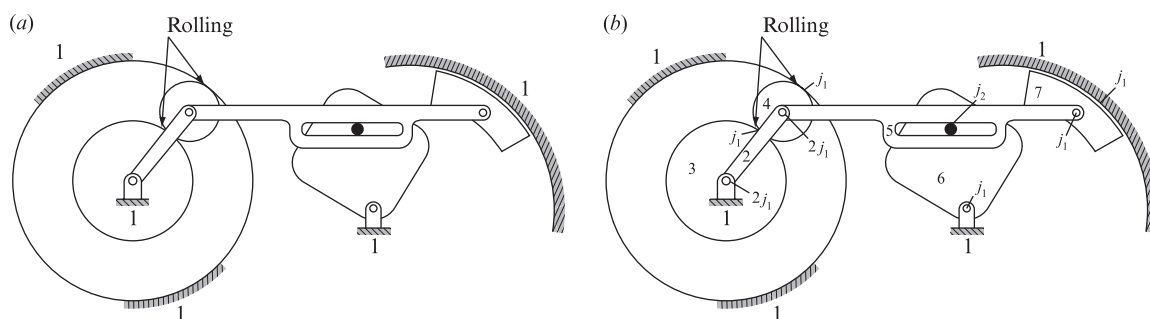


Figure 1.8 Planar mechanism.

SOLUTION

- (a) The links and the joint types of the mechanism are labeled in Fig. 1.8b. The number of links is $n = 7$, the number of lower pairs is $j_1 = 9$, and the number of higher pairs is

$$j_2 = 1. \quad \text{Ans.}$$

- (b) Substituting these values into the Kutzbach criterion, Eq. (1.1), the mobility of the mechanism is

$$m = 3(7 - 1) - 2(9) - 1(1) = -1. \quad \text{Ans.}$$

However, this answer is not correct; that is, the Kutzbach criterion does not give the correct mobility for this mechanism. The mobility of this mechanism is, in fact, $m = 1$; that is, a single input motion gives a unique output motion.

Reasoning: Links 3 and 4 are superfluous to the constraints of the mechanism. If links 3 and 4 were removed, the motion of the remaining links would be unaffected. With links 3 and 4 removed, the mobility of the mechanism using the Kutzbach criterion is $m = 1$. Note that if links 3 and 4 were attached with no special conditions—that is, not pinned at their centers, for example—then the mechanism would indeed be locked and the answer $m = -1$ would be correct.

EXAMPLE 1.3

For the mechanism shown in Fig. 1.9a, determine: (a) the number of lower pairs and the number of higher pairs; and (b) the mobility of the mechanism predicted by the Kutzbach criterion. Does this criterion provide the correct answer for this mechanism? Briefly explain why or why not.

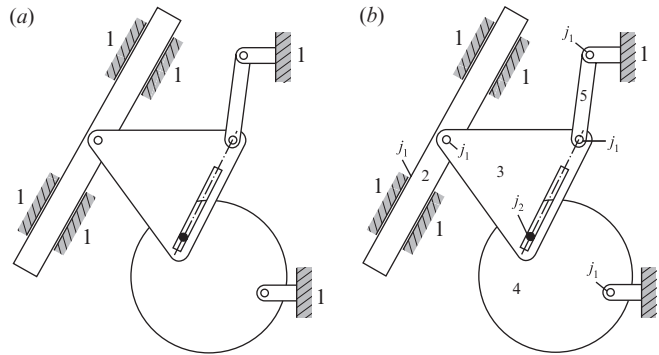


Figure 1.9 Planar mechanism.

SOLUTION

- (a) The links and the joints of the mechanism are labeled as shown in Fig. 1.9b. The number of links is $n = 5$, the number of lower pairs is $j_1 = 5$, and the number of higher pairs is

$$j_2 = 1. \quad \text{Ans.}$$

- (b) Substituting these values into the Kutzbach criterion, Eq. (1.1), the mobility of the mechanism is

$$m = 3(5 - 1) - 2(5) - 1(1) = 1. \quad \text{Ans.}$$

For this mechanism, the mobility is indeed 1, which indicates that the Kutzbach criterion gives the correct answer for this mechanism.

For a mechanism, or a linkage, with a mobility of $m = 1$, the input or driving link will, in general, be numbered as 2 in this text.

1.7 CHARACTERISTICS OF MECHANISMS

An ideal system for the classification of mechanisms would be a system that allows a designer to enter the system with a set of specifications and leave with one or more mechanisms that satisfy those specifications. Although history* demonstrates that

* For an excellent short history of the kinematics of mechanisms, see [4, Chap. 1].

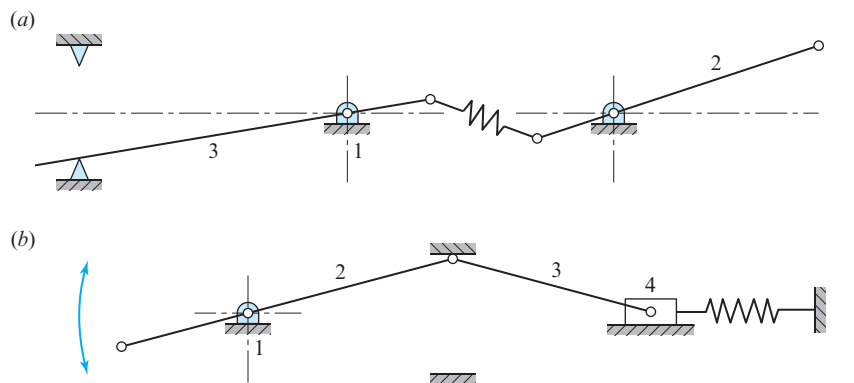


Figure 1.10 (a) Bistable mechanism; (b) true toggle mechanism.

many attempts have been made, few have been particularly successful in devising a satisfactory classification system. In view of the fact that the purpose of a mechanism is the transformation of motion, we will follow Torfason's lead [9] and classify mechanisms according to the type of motion transformation. In total, Torfason displays 262 mechanisms, each of which can have a variety of dimensions. His categories are as follows:

Snap-Action Mechanisms Snap-action, toggle, or flip-flop mechanisms are used for switches, clamps, or fasteners. Torfason also includes spring clips and circuit breakers. Fig. 1.10 shows examples of bistable and true toggle mechanisms.

Linear Actuators Linear actuators include stationary screws with traveling nuts, stationary nuts with traveling screws, and single-acting and double-acting hydraulic and pneumatic cylinders.

Fine Adjustments Fine adjustments may be obtained with screws, including differential screws, worm gearing, wedges, levers, levers in series, and various motion-adjusting mechanisms. For the differential screw shown in Fig. 1.11, you should be able to determine that the translation of the carriage resulting from one turn of the handle is 0.0069 in to the left (see Prob. 1.17).

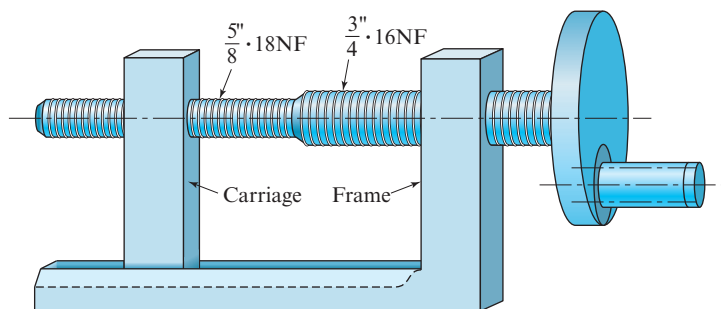


Figure 1.11 Differential screw.

Clamping Mechanisms Typical clamping mechanisms are the C-clamp, the wood worker's screw clamp, cam-actuated and lever-actuated clamps, vises, presses (such as the toggle press shown in Fig. 1.10*b*), collets, and stamp mills.

Locational Devices Torfason [9] shows 15 locational mechanisms. These are usually self-centering and locate either axially or angularly using springs and detents.

Ratchets and Escapements There are many different forms of ratchets and escapements, some quite clever. They are used in locks, jacks, clockwork, and other applications requiring some form of intermittent motion. Figure 1.12 shows four typical applications.

The ratchet in Fig. 1.12*a* allows only one direction of rotation of wheel 2. Pawl 3 is held against the wheel by gravity or by a spring. A similar arrangement is used for lifting jacks, which then employ a toothed rack for rectilinear motion.

Figure 1.12*b* is an escapement used for rotary adjustments.

Graham's escapement shown in Fig. 1.12*c* is used to regulate the movement of clockwork. Anchor 3 drives a pendulum whose oscillating motion is caused by the two clicks engaging wheel 2. One is a push click, the other is a pull click. The lifting and engaging of each click caused by oscillation of the pendulum results in a wheel motion

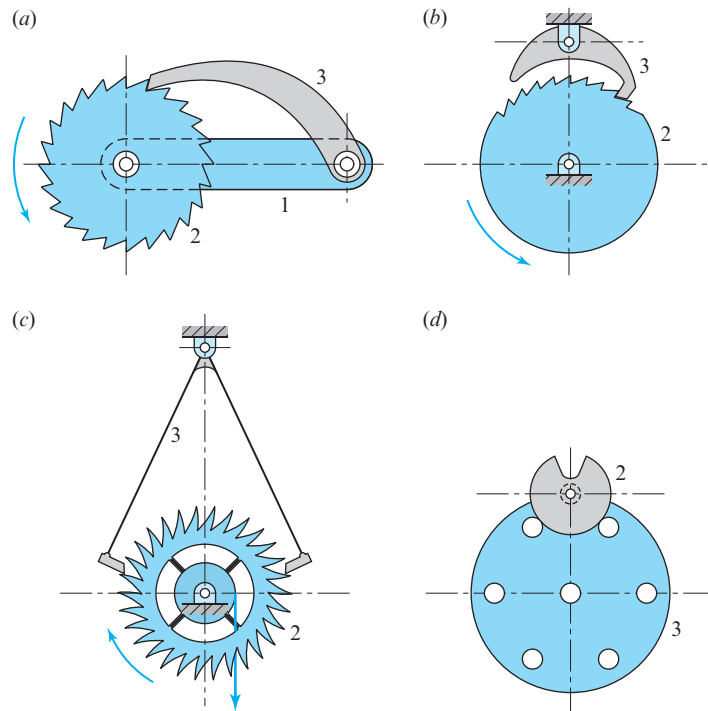
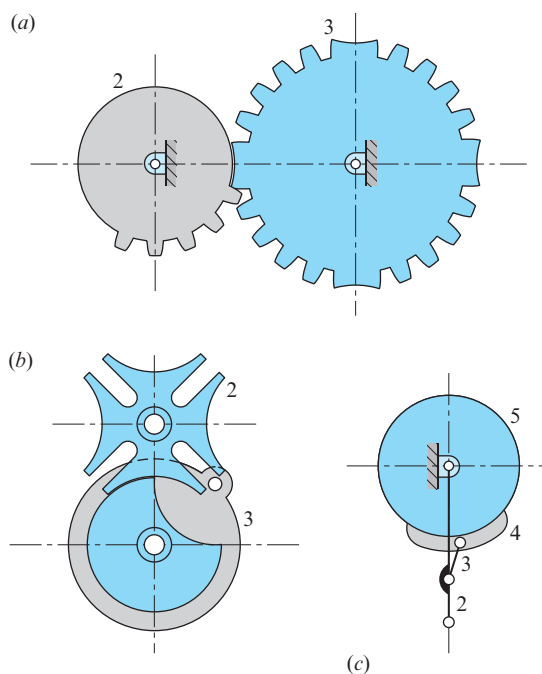


Figure 1.12 Ratchets and escapements.



 **Figure 1.13** Indexing mechanisms.

that, at the same time, presses each respective click and adds a gentle force to the motion of the pendulum.

The escapement shown in Fig. 1.12*d* has a control wheel, 2, that may rotate continuously to allow wheel 3 to be driven (by another source) in either direction.

Indexing Mechanisms The indexer shown in Fig. 1.13*a* uses standard gear teeth; for light loads, pins can be used in the input wheel 2 with corresponding slots in wheel 3, but neither form should be used if the shaft inertias are large.

Figure 1.13*b* shows a Geneva-wheel, sometimes called a “Maltese-cross,” indexer. Three or more slots may be used in the driven link, 2, which can be attached to, or geared to, the output to be indexed. High speeds and large inertias may cause problems with this indexer.

Toothless ratchet 5 in Fig. 1.13*c* is driven by the oscillating crank, 2, of variable throw. Note the similarity of this indexing mechanism to the ratchet of Fig. 1.12*a*.

Torfason [9] lists nine different indexing mechanisms, and many variations are possible.

Swinging or Rocking Mechanisms The category of swinging or rocking mechanisms is often termed *oscillators*; in each case, the output rocks or swings through an angle that is generally less than 360° . However, the output shaft can be geared to a successor shaft to produce a larger angle of oscillation.

Figure 1.14*a* is a mechanism consisting of a rotating crank 2 and a coupler 3 containing a rack, which meshes with output gear 4 to produce the oscillating motion.

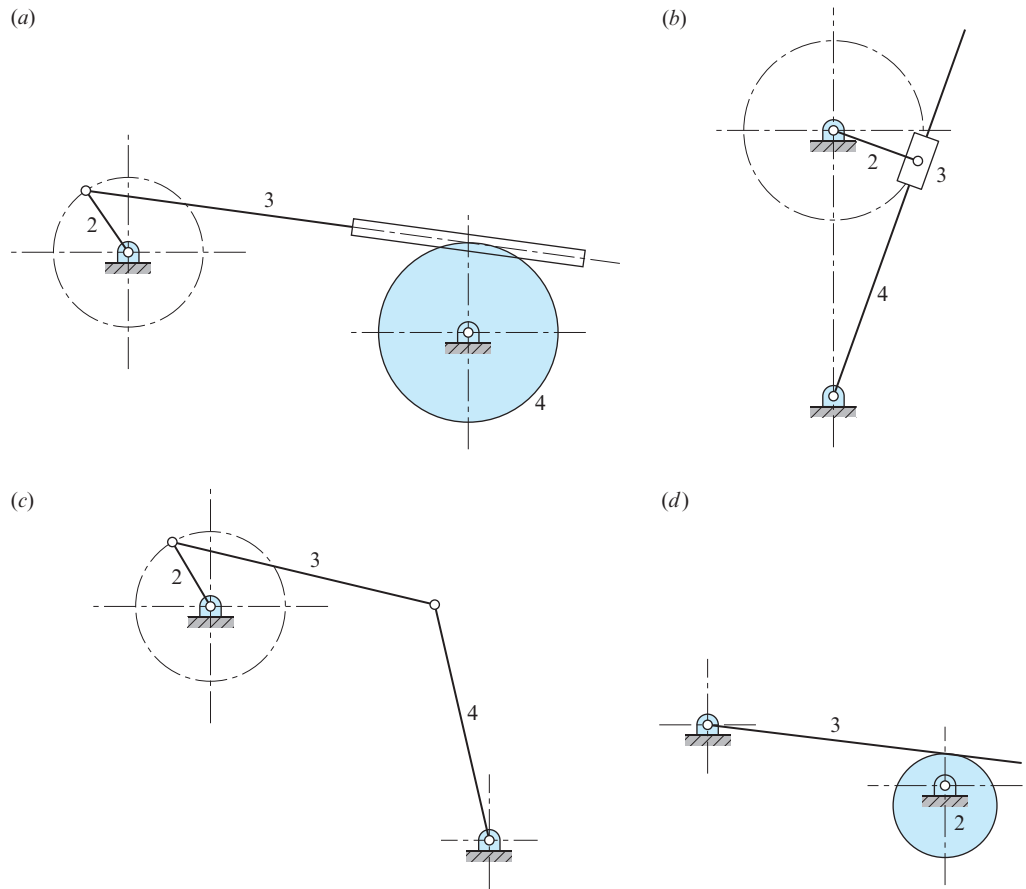


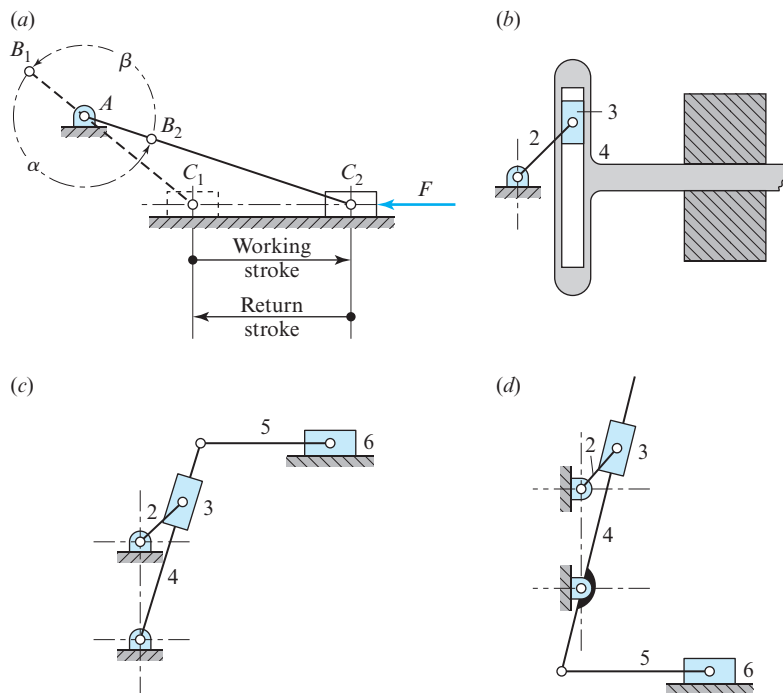
Figure 1.14 Oscillating mechanisms.

In Fig. 1.14*b*, crank 2 drives link 3, which slides on output link 4, producing a rocking motion. This mechanism is described as a *quick-return mechanism*, because crank 2 rotates through a larger angle on the forward stroke of link 4 than on the return stroke.

Figure 1.14*c* is a *four-bar linkage* called the *crank-rocker linkage* (Sec. 1.9). Crank 2 drives rocker 4 through the coupler 3. Of course, link 1 is the frame. The characteristics of the rocking motion depend on the dimensions of the links and the placement of the frame pivots.

Figure 1.14*d* shows a *cam-and-follower mechanism*, in which the rotating link 2, called the *cam*, drives link 3, called the *follower*, in a rocking motion. An endless variety of cam-and-follower mechanisms are possible, many of which are discussed in Chap. 6. In each case, the cam can be designed to produce an output motion with the desired characteristics.

Reciprocating Mechanisms Repeating straight-line motion is commonly obtained using either a pneumatic or hydraulic cylinder, a stationary screw with a traveling nut, a rectilinear drive using a reversible motor or reversing gears, or a cam-and-follower



 **Figure 1.15** Reciprocating linkages.

mechanism. A variety of typical linkages for obtaining reciprocating motion are shown in Figs. 1.15 and 1.16 [5].

The *offset slider-crank linkage* shown in Fig. 1.15a has kinematic characteristics that differ from the in-line (or on-center) slider-crank, shown in Fig. 1.13b. If the length of connecting rod 3 is long compared to the length of crank 2, then the resulting motion is nearly harmonic. Exact harmonic motion can be obtained from link 4 of the *Scotch-yoke linkage* shown in Fig. 1.15b.

The six-bar linkage shown in Fig. 1.15c is often called the *shaper linkage*, after the name of the machine tool in which it is used. Note that it is obtained from Fig. 1.14b by adding coupler 5 and slider 6. The stroke of the slider has a quick-return characteristic.

Figure 1.15d shows another version of the shaper linkage, which is termed the *Whitworth quick-return linkage*. The linkage is presented in an upside-down posture to illustrate its similarity to Fig. 1.15c.

Another example of a six-bar linkage is the Wanzer needle-bar linkage [5] shown in Fig. 1.16.

Figure 1.17a shows a six-bar linkage derived from the crank-rocker linkage of Fig. 1.14c by expanding coupler 3 and adding coupler 5 and slider 6. Coupler point C should be located to produce the desired motion characteristic for slider 6.

A crank-driven toggle linkage is shown in Fig. 1.17b. With this linkage, a high mechanical advantage is obtained at one end of the stroke of slider 6. (For a detailed discussion of the mechanical advantage of a mechanism, see Secs. 1.10 and 3.20).

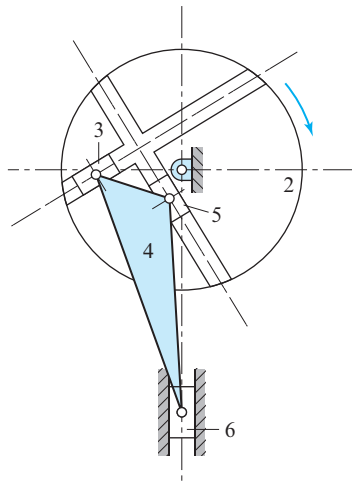


Figure 1.16 Wanzer needle-bar linkage.
(Richard Mott Wanzer, 1812–1900).

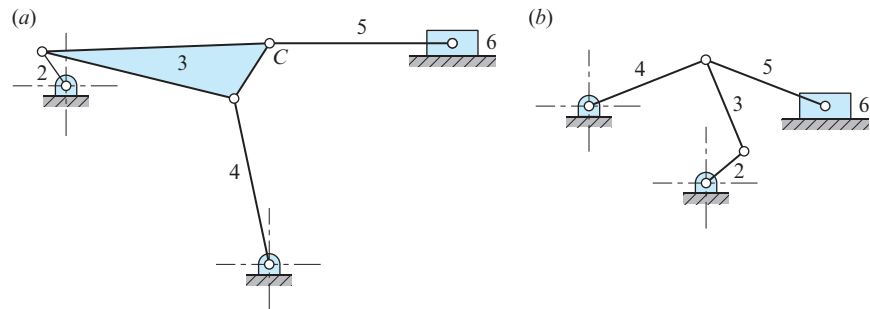


Figure 1.17 Additional six-bar reciprocating linkages.

In many applications, mechanisms are used to perform repetitive operations, such as pushing parts along an assembly line, clamping parts together while they are welded, or folding cardboard boxes in an automated packaging machine. In such applications it is often desirable to use a constant-speed motor; this leads us to a discussion of Grashof's law in Sec. 1.9. In addition, however, we should give some consideration to the power and timing requirements.

In such repetitive operations, there is usually a part of the cycle when the mechanism is under load, called the *advance* or *working stroke*, and a part of the cycle, called the *return stroke*, when the mechanism is not working but simply returning to repeat the operation. For example, consider the offset slider-crank linkage shown in Fig. 1.15a. Work may be required to overcome the load, F , while the piston moves to the right from position C_1 to position C_2 but not during its return to position C_1 , since the load may have been removed. In such situations, in order to keep the power requirement of the motor to a minimum and to avoid wasting valuable time, it is desirable to design a mechanism so that the piston moves much faster through the return stroke than it does during the advance (or working) stroke—that is, to use a higher portion of the cycle time for doing work than for returning.

A measure of the suitability of a mechanism from this viewpoint, called the *advance-to-return ratio* is defined as

$$Q = \frac{\text{cycle fraction for advance stroke}}{\text{cycle fraction for return stroke}}. \quad (a)$$

A mechanism for which the value of Q is high is more desirable for such repetitive operations than one in which the value of Q is lower. Certainly, any such operation would call for a mechanism for which Q is greater than unity. Because of this, mechanisms with Q greater than unity are called *quick-return mechanisms*.

As shown in Fig. 1.15a, the first step is to determine the two crank postures, AB_1 and AB_2 , that mark the beginning and the end of the working stroke. Next, noting the direction of rotation of the crank, we determine the crank angle α traveled through during the advance stroke and the remaining crank angle β of the return stroke. Then,

$$\text{cycle fraction for advance stroke} = \frac{\alpha}{2\pi}, \quad (b)$$

and

$$\text{cycle fraction for return stroke} = \frac{\beta}{2\pi}. \quad (c)$$

Finally, substituting Eqs. (b) and (c) into Eq. (a), the advance-to-return ratio can be written as

$$Q = \frac{\alpha}{\beta}. \quad (1.5)$$

Note that the advance-to-return ratio depends only on geometry (that is, on changes in the crank position); this ratio does not depend on the amount of work being done or on the speed of the driving motor. It is a kinematic property of the mechanism itself. Therefore, this ratio can be used for either design or analysis totally by graphic constructions. The following two examples illustrate applications in design.

EXAMPLE 1.4

The rocker of a crank-rocker four-bar linkage is required to have a length of 4 in and swing through a total angle of 45° . Also, the advance-to-return ratio of the linkage is required to be 2.0. Determine a suitable set of link lengths for the remaining three links.

SOLUTION

Equation (1.5) requires

$$Q = \frac{\alpha}{\beta} = 2.0, \quad (1)$$

where

$$\alpha = 180^\circ + \phi \quad (2)$$

and

$$\beta = 180^\circ - \phi. \quad (3)$$

Substituting Eqs. (2) and (3) into Eq. (1) allows us to solve for

$$\phi = 60^\circ, \quad \alpha = 240^\circ, \quad \text{and} \quad \beta = 120^\circ.$$

Now, referring to Fig. 1.18, we apply the following graphic procedure:

- Draw the rocker ($r_4 = 4.0$ in) to a suitable scale in its two extreme postures; that is, show the swing angle of the rocker of 45° . Label the ground pivot O_4 , and label pin B in the two positions B_1 and B_2 .
- Through point B_1 , draw an arbitrary line (labeled the X -line). Through B_2 , draw a line parallel to the X -line.
- Measure the angle $\phi = 60^\circ$ counterclockwise from the X -line through point B_1 . The intersection of this line with the line parallel to the X -line is the required position of the input crank pivot O_2 .
- The length O_2O_4 of the ground link can be measured from the drawing—that is

$$r_1 = 1.50 \text{ in.} \quad \text{Ans.}$$

- The lengths of crank r_2 and the coupler r_3 can be determined from the measurements

$$O_2B_1 = r_3 + r_2 = 3.50 \text{ in} \quad \text{and} \quad O_2B_2 = r_3 - r_2 = 2.50 \text{ in.}$$

That is,

$$r_2 = 0.5(O_2B_1 - O_2B_2) = 0.50 \text{ in} \quad \text{and} \quad r_3 = 0.5(O_2B_2 + O_2B_1) = 3.00 \text{ in.} \quad \text{Ans.}$$

The solution for the synthesized four-bar linkage is shown in Fig. 1.18.

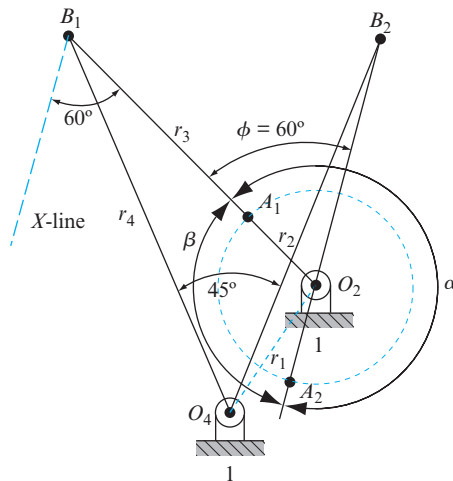


Figure 1.18 Synthesized four-bar linkage.

In the two limiting positions, B_1 and B_2 , the output is momentarily stopped, and, for this reason, these two postures of the linkage are referred to as *dead-center postures* (see Prob. 1.35). Also, note that this problem is an example of two-posture synthesis. For a detailed discussion on the general problems of two-, three-, and four-posture synthesis, see Chap. 9.

EXAMPLE 1.5

Determine a suitable set of link lengths for a slider-crank linkage such that the stroke is 2.50 in and the advance-to-return ratio is 1.4.

SOLUTION

Equation (1.5) requires

$$Q = \frac{\alpha}{\beta} = 1.40, \quad (1)$$

where

$$\alpha = 180^\circ + \phi \quad (2)$$

and

$$\beta = 180^\circ - \phi. \quad (3)$$

Substituting Eqs. (2) and (3) into Eq. (1) allows us to solve for

$$\phi = 30^\circ, \quad \alpha = 210^\circ, \quad \text{and} \quad \beta = 150^\circ.$$

Now, referring to Fig. 1.19, we apply the following graphic procedure:

- (a) Draw the stroke (shown horizontal) of 2.50 in of the slider-crank linkage to a suitable scale. Label pin B in its two extreme positions B_1 and B_2 .
- (b) Through point B_2 , draw an arbitrary line (labeled the X -line). Through point B_1 , draw a line parallel to the X -line.
- (c) Measure the angle $\phi = 30^\circ$ clockwise from the X -line. The intersection of this line with the line parallel to the X -line is the ground pivot O_2 .
- (d) The length of the ground link—that is, the offset or eccentricity (the perpendicular distance from the ground pivot O_2 to the line of travel of the slider)—can be measured from the drawing. That is,

$$r_1 = 2.17 \text{ in.} \quad \text{Ans.}$$

- (e) The lengths of crank r_2 and coupler r_3 can be determined from the measurements

$$O_2B_1 = r_3 + r_2 = 4.33 \text{ in} \quad \text{and} \quad O_2B_2 = r_3 - r_2 = 2.50 \text{ in.}$$

That is,

$$r_2 = 0.5(O_2B_1 - O_2B_2) = 0.92 \text{ in} \quad \text{and} \quad r_3 = 0.5(O_2B_2 + O_2B_1) = 3.42 \text{ in.}$$

Ans.

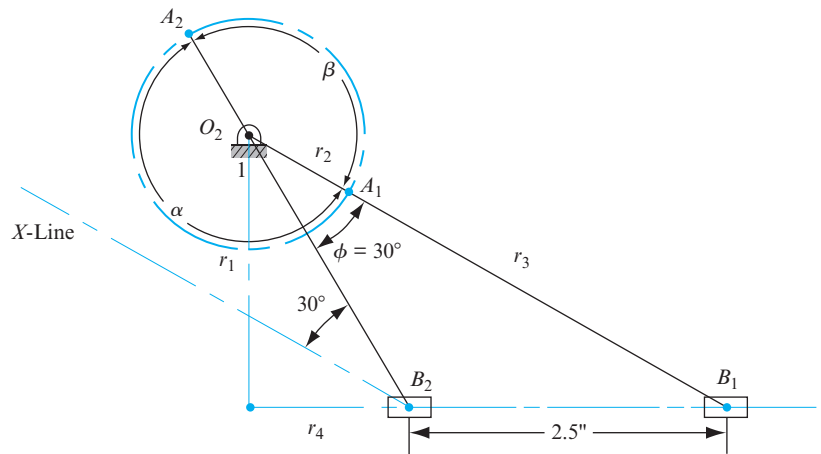


Figure 1.19 Synthesized slider-crank linkage.

The solution of the synthesized slider-crank linkage is shown in Fig. 1.19.

Note that there is a proper and an improper direction of rotation for the input of such a device. If the direction of crank rotation were reversed in the example of Fig. 1.19, the roles of α and β would also be reversed, and the advance-to-return ratio would be less than 1.0. Thus, the motor must rotate clockwise for this mechanism to have the quick-return property.

Other mechanisms with quick-return characteristics are the shaper linkage, shown in Fig. 1.15c, and the Whitworth linkage, shown in Fig. 1.15d. The synthesis of a quick-return mechanism, as well as mechanisms with other properties, is discussed in detail in Chap. 9.

Reversing Mechanisms When a mechanism capable of delivering output rotation in either direction is desired, some form of reversing mechanism is required. Many such devices make use of a two-way clutch that connects the output shaft to either of two driveshafts turning in opposite directions. This method is used in both gear and belt drives and does not require that the drive be stopped to change direction. Gear-shift devices, as in automotive transmissions, are also in common use.

Couplings and Connectors Couplings and connectors are used to transmit motion between coaxial, parallel, intersecting, or skewed shafts. Gears of one kind or another can be used for any of these situations. These are discussed in Chaps. 7 and 8.

Flat belts can be used to transmit motion between parallel shafts. They can also be used between intersecting or skewed shafts if guide pulleys are used, as shown in Fig. 1.20a. When parallel shafts are involved, the belts can be open or crossed, depending on the direction of rotation desired.

Figure 1.20b shows a *drag-link* (also referred to as a *double-crank*) four-bar linkage used to transmit rotary motion between parallel shafts. Here crank 2 is the driver and crank

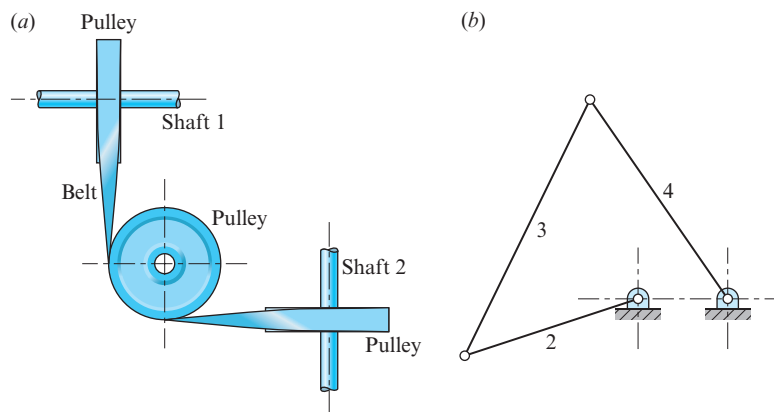


Figure 1.20 Two-shaft coupling mechanisms.

4 is the output. This is a very interesting linkage; you should try to construct one using cardboard strips and thumbtacks for joints to observe its motion. Can you devise a working model that allows complete rotation of both links 2 and 4 (see Prob. 1.14)?

The *Reuleaux coupling*, shown in Fig. 1.21a, for intersecting shafts is recommended only for light loads. The *Hooke joint*, shown in Fig. 1.21b, is also used for intersecting shafts. However, this joint can withstand heavy loads and is commonly used with a driveshaft in rear-wheel-drive vehicles. It is customary to use two of these joints in series for connecting parallel shafts.

Sliding Connectors Sliding connectors are used when one slider (the input) is to drive another slider (the output). The usual problem is that the two sliders operate in the same plane but in different directions. The possible solutions are:

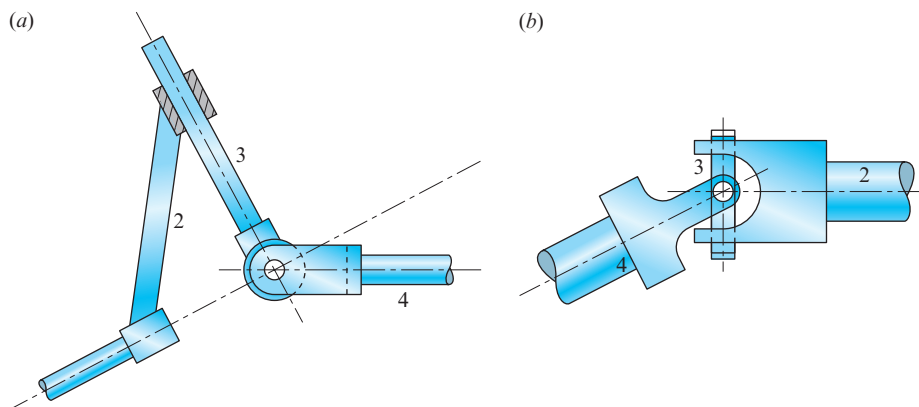


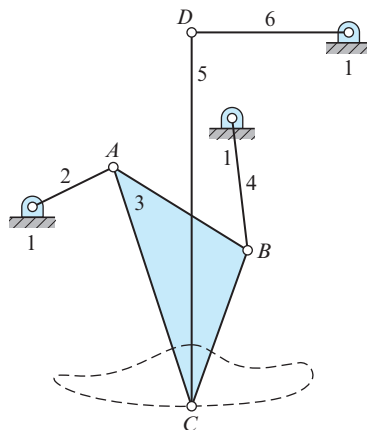
Figure 1.21 Coupling mechanisms for intersecting shafts.

1. A rigid link pivoted at each end to a slider.
2. A belt or chain connecting the two sliders with the use of a guide pulley or sprocket.
3. Rack gear teeth cut on each slider and the connection completed using one or more gears.
4. A flexible cable connector.

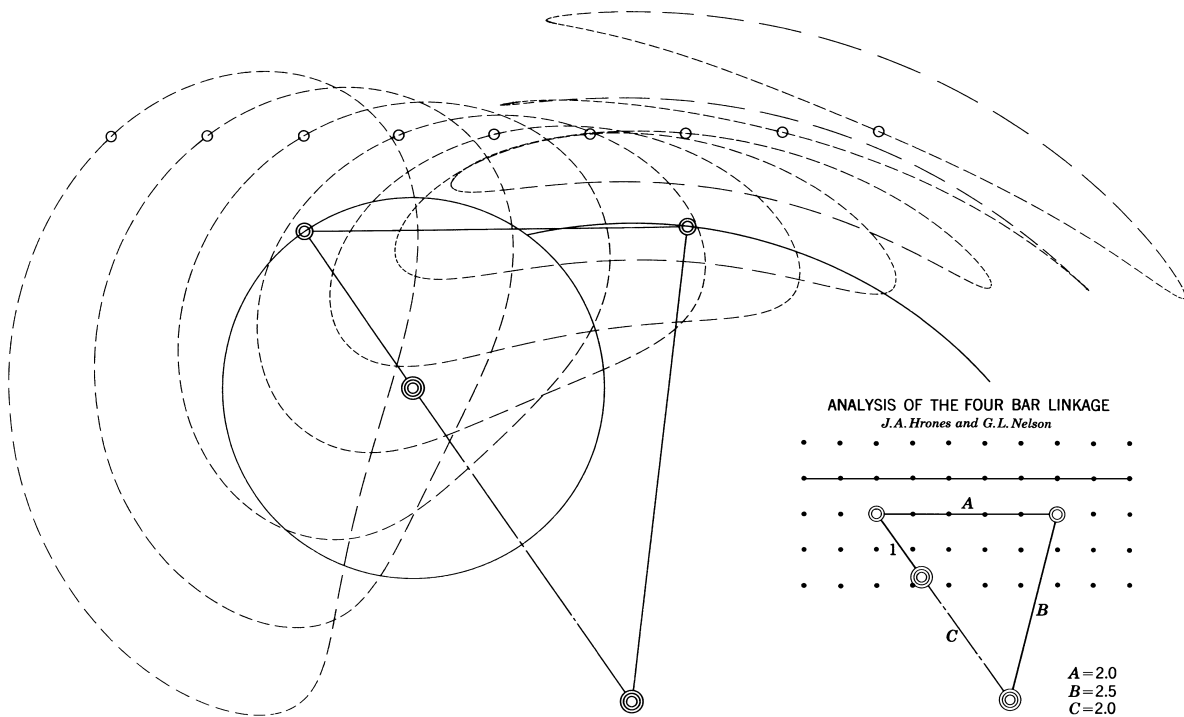
Stop, Pause, and Hesitation Mechanisms In an automotive engine a valve must open, remain open for a short period of time, and then close. A conveyor line may need to halt for an interval of time while an operation is being performed and then continue its motion. Many similar requirements occur in the design of machines. Torfason [9] classifies these as *stop-and-dwell*, *stop-and-return*, *stop-and-advance*, and so on. Such requirements can often be met using cam-and-follower mechanisms (Chap. 6), indexing mechanisms, including those of Fig. 1.13, ratchets, linkages at the limits of their motion, and gear-and-clutch mechanisms.

The six-bar linkage of Fig. 1.22 is a clever method to obtain a rocking motion (of link 6) containing a dwell. This linkage, an extension of the four-bar linkage, consisting of frame 1, crank 2, coupler 3, and rocker 4, can be designed such that point C on the coupler generates the curve shown by dashed lines. A portion of this curve will then fit closely to a circular arc whose radius is equal to the length of link 5—that is, distance DC . Thus, when point C traverses this portion of the coupler curve, link 6, the output rocker, is stationary.

Curve Generators The connecting rod, or coupler, of a planar four-bar linkage may be imagined as an infinite plane extending in all directions but pin-connected to the input and output cranks. Then, during motion of the linkage, any point attached to the plane of the coupler generates a path with respect to the fixed link; this path is called a *coupler curve*. Two of these paths, namely those generated by the pin connectors of the coupler, are true circles with centers at the two fixed pivots. However, other points can be found that trace much more complex curves.



 **Figure 1.22** Six-bar stop-and-dwell linkage.



 **Figure 1.23** A set of coupler curves [6].

One of the best sources of coupler curves for the four-bar linkage is the Hrones and Nelson atlas [6]. This book consists of a set of 11 in $\times$ 17 in drawings containing over 7,000 coupler curves of crank-rocker linkages. Figure 1.23 is a reproduction of a typical page of this atlas (by permission of the publishers). In each case, the crank has unit length, and the lengths of the remaining links vary from page to page to produce the different combinations. On each page a number of different coupler points are chosen, and their coupler curves are shown. This atlas of coupler curves has proven to be invaluable to the designer who needs a linkage to generate a curve with specified characteristics.

The algebraic equation of a four-bar linkage coupler curve is, in general, a sixth-order polynomial [1]; thus, it is possible to find coupler curves with a wide variety of shapes and many interesting features. Some coupler curves have sections that are nearly straight line segments; others have almost exact circular arc segments; still others have one or more cusps or cross over themselves like a figure eight. Therefore, it is often not necessary to use a mechanism with a large number of links to obtain a complex motion of a coupler point.

Yet the complexity of the coupler-curve equation is also a hinderance; it means that hand-calculation methods can become very cumbersome. Thus, over the years, many mechanisms have been designed by strictly intuitive procedures and proven with cardboard models, without the use of kinematic principles or procedures. Until quite recently,

those techniques that did offer a rational approach have been graphic, avoiding tedious computations. Finally, with the availability of digital computers, and particularly with computer graphics, useful design methods are now emerging that can deal directly with the complex calculations required without burdening the designer with the computational drudgery (Sec. 10.9 has details on some of these).

One of the more curious and interesting facts about the coupler-curve equation is that the same curve can always be generated by three different four-bar linkages. These are called *cognate linkages*, and the theory is developed in Sec. 9.10.

Straight-Line Generators In the late 17th century, before the development of the milling machine, it was extremely difficult to machine straight, flat surfaces. For this reason, good prismatic joints with close clearances were not available. During that era, much thought was given to the problem of attaining a straight line as a part of the coupler curve of a linkage having only revolute connections. Probably the best-known result of this search is the straight-line mechanism developed by Watt for guiding the piston of early steam engines. Figure 1.24a shows a four-bar linkage, known as *Watt's linkage*, which generates an approximate straight line as a part of its coupler curve. Although the coupler point (tracing point P) does not generate an exact straight line, a good approximation is achieved over a considerable distance of travel.

Another four-bar linkage in which the tracing point P generates an approximate straight-line coupler-curve segment is Roberts' linkage (Fig. 1.24b). The dashed lines BP and CP in the figure indicate that the linkage is defined by forming three congruent isosceles triangles; thus, $BC = AP = PD = AD/2$.

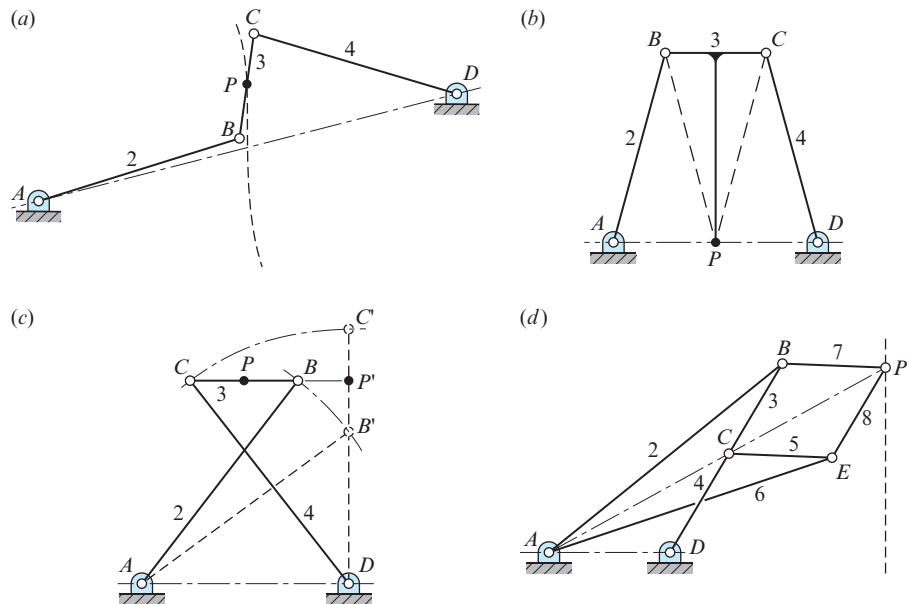


Figure 1.24 (a) Watt's linkage; (b) Roberts' linkage; (c) Chebyshev linkage; (d) Peaucellier invensor.

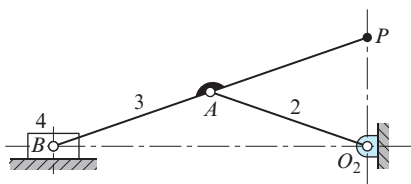


Figure 1.25 $AB = AP = O_2A$.

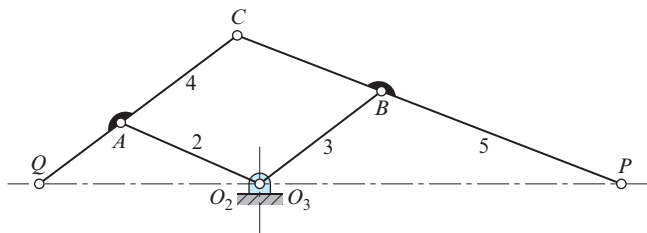


Figure 1.26 Pantograph linkage.

The tracing point P (the midpoint of coupler link BC) of the Chebyshev linkage shown in Fig. 1.24c also generates an approximate straight line. The linkage forms a 3:4:5 triangle when link 4 is in the vertical posture, as shown by the dashed lines; thus, $DB' = 3$ units, $AD = 4$ units, and $AB' = 5$ units. Note that $AB = DC$, $DC' = 5$ units, and tracing point P' is the midpoint of dashed line $B'C'$. Also, note that $DP'C$ forms another 3:4:5 triangle, and hence the line containing P and P' is parallel to the ground link AD .

A linkage that generates an exact straight line is the Peaucillier invisor shown in Fig. 1.24d. The conditions describing its geometry are that $BC = BP = EC = EP$ and $AB = AE$ such that, by symmetry, points A , C , and P always lie on a straight line passing through A . Under these conditions $AC \cdot AP = k$, a constant, and the curves generated by C and P are said to be *inverses* of each other. If we place the other fixed pivot D such that $AD = CD$, then point C must trace a circular arc while point P follows an exact straight line. Another interesting property is that if AD is not equal to CD , then point P traces a true circular arc of very large radius. This was the first straight-line generator, and it was important in the development of the steam engine.

Figure 1.25 shows another linkage that generates exact straight-line motion: the Scott-Russell linkage. However, note that it employs a slider.

The *pantograph* shown in Fig. 1.26 is used to trace figures at a larger or smaller size. If, for example, point P traces a map, then a pen at Q will draw a similar map at a smaller scale. The dimensions O_2A , AC , CB , and BO_3 must conform to an equal-sided parallelogram.

Torfason [9] also includes robots, speed-changing devices, computing mechanisms, function generators, loading mechanisms, and transportation devices in his classification. Many of these utilize arrangements of mechanisms already presented. Others appear in some of the chapters to follow.

1.8 KINEMATIC INVERSION

In Sec. 1.4 we noted that every mechanism has a fixed link called the frame. When different links are chosen as the frame, the *relative* motions between the various links are not altered, but their *absolute* motions (those measured with respect to the frame) may be changed significantly. The process of choosing different links for the frame is known as *kinematic inversion*. In an n -link mechanism, choosing each link in turn as the frame yields n distinct kinematic inversions—that is, n different mechanisms.

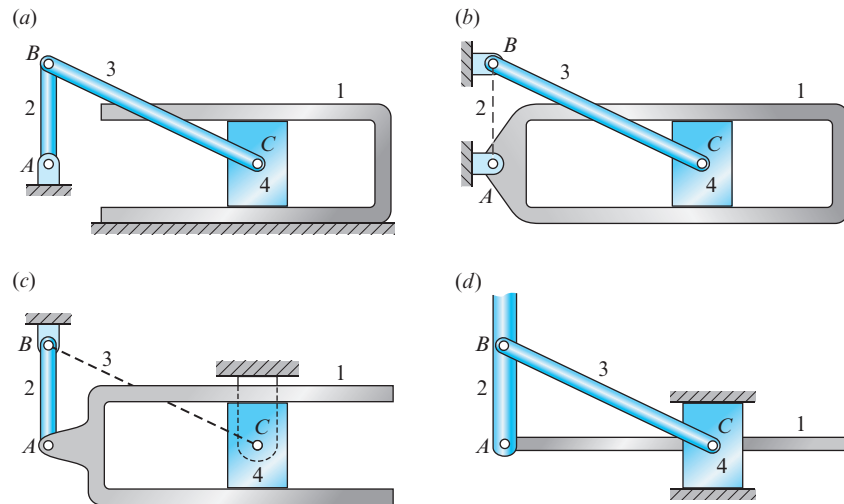


Figure 1.27 Four inversions of the slider-crank linkage.

For illustration, consider the slider-crank linkage. Figure 1.27a shows the first inversion, which is found in most internal combustion engines, where the frame is the cylinder block, link 1. Link 4, the piston, is driven by the expanding gases, and this movement provides the input. Link 3, the connecting rod, connects link 2, the crank, which rotates as the driven output. By reversing the roles of the input and output, this linkage can be used as a compressor.

Figure 1.27b shows another kinematic inversion, where link 2 is stationary. Link 1, formerly the frame, now rotates about revolute A. This inversion of the slider-crank linkage was the basis of the rotary engine found in many early aircraft.

Yet another inversion is shown in Fig. 1.27c; it has link 3, formerly the connecting rod, as the fixed link. This linkage was used to drive the wheels of early steam locomotives, with link 2 being attached to a wheel.

The final inversion, shown in Fig. 1.27d, has the piston, link 4, stationary. Although it is not found in engines, by rotating the figure 90° clockwise, this linkage is recognized as part of a garden water pump. Note in this inversion that the prismatic joint connecting links 1 and 4 is also inverted—that is, the “inside” and “outside” elements of the joint are reversed.

1.9 GRASHOF'S LAW

A very important consideration when designing a mechanism to be driven by a motor, obviously, is to ensure that the input crank can make a complete revolution. Mechanisms in which no link can make a complete revolution would not be useful in such applications. For the four-bar linkage, there is a very simple test of whether this is the case.

Grashof's law states that, *for a planar four-bar linkage, the sum of the shortest and longest link lengths cannot be greater than the sum of the remaining two link lengths if*

there is to be continuous relative rotation between two links. This is shown in Fig. 1.28, where the longest link has length l , the shortest link has length s , and the other two links have lengths p and q . In this notation, Grashof's law states that one of the links, in particular the shortest link, will rotate continuously relative to the other three links if and only if

$$s + l \leq p + q. \quad (1.6)$$

If this inequality is not satisfied, no link will make a complete revolution relative to another.

Attention is called to the fact that nothing in Grashof's law specifies the order in which the links are connected or which link of the four-bar linkage is fixed. We are free, therefore, to fix any of the four links. When we do so, we create variations of the four-bar linkage, four of which are shown in Fig. 1.28. All of these fit Grashof's law, and, in each, link s makes a complete revolution relative to the other links. The different variations are distinguished by the location of link s relative to the fixed link.

If the shortest link, s , is adjacent to the fixed link, as shown in Fig. 1.28a and Fig. 1.28b, we obtain what is called a *crank-rocker linkage*. Link s is, of course, the crank, since it is able to rotate continuously, and link p , which can only oscillate between limits, is the rocker.

The drag-link (or double-crank linkage) is obtained by fixing the shortest link, s , as the frame. In this variation, shown in Fig. 1.28c, both links adjacent to s can rotate continuously, and both are properly described as cranks; the shorter of the two is generally used as the input. Although this is a very common linkage, you will find it an interesting challenge to devise a practical working model that can operate through the full cycle.

By fixing the link opposite to link s , we obtain the final variation, the double-rocker linkage of Fig. 1.28d. Note that although link s is able to make a complete revolution, neither link adjacent to the frame can do so; both must oscillate between limits and, therefore, are rockers.

In each of the variations shown, the shortest link, s , is adjacent to the longest link, l . However, exactly the same types of linkages will occur if the longest link, l , is opposite the shortest link, s ; you should demonstrate this for your own satisfaction.

Reuleaux approaches the problem somewhat differently but, of course, obtains the same results. In his approach, and using Fig. 1.29, the links are labeled

$$\begin{array}{ll} s \text{ the crank,} & p \text{ the lever,} \\ l \text{ the coupler,} & q \text{ the frame,} \end{array}$$

where l need not be the longest link. Then the following conditions apply:

$$s + l + p \geq q, \quad (1.7a)$$

$$s + l - p \leq q, \quad (1.7b)$$

$$s + q + p \geq l, \quad (1.7c)$$

$$s + q - p \leq l. \quad (1.7d)$$

If these four conditions are not satisfied, then the results are as illustrated in Fig. 1.29.

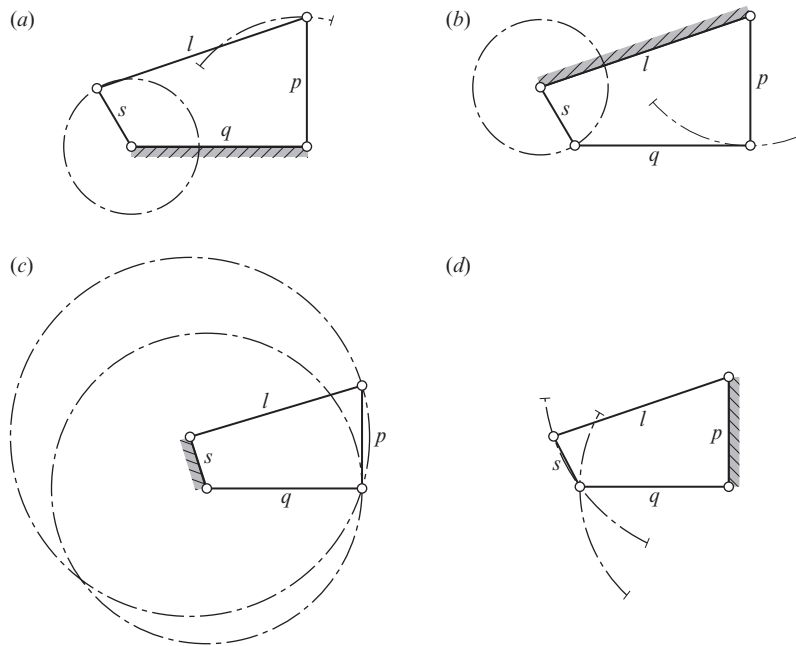


Figure 1.28 (a, b) Crank-rocker linkages; (c) drag-link linkage; and (d) double-rocker linkage.

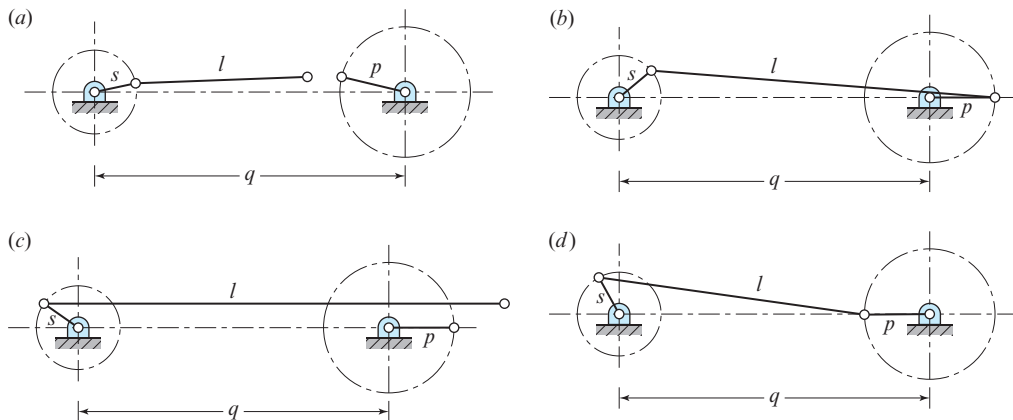


Figure 1.29 (a) Eq. (1.7a); $s + l + p < q$, and the links cannot be connected; (b) Eq. (1.7b); $s + l - p > q$, and s is incapable of rotation; (c) Eq. (1.7c); $s + q + p < l$, and the links cannot be connected; (d) Eq. (1.7d); $s + q - p > l$, and s is incapable of rotation.

EXAMPLE 1.6

Is the linkage shown in Fig. 1.30 a crank-rocker, a double-rocker, or a drag-link four-bar linkage?

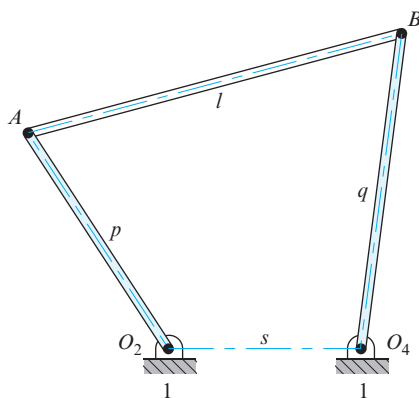


Figure 1.30 $p = 4$ in, $l = 6$ in, $q =$ in, and $s = 3$ in.

SOLUTION

Substituting the link lengths into Eq. (1.6) gives

$$3 \text{ in} + 6 \text{ in} \leq 4 \text{ in} + 5 \text{ in}$$

or

$$9 \text{ in} \leq 9 \text{ in}.$$

Therefore, the linkage satisfies Grashof's law; that is, the linkage is a Grashof four-bar linkage. Since the shortest link is the frame, the two links adjacent to the shortest link can both rotate continuously (Fig. 1.28c), and both are properly described as cranks. Therefore, Fig. 1.30 shows a drag-link linkage. *Ans.*

1.10 MECHANICAL ADVANTAGE

In general, the *mechanical advantage* of a mechanism is defined as the ratio of the force or torque exerted by the driven link to the necessary force or torque required at the driver. With the widespread use of the four-bar linkage, a few remarks are in order here that will help us judge the quality of such a linkage for its intended application.

Consider the crank-rocker four-bar linkage shown in Fig. 1.31, where link 2 is the driver and link 4 is the follower.

In Sec. 3.19, we will show that the mechanical advantage of the four-bar linkage can be written as

$$MA = \frac{R_{CD} \sin \gamma}{R_{BA} \sin \beta}. \quad (1.8)$$

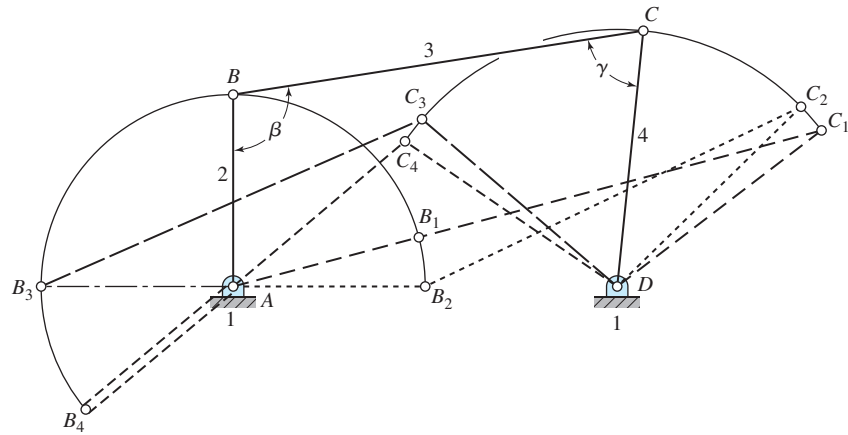


Figure 1.31 Crank-rocker four-bar linkage.

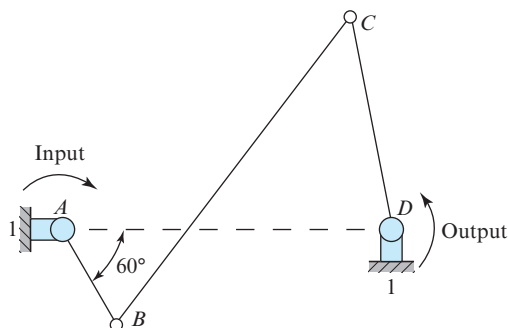
Note that this is directly proportional to the sine of the angle γ between the coupler and the follower, and is inversely proportional to the sine of angle β between the coupler and the driver. Of course, both these angles, and therefore the mechanical advantage, are continuously changing as the linkage moves.

When the sine of angle β becomes zero, the mechanical advantage becomes infinite; thus, at such a posture, only a small input torque is necessary to produce a very large output torque load. This is the case when the driver AB is directly in line with the coupler BC as shown in Fig. 1.31; it occurs when the crank is in posture AB_1 and again when the crank is in posture AB_4 . Note that these also define the extreme postures of travel of the rocker DC_1 and DC_4 . When the four-bar linkage is in either of these postures, the mechanical advantage is infinite—that is, $\beta = 0^\circ$ or $\beta = 180^\circ$ —and the linkage is said to be in a *toggle* (or *limit*) posture.

The angle γ between the coupler and the follower is called the *transmission angle*. As this angle becomes smaller, the mechanical advantage decreases and even a small amount of friction might cause the mechanism to lock or jam. A common rule of thumb is that a four-bar linkage should not be used to overcome a load in a region where the transmission angle is less than, say, 45° or 50° . The extreme values of the transmission angle occur when crank AB lies along the line of the frame, AD . The transmission angle is minimal when the crank is in posture AB_2 and is maximal when the crank is in posture AB_3 (see Fig. 1.31). Because of the ease with which it can be visually inspected, the transmission angle has become a commonly accepted measure of the quality of a design of the four-bar linkage. A double-rocker four-bar linkage has a *dead-center* posture when links 3 and 4 lie along a straight line. In a dead-center posture, the transmission angle is $\gamma = 0^\circ$ or $\gamma = 180^\circ$, and the linkage is locked. The designer must either avoid such a posture or provide an external force, such as a spring, to unlock the linkage.

EXAMPLE 1.7

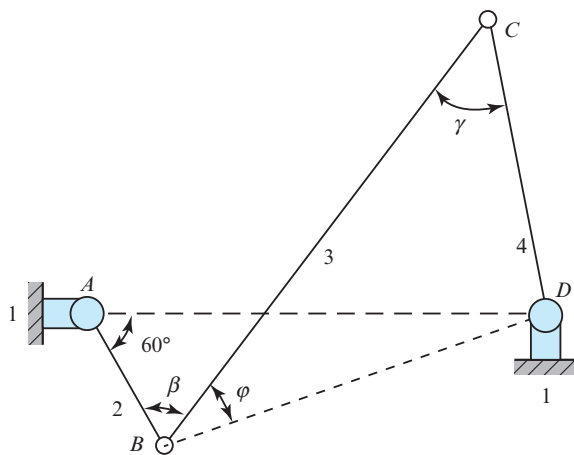
Determine the mechanical advantage of the four-bar linkage in the posture shown in Fig. 1.32.

**Figure 1.32**

$R_{DA} = 180$ mm, $R_{BA} = 60$ mm, $R_{CB} = 210$ mm, and $R_{CD} = 120$ mm.

SOLUTION

Angles γ and β for Eq. (1.8) are as shown in Fig. 1.33 and can be obtained from trigonometry.

**Figure 1.33** Angles γ and β .

Using the law of cosines,

$$R_{DB} = \sqrt{(60 \text{ mm})^2 + (180 \text{ mm})^2 - 2(60 \text{ mm})(180 \text{ mm}) \cos 60^\circ} = 158.745 \text{ mm}. \quad (1)$$

Also, the angles

$$\gamma = \cos^{-1} \left[\frac{(158.745 \text{ mm})^2 - (210 \text{ mm})^2 - (120 \text{ mm})^2}{-2(210 \text{ mm})(120 \text{ mm})} \right] = 48.65^\circ, \quad (2)$$

and

$$\phi = \cos^{-1} \left[\frac{(120 \text{ mm})^2 - (210 \text{ mm})^2 - (158.745 \text{ mm})^2}{-2(210 \text{ mm})(158.745 \text{ mm})} \right] = 34.57^\circ. \quad (3)$$

Finally, the sum of the angles

$$\beta + \phi = \cos^{-1} \left[\frac{(180 \text{ mm})^2 - (60 \text{ mm})^2 - (158.745 \text{ mm})^2}{-2(60 \text{ mm})(158.745 \text{ mm})} \right] = 100.90^\circ. \quad (4)$$

Subtracting Eq. (3) from Eq. (4) gives

$$\beta = 100.90^\circ - 34.57^\circ = 66.33^\circ. \quad (5)$$

Then, substituting Eqs. (2) and (5) into Eq. (1.8), the mechanical advantage of the four-bar linkage in the given posture is

$$MA = \frac{(120 \text{ mm}) \sin 48.65^\circ}{(60 \text{ mm}) \sin 66.33^\circ} = 1.64. \quad \text{Ans.}$$

Note that mechanical advantage, toggle posture, transmission angle, and dead-center posture depend on the choice of the driver and driven links. For example, in Fig. 1.31, if link 4 is used as the driver and link 2 as the driven link, then the roles of γ and β are reversed. In this case, the linkage has no toggle posture, and its mechanical advantage becomes zero when link 2 is in posture AB_1 or AB_4 , because the transmission angle is zero. These and other methods of rating the suitability of the four-bar linkage or other mechanisms are discussed more thoroughly in Sec. 3.19.

1.11 REFERENCES

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PROBLEMS

1.1 Sketch at least six different examples of the use of a planar four-bar linkage in practice. These can be found in the workshop, in domestic appliances, on vehicles, on agricultural machines, and so on.

†1.2 The link lengths of a planar four-bar linkage are 1 in, 3 in, 5 in, and 5 in. Assemble the links in all possible combinations and sketch the four inversions of each. Do these linkages satisfy Grashof's law? Describe each inversion by name (for example, a crank-rocker linkage or a drag-link linkage).

†1.3 A crank-rocker linkage has a 100-mm frame, a 25-mm crank, a 90-mm coupler, and a 75-mm rocker. Draw the linkage and find the maximum and minimum values of the transmission angle. Locate both toggle postures and record the corresponding crank angles and transmission angles.

†1.4 Plot the complete path of coupler point *C*.

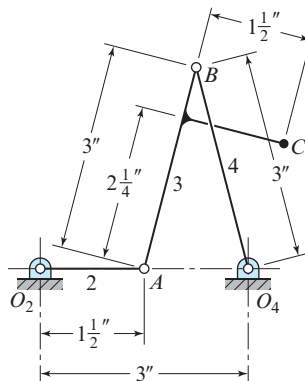


Figure P1.4

†1.5 Find the mobility of each mechanism.

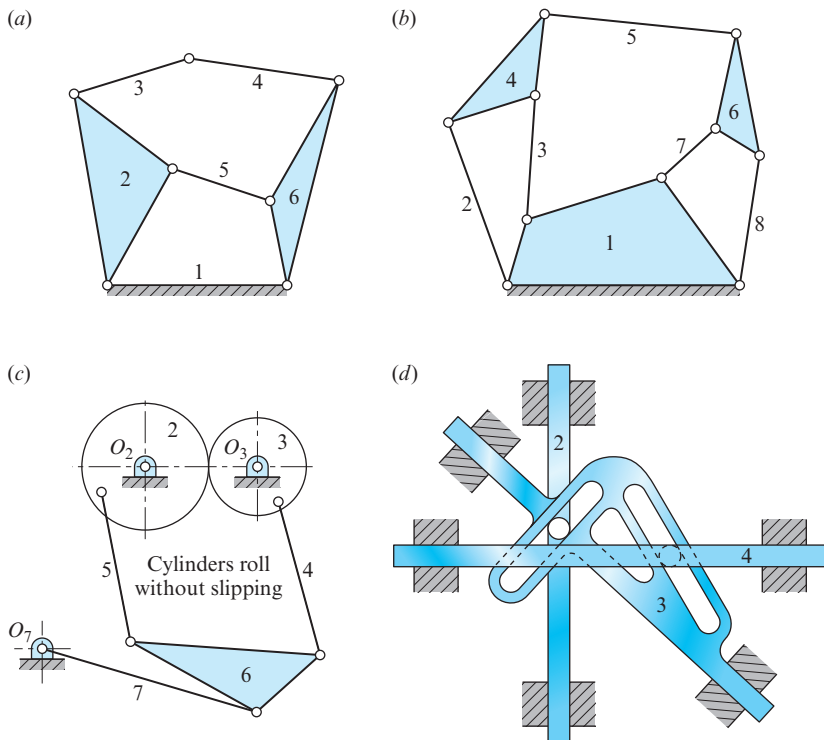


Figure P1.5

- 1.6** Use the Kutzbach criterion to determine the mobility of the mechanism.

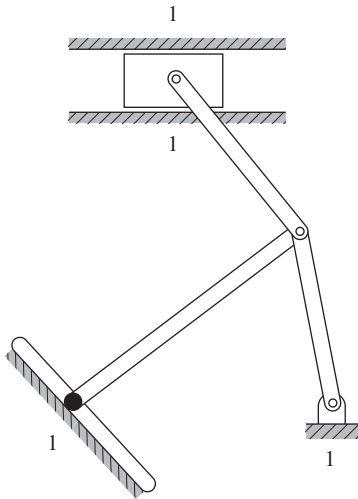


Figure P1.6

- 1.7** Sketch a planar linkage with only revolute joints and a mobility of $m = 1$ that contains a quaternary link. How many distinct variations of this linkage can you find?
- †1.8** Use the Kutzbach criterion to determine the mobility of the mechanism. Clearly number each link, and label the lower pairs (j_1 joints) and higher pairs (j_2 joints).

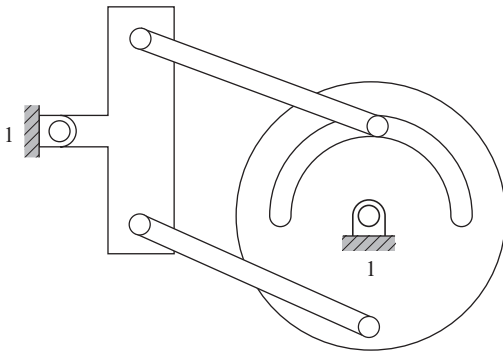


Figure P1.8

- †1.9** Determine the number of links, the number of lower pairs, and the number of higher pairs. Use

the Kutzbach criterion to determine the mobility of the mechanism. Is the answer correct? Briefly explain.

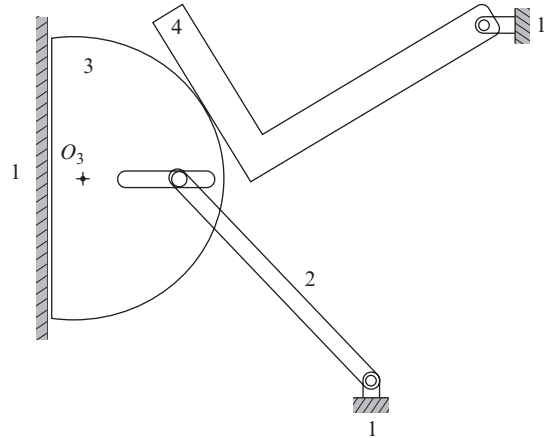


Figure P1.9

- 1.10** Use the Kutzbach criterion to determine the mobility of the mechanism. Clearly number each link, and label the lower pairs and higher pairs.

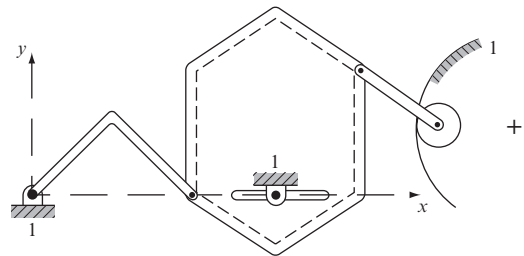


Figure P1.10

- 1.11** Determine the number of links, the number of lower pairs, and the number of higher pairs. Treat rolling contact to mean rolling with no slipping.

Using the Kutzbach criterion, determine the mobility. Is the answer correct? Briefly explain.

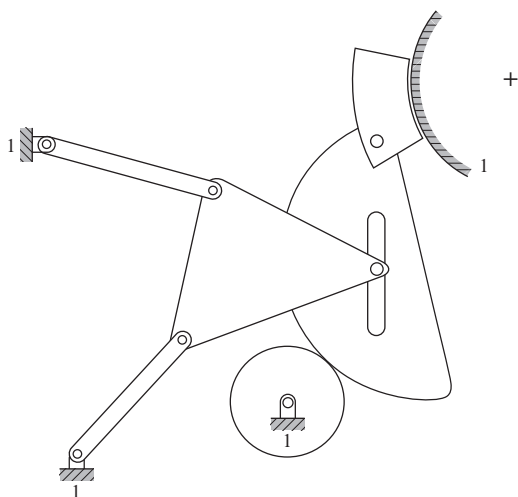


Figure P1.11

- 1.12** Does the Kutzbach criterion provide the correct result for this mechanism? Briefly explain why or why not.

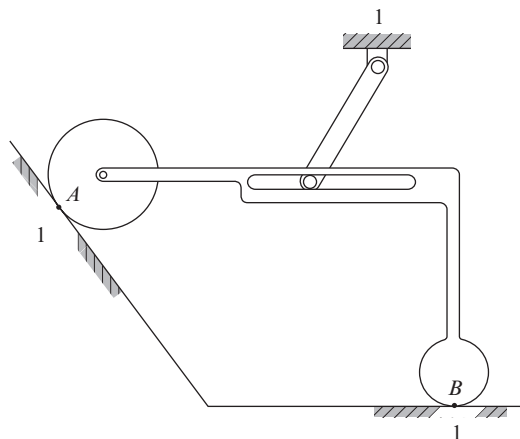


Figure P1.12

- 1.13** The mobility of the mechanism is $m = 1$. Use the Kutzbach criterion to determine the number of lower pairs and the number of higher pairs. Is the wheel rolling without slipping, or rolling and slipping, at point A on the wall?

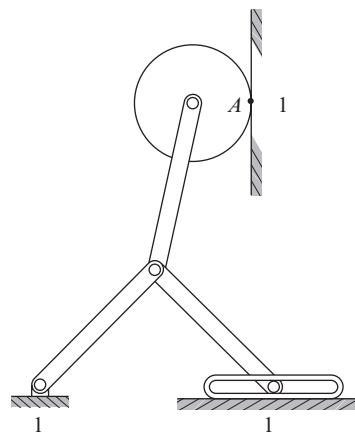


Figure P1.13

- 1.14** Devise a practical working model of the drag-link linkage.
- †1.15** Find the advance-to-return ratio of the linkage of Prob. 1.3.
- †1.16** Plot the complete coupler curve of the Roberts' linkage shown in Fig. 1.24b. Use $AB = CD = AD = 2.5$ in and $BC = 1.25$ in.
- 1.17** If the handle of the differential screw in Fig. 1.11 is turned 15 revolutions clockwise, how far and in what direction does the carriage move?
- †1.18** Show how the linkage of Fig. 1.15b can be used to generate a sine wave.
- †1.19** Devise a crank-rocker four-bar linkage, as in Fig. 1.14c, having a rocker angle of 60° . The rocker length is to be 0.50 m.
- 1.20** A crank-rocker four-bar linkage is required to have an advance-to-return ratio $Q = 1.2$. The rocker is to have a length of 2.5 in and oscillate through a total angle of 60° . Determine a suitable set of link lengths for the remaining three links of the four-bar linkage.

- 1.21** Determine the mobility of the mechanism. Number each link and label the lower pairs and the higher pairs. Identify a suitable input, or inputs, for the mechanism.

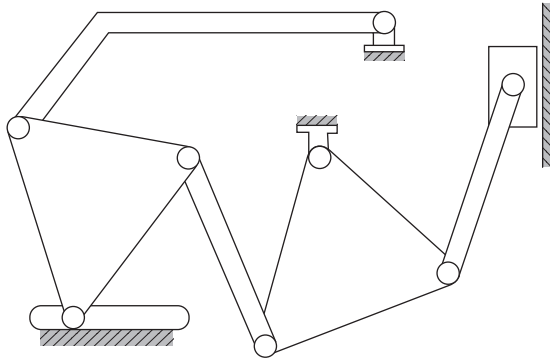


Figure P1.21

- 1.22** Determine the mobility of the mechanism. Number each link and label the lower pairs and the higher pairs. Identify a suitable input, or inputs, for the mechanism.

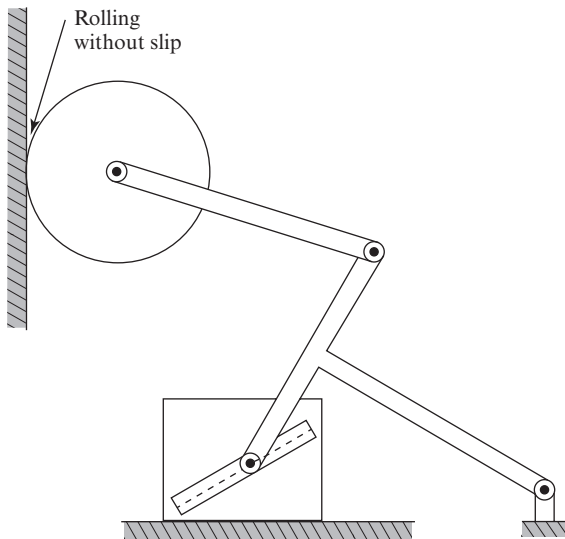


Figure P1.22

- 1.23** Determine the mobility of the mechanism. Number each link and label the lower pairs and the higher pairs. Identify a suitable input, or inputs, for the mechanism.

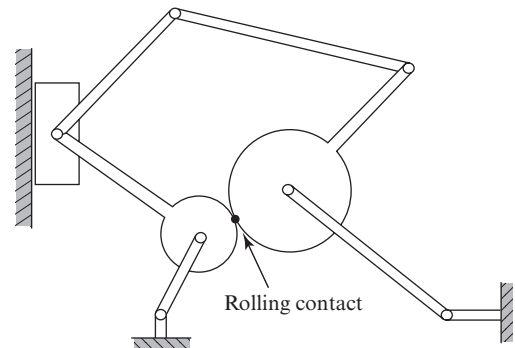


Figure P1.23

- 1.24** Determine the mobility of the mechanism. Number each link and label the lower pairs and the higher pairs. Identify a suitable input, or inputs, for the mechanism.

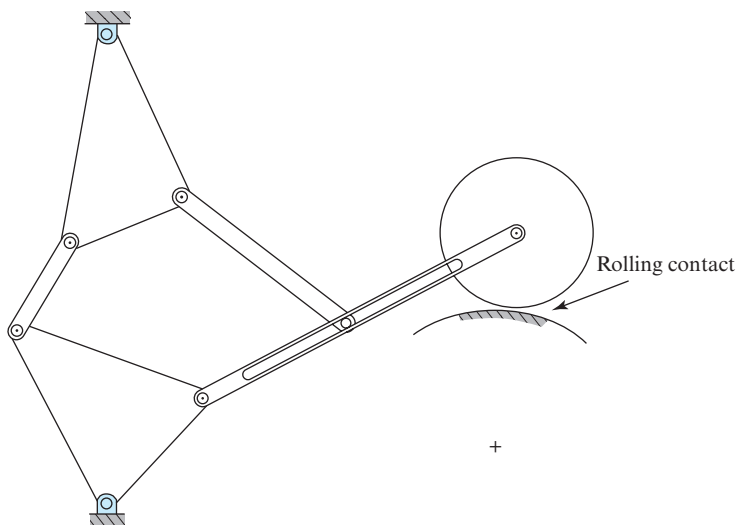


Figure P1.24

1.25 Determine the mobility of the mechanism. Number each link and label the lower pairs and the higher pairs. Identify a suitable input, or inputs, for the mechanism.

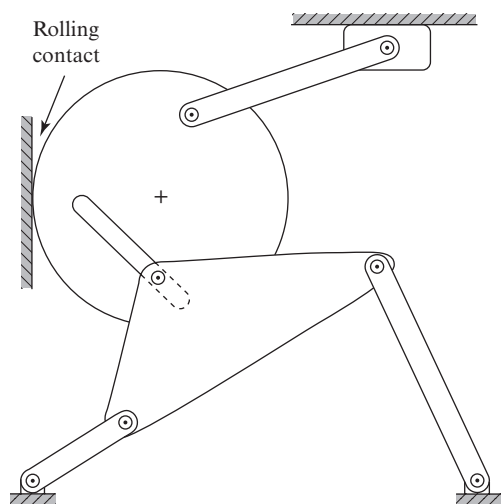


Figure P1.25

1.26 Determine the mobility of the mechanism. Number each link and label the lower pairs and the higher pairs. Identify a suitable input, or inputs, for the mechanism.

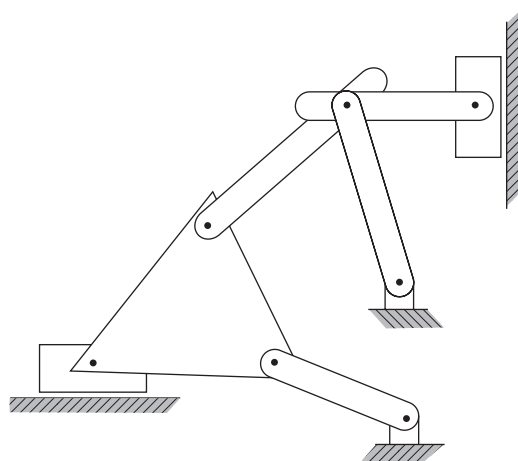


Figure P1.26

- 1.27** Determine the mechanical advantage of the four-bar linkage in the posture shown.

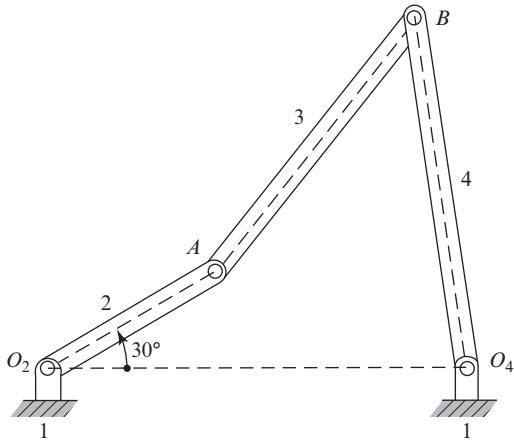


Figure P1.27 $O_2O_4 = 120$ mm, $O_2A = 60$ mm, $AB = 100$ mm, and $O_4B = 130$ mm.

- 1.28** Determine the mobility of the mechanism. Number each link and label the lower pairs and the higher pairs. Identify a suitable input, or inputs, for the mechanism.

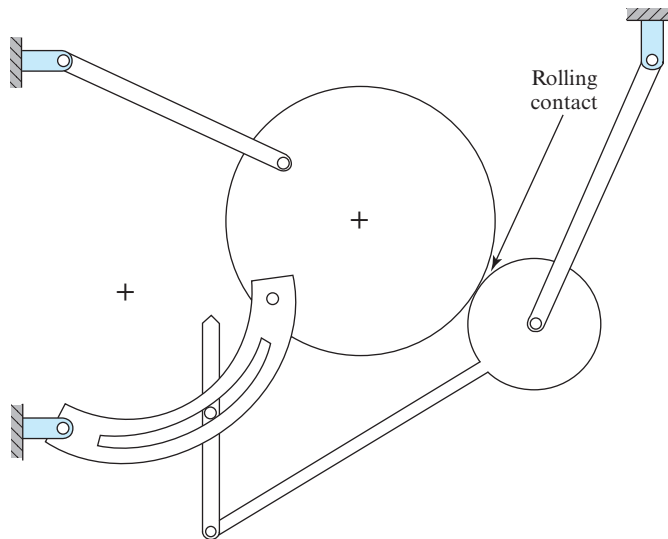


Figure P1.28

- 1.29** Determine the mobility of the mechanism. Number each link and label the lower pairs and the higher pairs. Identify a suitable input, or inputs, for the mechanism.

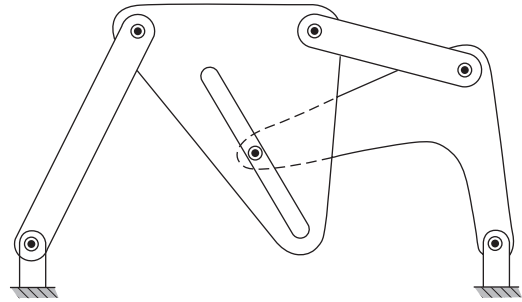


Figure P1.29

- 1.30** The rocker of a crank-rocker four-bar linkage is required to have a length of 6 in and swing through a total angle of 30° . Also, the advance-to-return ratio of the linkage is required to be 1.75. Determine a suitable set of link lengths for the remaining three links.
- 1.31** Determine a suitable set of link lengths for a slider-crank linkage such that the stroke will be 500 mm and the advance-to-return ratio will be 1.8.

- 1.32** Determine the transmission angle and the mechanical advantage of the four-bar linkage in the posture shown. What type of four-bar linkage is this?

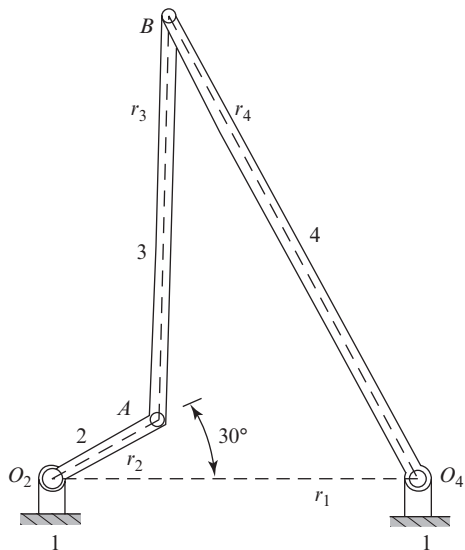


Figure P1.32 $R_{AO_2} = 20 \text{ mm}$, $R_{BA} = 70 \text{ mm}$, $R_{BO_4} = 90 \text{ mm}$, and $R_{O_4O_2} = 60 \text{ mm}$.

- 1.33** Determine the mobility of the mechanism. Number each link and label the lower pairs and the higher pairs. Identify a suitable input, or inputs, for the mechanism.

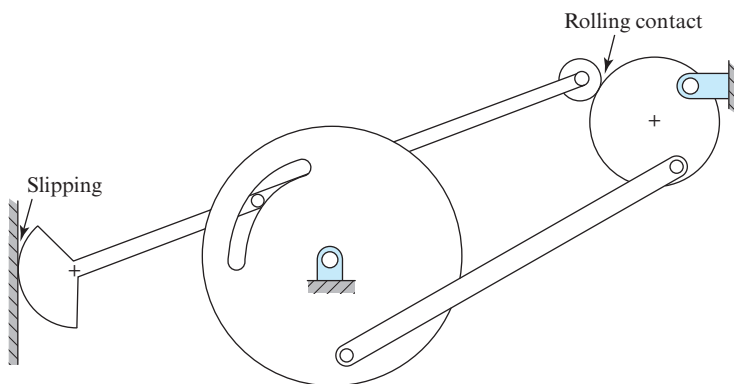


Figure P1.33

- 1.34** A crank-rocker four-bar linkage is shown in one of its two toggle postures. Find θ_2 and θ_4 corresponding to each toggle posture. What is the total rocking angle of link 4? What are the transmission angles at the extremes?

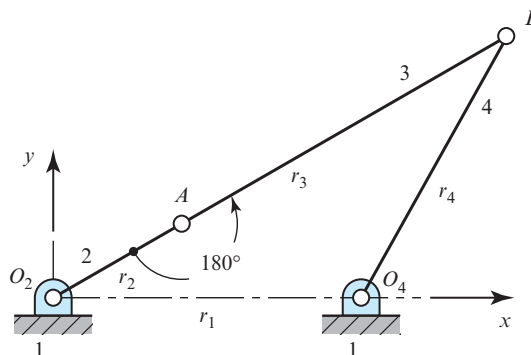


Figure P1.34 $R_{AO_2} = 8 \text{ in}$, $R_{BA} = 20 \text{ in}$, $R_{BO_4} = 16 \text{ in}$, and $R_{O_4O_2} = 16 \text{ in}$.

- 1.35** Find θ_2 and θ_4 corresponding to a dead-center posture. Is there a toggle posture?
- 1.36** Determine the advance-to-return ratio for the slider-crank linkage with the offset e . Also,

determine in which direction the crank should rotate to provide quick return.

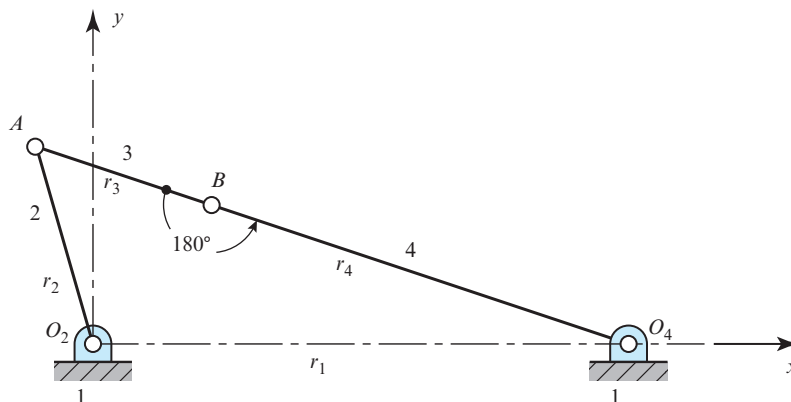


Figure P1.35 $R_{AO_2} = 110$ mm, $R_{BA} = 100$ mm, $R_{BO_4} = 240$ mm, and $R_{O_4O_2} = 280$ mm.

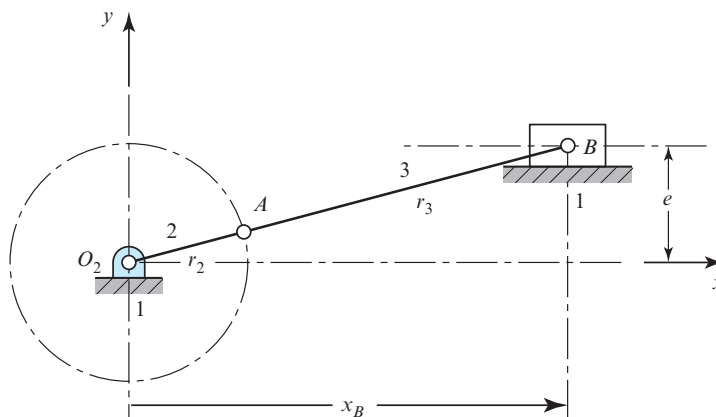
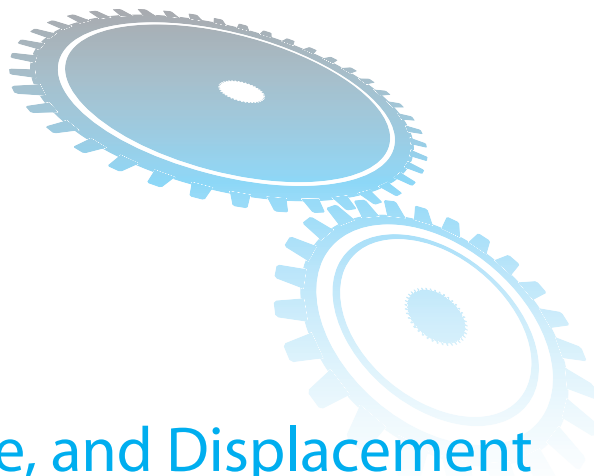


Figure P1.36 Offset slider-crank linkage in the two dead-center postures.

2

Position, Posture, and Displacement



In analyzing motion, the first and most basic problem encountered is that of defining and dealing with the concepts of position, posture, and displacement. Since motion can be thought of as a time series of displacements between successive positions of a point or postures of a body, it is important to understand exactly the meaning of the terms *position* and *posture*. Rules or conventions will be established here to make the definitions precise.

2.1 LOCUS OF A MOVING POINT

In speaking of the position of a point, or particle, we are really answering the question: Where is the point, or what is its location? We are speaking of something that exists in nature, and we are considering the question of how to express this (in words or symbols or numbers) in such a way that the meaning is clear. We soon discover that position cannot be defined on a truly absolute basis. We must define the position of a point in terms of some agreed-upon frame of reference (that is, some reference frame or reference coordinate system).

Once we have agreed upon the right-handed three-dimensional coordinate system xyz as the frame of reference, as shown in Fig. 2.1, we can say that point P is located x units along the x axis, y units along the y axis, and z units along the z axis *from the origin* O . In this statement, we see that three vitally important parts of the definition depend on the existence of the reference coordinate system:

1. The *origin* O of the coordinate system provides an agreed-upon location from which to measure the location of point P .

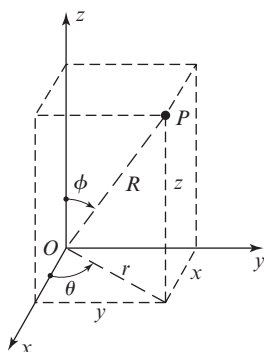


Figure 2.1 Right-handed three-dimensional coordinate system illustrating how point P is located algebraically.

2. The *coordinate axes* provide agreed-upon *directions* along which the measurements are to be made; they also provide known lines and planes for the definition and measurement of angles.
3. The unit distance along any of the axes provides a scale for quantifying distances.

These observations are not restricted to the Cartesian coordinates (x, y, z) of point P . All three properties of the coordinate system are also necessary in defining cylindric coordinates (r, θ, z) , spheric coordinates (R, θ, ϕ) , or any other coordinates of point P . The same properties are also required if point P is restricted to remain in a single plane and a two-dimensional coordinate system is used. No matter how it is defined, the concept of the position of a point cannot be realized without the definition of a reference coordinate system.

The direction cosines for locating point P are defined as

$$\cos \alpha = \frac{x}{R}, \quad \cos \beta = \frac{y}{R}, \quad \text{and} \quad \cos \gamma = \frac{z}{R}, \quad (2.1)$$

where α, β , and γ are the angles measured from the positive x -, y -, and z -coordinate axes, respectively, to the directed line OP .

One means of expressing the motion of a point, or particle, is to define its components along the reference axes as functions of some parameter, such as time:

$$x = x(t), \quad y = y(t), \quad z = z(t). \quad (2.2)$$

If these relations are known, then the position of the point can be found for any time t . This is the general case for the motion of a point, or particle, and is illustrated in the following example.

EXAMPLE 2.1

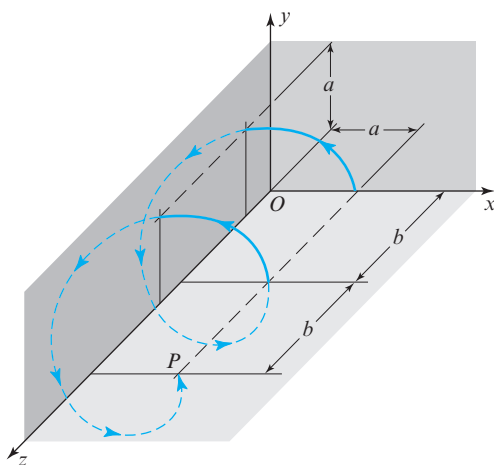
Describe the motion of a particle, P whose position changes with time according to the equations $x = a \cos 2\pi t$, $y = a \sin 2\pi t$, and $z = bt$.

SOLUTION

Substituting values for t from 0 to 2 s gives the coordinates listed in Table 2.1. As shown in Fig. 2.2, the particle moves with *helical motion* with radius a around the z axis and with a lead of b . Note that if $b = 0$, then $z(t) = 0$, the moving particle is confined to the xy plane, and the motion is a circle with its center at the origin.

Table 2.1 Helical Motion of a Particle

t	x	y	z
0	a	0	0
1/4	0	a	$b/4$
1/2	$-a$	0	$b/2$
3/4	0	$-a$	$3b/4$
1	a	0	b
5/4	0	a	$5b/4$
3/2	$-a$	0	$3b/2$
7/4	0	$-a$	$7b/4$
2	a	0	$2b$

**Figure 2.2** Helical motion of a particle, P .

So far, we have been using the words *point* and *particle* interchangeably. Henceforth, when the word *point* is used, we have in mind something that has no dimensions—that is, something with zero length, zero width, and zero thickness. When the word *particle* is used, we have in mind something whose dimensions are unimportant—that is, a material body whose dimensions are negligible or, in other words, a body whose dimensions have no effect on the analysis to be performed.

The successive positions of a moving point define a line or curve. This curve has no thickness, since the point has no dimensions. However, the curve does have length,

since the point occupies different positions as time changes. This curve, representing the successive positions of the point, is called the *path* or *locus* of the moving point in the reference coordinate system.

If three coordinates are necessary to describe the path of a moving point, the point is said to have *spatial motion*. If the path can be described by only two coordinates—that is, if the coordinate axes can be chosen such that one coordinate is always zero or constant—the path lies in a single plane, and the point is said to have *planar motion*. Sometimes it happens that the path of a point can be described by a single coordinate—that is, two of the spatial position coordinates are zero or constant. In this case, the path of the point is a straight line, and the point is said to have *rectilinear motion*.

In each of these three cases, it is assumed that the coordinate system is chosen so as to obtain the least number of coordinates necessary to describe the motion of the point. Thus, a point whose locus is a *spatial curve*, sometimes called a *skew curve*, requires three position coordinates, a point whose locus is a *planar curve* requires two coordinates, and a point whose locus is a straight line (rectilinear motion) requires only one coordinate.

2.2 POSITION OF A POINT

The physical process involved in observing the position of a point, as shown in Fig. 2.3, implies that the observer is actually keeping track of the relative location of two points, P and O . The observer looks at both points, performs a mental comparison, and recognizes that point P has a certain location *with respect to point O* . In this determination, two properties are noted: the distance from O to P (based on the unit distance or grid size of the reference coordinate system), and the *relative angular orientation* of line OP in the coordinate system. Note that these properties, magnitude and direction, are precisely those required for a vector. Therefore, we define the *position of a point* as the *vector from the origin of a specified reference coordinate system to the point*. We choose the symbol $\mathbf{R}_{PO}$

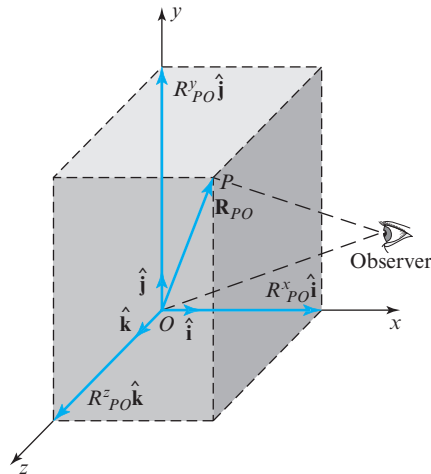


Figure 2.3 Position of a point defined by a vector.

to denote the vector position of point P relative to point O , which points from O to P and is read as *the position of P with respect to O* .*

The coordinate system is related in a very special way to what is seen by a specific observer. This raises the important question: What properties must this coordinate system have to ensure that position measurements made in this system are actually those of the observer? The key to this question is that the coordinate system *must* be stationary with respect to this particular observer. To phrase this another way, the observer is always *stationary* in this coordinate system. This means that if the observer moves, the coordinate system also moves—through a rotation, a distance, or both. If points or bodies are fixed in this coordinate system, then they always appear stationary to the observer, regardless of what movements the observer (and the coordinate system) may execute. Their positions with respect to the observer do not change, and hence their position vectors remain unchanged. The actual location of the observer within the frame of reference has no meaning because the positions of observed points are always defined with respect to the origin of the coordinate system.

Often it is convenient to express the position vector in terms of its components along the coordinate axes:

$$\mathbf{R}_{PO} = R_{PO}^x \hat{\mathbf{i}} + R_{PO}^y \hat{\mathbf{j}} + R_{PO}^z \hat{\mathbf{k}}, \quad (2.3)$$

where superscripts are used to denote the direction of each component. As in the remainder of this text, $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$ are used to designate unit vectors in the directions of the x -axis, y -axis, and z -axis, respectively.

Whereas vectors are denoted throughout this text by boldface symbols, the scalar magnitude of a vector is signified by the same symbol in italics, without boldface.

For example, the magnitude of the position vector is

$$R_{PO} = |\mathbf{R}_{PO}| = \sqrt{\mathbf{R}_{PO} \cdot \mathbf{R}_{PO}} = \sqrt{(R_{PO}^x)^2 + (R_{PO}^y)^2 + (R_{PO}^z)^2}. \quad (2.4)$$

The unit vector in the direction of $\mathbf{R}_{PO}$ is denoted by the same boldface symbol with a caret; that is

$$\hat{\mathbf{R}}_{PO} = \frac{\mathbf{R}_{PO}}{R_{PO}}. \quad (2.5)$$

A distinction can be made between the direction of a line and the orientation of a directed line—that is, a line that is assigned a positive or a negative sense. In general, a vector has a magnitude, a direction, and a sense. The sense defines the positive or negative attribute of the vector and can be used to distinguish between the direction of a vector and its orientation.

* Note that we do not use the slash notation of some other texts; that is, $\mathbf{R}_{PO} \neq \mathbf{R}_{P/O}$. The slash notation is reserved for a different meaning and is described in Sec. 2.4.

2.3 POSITION DIFFERENCE BETWEEN TWO POINTS

We now investigate the relationship between the position vectors of two different points. For purposes of illustration, consider points P and Q in Fig. 2.4. An observer fixed in the xyz coordinate system would observe the positions of P and Q by comparing each with the position of the origin O ; see Eq. (2.3). The positions of the two points are defined by vectors $\mathbf{R}_{PO}$ and $\mathbf{R}_{QO}$. These two vectors are related by a third vector, $\mathbf{R}_{PQ}$, the *position difference* to point P from point Q ; that is,

$$\mathbf{R}_{PQ} = \mathbf{R}_{PO} - \mathbf{R}_{QO}. \quad (2.6)$$

The physical interpretation of this equation is different from that of the position vector itself. The observer is no longer defining the position of P with respect to O but is now defining the position of P with respect to Q . Put another way, the position of P is being defined as if it were in the $x'y'z'$ coordinate system with origin at Q and axes directed parallel to the xyz coordinate system. Either point of view can be used for the interpretation, but we should understand both of them since both are used in future developments.

Finally, it is worth remarking that having the $x'y'z'$ axes parallel to the x,y,z axes is only a matter of convenience and not a necessary condition; it is only necessary that $x'y'z'$ coordinate system does not rotate with respect to the xyz coordinate system. This parallelism causes no loss of generality and it simplifies the visualization when the coordinate systems are in motion.

Having now generalized our concept of position to include the position difference between any two points, we reflect again on the previous discussion of the position vector itself. We note that it is merely the special case where we agree to use the origin of coordinates as the second point. Thus, to be consistent in notation, we have denoted the position vector of a single point P by the dual subscripted symbol $\mathbf{R}_{PO}$. However, in the interest of brevity, we will henceforth agree that when the second subscript is not given explicitly, it is understood to be the origin of the observer's coordinate system; that is,

$$\mathbf{R}_P = \mathbf{R}_{PO}. \quad (2.7)$$

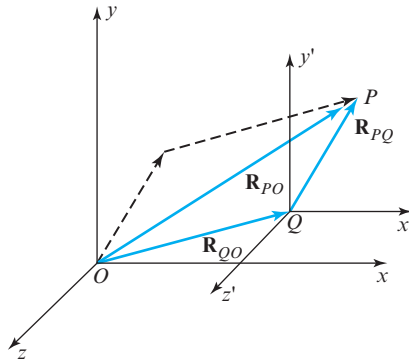


Figure 2.4 Definition of the position-difference vector $\mathbf{R}_{PQ}$.

2.4 APPARENT POSITION OF A POINT

In discussing the position vector, our point of view, up to now, has been entirely that of a single observer in a single coordinate system. However, it is sometimes desirable to make observations in a secondary coordinate system—that is, as seen by a second observer in a different coordinate system, and then to convert this information into the primary coordinate system. Such a situation is shown in Fig. 2.5.

If observer 1, using the primary coordinate system $x_1y_1z_1$, and observer 2, using the secondary coordinate system $x_2y_2z_2$, were asked to give the location of a point at P , they would report different results. Observer 1 would observe the vector $\mathbf{R}_{PO_1}$, whereas observer 2 would report the position vector $\mathbf{R}_{PO_2}$. These position vectors are related by

$$\mathbf{R}_{PO_1} = \mathbf{R}_{O_2O_1} + \mathbf{R}_{PO_2}. \quad (2.8)$$

The difference in the positions of the two origins is not the only discrepancy between the two observations of the position of point P . Since the two coordinate systems are not aligned, the two observers are using different reference lines for their measurements of direction; observer 1 measures along the x_1, y_1, z_1 axes while observer 2 measures along the x_2, y_2, z_2 axes.

Another very important distinction between these two observations arises when we consider that the two coordinate systems could be moving with respect to each other. Whereas point P may appear stationary with respect to one observer, while it is in motion with respect to the other observer; that is, position vector $\mathbf{R}_{PO_1}$ may appear constant to observer 1 while $\mathbf{R}_{PO_2}$ appears to vary as seen by observer 2.

When any of these conditions exists, it is convenient to add an additional subscript to our notation that will distinguish which observer is being considered. When we are considering the position of P as seen by observer 1 using coordinate system $x_1y_1z_1$, we denote this by the symbol $\mathbf{R}_{PO_1/1}$, or, since O_1 is the origin for this observer,* by $\mathbf{R}_{P/1}$. The observations made by observer 2, in coordinate system $x_2y_2z_2$, are denoted as $\mathbf{R}_{PO_2/2}$

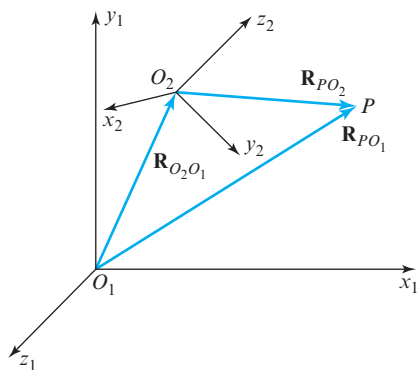


Figure 2.5 Definition of the apparent-position vector $\mathbf{R}_{PO_2}$ of point P .

* Note that $\mathbf{R}_{PO_2/1}$ cannot be abbreviated as $\mathbf{R}_{P/1}$, since O_2 is not the origin used by observer 1.

or $\mathbf{R}_{P/2}$. With this extension of the notation, Eq. (2.8) can be written as

$$\mathbf{R}_{P/1} = \mathbf{R}_{O_2/1} + \mathbf{R}_{P/2}. \quad (2.9)$$

We refer to $\mathbf{R}_{P/2}$ as *the apparent position* of point P to an *observer in coordinate system 2*, and we note that it is by no means equal to the apparent position vector $\mathbf{R}_{P/1}$ seen by observer 1.

We have now made note of certain intrinsic differences between $\mathbf{R}_{P/1}$ and $\mathbf{R}_{P/2}$ and found Eq. (2.9) to relate them. However, there is no reason why components of either vector must be taken along the natural axes of the observer's coordinate system. As with all vectors, components can be found along any desired set of axes; in applying Eq. (2.9), we must use a single consistent set of axes during the numeric evaluation. Although the observer in coordinate system 2 may find it natural to measure the components of $\mathbf{R}_{P/2}$ along the $x_2y_2z_2$ axes, these components must be transformed into the equivalent components along the $x_1y_1z_1$ axes before the addition is actually performed; that is,

$$\begin{aligned} \mathbf{R}_{P/1} &= \mathbf{R}_{O_2/1} + \mathbf{R}_{P/2} \\ &= R_{O_2/1}^{x_1} \hat{\mathbf{i}}_1 + R_{O_2/1}^{y_1} \hat{\mathbf{j}}_1 + R_{O_2/1}^{z_1} \hat{\mathbf{k}}_1 + R_{P/2}^{x_1} \hat{\mathbf{i}}_1 + R_{P/2}^{y_1} \hat{\mathbf{j}}_1 + R_{P/2}^{z_1} \hat{\mathbf{k}}_1 \\ &= (R_{O_2/1}^{x_1} + R_{P/2}^{x_1}) \hat{\mathbf{i}}_1 + (R_{O_2/1}^{y_1} + R_{P/2}^{y_1}) \hat{\mathbf{j}}_1 + (R_{O_2/1}^{z_1} + R_{P/2}^{z_1}) \hat{\mathbf{k}}_1 \\ &= R_{P/1}^{x_1} \hat{\mathbf{i}}_1 + R_{P/1}^{y_1} \hat{\mathbf{j}}_1 + R_{P/1}^{z_1} \hat{\mathbf{k}}_1. \end{aligned}$$

The addition can be performed equally well if all vector components are transformed into the $x_2y_2z_2$ coordinate system or, for that matter, into any other consistent coordinate system. However, they cannot be added algebraically when they have been evaluated along inconsistent coordinate systems. The additional subscript in the apparent-position vector, therefore, does not specify a set of directions to be used in the evaluation of components; it merely states the coordinate system in which the vector is defined, the coordinate system in which the observer is stationary.

2.5 ABSOLUTE POSITION OF A POINT

In Sec. 2.2 we learned that every position vector is defined relative to a second point, the origin of the observer's coordinate reference frame. It is one particular case of the position-difference vector studied in Sec. 2.3, where the reference point is the origin of the coordinate system. Then in Sec. 2.4, we noted that, for certain problems, it may be convenient to consider the apparent positions of a single point as viewed by more than one observer using different coordinate systems. When a particular problem leads us to consider multiple coordinate systems, the application will lead us to choose one of the coordinate systems as primary. This primary coordinate system is commonly referred to as the *absolute coordinate system*. Most often this is the coordinate system in which the final result is to be expressed, and this coordinate system is usually stationary. The absolute position of a point is defined as the apparent position of the point as seen by an observer in the *absolute coordinate system*.

Which coordinate system is designated absolute is an arbitrary decision and unimportant in the study of kinematics. Whether the absolute coordinate system is truly stationary is also a moot point, since, as we have seen, all position (and motion) information is measured relative to something else; nothing is truly absolute in the strict sense. When analyzing the kinematics of an automobile suspension, for example, it may be convenient to choose an “absolute” coordinate system attached to the frame of the car and to study the motion of the suspension relative to this coordinate system. It is unimportant whether the car is moving; motions of the suspension relative to the frame are then defined as absolute.

It is a common convention to number the absolute coordinate system 1 and to use other numbers for other coordinate systems. Since we adopt this convention throughout this text, absolute-position vectors are those apparent-position vectors viewed by an observer in coordinate system 1 and carry symbols of the form $\mathbf{R}_{P/1}$. In the interest of brevity and to reduce complexity, we will agree that when the coordinate system number is not given explicitly, it is assumed to be 1; thus, $\mathbf{R}_{P/1}$ can be abbreviated as $\mathbf{R}_P$. Similarly, the apparent-position equation, Eq. (2.9), can be written* as

$$\mathbf{R}_P = \mathbf{R}_{O_2} + \mathbf{R}_{P/2}. \quad (2.10)$$

EXAMPLE 2.2

The path of a moving point is defined by the equation $y = 2x^2 - 28$. Find the position difference from point P to point Q on the path where $R_P^x = 4$ and $R_Q^x = -3$.

SOLUTION

The y components of the two vectors can be written as

$$R_P^y = 2(4)^2 - 28 = 4 \quad \text{and} \quad R_Q^y = 2(-3)^2 - 28 = -10.$$

Therefore, the two vectors can be written as

$$\mathbf{R}_P = 4\hat{\mathbf{i}} + 4\hat{\mathbf{j}} \quad \text{and} \quad \mathbf{R}_Q = -3\hat{\mathbf{i}} - 10\hat{\mathbf{j}}.$$

The position difference from point P to point Q is

$$\mathbf{R}_{QP} = \mathbf{R}_Q - \mathbf{R}_P = -7\hat{\mathbf{i}} - 14\hat{\mathbf{j}} = 15.65\angle -116.6^\circ. \quad \text{Ans.}$$

2.6 POSTURE OF A RIGID BODY

Consider the term *position* when applied to something other than a point. In order to specify the location of a rigid body, for example, it is necessary to specify more than just three coordinates. It is necessary to specify coordinates that uniquely determine the location

* Reviewing Secs. 2.1 through 2.3 will verify that the position-difference vector $\mathbf{R}_{PQ}$ was treated entirely from the absolute coordinate system and is an abbreviation of the notation $\mathbf{R}_{PQ/1}$. We have no need to treat the completely general case $\mathbf{R}_{PQ/2}$, the apparent-position-difference vector.

of every point of the body. If all of these coordinates are grouped into a single quantity according to some agreed-upon set of conventions, then the result describes the location of the body.

The location and orientation of a coordinate system fixed to the body with respect to a stationary reference, or world coordinate system, describes the *posture* of that body. Posture is described in terms of the location of the origin of the coordinate system fixed to the body as well as a description of the orientation of this coordinate system, both specified with respect to the world coordinate system.

In the robotics literature, for example, a matrix is often used to describe the location and orientation of a coordinate system attached to the end-effector with respect to a base or world coordinate system (Chap. 10). In some robotics literature, the word *position* is used loosely to describe the location of only a single point (such as the origin) of a coordinate system attached to the end-effector. In such literature, the end-effector is said to have a certain position; the orientation may then be added and the term *pose* is sometimes used for the combination of the two. It should be pointed out, however, that the term *posture* is more appropriate* and is utilized throughout this text.

The term *posture* becomes even more suitable when dealing with a mechanism or multibody mechanical system, because we are not concerned with the position of only a single point or the position and orientation of only a single rigid body, but we wish to describe the positions and orientations of the assembly of rigid bodies. We use the term *posture* to describe the location and orientation of a rigid body or for the configuration of a mechanism, including both the locations and orientations of every link, all at a particular instant in time.

The problem of posture analysis is to determine the values of all position variables (the positions of all points and joints) and the postures of all links, given the dimensions of each link and the value(s) of the independent variable(s)—that is, the variables chosen to represent the degree(s) of freedom of the mechanism.

2.7 LOOP-CLOSURE EQUATIONS

Our discussion of the position-difference and apparent-position vectors has been somewhat abstract so far, the intent being to develop a rigorous foundation for the analysis of motion in mechanical systems. Certainly, precision is not without merit, since it is rigor that permits science to predict a correct result despite the personal prejudices and emotions of the analyst. However, tedious developments are not interesting unless they lead to applications in real-life problems. Although many fundamental principles are yet to be discovered, it may be worthwhile at this point to show the relationship between the relative-position vectors discussed earlier and some typical linkages encountered in real machines.

* The *Webster Comprehensive Dictionary: International Edition*, states under its definition of attitude, “Synonyms: pose, position, posture. A posture is assumed without any special reference to expression of feeling. . . . A pose is a position studied for artistic effect or considered with reference to such an effect.”

As pointed out in Chap. 1, one of the most common and most useful of all mechanisms is the planar four-bar linkage. A practical example of this linkage is the clamping device shown in Figs 2.6 and 2.7. A brief study of the assembly drawing indicates that, as the handle of the clamp is lifted, the clamping bar swings away from the clamping surface, thereby opening the clamp. As the handle is pressed down, the clamping bar swings down and the clamp closes again. If we wish to design such a clamp, however, things are not quite this simple. It may be desirable, for example, for the clamp to open at a given rate for a specified rate of lift of the handle. Such relationships are not obvious; they depend on the exact dimensions of the various parts and the relationships, or interactions, between the parts. To discover these relationships, a rigorous description of the essential features of the device is required. The position-difference and apparent-position vectors can be used to provide a description of any posture of the clamp.

Figure 2.7 shows the detail drawings of the individual links of the disassembled clamp. Although not shown here, the detail drawings would be completely dimensioned, thus fixing the complete geometry of each link. The assumption that each link is rigid ensures

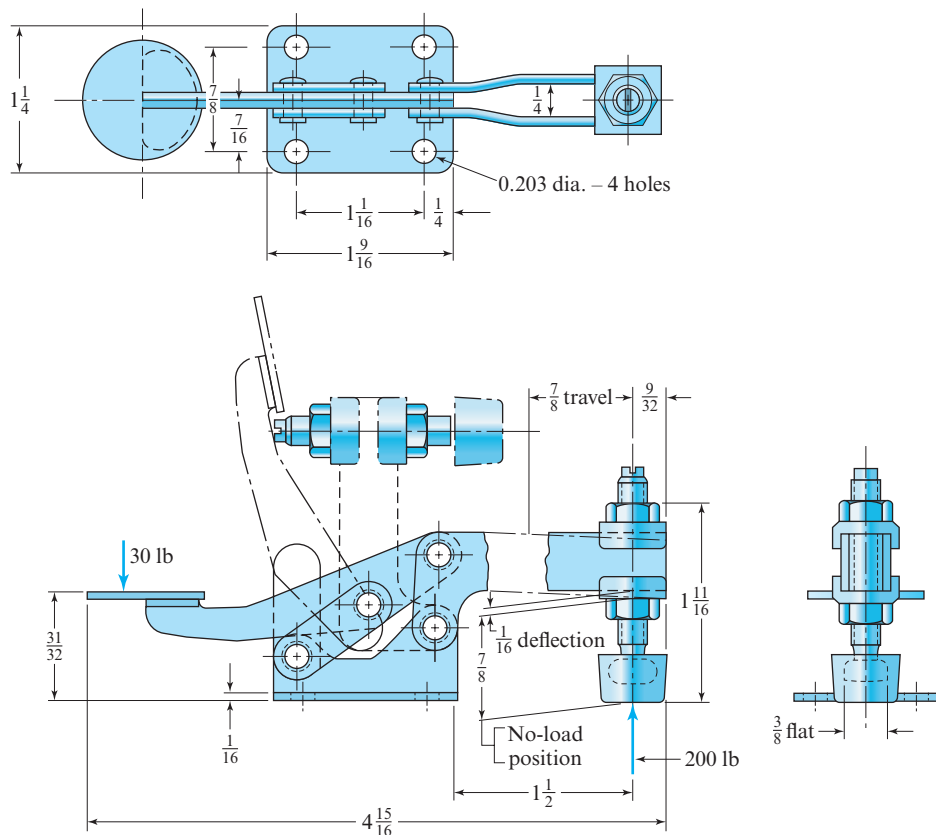


Figure 2.6 Assembly drawing of a hand-operated clamp.

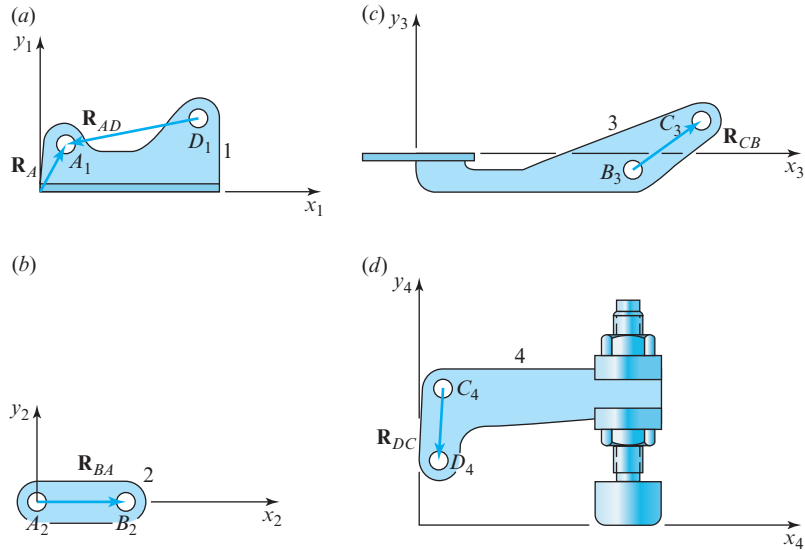


Figure 2.7 (a) Frame link; (b) connecting link; (c) handle; (d) clamping bar.

that the position of any point is determined precisely relative to any other point on the same link. However, the detail drawings do not provide the interrelationships between the individual parts—that is, the constraints that ensure each link moves relative to its neighboring links in a prescribed fashion. These constraints are, of course, provided by the four pinned joints. Anticipating that they will be of importance in any description of the linkage, we label the pin centers as A , B , C , and D , and we identify the appropriate points on link 1 as A_1 and D_1 , those on link 2 as A_2 and B_2 , and so on. We also choose a coordinate system fixed to each link, as shown in Fig. 2.7.

Since it is necessary to relate the positions of the successive pin centers, we define the position-difference vectors $\mathbf{R}_{AD}$ on link 1, $\mathbf{R}_{BA}$ on link 2, $\mathbf{R}_{CB}$ on link 3, and $\mathbf{R}_{DC}$ on link 4. We note that each vector appears constant to an observer fixed in the coordinate system of that link; the magnitude and direction of each vector can be obtained from the constant dimensions of that link.

A vector equation can also be written to describe the constraints provided by each of the pinned joints. Note that no matter which position or which observer is chosen, the two points describing each pin center (for example, A_1 and A_2) remain coincident. Thus,

$$\mathbf{R}_{A_2A_1} = \mathbf{R}_{B_3B_2} = \mathbf{R}_{C_4C_3} = \mathbf{R}_{D_1D_4} = \mathbf{0}. \quad (2.11)$$

Let us now develop vector equations for the absolute position of each pin center. Since link 1 is the frame, absolute positions are those defined relative to an observer in coordinate system 1. Point A_1 is, of course, at the position described by $\mathbf{R}_A$. Next, we connect link 2

to link 1 (mathematically) by writing

$$\mathbf{R}_{A_2} = \mathbf{R}_{A_1} + \mathbf{R}_{A_2A_1}^0 = \mathbf{R}_A. \quad (a)$$

Transferring to the other end of link 2, we attach link 3. Therefore,

$$\mathbf{R}_B = \mathbf{R}_A + \mathbf{R}_{BA}. \quad (b)$$

Connecting joints C and D in the same manner, we obtain

$$\mathbf{R}_C = \mathbf{R}_B + \mathbf{R}_{CB} = \mathbf{R}_A + \mathbf{R}_{BA} + \mathbf{R}_{CB}, \quad (c)$$

$$\mathbf{R}_D = \mathbf{R}_C + \mathbf{R}_{DC} = \mathbf{R}_A + \mathbf{R}_{BA} + \mathbf{R}_{CB} + \mathbf{R}_{DC}. \quad (d)$$

Then, transferring back across link 1 to point A , we have

$$\mathbf{R}_A = \mathbf{R}_D + \mathbf{R}_{AD} = \mathbf{R}_A + \mathbf{R}_{BA} + \mathbf{R}_{CB} + \mathbf{R}_{DC} + \mathbf{R}_{AD}. \quad (e)$$

Finally, rearranging Eq. (e), we obtain

$$\mathbf{R}_{BA} + \mathbf{R}_{CB} + \mathbf{R}_{DC} + \mathbf{R}_{AD} = \mathbf{0}. \quad (2.12)$$

This important equation is called the *loop-closure* equation, or the *vector loop* equation, for the clamp. As shown in Fig. 2.8, it expresses the fact that this linkage forms a closed loop. Therefore, the polygon formed by the position-difference vectors through successive links and joints must remain closed as the linkage moves. The constant lengths of these vectors ensure that the joint centers remain separated by constant distances, the requirement for rigid links. The rotations between successive vectors indicate the motions within the pinned joints. Recall that the rotation of a position-difference vector shows the rotation of a particular link. Thus, the loop-closure equation holds within it all the important constraints that illustrate the operation of the clamp. This equation forms a mathematical description, or *model*, of the linkage, and many of the later developments are based on this loop-closure model as a starting point.

Of course, the form of the loop-closure equation depends on the type of mechanism. This is demonstrated by another example, the *Geneva* mechanism or *Maltese cross*, shown in Fig. 2.9. One early application of this mechanism was to prevent overwinding a watch.

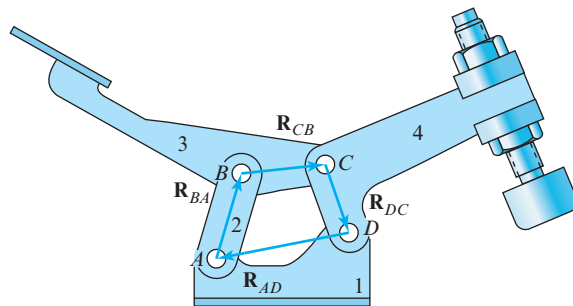


Figure 2.8 Loop-closure equation.

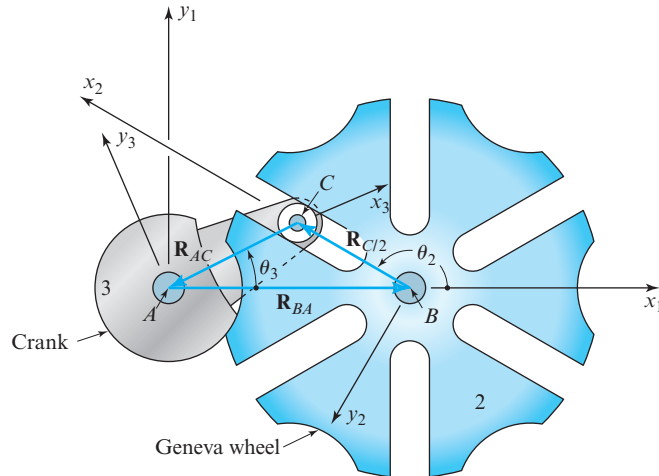


Figure 2.9 Geneva mechanism or Maltese cross.

Today the mechanism finds wide use as an indexing device, for example, in a milling machine with an automatic tool changer.

First, we define vectors for the absolute posture of each link. Although the frame of the mechanism, link 1, is not shown in Fig. 2.9, it is an important part of the mechanism, since it holds the two shafts, with centers A and B , a constant distance apart. Thus we define the vector $\mathbf{R}_{BA}$ to show this dimension. The left crank, link 3, is attached to a shaft, usually rotating at a constant speed, and carries a roller at C , running in the slot of the Geneva wheel. The vector $\mathbf{R}_{AC}$ has a constant magnitude equal to the crank length, the distance from the center of the roller C to the shaft center A . The rotation of this vector relative to link 1 is used later to describe the angular speed of the crank. The x_2 axis is aligned along one slot of the wheel; thus the roller is constrained to ride along this slot. The vector $\mathbf{R}_{C/2}$ has the same rotation as the wheel, link 2; also, the change in length, $\Delta \mathbf{R}_{C/2}$, demonstrates the relative sliding motion taking place between the roller C and the slot in link 2. Note that $\mathbf{R}_{C/2}$ is equivalent to $\mathbf{R}_{CB/2}$, since point B is the origin of coordinate system 2. Therefore, the loop-closure equation for this mechanism can be written as

$$\mathbf{R}_{BA} + \mathbf{R}_{C/2} + \mathbf{R}_{AC} = \mathbf{0}. \quad (2.13)$$

This form of the loop-closure equation is a valid mathematical model only while roller C remains in the slot along the x_2 axis. However, this condition does not hold throughout the entire cycle of motion. Once the roller leaves the slot, the motion is controlled by the two mating circular arcs on links 2 and 3. A new form of the loop-closure equation rules that portion of the cycle.

A mechanism can, of course, be composed of a multiple-loop kinematic chain. In such a case, more than one loop-closure equation is required to completely model the mechanism. The procedures for obtaining the loop-closure equations, however, are identical to those shown in the previous two examples.

2.8 GRAPHIC POSTURE ANALYSIS

When the paths of the points in the moving links of a mechanism lie in a single plane or in parallel planes, the mechanism is called a *planar* mechanism. Since a substantial portion of the study in this book deals with planar mechanisms, the development of methods specific to the analysis of planar mechanisms is justified. As we will see in the following section, the nature of the loop-closure equation approached analytically often leads to the solution of simultaneous nonlinear equations and can become quite cumbersome. Yet, particularly for planar mechanisms, the solution is usually straightforward when approached graphically.

First, let us briefly review the graphic process of vector addition. Any two known vectors **A** and **B** can be added graphically as shown in Fig. 2.10*a*. After a scale is chosen, the vectors are drawn tip to tail in either order and their vector sum **C** is identified:

$$\mathbf{C} = \mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}. \quad (2.14)$$

Note that the magnitudes and orientations of both vectors **A** and **B** are used in performing the addition, and that both the magnitude and orientation of the sum **C** are found as a result.

The operation of graphic vector subtraction is shown in Fig. 2.10*b*, where the vectors are drawn tip to tip in solving the equation

$$\mathbf{A} = \mathbf{C} - \mathbf{B}. \quad (2.15)$$

These graphic vector operations should be studied carefully and understood, since they are used extensively throughout the book.

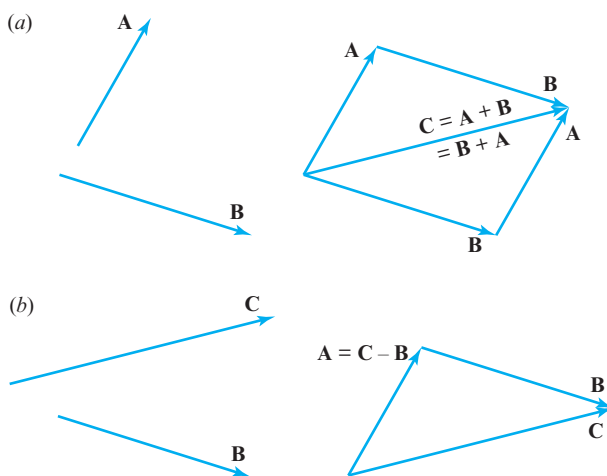


Figure 2.10 (a) Vector addition; (b) vector subtraction.

A spatial (three-dimensional) vector equation, such as

$$\mathbf{C} = \mathbf{D} + \mathbf{E} + \mathbf{B}, \quad (a)$$

can be divided into components along any convenient axes, leading to the three scalar equations

$$C^x = D^x + E^x + B^x, \quad C^y = D^y + E^y + B^y, \quad \text{and} \quad C^z = D^z + E^z + B^z. \quad (b)$$

Since they are components of the same vector equation, these three scalar equations must be consistent. If the three equations are also linearly independent, they can be solved simultaneously for three unknowns, which may be three magnitudes, three directions, or any combination of three magnitudes and directions. For some combinations, however, the problem is highly nonlinear and quite difficult to solve. Therefore, we shall delay consideration of the three-dimensional case until it is needed in Chap. 10.

A planar (two-dimensional) vector equation can be solved for two unknowns—namely, two magnitudes, two directions, or one magnitude and one direction. In general, it is desirable to indicate the known ($\surd$) and unknown (?) quantities above each vector in an equation like this:

$${}^? \surd \mathbf{C} = \surd \surd \mathbf{D} + \surd \surd \mathbf{E} + {}^? \surd \mathbf{B}, \quad (c)$$

where the first symbol ($\surd$ or $?$) above each vector indicates the state of its magnitude and the second symbol indicates the state of its direction. Another, equivalent form is

$${}^? \hat{\surd} \mathbf{C} = \surd \hat{\surd} \mathbf{D} + \surd \hat{\surd} \mathbf{E} + {}^? \hat{\surd} \mathbf{B}. \quad (d)$$

Either of Eqs. (c) or (d) clearly identifies the unknowns and indicates whether a solution is possible. In Eq. (c), vectors $\mathbf{D}$ and $\mathbf{E}$ are completely defined and can be replaced by their sum

$$\mathbf{A} = \mathbf{D} + \mathbf{E}, \quad (e)$$

giving

$$\mathbf{C} = \mathbf{A} + \mathbf{B}. \quad (2.16)$$

Note that any planar vector equation, if it is solvable, can be reduced to a three-term equation with two unknowns. Depending on the forms of the two unknowns, four distinct cases occur. These cases and the corresponding unknowns are presented in Table 2.2.

We will show the solutions of these four cases graphically.

In case 1, the two unknowns are the magnitude and the direction of the same vector (say, C and θ_C). This case can be solved by graphic addition or subtraction of the remaining two vectors, which are completely defined. This case was illustrated in Fig. 2.10.

Table 2.2 Unknowns in Planar Vector Equations

Case	Unknowns		
1	C, θ_C	Fig. 2.10	Eq. (2.14)
2	A, θ_B	Fig. 2.11	Eq. (2.17)
3	A, B	Fig. 2.12	Eq. (2.18)
4	θ_A, θ_B	Fig. 2.13	Eq. (2.19)

In case 2, the two unknowns are a magnitude and a direction from different vectors (say, A and θ_B). The vector equation is

$$\overset{\sqrt{\sqrt{}}}{\mathbf{C}} = \overset{? \sqrt{}}{\mathbf{A}} + \overset{\sqrt{?}}{\mathbf{B}}. \quad (2.17)$$

The solution, illustrated in Fig. 2.11, is obtained as follows:

1. Choose a convenient scale factor and draw vector $\mathbf{C}$.
2. Construct a line through the origin of $\mathbf{C}$ parallel to $\hat{\mathbf{A}}$.
3. Adjust a compass to the scaled magnitude B and construct a circular arc with the center at the terminus of $\mathbf{C}$.
4. The two intersections of the line and the arc define the two sets of solutions $A, \hat{\mathbf{B}}$ and $A', \hat{\mathbf{B}}'$.

In case 3, the two unknowns are two magnitudes (say, A and B). The vector equation is

$$\overset{\sqrt{\sqrt{}}}{\mathbf{C}} = \overset{? \sqrt{}}{\mathbf{A}} + \overset{? \sqrt{}}{\mathbf{B}}. \quad (2.18)$$

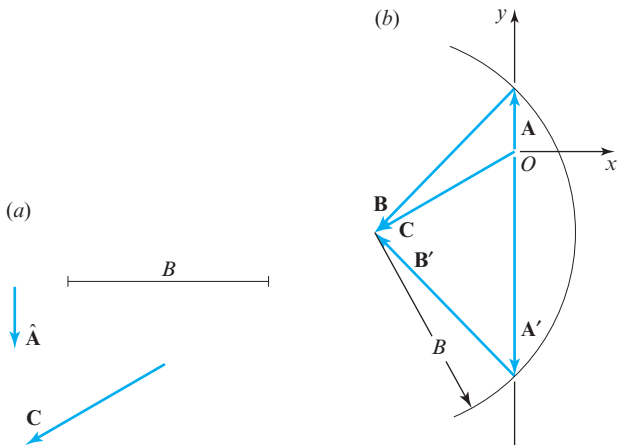


Figure 2.11 Case 2: (a) given $\mathbf{C}$, θ_A , and B ; (b) solutions for A, θ_B , and A', θ_B' .

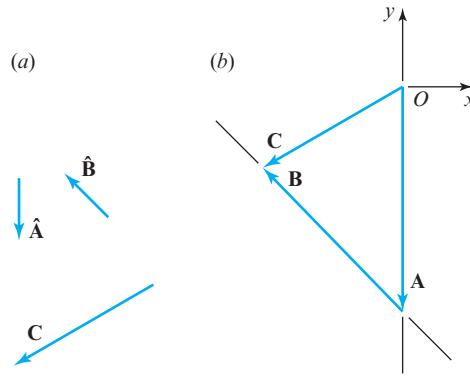


Figure 2.12 Case 3: (a) given $\mathbf{C}$, θ_A , and θ_B . (b) solution for A and B .

The solution, illustrated in Fig. 2.12, is obtained as follows:

1. Choose a convenient scale factor and draw vector $\mathbf{C}$.
2. Construct a line through the origin of $\mathbf{C}$ parallel to $\hat{\mathbf{A}}$.
3. Construct another line through the terminus of $\mathbf{C}$ parallel to $\hat{\mathbf{B}}$.
4. The intersection of these two lines defines both magnitudes, A and B , each of which may be either positive or negative.

Note that case 3 has a unique solution unless the lines are collinear or parallel. If the lines are collinear, the magnitudes A and B are both indeterminate. If the lines are parallel but distinct, magnitudes A and B are both infinite.

Finally, in case 4, the two unknowns are the directions of two vectors, $\hat{\mathbf{A}}$ and $\hat{\mathbf{B}}$. The vector equation is

$$\sqrt{\sqrt{\mathbf{C}}} = \sqrt{?} \sqrt{\mathbf{A}} + \sqrt{?} \sqrt{\mathbf{B}}. \quad (2.19)$$

The solution, illustrated in Fig. 2.13, is obtained as follows:

1. Choose a convenient scale factor and draw vector $\mathbf{C}$.
2. Construct a circular arc of radius A centered at the origin of $\mathbf{C}$.
3. Construct a circular arc of radius B centered at the terminus of $\mathbf{C}$.
4. The two intersections of these arcs define the two sets of solutions $\hat{\mathbf{A}}, \hat{\mathbf{B}}$ and $\hat{\mathbf{A}}', \hat{\mathbf{B}}'$.

Note that a real solution is possible only if $A + B \geq C$.

The graphic procedures of this section are now applied to obtain solutions of the loop-closure equations for the slider-crank linkage and the four-bar linkage.

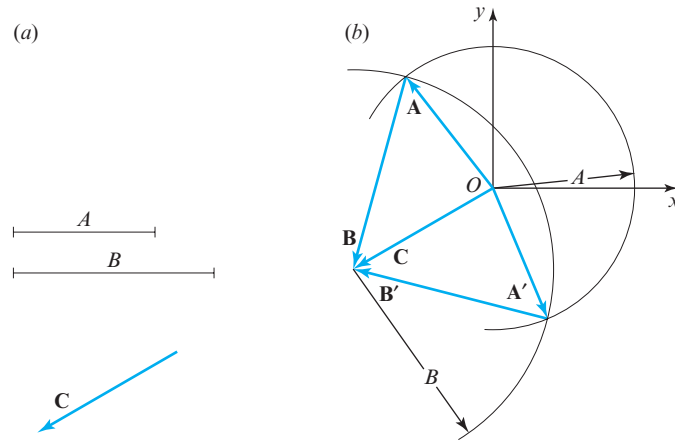


Figure 2.13 Case 4: (a) given C, A, and B; (b) solution for θ_A , θ_B and θ'_A , θ'_B .

Slider-Crank Linkage Consider the slider-crank linkage shown in Fig. 2.14a. Given the position of link 4, the problem is to determine the postures of links 2 and 3.

Replace the link centerlines with vectors $\mathbf{R}_A$ for the crank, $\mathbf{R}_B$ for the slider, and $\mathbf{R}_{BA}$ for the connecting rod, as shown in Fig. 2.14b. With a known location of the slider (that is, the distance R_B) and the angles θ_2 and θ_3 unknown, the loop-closure equation for the linkage is

$$\overset{\sqrt{?}}{\mathbf{R}}_A + \overset{\sqrt{?}}{\mathbf{R}}_{BA} - \overset{\sqrt{I}}{\mathbf{R}}_B = \mathbf{0},$$

which can be written as

$$\overset{\sqrt{I}}{\mathbf{R}}_B = \overset{\sqrt{?}}{\mathbf{R}}_A + \overset{\sqrt{?}}{\mathbf{R}}_{BA}, \quad (a)$$

where the “I” symbol, for the direction of vector $\mathbf{R}_B$, denotes that the location of the slider is the given input.

We recognize Eq. (a) as case 4 in Table 2.2.

The graphic solution procedure explained in Fig. 2.13 is now carried out in Fig. 2.14c. We note that there are two possible solutions (θ_2 and θ_3 , and θ'_2 and θ'_3), which correspond to two different postures of the linkage (that is, two configurations of the links), both of which are consistent with the given position of the slider. We know in advance which of the two solutions is desired; both sets of results are equally valid roots to the loop-closure equation, and the choice is based on the application.

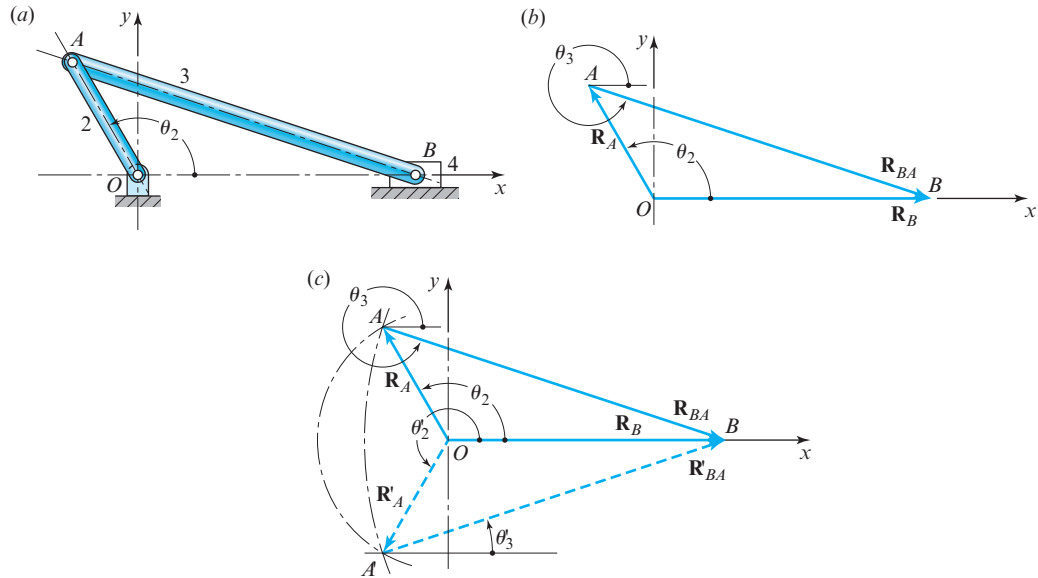


Figure 2.14 (a) Slider-crank linkage; (b) vectors replace the link centerlines; (c) graphic posture analysis.

Four-Bar Linkage Consider the four-bar linkage shown in Fig. 2.15. Given the posture of link 2, the problem is to define the postures of links 3 and 4 and the position of coupler point P .

Replace the link centerlines with vectors $\mathbf{R}_A$ for the crank, $\mathbf{R}_{BA}$ for the connecting rod, $\mathbf{R}_{BC}$ for the output link, and $\mathbf{R}_C$ for the frame, as shown in Fig. 2.16a. With a known posture of link 2, that is, the angle θ_2 , and the angles θ_3 and θ_4 unknown, the loop-closure equation for the linkage is

$$\sqrt{I} \mathbf{R}_A + \sqrt{?} \mathbf{R}_{BA} - \sqrt{?} \mathbf{R}_{BC} - \sqrt{?} \mathbf{R}_C = \mathbf{0}, \quad (a)$$

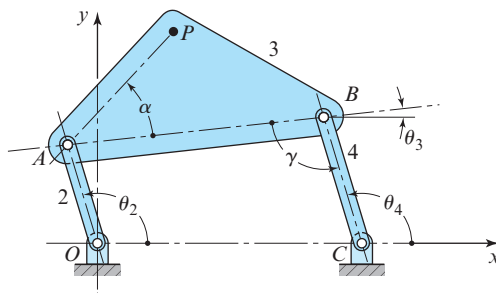


Figure 2.15 Four-bar linkage and coupler point P .

which can be rearranged as

$$\overset{\sqrt{I}}{\mathbf{R}}_A + \overset{\sqrt{?}}{\mathbf{R}}_{BA} = \overset{\sqrt{\sqrt{?}}}{\mathbf{R}}_C + \overset{\sqrt{?}}{\mathbf{R}}_{BC}, \quad (b)$$

where the “ I ” symbol, for the direction of vector $\mathbf{R}_A$, denotes that the orientation of the crank is the given input.

Also, the position of coupler point P is given by the position-difference equation

$$\overset{??}{\mathbf{R}}_P = \overset{\sqrt{I}}{\mathbf{R}}_A + \overset{\sqrt{?}}{\mathbf{R}}_{PA}. \quad (c)$$

Note that this equation contains three unknowns, but this can be reduced to two after Eq. (b) is solved by noting the constant angular relationship between $\hat{\mathbf{R}}_{PA}$ and $\hat{\mathbf{R}}_{BA}$ —namely,

$$\theta_5 = \theta_3 + \alpha, \quad (d)$$

where the constant coupler angle $\alpha = \angle BAP$. Therefore, Eq. (c) reduces to

$$\overset{??}{\mathbf{R}}_P = \overset{\sqrt{I}}{\mathbf{R}}_A + \overset{\sqrt{C}}{\mathbf{R}}_{PA}, \quad (e)$$

where the “ C ” symbol, for the direction of $\mathbf{R}_{PA}$, denotes that this direction is known from the constraint equation—that is, Eq. (d)—and, therefore, is not a third unknown.

We begin the graphic solution by combining the two unknown terms in Eq. (b), thus locating the positions of points A and C as shown in Figs. 2.16a and 2.16b:

$$\overset{\sqrt{\sqrt{?}}}{\mathbf{S}} = \overset{\sqrt{\sqrt{?}}}{\mathbf{R}}_C - \overset{\sqrt{I}}{\mathbf{R}}_A = \overset{\sqrt{?}}{\mathbf{R}}_{BA} - \overset{\sqrt{?}}{\mathbf{R}}_{BC}. \quad (f)$$

We recognize this equation as case 4 in Table 2.2. The solution procedure (that is, two unknown orientations) is then used to locate point B . Note that two solutions are possible, which indicates that the four-bar linkage can be assembled in two different postures. The two postures will be referred to here as (a) the open posture shown in Fig. 2.16a and (b) the crossed posture shown in Fig. 2.16b.

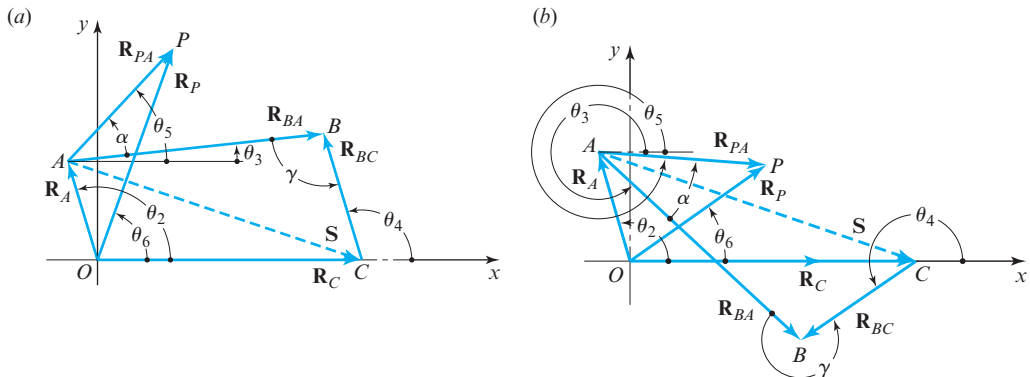


Figure 2.16 Vector diagrams illustrating the graphic solutions for (a) open posture; (b) crossed posture.

To determine the position of coupler point P , first we obtain the two orientations of $\hat{\mathbf{R}}_{PA}$ from Eq. (d). Then we solve Eq. (c) by the procedure for case 1. Two solutions are obtained for the position of point P , as shown in Figs. 2.16a and 2.16b. Both are valid solutions to Eqs. (b) through (d), although, in this instance, the crossed posture shown could not be reached from the open posture without first disassembling the linkage.

From the two examples presented here it is clear that graphic posture analysis usually requires precisely the same constructions that would be chosen naturally in drafting scale drawings of a mechanism at the posture under consideration. For this reason, the procedure may seem trivial and not truly worthy of the title “analysis.” Yet this is highly misleading. As we shall see in later sections of Chap. 2, the posture analysis of a mechanism is a nonlinear algebraic problem when approached by analytic or numeric methods. It is, in fact, the most difficult problem in kinematic analysis, and this is one primary reason why graphic solution techniques have retained their attraction.

2.9 ALGEBRAIC POSTURE ANALYSIS

To illustrate the classic algebraic approach for the posture analysis of planar mechanisms, this section presents the same two examples of the previous section—namely, the slider-crank linkage and the four-bar linkage.

Slider-Crank Linkage For generality, the offset version of this linkage, shown in Fig. 2.17, is chosen for the analysis. By making the offset or eccentricity $e = R_{O_2O} = 0$, the resulting equations can be used for the in-line (on-center or symmetric) version. The notation in Fig. 2.17 shows that the input crank angle θ_2 is measured around the origin of vector $\mathbf{r}_2 = \mathbf{R}_{AO_2}$, and the connecting rod angle θ_3 is measured around the origin of vector $\mathbf{r}_3 = \mathbf{R}_{AB}$. Both measurements are taken as positive (counterclockwise) angles.

There are two problems that occur in the posture analysis of the slider-crank linkage—namely:

Problem 1: Given the input crank angle θ_2 , find the connecting rod angle θ_3 and the position x_B .

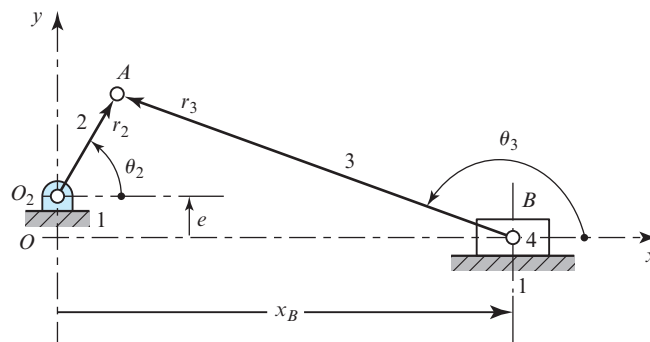


Figure 2.17 Notation for the offset slider-crank linkage.

Problem 2: Given the position x_B , find the input crank angle θ_2 and connecting rod angle θ_3 . This is the problem that we addressed in the previous section.

Solution to Problem 1: First, we define the position of point A by the equations

$$x_A = r_2 \cos \theta_2 \quad \text{and} \quad y_A = e + r_2 \sin \theta_2. \quad (2.20)$$

Next, we note that

$$e + r_2 \sin \theta_2 = r_3 \sin \theta_3, \quad (a)$$

so that

$$\sin \theta_3 = \frac{1}{r_3} (e + r_2 \sin \theta_2). \quad (2.21)$$

From the geometry of Fig. 2.17, we see that

$$x_B = r_2 \cos \theta_2 - r_3 \cos \theta_3. \quad (b)$$

Substituting Eq. (2.21) into the trigonometric identity $\cos \theta_3 = \pm \sqrt{1 - \sin^2 \theta_3}$ gives

$$\cos \theta_3 = -\frac{1}{r_3} \sqrt{r_3^2 - (e + r_2 \sin \theta_2)^2}. \quad (c)$$

Here the negative sign corresponds to an obtuse angle for θ_3 when measured around point B, as shown in Fig. 2.17. (A positive sign would imply an acute angle for θ_3 .)

Finally, substituting Eq. (c) into Eq. (b) gives

$$x_B = r_2 \cos \theta_2 + \sqrt{r_3^2 - (e + r_2 \sin \theta_2)^2}. \quad (2.22)$$

Thus, with the angle θ_2 given, the unknowns θ_3 and x_B can be obtained by solving Eqs. (2.21), (c), and (2.22).

Solution to Problem 2: Given x_B , we must solve Eq. (2.22) for the angle θ_2 . This requires the use of a calculator or a computer together with a root-finding technique. Here we select the well-known Newton-Raphson method.* This method can be explained by reference to Fig. 2.18, which is a graph of some function $f(x)$ versus x .

Let x_n be a first approximation (a rough estimate) of the root that we wish to find and for which $f(x) = 0$. A tangent line to the curve at $x = x_n$ intersects the x axis at x_{n+1} , which is a better approximation to the root. The slope of the tangent line is equal to the derivative of the function at $x = x_n$ and is

$$f'(x_n) = \frac{f(x_n)}{x_n - x_{n+1}}. \quad (d)$$

* See, for example, [1].

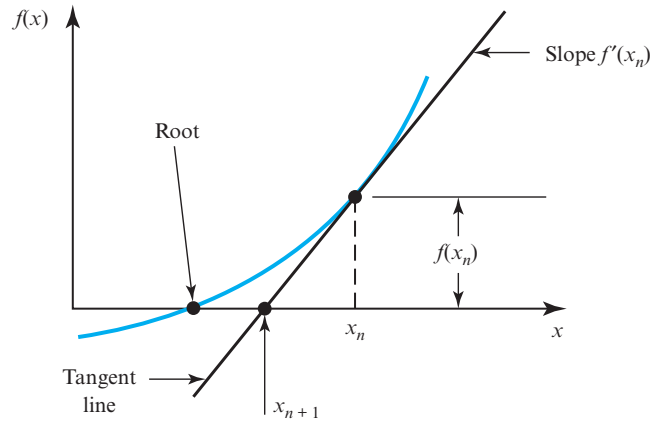


Figure 2.18 Newton-Raphson method.

Solving this for x_{n+1} gives

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}. \quad (2.23)$$

Using a calculator, for example, to start a solution, we enter an estimate of x_n , solve Eq. (2.23) for x_{n+1} , use this as the next estimate, and repeat this process as many times as needed to obtain the result with satisfactory accuracy. The accuracy is evaluated by comparing $(x_{n+1} - x_n)$ with a small number ε after each repetition and continuing until $(x_{n+1} - x_n) < \varepsilon$.

We note that the root-finding programs built into many calculators utilize an approximation to obtain $f'(x_n)$. These will sometimes be of little value in solving Eq. (2.22), so we proceed as follows. In Eq. (2.22), we replace the angle θ_2 with the symbol x , and let r_2 , r_3 , e , and x_B be given constants. Then

$$f(x) = r_2 \cos x + \sqrt{r_3^2 - (e + r_2 \sin x)^2} - x_B, \quad (e)$$

and

$$f'(x) = -r_2 \sin x - \frac{(e + r_2 \sin x)r_2 \cos x}{\sqrt{r_3^2 - (e + r_2 \sin x)^2}}. \quad (f)$$

These two equations can now be programmed with Eq. (2.23) to solve for the unknown value of the angle θ_2 when x_B is given. We note that the angle θ_2 will have two possible values. These may be found separately by using appropriate initial estimates.

A closed-form algebraic solution is possible if the eccentricity e is zero—that is, if the linkage is centered. For this case, we take Eqs. (a) and (b), square them, and add them together. Noting a trigonometric identity, the result is

$$x_B^2 - 2x_B r_2 \cos \theta_2 + r_2^2 = r_3^2. \quad (g)$$

Solving for θ_2 gives

$$\theta_2 = \cos^{-1} \frac{x_B^2 + r_2^2 - r_3^2}{2x_B r_2}. \quad (2.24)$$

The solution of Problem 1 for the centered version is, of course, obtained directly from Eq. (2.22) by setting $e = 0$.

Four-Bar Linkage To obtain the analytic solution, the distance R_{AO_4} is designated as S in Fig. 2.19 (as it was in Figs. 2.16a and 2.16b). The cosine law for each of the two triangles AO_2O_4 and ABO_4 can then be written in terms of the angles and link lengths; that is,

$$S = \sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos \theta_2}, \quad (2.25)$$

$$\beta = \cos^{-1} \frac{r_1^2 + S^2 - r_2^2}{2r_1 S}, \quad (2.26)$$

$$\psi = \cos^{-1} \frac{r_3^2 + S^2 - r_4^2}{2r_3 S}, \quad (2.27)$$

$$\lambda = \cos^{-1} \frac{r_4^2 + S^2 - r_3^2}{2r_4 S}. \quad (2.28)$$

Note that Eqs. (2.26) through (2.28) can yield double values, since they involve inverse cosines. Here we will take β , ψ , and λ as positive numeric values. Then a careful study of the figure shows that, when θ_2 is in the range $0 \leq \theta_2 \leq \pi$, the unknown angles are

$$\theta_3 = -\beta \pm \psi, \quad (2.29)$$

$$\theta_4 = \pi - \beta \mp \lambda. \quad (2.30)$$

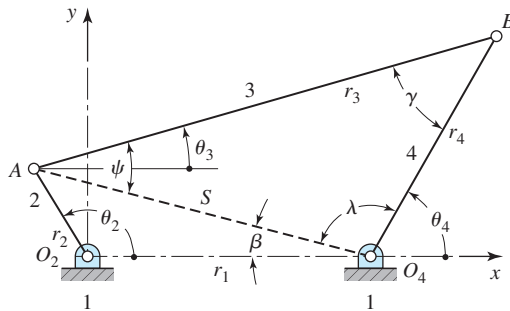


Figure 2.19 Four-bar linkage.

However, when θ_2 is in the range $\pi \leq \theta_2 \leq 2\pi$, then

$$\theta_3 = \beta \pm \psi, \quad (2.31)$$

$$\theta_4 = \pi + \beta \mp \lambda. \quad (2.32)$$

In these results, the upper signs correspond to the open posture, whereas the lower signs correspond to the crossed posture.

In Sec. 1.10, the concept of transmission angle was discussed in connection with the subject of mechanical advantage. Now, with Fig. 2.19 and using the cosine law, this angle can be written as

$$\gamma = \pm \cos^{-1} \frac{r_3^2 + r_4^2 - S^2}{2r_3r_4}. \quad (2.33)$$

2.10 COMPLEX-ALGEBRAIC SOLUTIONS OF PLANAR VECTOR EQUATIONS

In planar problems, it is often desirable to express a vector by specifying its magnitude and orientation in *polar notation*:

$$\mathbf{R} = R \angle \theta. \quad (a)$$

In Fig. 2.20a, the two-dimensional vector

$$\mathbf{R} = R^x \hat{\mathbf{i}} + R^y \hat{\mathbf{j}} \quad (2.34)$$

has two rectangular components of magnitudes,

$$R^x = R \cos \theta \quad \text{and} \quad R^y = R \sin \theta, \quad (2.35)$$

with

$$R = \sqrt{(R^x)^2 + (R^y)^2} \quad \text{and} \quad \theta = \tan^{-1} \frac{R^y}{R^x}. \quad (2.36)$$

Note that we have made the arbitrary choice here of accepting the positive square root for magnitude R when calculating from the components of $\mathbf{R}$. Therefore, we must be careful

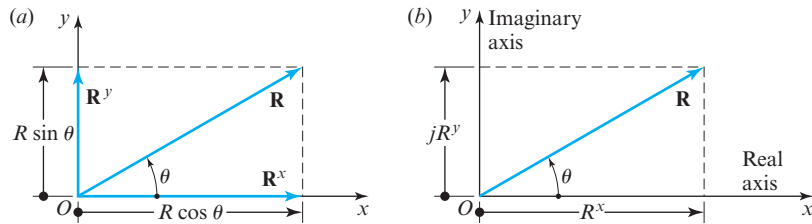


Figure 2.20 Correlation of planar vectors and complex values.

to interpret the signs of R^x and R^y individually when deciding upon the quadrant of the angle θ . Note that θ is defined as the angle from the positive x axis to the positive end of vector $\mathbf{R}$, measured about the origin of the vector, and this angle is positive when measured counterclockwise.

EXAMPLE 2.3

Express vectors $\mathbf{A} = 10\angle 30^\circ$ and $\mathbf{B} = 8\angle -15^\circ$ in rectangular notation* and find their sum.

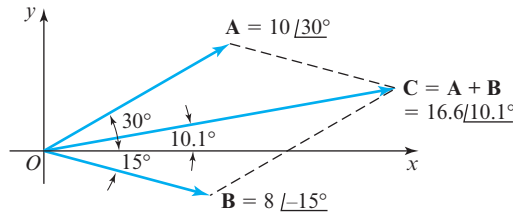


Figure 2.21 Addition of vectors.

SOLUTION

The vectors are shown in Fig. 2.21 and can be written as

$$\mathbf{A} = 10\cos 30^\circ \hat{\mathbf{i}} + 10\sin 30^\circ \hat{\mathbf{j}} = 8.66\hat{\mathbf{i}} + 5.00\hat{\mathbf{j}} \quad \text{Ans.}$$

$$\mathbf{B} = 8\cos(-15^\circ) \hat{\mathbf{i}} + 8\sin(-15^\circ) \hat{\mathbf{j}} = 7.73\hat{\mathbf{i}} - 2.07\hat{\mathbf{j}} \quad \text{Ans.}$$

$$\mathbf{C} = \mathbf{A} + \mathbf{B} = (8.66 + 7.73)\hat{\mathbf{i}} + (5.00 - 2.07)\hat{\mathbf{j}} = 16.39\hat{\mathbf{i}} + 2.93\hat{\mathbf{j}}.$$

The magnitude of the resultant is determined from Eq. (2.36),

$$C = \sqrt{16.39^2 + 2.93^2} = 16.65,$$

and the angle is

$$\theta = \tan^{-1} \frac{2.93}{16.39} = 10.1^\circ.$$

The final result in polar notation is

$$\mathbf{C} = 16.6\angle 10.1^\circ. \quad \text{Ans.}$$

2.11 COMPLEX POLAR ALGEBRA

Another method of treating two-dimensional vector problems analytically makes use of complex algebra. Although complex numbers are not vectors, they can be used to represent

*Many calculators are equipped to perform polar-rectangular and rectangular-polar conversions directly.

vectors in a plane by choosing an origin and real and imaginary axes. In two-dimensional kinematic problems, these axes can conveniently be chosen coincident with the x_1y_1 axes of the absolute coordinate system.

As shown in Fig 2.20*b*, the location of any point in the plane can be specified either by its absolute-position vector or by its corresponding real and imaginary coordinates,

$$\mathbf{R} = R^x + jR^y, \quad (2.37)$$

where the operator j is defined as the unit imaginary number,

$$j = \sqrt{-1}. \quad (2.38)$$

Complex numbers are commonly used in planar analysis because of the ease with which they can be changed into polar form. Employing complex rectangular notation for vector $\mathbf{R}$, we can write

$$\mathbf{R} = R\angle\theta = R\cos\theta + jR\sin\theta. \quad (2.39)$$

But using the well-known Euler equation from trigonometry,

$$e^{j\theta} = \cos\theta + j\sin\theta, \quad (2.40)$$

we can also write $\mathbf{R}$ in complex polar form as

$$\mathbf{R} = Re^{j\theta}, \quad (2.41)$$

where the magnitude and direction of the vector appear explicitly. As we will see in Chaps. 3 and 4, expressing a vector in this form is also especially useful when differentiation is required.

Some familiarity with useful manipulation techniques for vectors written in complex polar forms can be gained by again solving the four cases of the loop-closure equation of Table 2.2. Writing Eq. (2.16) in complex polar form, we have

$$Ce^{j\theta_C} = Ae^{j\theta_A} + Be^{j\theta_B}. \quad (2.42)$$

In case 1, the two unknowns are C and θ_C . We begin the solution by substituting Euler's equation, Eq. (2.40), into Eq. (2.42), which gives

$$C(\cos\theta_C + j\sin\theta_C) = A(\cos\theta_A + j\sin\theta_A) + B(\cos\theta_B + j\sin\theta_B). \quad (a)$$

Then, equating the real terms and the imaginary terms separately, we obtain two real equations corresponding to the horizontal and vertical components of the two-dimensional vector equation:

$$C\cos\theta_C = A\cos\theta_A + B\cos\theta_B, \quad (b)$$

$$C\sin\theta_C = A\sin\theta_A + B\sin\theta_B. \quad (c)$$

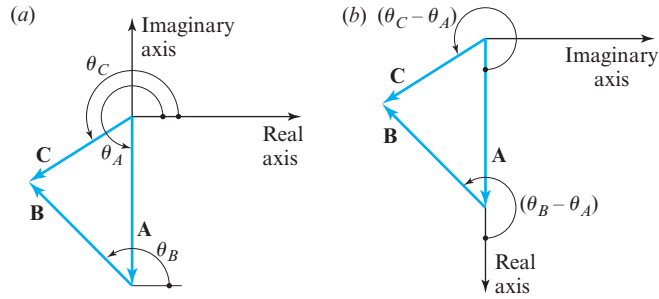


Figure 2.22 (a) Original axes; (b) rotated axes.

The unknown angle θ_C can be eliminated by squaring and adding these two equations. The result is

$$C = \sqrt{A^2 + B^2 + 2AB\cos(\theta_B - \theta_A)}. \quad (2.43)$$

The positive square root is chosen arbitrarily; the negative square root would yield a negative solution for C with θ_C differing by 180° . Dividing Eq. (c) by Eq. (b), and rearranging, the angle θ_C can be written as

$$\theta_C = \tan^{-1} \frac{A\sin\theta_A + B\sin\theta_B}{A\cos\theta_A + B\cos\theta_B}, \quad (2.44)$$

where the signs of the numerator and the denominator must be considered separately in determining the proper quadrant of θ_C .* Only a single solution is found for case 1, as previously illustrated in Fig. 2.10.

In case 2, the two unknowns are A and θ_B . One convenient way of solving this case, in complex polar form, is to first divide Eq. (2.42) by $e^{j\theta_A}$. The resulting equation can be written as

$$Ce^{j(\theta_C - \theta_A)} = A + Be^{j(\theta_B - \theta_A)}. \quad (d)$$

Comparing this equation with Fig. 2.22, we see that division by the complex polar form of a unit vector $e^{j\theta_A}$ has the effect of rotating the real and imaginary axes by the angle θ_A *clockwise* such that the real axis lies along vector **A**.

* When writing computer programs it will be noted that most programming languages, including both ANSI/ISO standard FORTRAN and C, provide a library function named ATAN2(y, x) that accepts the numerator and denominator separately and provides the solution (in radians) in the proper quadrant. If such a function is not available, then a solution (in radians) in either the first or fourth quadrant is usually provided, and π radians (180°) should be added to the angle if the denominator is negative.

We can now use Euler's equation, Eq. (2.40), to separate the real and imaginary components; that is,

$$C\cos(\theta_C - \theta_A) = A + B\cos(\theta_B - \theta_A), \quad (e)$$

$$C\sin(\theta_C - \theta_A) = B\sin(\theta_B - \theta_A). \quad (f)$$

The solutions are then obtained directly from Eqs. (f) and (e), respectively, as

$$\theta_B = \theta_A + \sin^{-1} \frac{C\sin(\theta_C - \theta_A)}{B}. \quad (2.45)$$

$$A = C\cos(\theta_C - \theta_A) - B\cos(\theta_B - \theta_A). \quad (2.46)$$

The solutions given by Eqs. (2.45) and (2.46) are intentionally presented in this order since Eq. (2.46) cannot be evaluated numerically until after θ_B is found. We also note that the arcsine term in Eq. (2.45) is double-valued. Therefore, case 2 has two distinct solutions, A, θ_B and A', θ'_B ; these are shown in Fig. 2.11b.

In case 3, the two unknowns are the magnitudes A and B . As in case 2, we obtain Eqs. (e) and (f). Therefore, from Eq. (f), the solution for B is

$$B = C \frac{\sin(\theta_C - \theta_A)}{\sin(\theta_B - \theta_A)}. \quad (2.47)$$

The solution for the unknown magnitude, A , is obtained by dividing Eq. (2.42) by $e^{j\theta_B}$. The resulting equation is then separated into real and imaginary parts and yields

$$A = C \frac{\sin(\theta_C - \theta_B)}{\sin(\theta_A - \theta_B)}. \quad (2.48)$$

Note that this case yields a unique solution, as illustrated in Fig. 2.12.

In case 4 the two unknowns are θ_A and θ_B . We begin by dividing Eq. (2.42) by $e^{j\theta_C}$ to align the real axis along vector $\mathbf{C}$:

$$C = Ae^{j(\theta_A - \theta_C)} + Be^{j(\theta_B - \theta_C)}. \quad (g)$$

Using Euler's equation to separate components and then rearranging terms, we obtain

$$A\cos(\theta_A - \theta_C) = C - B\cos(\theta_B - \theta_C), \quad (h)$$

$$A\sin(\theta_A - \theta_C) = -B\sin(\theta_B - \theta_C). \quad (i)$$

Squaring both equations and adding the results gives

$$A^2 = C^2 + B^2 - 2BC\cos(\theta_B - \theta_C). \quad (j)$$

We recognize this as the law of cosines for the vector triangle. Rearranging this equation, the angle θ_B can be written as

$$\theta_B = \theta_C \pm \cos^{-1} \frac{C^2 + B^2 - A^2}{2BC}. \quad (2.49)$$

Moving C to the left-hand side of Eq. (h) before the squaring and adding operations results in another form of the law of cosines, from which

$$\theta_A = \theta_C \pm \cos^{-1} \frac{C^2 + A^2 - B^2}{2AC}. \quad (2.50)$$

The plus or minus signs in these two equations are a reminder that the arccosines are each double-valued and, therefore, θ_B and θ_A each have two solutions. These two pairs of angles can be paired naturally together as θ_A, θ_B and θ'_A, θ'_B under the restriction of Eq. (i). Therefore, case 4 has two distinct solutions, as shown in Fig. 2.13.

2.12 POSTURE ANALYSIS TECHNIQUES

There are many different approaches to the posture analysis of mechanisms. Here, we identify the following five approaches:

- (a) Graphic approach.
- (b) Analytic approach.
- (c) Complex-algebraic approach.
- (d) Vector-algebraic approach.
- (e) Numeric approach.

These approaches are illustrated in the following two examples.

EXAMPLE 2.4

Consider a sliding-block linkage in the posture illustrated in Fig. 2.23. Perform the posture analysis; that is, find θ_4 and distance R_{AO_4} .

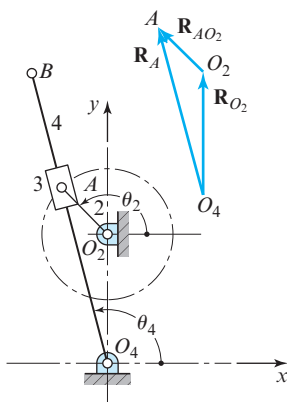


Figure 2.23 $R_{O_2O_4} = 9.0$ in, $R_{AO_2} = 4.5$ in, and $\theta_2 = 135^\circ$.

SOLUTION

Using the vector diagram shown in Fig. 2.23 we recognize this as a case 1 problem, and the vector loop equation can be written as

$$\mathbf{R}_{AO_4} = \mathbf{R}_{O_2O_4}^{\sqrt{\sqrt{\quad}}} + \mathbf{R}_{AO_2}^{\sqrt{I}}, \quad (1)$$

where the “ I ” symbol, for the direction of vector $\mathbf{R}_{AO_2}$, denotes that the orientation of link 2 is the given input.

(a) *Graphic approach* When Fig. 2.23 is drawn to a suitable scale (for example, 6 in/in), then, by direct measurement of this figure, we find that

$$\theta_4 = 105.3^\circ \quad \text{Ans.}$$

$$R_{AO_4} = 2.08 \text{ in (6 in/in)} = 12.48 \text{ in.} \quad \text{Ans.}$$

(b) *Analytic approach* For case 1, we employ Eqs. (2.43) and (2.44); therefore,

$$\begin{aligned} R_{AO_4} &= \sqrt{R_{O_2O_4}^2 R_{AO_2}^2 + 2R_{O_2O_4} R_{AO_2} \cos(\theta_2 - 90^\circ)} \\ &= \sqrt{(9 \text{ in})^2 + (4.5 \text{ in})^2 + 2(9 \text{ in})(4.5 \text{ in}) \cos(135^\circ - 90^\circ)} \\ &= 12.59 \text{ in,} \quad \text{Ans.} \\ \theta_4 &= \tan^{-1} \frac{R_{O_2O_4} \sin 90^\circ + R_{AO_2} \sin \theta_2}{R_{O_2O_4} \cos 90^\circ + R_{AO_2} \cos \theta_2} \\ &= \tan^{-1} \frac{(9 \text{ in}) \sin 90^\circ + (4.5 \text{ in}) \sin 135^\circ}{(9 \text{ in}) \cos 90^\circ + (4.5 \text{ in}) \cos 135^\circ} = \tan^{-1} \left(\frac{12.182 \text{ in}}{-3.182 \text{ in}} \right) = -75.36^\circ. \end{aligned}$$

However, note that the calculator used here did not recognize for the arctangent function that the negative sign in the denominator indicates an angle in the second or third quadrant. Therefore, the proper value is

$$\theta_4 = -75.36^\circ \pm 180^\circ = 104.64^\circ. \quad \text{Ans.}$$

(c) *Complex-algebraic approach* The terms for Eq. (1) are

$$\begin{aligned} \mathbf{R}_{AO_4} &= R_{AO_4} \angle \theta_4 = R_{AO_4} \cos \theta_4 + j R_{AO_4} \sin \theta_4, \\ \mathbf{R}_{O_2O_4} &= 9 \text{ in} \angle 90^\circ = (9 \text{ in}) \cos 90^\circ + j(9 \text{ in}) \sin 90^\circ = 0 + j9 \text{ in}, \\ \mathbf{R}_{AO_2} &= 4.5 \text{ in} \angle 135^\circ = (4.5 \text{ in}) \cos 135^\circ + j(4.5 \text{ in}) \sin 135^\circ \\ &= -3.182 \text{ in} + j3.182 \text{ in}. \end{aligned}$$

Substituting these into Eq. (1) yields

$$\mathbf{R}_{AO_4} = (0 + j9 \text{ in}) + (-3.182 \text{ in} + j3.182 \text{ in}) = -3.182 \text{ in} + j12.182 \text{ in}.$$

Thus,

$$R_{AO_4} = \sqrt{(-3.182 \text{ in})^2 + (12.182 \text{ in})^2} = 12.59 \text{ in} \quad \text{Ans.}$$

and

$$\theta_4 = \tan^{-1} \frac{R_{AO_4}^y}{R_{AO_4}^x} = \tan^{-1} \frac{12.182 \text{ in}}{-3.182 \text{ in}} = 104.64^\circ. \quad \text{Ans.}$$

Again, note the comment in (b) regarding arctangent results from the calculator.

(d) Vector-algebraic approach Here we employ a scientific hand calculator that will add and subtract complex numbers in rectangular notation and will convert from rectangular to polar notation or vice versa. Thus,

$$\mathbf{R}_{O_2O_4} = 9 \text{ in} \angle 90^\circ = 0 + j9 \text{ in},$$

$$\mathbf{R}_{AO_2} = 4.5 \text{ in} \angle 135^\circ = -3.182 \text{ in} + j3.182 \text{ in},$$

$$\begin{aligned} \mathbf{R}_{AO_4} &= \mathbf{R}_{O_2O_4} + \mathbf{R}_{AO_2} = (0 + j9 \text{ in}) + (-3.182 \text{ in} + j3.182) \\ &= -3.182 \text{ in} + j12.182 \text{ in} = 12.59 \text{ in} \angle 104.64^\circ. \end{aligned} \quad \text{Ans.}$$

(e) Numeric approach Some scientific hand calculators permit equations containing complex numbers to be solved in either rectangular or polar form or both, with the results displayed in either mode. Using such a calculator, we enter

$$\mathbf{R}_{AO_4} = 9 \text{ in} \angle 90^\circ + 4.5 \text{ in} \angle 135^\circ = 12.59 \text{ in} \angle 104.64^\circ. \quad \text{Ans.}$$

Note that all algebraic methods give results that are in exact agreement and are more accurate than graphic results obtained manually.

EXAMPLE 2.5

A crank-rocker four-bar linkage with the given input angle, is shown in both the open and crossed postures in Fig. 2.24. Perform a posture analysis of the linkage; that is, find θ_3 and θ_4 for both postures.

SOLUTION

Note that the line connecting the ground pins, O_2 and O_4 (that is, the ground link), is not horizontal. In the previous examples, the ground link was chosen to be coincident with the x axis; that is, $\theta_1 = 0$. In this example, however, the angle $\theta_1 = 15^\circ$ and must be accounted for when using Eqs. (2.25) through (2.30).

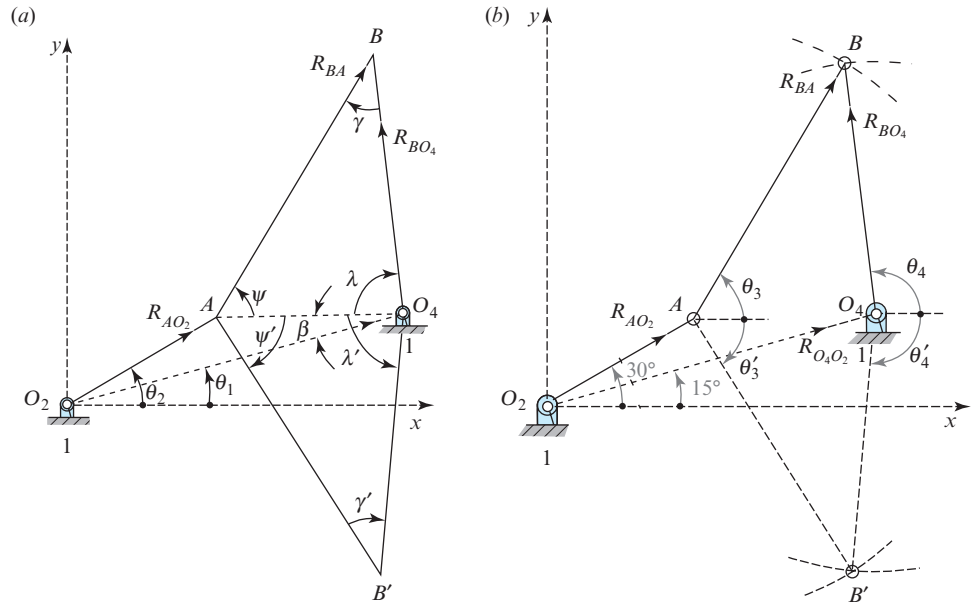


Figure 2.24 $\mathbf{R}_{AO_2} = 200 \text{ mm} \angle 30^\circ$, $R_{BA} = 350 \text{ mm}$, $R_{BO_4} = 300 \text{ mm}$, and $\mathbf{R}_{O_4O_2} = 400 \text{ mm} \angle 15^\circ$.

For the input angle $\theta_2 = 30^\circ$, we observe that

$$\mathbf{R}_{AO_2} = 0.200 \text{ m} \angle 30^\circ \quad \text{and} \quad \mathbf{R}_{O_4O_2} = 0.400 \text{ m} \angle 15^\circ.$$

Therefore,

$$\mathbf{S} = \mathbf{R}_{O_4O_2} - \mathbf{R}_{AO_2} = 0.400 \text{ m} \angle 15^\circ - 0.200 \text{ m} \angle 30^\circ = 0.213 \text{ m} \angle 0.95^\circ.$$

Note that the vectors in triangle ABO_4 are related by the equation

$$\sqrt{\sqrt{}} = \sqrt{?} - \sqrt{?} \quad \mathbf{S} = \mathbf{R}_{BA} - \mathbf{R}_{BO_4}. \quad (1)$$

There are two unknown orientations in this equation, and so we identify this as case 4 (Table 2.2).

(a) Graphic approach When Fig. 2.24 is drawn at a chosen scale, then, by direct measurement of the figure in the open posture, we find

$$\theta_3 = 60^\circ \quad \text{and} \quad \theta_4 = 97^\circ, \quad \text{Ans.}$$

and in the crossed posture, we find

$$\theta'_3 = -57^\circ \quad \text{and} \quad \theta'_4 = -95^\circ. \quad \text{Ans.}$$

(b) *Analytic approach* Comparing the notation in Fig. 2.24 with Fig. 2.19, we set $r_1 = R_{O_4O_2} = 0.400$ m, $r_2 = R_{AO_2} = 0.200$ m, $r_3 = R_{BA} = 0.350$ m, and $r_4 = R_{BO_4} = 0.300$ m. Also, note that the ground link of Fig. 2.19 must be rotated by $\theta_1 = 15^\circ$ ccw, and $\theta_2 = 30^\circ - 15^\circ = 15^\circ$; therefore, Eqs. (2.25) through (2.30) give

$$S = \sqrt{(0.400 \text{ m})^2 + (0.200 \text{ m})^2 - 2(0.400 \text{ m})(0.200 \text{ m})\cos(30^\circ - 15^\circ)} \\ = 0.213 \text{ m},$$

$$\beta = \cos^{-1} \frac{(0.400 \text{ m})^2 + (0.213 \text{ m})^2 - (0.200 \text{ m})^2}{2(0.400 \text{ m})(0.213 \text{ m})} = 14.05^\circ,$$

$$\psi = \cos^{-1} \frac{(0.350 \text{ m})^2 + (0.213 \text{ m})^2 - (0.300 \text{ m})^2}{2(0.350 \text{ m})(0.213 \text{ m})} = 58.51^\circ,$$

$$\lambda = \cos^{-1} \frac{(0.300 \text{ m})^2 + (0.213 \text{ m})^2 - (0.350 \text{ m})^2}{2(0.300 \text{ m})(0.213 \text{ m})} = 84.19^\circ.$$

Then, in the open posture, remembering that $\theta_1 = 15^\circ$,

$$\theta_3 = 15^\circ - 14.05^\circ + 58.51^\circ = 59.46^\circ, \quad \text{Ans.}$$

$$\theta_4 = 15^\circ + 180^\circ - 14.05^\circ - 84.19^\circ = 96.76^\circ, \quad \text{Ans.}$$

and, in the crossed posture,

$$\theta'_3 = 15^\circ - 14.05^\circ - 58.51^\circ = -57.56^\circ, \quad \text{Ans.}$$

$$\theta'_4 = 15^\circ + 180^\circ - 14.05^\circ + 84.19^\circ = -94.86^\circ. \quad \text{Ans.}$$

(c) *Complex-algebraic approach* Substituting S for C , R_{BA} for A , $-R_{BO_4}$ for B , θ_S for θ_C , and θ_4 for θ_B , into Eq. (2.49) gives

$$\theta_4 = \theta_S \pm \cos^{-1} \frac{S^2 + R_{BO_4}^2 - R_{BA}^2}{2(-R_{BO_4})S} = (\theta_1 - \beta) \pm \cos^{-1} \frac{S^2 + R_{BO_4}^2 - R_{BA}^2}{2(-R_{BO_4})S} \\ = 0.95^\circ \pm \cos^{-1} \frac{(0.213 \text{ m})^2 + (-0.300 \text{ m})^2 - (0.350 \text{ m})^2}{2(-0.300 \text{ m})(0.213 \text{ m})} \\ = 0.95^\circ \pm 95.81^\circ = 96.76^\circ \text{ or } -94.86^\circ. \quad \text{Ans.}$$

In addition, substituting θ_3 for θ_A into Eq. (2.50) gives

$$\theta_3 = \theta_S \pm \cos^{-1} \frac{S^2 + R_{BA}^2 - R_{BO_4}^2}{2R_{BA}S} \\ = 0.95^\circ \pm \cos^{-1} \frac{(0.213 \text{ m})^2 + (0.350 \text{ m})^2 - (-0.300 \text{ m})^2}{2(0.350 \text{ m})(0.213 \text{ m})} \\ = 0.95^\circ \pm 58.51^\circ = 59.46^\circ \text{ or } -57.56^\circ, \quad \text{Ans.}$$

where the first result corresponds to the open posture and the second result is for the crossed posture. We note that the answers in Parts (b) and (c) are in good agreement with those of Part (a), but are of higher accuracy than measurements taken from the scaled drawing.

(d) *Vector-algebraic approach* The vector loop equation for the four-bar-linkage can be written as

$$\mathbf{R}_{BA} = \mathbf{R}_{O_4O_2} + \mathbf{R}_{BO_4} - \mathbf{R}_{AO_2}. \quad (2)$$

Separating this equation into horizontal and vertical components,

$$R_{BA} \cos \theta_3 = R_{O_4O_2} \cos \theta_1 + R_{BO_4} \cos \theta_4 - R_{AO_2} \cos \theta_2, \quad (3a)$$

$$R_{BA} \sin \theta_3 = R_{O_4O_2} \sin \theta_1 + R_{BO_4} \sin \theta_4 - R_{AO_2} \sin \theta_2. \quad (3b)$$

By squaring and adding these two equations, we can eliminate the variable θ_3 :

$$\begin{aligned} R_{BA}^2 = & R_{O_4O_2}^2 + R_{BO_4}^2 + R_{AO_2}^2 + 2R_{O_4O_2}R_{BO_4}[\cos \theta_1 \cos \theta_4 + \sin \theta_1 \sin \theta_4] \\ & - 2R_{O_4O_2}R_{AO_2}[\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2] - 2R_{BO_4}R_{AO_2}[\cos \theta_4 \cos \theta_2 + \sin \theta_4 \sin \theta_2]. \end{aligned}$$

Then we can write this equation as

$$A \cos \theta_4 + B \sin \theta_4 = C, \quad (4)$$

where

$$A = 2R_{O_4O_2}R_{BO_4} \cos \theta_1 - 2R_{BO_4}R_{AO_2} \cos \theta_2, \quad (5a)$$

$$B = 2R_{O_4O_2}R_{BO_4} \sin \theta_1 - 2R_{BO_4}R_{AO_2} \sin \theta_2, \quad (5b)$$

$$C = R_{BA}^2 - R_{O_4O_2}^2 - R_{BO_4}^2 - R_{AO_2}^2 + 2R_{O_4O_2}R_{AO_2} \cos(\theta_2 - \theta_1). \quad (5c)$$

Equation (4) is one form of Freudenstein's equation (Sec. 9.11 shows another variation that is more suitable for kinematic synthesis).

Substituting the known data into Eqs. (5) gives

$$A = 2(0.400 \text{ m})(0.300 \text{ m}) \cos 15^\circ - 2(0.300 \text{ m})(0.200 \text{ m}) \cos 30^\circ = 0.127 \text{ 90 m}^2,$$

$$B = 2(0.400 \text{ m})(0.300 \text{ m}) \sin 15^\circ - 2(0.300 \text{ m})(0.200 \text{ m}) \sin 30^\circ = 0.002 \text{ 12 m}^2,$$

$$\begin{aligned} C = & (0.350 \text{ m})^2 - (0.400 \text{ m})^2 - (0.300 \text{ m})^2 - (0.200 \text{ m})^2 \\ & + 2(0.400 \text{ m})(0.200 \text{ m}) \cos(30^\circ - 15^\circ) \\ = & -0.012 \text{ 95 m}^2. \end{aligned}$$

Therefore, Eq. (4) can be written as

$$(0.127 \text{ 90 m}^2) \cos \theta_4 + (0.002 \text{ 12 m}^2) \sin \theta_4 = -0.012 \text{ 95 m}^2. \quad (6)$$

To determine the output angle θ_4 , this transcendental equation can be written as a quadratic equation. The procedure is to use the half-angle relationships; that is, to define

$$Z = \tan(\theta_4/2), \quad (7a)$$

which gives

$$\sin \theta_4 = \frac{2Z}{1+Z^2} \quad \text{and} \quad \cos \theta_4 = \frac{1-Z^2}{1+Z^2}. \quad (7b)$$

Substituting Eqs. (7b) into Eq. (6), multiplying through by $(1+Z^2)$, and rearranging, gives the quadratic equation

$$(0.114\,95\text{ m}^2)Z^2 + (-0.004\,24\text{ m}^2)Z + (-0.140\,85\text{ m}^2) = 0.$$

The solutions to this equation can be written as

$$\begin{aligned} Z &= \frac{-(-0.004\,24\text{ m}) \pm \sqrt{(-0.004\,24\text{ m})^2 - 4(0.114\,95\text{ m})(-0.140\,85\text{ m})}}{2(0.114\,95\text{ m})} \\ &= 0.018\,41 \pm 1.106\,80 = 1.125\,21 \text{ or } -1.088\,39. \end{aligned} \quad (8)$$

Substituting Eq. (8) into Eq. (7a), gives the two output angles corresponding to the open and crossed postures, namely

$$\theta_4 = 96.76^\circ \quad \text{and} \quad \theta'_4 = -94.86^\circ. \quad \text{Ans.}$$

Substituting these results into Eqs. (3a) and (3b) gives the corresponding coupler angles

$$\theta_3 = 59.46^\circ \quad \text{and} \quad \theta'_3 = -57.56^\circ. \quad \text{Ans.}$$

(e) *Numeric approach* Rearrange the vector loop equation, Eq. (2), into the form

$$\mathbf{f} = \mathbf{R}_{AO_2} + \mathbf{R}_{BA} - \mathbf{R}_{O_4O_2} - \mathbf{R}_{BO_4} = \mathbf{0}, \quad (9)$$

and separate the horizontal and vertical components:

$$f^x = R_{AO_2} \cos \theta_2 + R_{BA} \cos \theta_3 - R_{O_4O_2} \cos \theta_1 - R_{BO_4} \cos \theta_4 = 0, \quad (10a)$$

$$f^y = R_{AO_2} \sin \theta_2 + R_{BA} \sin \theta_3 - R_{O_4O_2} \sin \theta_1 - R_{BO_4} \sin \theta_4 = 0. \quad (10b)$$

We recognize these as two equations in two unknowns, θ_3 and θ_4 , and we solve them by the Newton-Raphson method introduced in Sec. 2.9 to an accuracy of 0.01° . To do this, we expand Eqs. (10) to first-order in Taylor series in the two unknowns as follows:

$$R_{AO_2} \cos \theta_2 + R_{BA} \cos \theta_3 - R_{BA} \sin \theta_3 \Delta \theta_3 - R_{O_4O_2} \cos \theta_1 - R_{BO_4} \cos \theta_4 + R_{BO_4} \sin \theta_4 \Delta \theta_4 = 0,$$

$$R_{AO_2} \sin \theta_2 + R_{BA} \sin \theta_3 + R_{BA} \cos \theta_3 \Delta \theta_3 - R_{O_4O_2} \sin \theta_1 - R_{BO_4} \sin \theta_4 - R_{BO_4} \cos \theta_4 \Delta \theta_4 = 0.$$

Then, writing these equations in matrix form,

$$\begin{bmatrix} R_{BA} \sin \theta_3 & -R_{BO_4} \sin \theta_4 \\ -R_{BA} \cos \theta_3 & R_{BO_4} \cos \theta_4 \end{bmatrix} \begin{bmatrix} \Delta \theta_3 \\ \Delta \theta_4 \end{bmatrix} = \begin{bmatrix} R_{AO_2} \cos \theta_2 + R_{BA} \cos \theta_3 - R_{O_4O_2} \cos \theta_1 - R_{BO_4} \cos \theta_4 \\ R_{AO_2} \sin \theta_2 + R_{BA} \sin \theta_3 - R_{O_4O_2} \sin \theta_1 - R_{BO_4} \sin \theta_4 \end{bmatrix}.$$

Substituting the given data, these matrices become

$$\begin{bmatrix} (0.350 \text{ m}) \sin \theta_3 & -(0.300 \text{ m}) \sin \theta_4 \\ -(0.350 \text{ m}) \cos \theta_3 & (0.300 \text{ m}) \cos \theta_4 \end{bmatrix} \begin{bmatrix} \Delta \theta_3 \\ \Delta \theta_4 \end{bmatrix} = \begin{bmatrix} (-0.213 \text{ m}) + (0.350 \text{ m}) \cos \theta_3 - (0.300 \text{ m}) \cos \theta_4 \\ (-0.003 \text{ m}) + (0.350 \text{ m}) \sin \theta_3 - (0.300 \text{ m}) \sin \theta_4 \end{bmatrix}. \quad (11)$$

Note that $\Delta \theta_3$ and $\Delta \theta_4$ are called the *corrections* and are dimensionless values, meaning values in radians. For the iteration procedure, we must be careful when combining the corrections with θ_3 and θ_4 if the two angles have units of degrees; therefore,

$$(\theta_3^\circ)_{\text{new}} = (\theta_3^\circ)_{\text{old}} + \frac{180^\circ}{\pi \text{ rad}} \Delta \theta_3, \quad (12a)$$

$$(\theta_4^\circ)_{\text{new}} = (\theta_4^\circ)_{\text{old}} + \frac{180^\circ}{\pi \text{ rad}} \Delta \theta_4. \quad (12b)$$

To use the iteration procedure, we begin with initial estimates of the unknowns θ_3 and θ_4 . From a scale drawing of the linkage, such that, for the open posture, our starting estimates are

$$\theta_3 = 60^\circ \quad \text{and} \quad \theta_4 = 100^\circ. \quad (13a)$$

Inserting these estimates into Eq. (11) gives

$$\begin{bmatrix} 0.303 \text{ m} & -0.295 \text{ m} \\ -0.175 \text{ m} & -0.052 \text{ m} \end{bmatrix} \begin{bmatrix} \Delta \theta_3 \\ \Delta \theta_4 \end{bmatrix} = \begin{bmatrix} 0.013 \text{ m} \\ 0.004 \text{ m} \end{bmatrix}.$$

Using Cramer's rule, the corrections are

$$\Delta \theta_3 = -0.007 \text{ rad} \quad \text{and} \quad \Delta \theta_4 = -0.005 \text{ rad}.$$

Substituting these corrections into Eqs. (12) gives

$$\theta_3 = 59.58^\circ \quad \text{and} \quad \theta_4 = 96.87^\circ. \quad (13b)$$

Inserting these improved estimates into Eq. (11) gives

$$\begin{bmatrix} 0.301 \text{ m} & -0.297 \text{ m} \\ -0.177 \text{ m} & -0.035 \text{ m} \end{bmatrix} \begin{bmatrix} \Delta \theta_3 \\ \Delta \theta_4 \end{bmatrix} = \begin{bmatrix} -0.577 \text{ m} (10)^{-4} \\ 0.441 \text{ m} (10)^{-3} \end{bmatrix}.$$

Solving again by Cramer's rule, the updated corrections are

$$\Delta \theta_3 = -0.002 \text{ rad} \quad \text{and} \quad \Delta \theta_4 = -0.001 \text{ rad}.$$

Substituting these corrections into Eqs. (12), gives

$$\theta_3 = 59.46^\circ \quad \text{and} \quad \theta_4 = 96.76^\circ. \quad (13c)$$

Inserting these improved estimates into Eq. (11) gives us

$$\begin{bmatrix} 0.30145 \text{ m} & -0.29791 \text{ m} \\ -0.17785 \text{ m} & -0.03531 \text{ m} \end{bmatrix} \begin{bmatrix} \Delta\theta_3 \\ \Delta\theta_4 \end{bmatrix} = \begin{bmatrix} 0.21351(10)^{-5} \text{ m} \\ 0.17394(10)^{-5} \text{ m} \end{bmatrix}.$$

Solving again by Cramer's rule, the updated corrections are

$$\Delta\theta_3 = -0.00006959 \text{ rad} \quad \text{and} \quad \Delta\theta_4 = -0.0000142 \text{ rad}.$$

Substituting the corrections into Eqs. (12), gives

$$\theta_3 = 59.46^\circ \quad \text{and} \quad \theta_4 = 96.76^\circ. \quad \text{Ans.}$$

Since the corrections are now both smaller than the desired accuracy of $0.00017 \text{ rad} = 0.01^\circ$, we accept these values as final results for the open posture.

For the crossed posture, we begin with initial estimates from our drawing of $\theta_3 = -60^\circ$ and $\theta_4 = -100^\circ$. Then, using these as starting estimates with Eq. (11), we repeat the same process and iterate until we converge to a result of

$$\theta_3 = -57.56^\circ \quad \text{and} \quad \theta_4 = -94.86^\circ. \quad \text{Ans.}$$

A computer code could be written using a commercially available language such as MATLAB or JAVA.

2.13 COUPLER-CURVE GENERATION

In Chap. 1, Fig. 1.23, we learned of the vast variety of useful coupler curves that can be generated by a planar four-bar linkage. These curves are quite easy to obtain graphically, but computer-generated curves can be obtained more quickly and are easier to vary to obtain desired curve characteristics. Here we present the basic equations but omit the programming details required for display of a curve on an electronic monitor.

Considering the four-bar linkage of Fig. 2.16, we first write the vector equation

$$\mathbf{S} = \mathbf{R}_C - \mathbf{R}_A.$$

This is case 1, where S and θ_S are the two unknowns. The solutions are given by Eqs. (2.43) and (2.44). Substituting gives

$$S = \sqrt{R_C^2 + R_A^2 - 2R_C R_A \cos(\theta_A - \theta_C)}, \quad (2.51)$$

$$\theta_S = \tan^{-1} \frac{R_C \sin \theta_C - R_A \sin \theta_A}{R_C \cos \theta_C - R_A \cos \theta_A}. \quad (2.52)$$

Then we note that

$$\sqrt{\sqrt{}} \mathbf{S} = \mathbf{R}_{BA} - \mathbf{R}_{BC}.$$

This is case 4, where θ_3 and θ_4 are the two unknowns. The solutions are given by Eqs. (2.49) and (2.50). Substituting gives

$$\theta_4 = \theta_S \pm \cos^{-1} \frac{S^2 + R_{BC}^2 - R_{BA}^2}{2R_{BC}S}, \quad (2.53)$$

$$\theta_3 = \theta_S \mp \cos^{-1} \frac{S^2 + R_{BA}^2 - R_{BC}^2}{2R_{BA}S}. \quad (2.54)$$

The solution for the open posture of the linkage shown in Fig. 2.16a corresponds to the lower set of signs; the crossed posture of the linkage shown in Fig. 2.16b corresponds to the upper set.

Coupler point P generates a coupler curve when crank 2 is rotated. From Fig. 2.16, we see that

$$\mathbf{R}_P = R_P e^{j\theta_6} = R_A e^{j\theta_2} + R_{PA} e^{j(\theta_3 + \alpha)}. \quad (2.55)$$

We recognize this as case 1, where R_P and θ_6 are the two unknowns. The solutions are given by Eqs. (2.43) and (2.44); that is,

$$R_P = \sqrt{R_A^2 + R_{PA}^2 + 2R_A R_{PA} \cos(\theta_3 + \alpha - \theta_2)} \quad (2.56)$$

and

$$\theta_6 = \tan^{-1} \frac{R_A \sin \theta_2 + R_{PA} \sin(\theta_3 + \alpha)}{R_A \cos \theta_2 + R_{PA} \cos(\theta_3 + \alpha)}. \quad (2.57)$$

Note that each of these equations give two solutions because of the double values for θ_3 corresponding to the two postures of the linkage. The following example uses Eqs. (2.56) and (2.57) to plot the coupler curve of a four-bar linkage.

EXAMPLE 2.6

Consider the crank-rocker four-bar linkage shown in Fig. 2.25 with $R_{BA} = 100$ mm, $R_{CB} = 250$ mm, $R_{CD} = 300$ mm, and $R_{DA} = 200$ mm. The paths of coupler pins B and C are shown by the circle and the circular arc, respectively. The location of coupler point P is given by $R_{PB} = 150$ mm and $\alpha = \angle CBP = -45^\circ$. Calculate the coordinates of coupler

point P and plot the path of this point (the coupler curve) for a complete rotation of the crank.

SOLUTION

For each value of crank angle θ_2 , the angles β , ψ , and γ are calculated from Eqs. (2.26), (2.27), and (2.33), respectively. This notation corresponds to that of Figs. 2.15 and 2.16. Next, Eqs. (2.29) and (2.31) are applied to give the posture of link 3, θ_3 , as displayed in Table 2.3. Note that the same result can be obtained from Eq. (2.50).

Finally, the polar coordinates of coupler point P are calculated from Eqs. (2.56) and (2.57). From these, we can obtain the Cartesian coordinates of point P . The solutions for crank angles from 0° to 90° for the open posture are displayed in Table 2.3. The coupler curve is shown in Fig. 2.25. The interested reader can repeat the solution to this problem for the crossed posture.

Table 2.3 Posture of Link 3 and Coordinates of Coupler Point P

θ_2, deg	θ_3, deg	R_P, mm	θ_6, deg	R_P^x, mm	R_P^y, mm
0.0	110.5	212.0	40.1	162.2	136.5
10.0	99.4	232.2	36.9	185.8	139.3
20.0	87.8	245.3	33.7	204.0	136.1
30.0	77.5	249.9	31.5	213.1	130.6
40.0	69.2	247.7	30.5	213.4	125.8
50.0	62.9	240.7	30.7	207.0	122.7
60.0	58.3	230.4	31.7	196.0	121.1
70.0	55.1	218.0	33.5	181.9	120.3
80.0	53.0	204.4	35.7	165.9	119.4
90.0	51.8	189.9	38.3	148.9	117.8

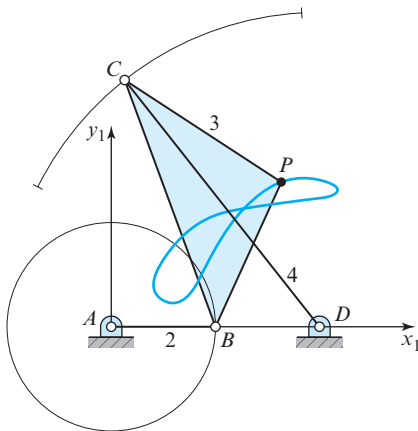


Figure 2.25 The coupler curve traced by point P .

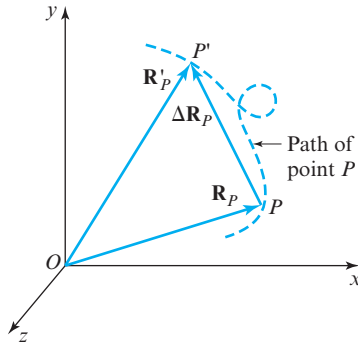


Figure 2.26 Displacement of a moving point.

2.14 DISPLACEMENT OF A MOVING POINT

In the study of motion, we must be concerned with the relationship between successive positions of a point and postures of a link. In Fig. 2.26, a point, originally at position P , is moving along the path shown and, sometime later, arrives at position P' . The *displacement* of the point during the time interval is defined as the *net change in position*; that is,

$$\Delta \mathbf{R}_P = \mathbf{R}'_P - \mathbf{R}_P. \quad (2.58)$$

Note that this displacement is a vector quantity having the magnitude and direction of the vector from point P to point P' .

It is important to note that the displacement $\Delta \mathbf{R}_P$ is the net change in position and does not depend on the particular path taken from P to P' . The magnitude of this vector is *not* necessarily equal to the length of the path (the distance traveled), and its direction is *not* necessarily along the tangent to the path, although both are true when the displacement is infinitesimally small. Knowledge of the path actually traveled from P to P' is not even necessary to find the displacement vector, providing the initial and final positions of the point are known.

2.15 DISPLACEMENT DIFFERENCE BETWEEN TWO POINTS

In this section we consider the difference in the displacements of two moving points. In particular, we are concerned with the case where the two moving points are both fixed in the same rigid body. The situation is shown in Fig. 2.27, where a rigid body moves from an initial posture defined by $x_2y_2z_2$ to a later posture defined by $x'_2y'_2z'_2$.

From Eq. (2.6), the position difference between the two points P and Q at the initial instant is

$$\mathbf{R}_{PQ} = \mathbf{R}_P - \mathbf{R}_Q. \quad (a)$$

After the displacement of this body, the two points are located at P' and Q' . At that time, the position difference is

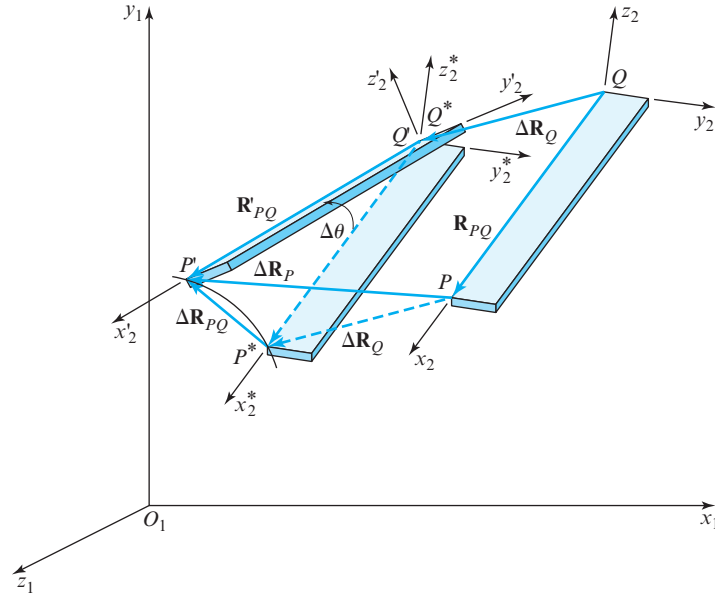


Figure 2.27 Displacement difference between two points fixed in the same rigid body.

$$\mathbf{R}'_{PQ} = \mathbf{R}'_P - \mathbf{R}'_Q. \quad (b)$$

During the time interval of the movement, the two points have undergone individual displacements of $\Delta\mathbf{R}_P$ and $\Delta\mathbf{R}_Q$, respectively.

As the name implies, the *displacement difference* between the two points is defined as the net difference between their respective displacements and can be written as

$$\Delta\mathbf{R}_{PQ} = \Delta\mathbf{R}_P - \Delta\mathbf{R}_Q. \quad (2.59)$$

Note that this equation corresponds to the vector triangle PP^*P' in Fig. 2.27. As stated in the previous section, the displacement depends only on the net change in position and not on the particular path taken. Therefore, no matter how the body (containing points P and Q) was *actually* displaced, we are free to visualize the path as we choose. Equation (2.59) leads us to visualize the displacement as taking place in two stages. First, the body translates (slides without rotation) from $x_2y_2z_2$ to $x_2^*y_2^*z_2^*$; during this movement, all points, including P and Q , have the same displacement $\Delta\mathbf{R}_Q$. Second, the body rotates about point Q' to its final posture $x_2'y_2'z_2'$.

A different interpretation can be obtained by manipulating Eq. (2.59) as follows:

$$\begin{aligned} \Delta\mathbf{R}_{PQ} &= (\mathbf{R}'_P - \mathbf{R}_P) - (\mathbf{R}'_Q - \mathbf{R}_Q) \\ &= (\mathbf{R}'_P - \mathbf{R}'_Q) - (\mathbf{R}_P - \mathbf{R}_Q), \end{aligned} \quad (c)$$

and then, from Eqs. (a) and (b),

$$\Delta \mathbf{R}_{PQ} = \mathbf{R}'_{PQ} - \mathbf{R}_{PQ}. \quad (2.60)$$

This equation corresponds to the vector triangle $Q'P^*P'$ in Fig. 2.27 and demonstrates that the displacement difference, defined as the difference between two displacements, is equal to the net change between the position-difference vectors.

For either interpretation, we are illustrating that *any displacement of a rigid body is equivalent to the sum of a net translation of one point and a net rotation of the body about that point*. We also see that only the rotation contributes to the displacement difference between two points fixed in the same rigid body; that is, *there is no difference between the displacements of any two points fixed in the same rigid body as the result of a translation*. (The following section gives the definition of the term *translation*.)

In view of the previous discussion, we can visualize the displacement difference $\Delta \mathbf{R}_{PQ}$ as the displacement that would be seen for point P by a moving observer who travels along, always staying coincident with point Q but *not rotating* with the moving body—that is, always using the absolute coordinate axes $x_1y_1z_1$ for the measurement of direction. It is important to understand the difference between the interpretation of an observer moving with point Q but not rotating and the case of the observer *on* the moving body. To an observer *on* the moving body, both points P and Q would appear stationary; neither would be seen to have a displacement because they do not move relative to the observer, and the displacement difference seen by such an observer would be zero.

2.16 TRANSLATION AND ROTATION

Using the concept of displacement difference between two points fixed in the same rigid body, we are now able to define translation and rotation.

Translation is defined as *a motion of a body for which the displacement difference between any two points P and Q in the body is zero*; that is, from the displacement-difference equation, Eq. (2.60),

$$\Delta \mathbf{R}_{PQ} = \Delta \mathbf{R}_P - \Delta \mathbf{R}_Q = \mathbf{0}.$$

That is,

$$\Delta \mathbf{R}_P = \Delta \mathbf{R}_Q, \quad (2.61)$$

which states that *the displacements of any two points in the body are equal*.

Rotation is a motion of a body for which different points of the body exhibit different displacements.

Figure 2.28a shows a situation where the body has moved along a curved path from posture x_2y_2 to posture $x'_2y'_2$. Despite the fact that the point paths are curved,* $\Delta \mathbf{R}_P$ is still

* Translation in which the point paths are straight lines is called *rectilinear* translation; translation in which the point paths are identical curves is referred to as *curvilinear* translation.

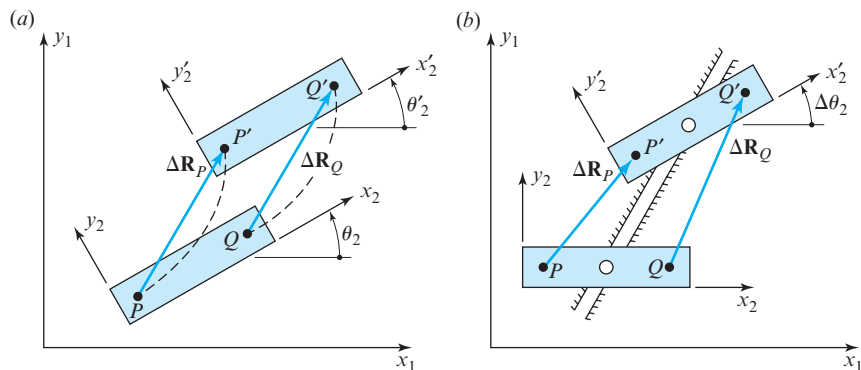


Figure 2.28 (a) Translation: $\Delta \mathbf{R}_P = \Delta \mathbf{R}_Q$, $\Delta \theta_2 = 0$; (b) rotation: $\Delta \mathbf{R}_P \neq \Delta \mathbf{R}_Q$, $\Delta \theta_2 \neq 0$.

equal to $\Delta \mathbf{R}_Q$, and the body has undergone a translation. Note that in translation, the point paths described by any two points on the body are identical, and there is no change of angular orientation between the moving coordinate system and the coordinate system of the observer; that is, $\Delta \theta_2 = \theta'_2 - \theta_2 = 0$.

Now consider the midpoint of a body that is *constrained* to move along a straight-line path, as shown in Fig. 2.28b. As it does so, the body rotates so that $\Delta \theta_2 = \theta'_2 - \theta_2 \neq 0$ and the displacements $\Delta \mathbf{R}_P$ and $\Delta \mathbf{R}_Q$ are not equal. Although there is no obvious point on the body about which it has rotated, the coordinate system $x'_2y'_2$ has changed angular orientation relative to x_1y_1 , and the body is said to have undergone a rotation.

In conclusion, we note that translation and rotation of a body cannot be defined from the motion of a single point. These are characteristics of a body or a coordinate system. It is improper to speak of “rotation of a point” since there is no meaning for angular orientation of a point. It is also improper to associate the terms “translation” and “rotation” with the path of a single point of a moving body. Although it does not matter which points of the body are chosen, the motions of two or more points must be compared before meaningful definitions exist for these terms.

2.17 APPARENT DISPLACEMENT

We have stated in previous sections that the displacement of a point does not depend on the particular path on which it travels. However, since displacement is computed from the position vectors of the endpoints of the path, knowledge of the coordinate system of the observer is essential.

In Fig. 2.29 we identify three bodies: body 1 is a fixed or stationary body containing the absolute reference system $x_1y_1z_1$; body 2 is a moving body containing the reference system $x_2y_2z_2$; and body 3, which moves with respect to body 2. We also identify two observers. We designate that HE is an observer fixed to body 1, the stationary system, and we designate that SHE is another observer on body 2, fixed to the moving system $x_2y_2z_2$.

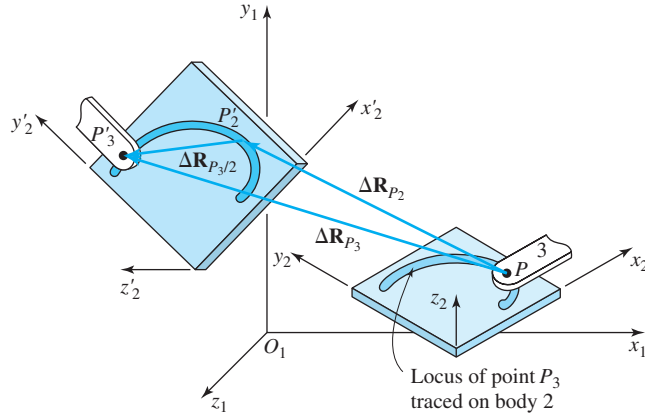


Figure 2.29 Apparent displacement of a point.

Thus, we might say that HE is on the ground observing whereas SHE is going for a ride on body 2.

Now consider a point P_3 , fixed in body 3 and moving along a known path on body 2. HE and SHE are both observing the motion of P_3 , and we want to compare what they see. We must define another point, P_2 , which is fixed in (or rigidly attached to) body 2 and is instantaneously coincident with P_3 .

Now let body 2 and the axes $x_2y_2z_2$ be displaced to a new posture $x'_2y'_2z'_2$. While this motion is taking place, let P_3 move to another position on body 2, which we identify as P'_3 . But P_2 moves with body 2 and is now in the new position identified as P'_2 . SHE, on body 2, reports the motion of P_3 as vector $\Delta\mathbf{R}_{P_3/2}$, which is read as the displacement of P_3 as it appears to an observer on body 2. This is called the apparent-displacement vector. Note that SHE sees no motion of P_2 , since it appears to HER to be stationary on body 2. Therefore, $\Delta\mathbf{R}_{P_2/2} = \mathbf{0}$.

However, HE, on the stationary body, reports the displacement of P_3 as vector $\Delta\mathbf{R}_{P_3}$. Note that HE also reports the displacement of P_2 as vector $\Delta\mathbf{R}_{P_2}$. From the vector triangle in Fig. 2.29, we see that the observations of the two observers are related by the *apparent-displacement equation*

$$\Delta\mathbf{R}_{P_3} = \Delta\mathbf{R}_{P_2} + \Delta\mathbf{R}_{P_3/2}. \quad (2.62)$$

We can take this equation as the definition of the apparent-displacement vector, although it is important also to understand the physical concepts involved. Note that the apparent-displacement vector relates the absolute displacements of two *coincident points* that are points of *different* moving bodies. Note also that there is no restriction on the actual location of the observer moving with coordinate system 2, only that SHE be fixed in that coordinate system so that SHE senses no displacement for point P_2 .

One primary use of the apparent displacement is to determine an absolute displacement. It is not uncommon in machines to find a point, such as P_3 , which is constrained to move along a known slot or path or guideway defined by the shape of another moving

link 2. In such cases, it may be much more convenient to measure or calculate $\Delta \mathbf{R}_{P_2}$ and $\Delta \mathbf{R}_{P_3/2}$ and to use Eq. (2.62) than to measure the absolute displacement $\Delta \mathbf{R}_{P_3}$ directly.

2.18 ABSOLUTE DISPLACEMENT

In reflecting on the definition and concept of the apparent-displacement vector, we conclude that the absolute displacement of a moving point, say $\Delta \mathbf{R}_{P_3/1}$, is the special case of an apparent displacement where the observer is fixed in the absolute coordinate system. As explained for the position vector, the notation is often abbreviated to read $\Delta \mathbf{R}_{P_3}$ or just $\Delta \mathbf{R}_P$ and an absolute observer is implied whenever this is not noted explicitly.

Perhaps a better physical understanding of apparent displacement can be achieved by relating it to absolute displacement. Imagine a car, P_3 , traveling along a roadway and under observation by an absolute observer some distance to one side. Consider how this observer visually senses the motion of the car. Although HE may not be conscious of all of the following steps, the idea here is that the observer first imagines a fixed point P_1 coincident with P_3 , which HE defines in his mind as stationary; HE may relate to a fixed point of the roadway or a nearby tree or road sign, for example. HE then compares his later observation of the car P_3 with that of P_1 to sense displacement. Note that HE does not compare with his own location but with the initially coincident point P_1 . In this instance, the apparent-displacement equation becomes an identity:

$$\Delta \mathbf{R}_{P_3} = \Delta \mathbf{R}_{P_1}^0 + \Delta \mathbf{R}_{P_3/1}.$$

2.19 APPARENT ANGULAR DISPLACEMENT

In general, rotations cannot be treated as vectors (this is explained in some detail in Sec. 3.2). However, for planar motion, the concept of apparent displacement also extends to include rotation. For example, suppose we consider the rotations of the two gears 2 and 3 connected by an arm, body 4, as shown in Fig. 2.30. In Fig. 2.30a, we see the rotations as they would appear in HER coordinate system attached to the arm. In Fig. 2.30b, we see the same situation, but as it might appear to HIM in the absolute frame of reference.

Although it may not be easy to write equations relating the absolute rotations of the gears shown in Fig. 2.30, it is quite easy to relate their apparent rotations if we take HER point of view as an observer riding on the moving coordinate system attached to the arm. From HER vantage point it is obvious that, if there is no slipping between bodies 2 and 3, then the arc lengths of the two gear sectors that pass the arm at the point of contact must be equal. That is, if ρ_2 is the radius of gear 2 and ρ_3 is the radius of gear 3, then

$$\rho_2 \Delta \theta_{2/4} = -\rho_3 \Delta \theta_{3/4}, \quad (a)$$

where $\Delta \theta_{2/4}$ and $\Delta \theta_{3/4}$ are the angular displacements of gears 2 and 3 as they appear to HER in coordinate system 4, and the negative sign accounts for the difference in the senses of the two apparent rotations.

When these angular displacements are replaced by those seen by HIM from the absolute coordinate system, Eq. (a) becomes

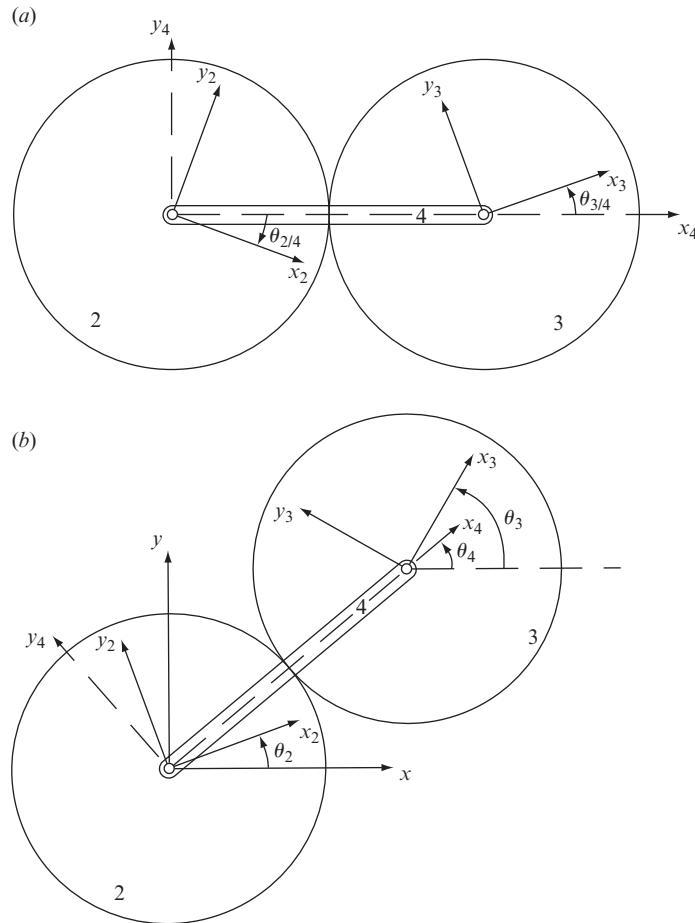


Figure 2.30 (a) Apparent rotations observed from the arm; (b) absolute rotations.

$$\rho_2(\Delta\theta_2 - \Delta\theta_4) = -\rho_3(\Delta\theta_3 - \Delta\theta_4). \quad (b)$$

Using these ideas, the loop-closure equations for mechanisms having rolling contact without slipping can be solved. The following two examples will make the procedure clear.

EXAMPLE 2.7

For the cam-and-follower mechanism in the posture shown in Fig. 2.31, define a set of vectors that is suitable for a complete kinematic analysis. Label and show the sense and orientation of each vector on the mechanism. Write the vector loop equation(s) and identify suitable input(s) for the mechanism. Identify the known quantities, the unknown variables, and any constraints. If you identify constraints, then write the constraint equation(s).

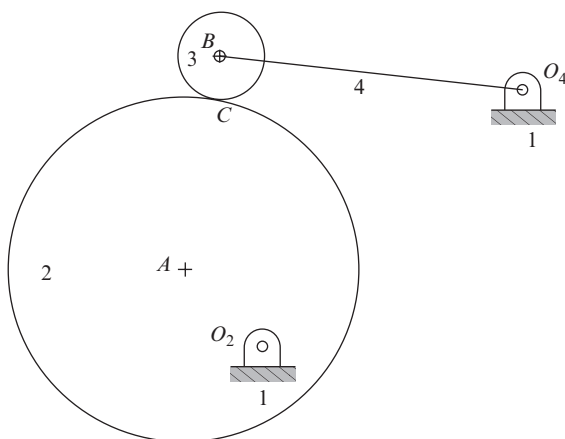


Figure 2.31
Cam-and-follower mechanism.

SOLUTION

Since the circular cam, link 2, and the roller follower, link 3, are in rolling contact, let us consider them linked together by a (fictitious) arm, link 23. This leads us to consider the set of vectors shown in Fig. 2.32.

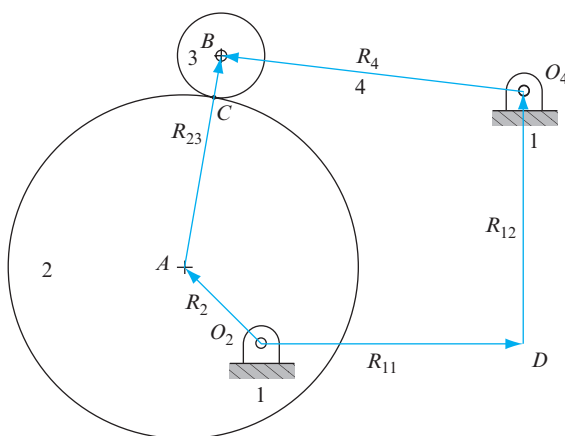


Figure 2.32 Vectors chosen
for analysis of the mechanism.

The vector loop equation for the mechanism can be written as

$$\mathbf{R}_2 + \mathbf{R}_{23} - \mathbf{R}_4 - \mathbf{R}_{12} - \mathbf{R}_{11} = \mathbf{0}. \quad \text{Ans. (1)}$$

This is a valid loop-closure equation and, given the angle θ_2 as input, it can be solved for the postures of link 4 and of the fictitious link 23. However, this does not yet allow a solution for the rotation of the roller, link 3.

Since there is rolling contact between cam 2 and roller 3 at point C, let us consider the apparent angular displacements of links 2 and 3 as they would appear to an observer riding on the fictitious arm, link 23. Such an observer would observe

$$\rho_3 \Delta\theta_{3/23} = -\rho_2 \Delta\theta_{2/23}, \quad (2)$$

which gives the following relationship between the absolute angular displacements:

$$\rho_3(\Delta\theta_3 - \Delta\theta_{23}) = -\rho_2(\Delta\theta_2 - \Delta\theta_{23}). \quad (3)$$

Rearranging this equation gives

$$\rho_3 \Delta\theta_3 = (\rho_2 + \rho_3) \Delta\theta_{23} - \rho_2 \Delta\theta_2. \quad \text{Ans. (4)}$$

Since the angular displacement of link 2 is known (the input) and $\Delta\theta_{23}$ is determined from Eq. (1), the angular displacement, $\Delta\theta_3$, can be determined from Eq. (4). It is worth noting that we can obtain a solution for the posture of link 3 even though no vector is attached to or rotates with link 3.

EXAMPLE 2.8

For the rack-and-pinion mechanism shown in Fig. 2.33, define a set of vectors that is suitable for a complete kinematic analysis. Label and show the sense and orientation of each vector on the mechanism. Write the vector loop equation(s) for the mechanism and identify suitable input(s) for the mechanism. Identify the known quantities, the unknown variables, and any constraints. If you identify constraints, then write the constraint equation(s).

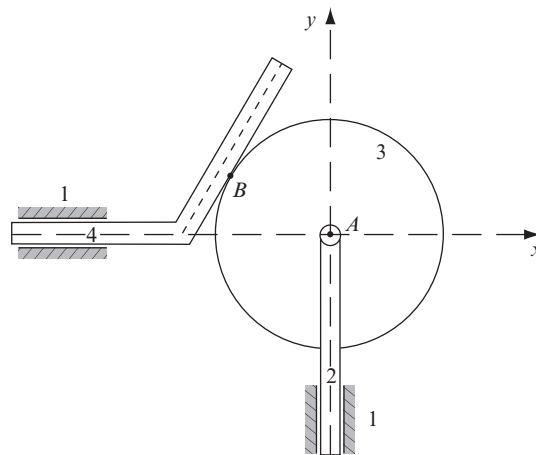


Figure 2.33 Rack-and-pinion mechanism.

SOLUTION

Let us consider the set of vectors shown in Fig. 2.34.

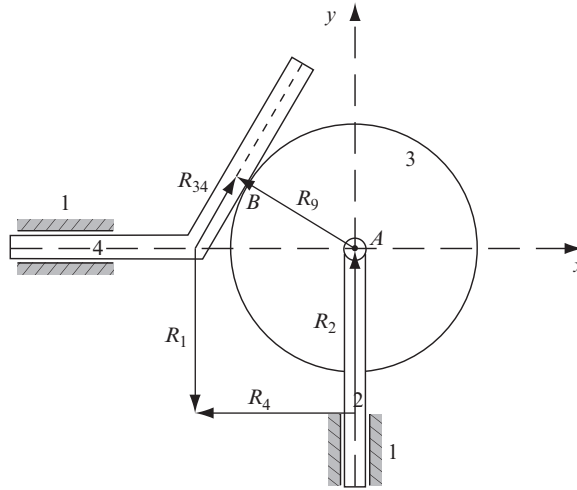


Figure 2.34 Vectors chosen for analysis of the mechanism.

The length of vector $\mathbf{R}_2$ represents the input, and the length of vector $\mathbf{R}_4$ indicates the output. The loop-closure equation is

$$\overset{I\sqrt{}}{\mathbf{R}_2} + \overset{\sqrt{C}}{\mathbf{R}_9} - \overset{?\sqrt{}}{\mathbf{R}_{34}} + \overset{\sqrt{\sqrt{}}}{\mathbf{R}_1} - \overset{?\sqrt{}}{\mathbf{R}_4} = \mathbf{0}. \quad \text{Ans.}$$

The constraint that gives the orientation of vector $\mathbf{R}_9$ is

$$\theta_9 = \theta_{34} + 90^\circ. \quad \text{Ans.}$$

The angular displacement of pinion 3 can be determined by considering the apparent rotation of the pinion to an observer riding on the rack, link 4. As such an observer watches the changing position of the point of contact B , SHE must see the same increment of surface on both links 3 and 4. The constraint relationship can be written as

$$\Delta R_{34} = \rho_3 \Delta \theta_{3/4} = \rho_3 (\Delta \theta_3 - \overset{0}{\Delta \theta_4}) = \rho_3 \Delta \theta_3.$$

Therefore, the angular displacement of the pinion is

$$\Delta \theta_3 = \Delta R_{34} / \rho_3. \quad \text{Ans.}$$

2.20 REFERENCES

- [1] Mischke, C. R., 1980. *Mathematical Model Building*, Ames, IA: Iowa State University Press, 86.

PROBLEMS*

- †2.1** Describe and sketch the locus of a point A , which moves according to the equations $R_A^x = at\cos(2\pi t)$, $R_A^y = at\sin(2\pi t)$, and $R_A^z = 0$.
- †2.2** Find the position difference from point P to point Q on the curve $y = x^2 + x - 16$, where $R_P^x = 2$ and $R_Q^x = 4$.
- †2.3** The path of a moving point is defined by the equation $y = 2x^2 - 28$. Find the position difference from point P to point Q if $R_P^x = 4$ and $R_Q^x = -3$.
- †2.4** The path of a moving point P is defined by the equation $y = 60 - x^3/3$. What is the displacement of the point if its motion begins at $R_P^x = 0$ and ends at $R_P^x = 3$?
- †2.5** If point A moves on the locus of Problem 2.1, find its displacement from $t = 2$ to $t = 2.5$.
- †2.6** The position of a point is given by the equation $\mathbf{R} = 100e^{j2\pi t}$. What is the path of the point? Determine the displacement of the point from $t = 0.10$ to $t = 0.40$.
- †2.7** The equation $\mathbf{R} = (t^2 + 4)e^{-j\pi t/10}$ defines the position of a point. In which direction is the position vector rotating? Where is the point located when $t = 0$? What is the next value t can have if the orientation of the position vector is to be the same as it is when $t = 0$? What is the displacement from the first position of the point to the second?
- †2.8** The location of a point is defined by the equation $\mathbf{R} = (4t + 2)e^{j\pi t^2/30}$. Motion of the point is initiated when $t = 0$. What is the displacement until $t = 3$? Find the change in angular orientation of the position vector during the same time interval.
- †2.9** Link 2 rotates according to the equation $\theta = \pi t/4$. Block 3 slides outward on link 2 according to the equation $r = t^2 + 2$. What is the absolute displacement $\Delta \mathbf{R}_{P_3}$ from $t = 1$ to $t = 2$? What is the apparent displacement $\Delta \mathbf{R}_{P_{3/2}}$?

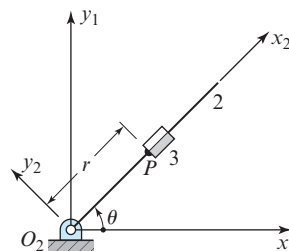


Figure P2.9

- †2.10** A wheel with center at O rolls without slipping on the ground at point P . If point O is displaced 10 in to the right, determine the displacement of point P during this interval.

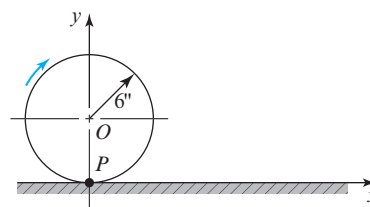
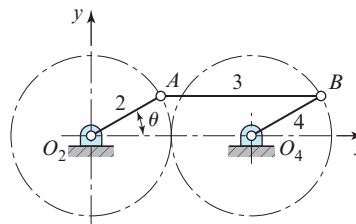


Figure P2.10 Rolling wheel.

- †2.11** A point Q moves from A to B along link 3 while link 2 rotates from $\theta_2 = 30^\circ$ to $\theta'_2 = 120^\circ$. Find the absolute displacement of Q .

Figure P2.11 $R_{AO_2} = R_{BO_4} = 3$ in and $R_{BA} = R_{O_4O_2} = 6$ in.

- †2.12** The double-slider linkage is driven by moving sliding block 2. Write the loop-closure equation. Solve analytically for the position of sliding block 4. Check the result graphically for the position where $\phi = -45^\circ$.

* When assigning problems, the instructor may wish to specify the method of solution to be used, since a variety of approaches are presented in the text.

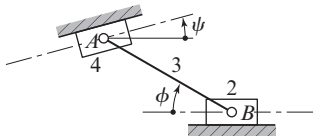


Figure P2.12 $R_{AB} = 200$ mm and $\psi = 15^\circ$.

- †2.13 The offset slider-crank linkage is driven by crank 2. Write the loop-closure equation. Solve for the position of slider 4 as a function of θ_2 .

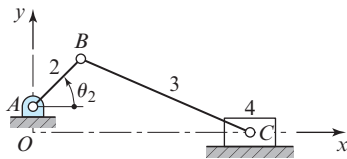


Figure P2.13 $R_{AO} = 1$ in, $R_{BA} = 2.5$ in, and $R_{CB} = 7$ in.

- 2.14 Define a set of vectors that is suitable for a complete kinematic analysis of the mechanism. Label and show the sense and orientation of each vector. Write the vector loop equation(s) for the mechanism. Identify suitable input(s), known quantities, unknown variables, and any constraints. If you identify constraints, then write the constraint equation(s).

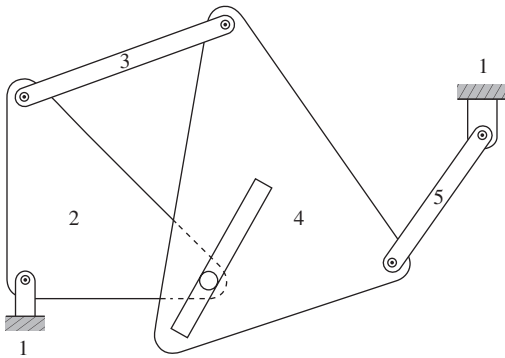


Figure P2.14

- 2.15 Define a set of vectors that is suitable for a complete kinematic analysis of the mechanism. Label and show the sense and orientation of each vector. Assuming rolling with no slip between rack 4 and pinion 5, write the vector loop equation(s) for the mechanism. Identify suitable input(s), known quantities, unknown

variables, and any constraints. If you identify constraints, then write the constraint equation(s).

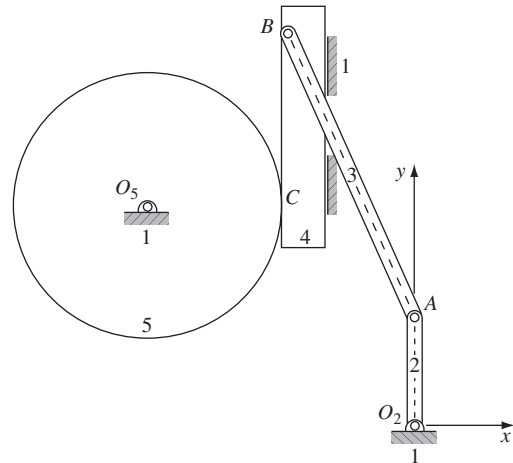


Figure P2.15 Rack-and-pinion mechanism.

- 2.16 Define a set of vectors that is suitable for a complete kinematic analysis of the mechanism. Label and show the sense and orientation of each vector. Assuming rolling with no slipping between gears 2 and 5, write the vector loop equation(s) for the mechanism. Identify suitable input(s), known quantities, unknown variables, and any constraints. If you identify constraints, then write the constraint equation(s).

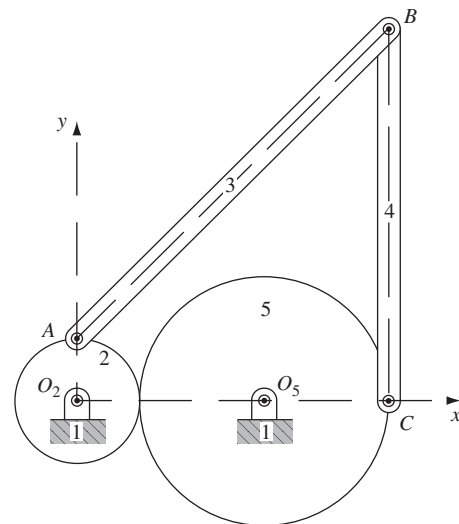


Figure P2.16 Geared five-bar mechanism.

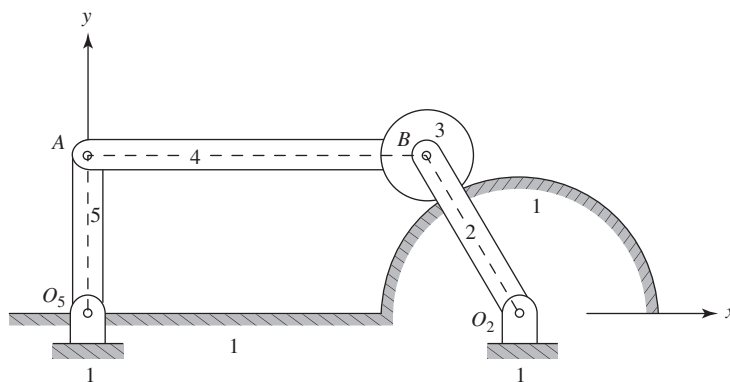


Figure P2.17

- 2.17** Gear 3, which is pinned to link 4 at point B , is rolling without slipping on semicircular ground link 1. The radius of gear 3 is ρ_3 , and the radius of the ground link is ρ_1 . Define a set of vectors that is suitable for a complete kinematic analysis of the mechanism. Label and show the sense and orientation of each vector. Write the vector loop equation(s) for the mechanism. Identify suitable input(s), known quantities, unknown variables, and any constraints. If you identify constraints, then write the constraint equation(s).
- 2.18** For the mechanism in Fig. P1.6, define a set of vectors that is suitable for a complete kinematic analysis of the mechanism. Label and show the sense and orientation of each vector. Write the vector loop equation(s) for the mechanism. Identify suitable input(s), known quantities, unknown variables, and any constraints. If you identify constraints, then write the constraint equation(s).
- 2.19** For the mechanism in Fig. P1.8, define a set of vectors that is suitable for a complete kinematic analysis of the mechanism. Label and show the sense and orientation of each vector. Write the vector loop equation(s) for the mechanism. Identify suitable input(s), known quantities, unknown variables, and any constraints. If you identify constraints, then write the constraint equation(s).
- 2.20** For the mechanism in Fig. P1.9, define a set of vectors that is suitable for a complete kinematic analysis of the mechanism. Label and show the sense and orientation of each vector. Write the vector loop equation(s) for the mechanism. Identify suitable input(s), known quantities, unknown variables, and any constraints. If you identify constraints, then write the constraint equation(s).
- 2.21** For the mechanism in Fig. P1.10, define a set of vectors that is suitable for a complete kinematic analysis of the mechanism. Label and show the sense and orientation of each vector. Write the vector loop equation(s) for the mechanism. Identify suitable input(s), known quantities, unknown variables, and any constraints. If you identify constraints, then write the constraint equation(s).
- †2.22** Write a calculator program to find the sum of any number of two-dimensional vectors expressed in mixed rectangular or polar forms. The result should be obtainable in either form with the magnitude and angle of the polar form having only positive values.
- †2.23** Write a computer program to plot the coupler curve of any crank-rocker or double-crank form of the four-bar linkage. The program should accept four link lengths and either rectangular or polar coordinates of the coupler point with respect to the coupler.

- †2.24 Plot the path of point P for: (a) inverted slider-crank linkage; (b) second inversion of the slider-crank linkage; (c) Scott-Russell straight-line linkage; and (d) drag-link linkage.

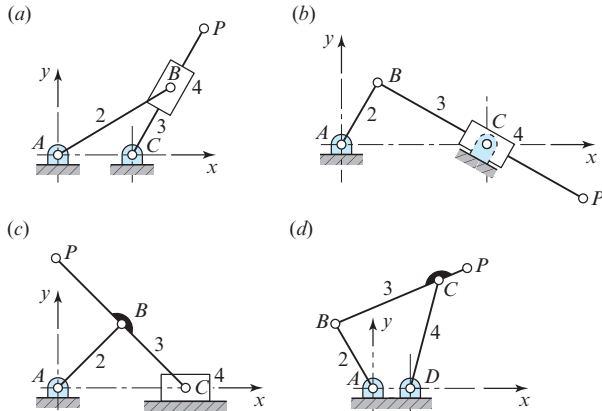


Figure P2.24 (a) $R_{CA} = 2$ in, $R_{BA} = 3.5$ in, and $R_{PC} = 4$ in; (b) $R_{CA} = 40$ mm, $R_{BA} = 20$ mm, and $R_{PB} = 65$ mm; (c) $R_{BA} = R_{CB} = R_{PB} = 25$ mm; (d) $R_{DA} = 1$ in, $R_{BA} = 2$ in, $R_{CB} = R_{CD} = 3$ in, and $R_{PB} = 4$ in.

- †2.25 Using the offset slider-crank linkage in Fig. P2.13, find the crank angles corresponding to the extreme values of the transmission angle.

2.26 Section 1.10 states that the transmission angle reaches an extreme value for the four-bar linkage when the crank lies on the line between the fixed pivots. Referring to Fig. 2.19, this means that γ reaches a maximum or minimum when crank 2 is collinear with line O_2O_4 . Show analytically that this statement is true.

- 2.27 Define a set of vectors that is suitable for a complete kinematic analysis of the mechanism. Label and show the sense and orientation of each vector. Write the vector loop equation(s) for

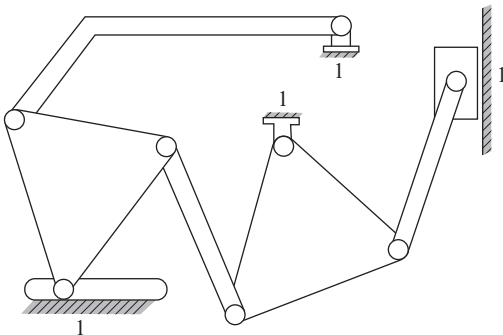


Figure P2.27

the mechanism. Identify suitable input(s), known quantities, unknown variables, and any constraints. If you identify constraints, then write the constraint equation(s).

- 2.28 Define a set of vectors that is suitable for a complete kinematic analysis of the mechanism. Label and show the sense and orientation of each vector. Write the vector loop equation(s) for the mechanism. Identify suitable input(s), known quantities, unknown variables, and any constraints. If you identify constraints, then write the constraint equation(s).

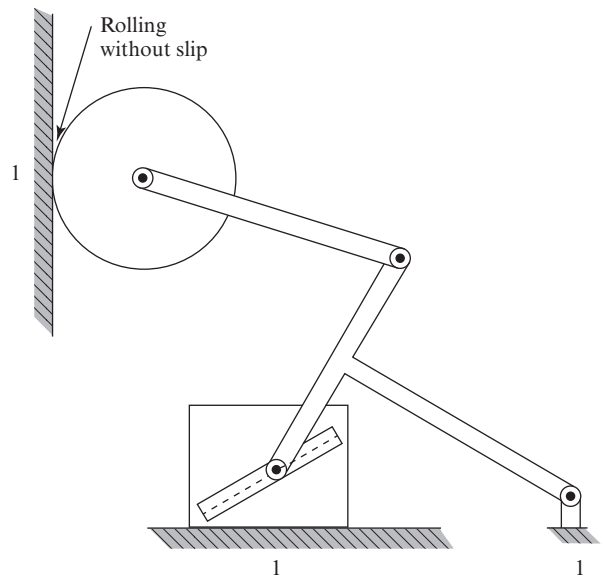


Figure P2.28

- 2.29 Define a set of vectors that is suitable for a complete kinematic analysis of the mechanism. Label and show the sense and orientation of each vector. Write the vector loop equation(s) for the mechanism. Identify suitable input(s), known quantities, unknown variables, and any constraints. If you identify constraint(s), then write the constraint equation(s).

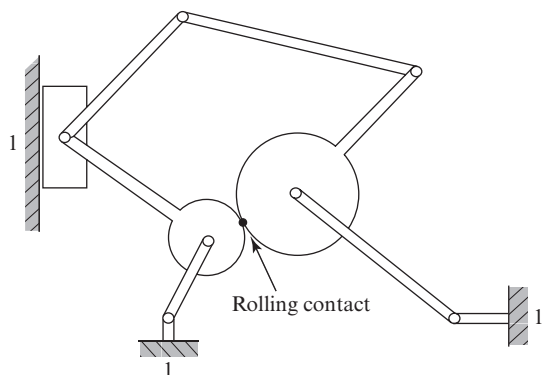


Figure P2.29

- 2.30** Define a set of vectors that is suitable for a complete kinematic analysis of the mechanism. Label and show the sense and orientation of each vector. Write the vector loop equation(s) for the mechanism. Identify suitable input(s), known quantities, unknown variables, and any constraints. If you identify constraint(s), then write the constraint equation(s).

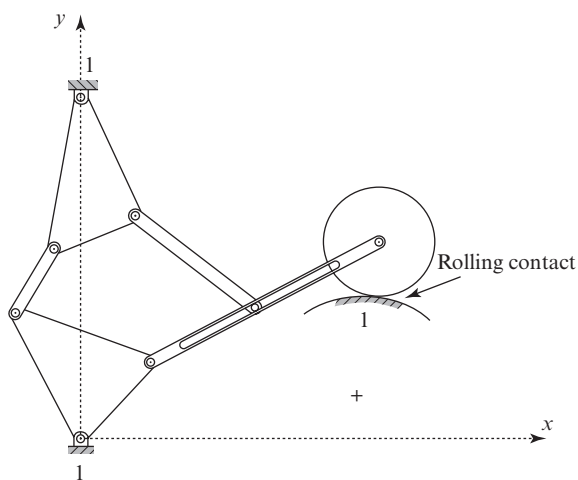


Figure P2.30

- 2.31** For the input angle $\theta_2 = 300^\circ$, measured counterclockwise from the x axis, determine the two postures of link 4.

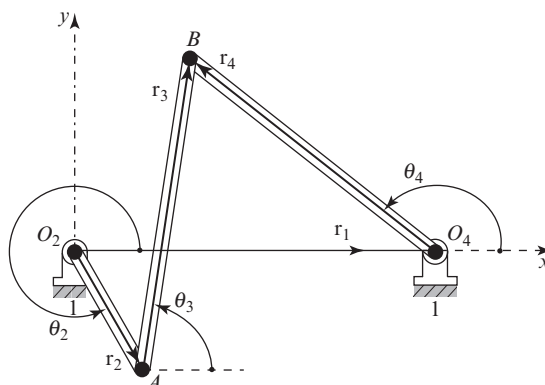


Figure P2.31 $r_2 = 60$ mm, $r_3 = 140$ mm, $r_4 = 140$ mm, and $r_1 = 160$ mm.

- 2.32** For the input angle $\theta_2 = 60^\circ$, measured counterclockwise from the x axis, determine the two postures of link 4.

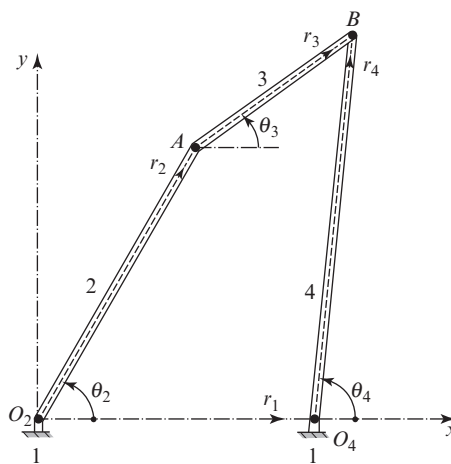


Figure P2.32 $r_2 = 80$ mm, $r_3 = 50$ mm, $r_4 = 100$ mm, and $r_1 = 70$ mm.

2.33 Consider a four-bar linkage for which ground link 1 is 14 in, input link 2 is 7 in, coupler link 3 is 10 in, and output link 4 is 8 in. The fixed x and y axes are specified as horizontal and vertical, respectively. The origin of this reference frame is coincident with the ground pivot of link 2, and the ground link is aligned with the x axis. For input angle $\theta_2 = 60^\circ$ (counterclockwise from the x axis): (a) Using a suitable scale, draw the linkage in the open and crossed postures and measure the values of variables θ_3 and θ_4 for each posture. (b) Use trigonometry (that is, the laws of sines and cosines) to determine θ_3 and θ_4 for the open posture. (c) Use Freudenstein's equation to determine θ_3 and θ_4 for both postures. (d) Use the Newton-Raphson iteration procedure to determine θ_3 and θ_4 for the open posture. Using the measurements in (a) as initial estimates for θ_3 and θ_4 , iterate until the two variables converge to within 0.01° .

2.34 A crank-rocker four-bar linkage is shown in two different postures for which $\theta_2 = 150^\circ$ and $\theta'_2 = 240^\circ$. Determine θ_3 and θ_4 for the open posture and θ'_3 and θ'_4 for the crossed posture.

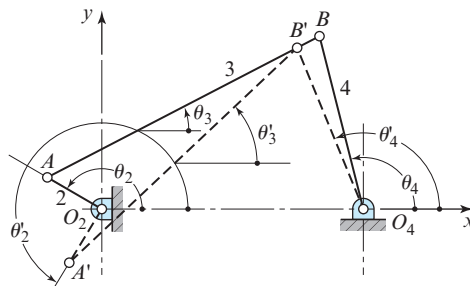
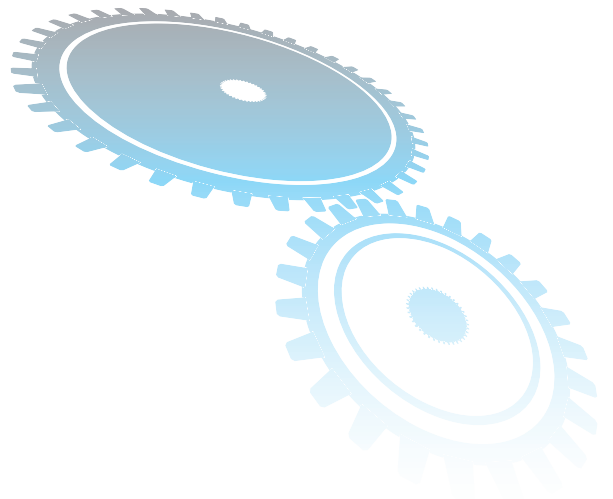


Figure P2.34 $r_{O_4O_2} = 600$ mm, $r_{AO_2} = 140$ mm, $r_{BA} = 690$ mm, and $r_{BO_4} = 400$ mm.

3

Velocity



3.1 DEFINITION OF VELOCITY

In Fig. 3.1, a moving point is first observed at location P , defined by the absolute position vector $\mathbf{R}_P$. After a small time increment Δt , its location is observed to have changed to P' , defined by $\mathbf{R}'_P$. Recall from Eq. (2.58) that the displacement of the point during this time increment is defined as

$$\Delta \mathbf{R}_P = \mathbf{R}'_P - \mathbf{R}_P.$$

The *average velocity* of the point during the time increment Δt is defined by the ratio $\Delta \mathbf{R}_P / \Delta t$. The *instantaneous velocity* (hereafter simply called *velocity*) is defined by the limit of this ratio as the time increment goes to zero and is given by

$$\mathbf{V}_P = \lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{R}_P}{\Delta t} = \frac{d\mathbf{R}_P}{dt}. \quad (3.1)$$

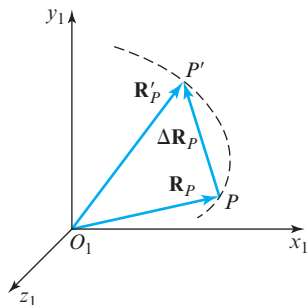


Figure 3.1 Displacement of a moving point.

Since $\Delta \mathbf{R}_P$ is a vector, there are two convergences in taking this limit, the magnitude and the direction. Therefore, the velocity of a point is a vector quantity equal to the time rate of change of its position. Like the position and displacement vectors, the velocity vector is defined for a specific point. The term “velocity” should not be applied to a line, coordinate system, volume, or other collection of points, since the velocity of each point may be different.

We recall from Chap. 2 that, for their definitions, the position vectors $\mathbf{R}_P$ and $\mathbf{R}'_P$ depend on the posture of the coordinate system of the observer. The displacement vector $\Delta \mathbf{R}_P$ and the velocity vector $\mathbf{V}_P$, on the other hand, are independent of the location of the coordinate system or the location of the observer within the coordinate system. However, the velocity vector $\mathbf{V}_P$ does depend on the motion, if any, of the observer or the coordinate system during the time interval; it is for this reason that the observer is assumed to remain stationary within the coordinate system. If the coordinate system involved is the absolute coordinate system, the velocity is referred to as an *absolute velocity* and is denoted by $\mathbf{V}_{P/1}$ or simply $\mathbf{V}_P$. This is consistent with the notation used for absolute position and absolute displacement.

3.2 ROTATION OF A RIGID BODY

When a rigid body translates, the motion of any particular point is equal to the motion of every other point of the body. When a rigid body rotates, however, two arbitrarily chosen points, P and Q , do not undergo the same motion, and a coordinate system attached to the body does not remain parallel to its initial orientation; that is, the body undergoes some angular displacement $\Delta\theta$. This was discussed in Sec. 2.16.

Angular displacements were not treated in detail in Chap. 2, since, in general, they cannot be treated as vectors. The reason is that they do not obey the laws of vector addition. If a rigid body undergoes multiple finite angular displacements in succession, in three dimensions, the result depends on the order in which the displacements take place. To show this, consider the rectangular body $ABCO$ in Fig. 3.2a. If the body is first rotated by -90° (that is, clockwise) about the y axis and then rotated by $+90^\circ$ (that is, counterclockwise) about the x axis, the final posture of the body is in the yz plane. In Fig. 3.2b, the body occupies the same starting posture and is again rotated about the same axes, through the same angles, and in the same directions; however, in this case, the first rotation is about the x axis and the second is about the y axis. The order of the rotations is reversed and the final posture of the body is now in the zx plane rather than in the yz plane, as it was before. Since this characteristic does not correspond to the commutative law of vector addition, three-dimensional angular displacements cannot be treated as vectors.

Angular displacements that occur about the same axis or parallel axes, on the other hand, do follow the commutative law. Also, infinitesimally small angular displacements are commutative. To avoid confusion, we will treat all finite angular displacements as scalar quantities. However, we will have occasion to treat infinitesimal angular displacements as vectors.

Recall from Sec. 2.15 the definition of the displacement difference between two points, P and Q , both attached to the same rigid body, as shown again in Fig. 3.3. The displacement-difference vector is entirely attributable to the rotation of the body; there is no displacement difference between points in a rigid body undergoing a translation. We

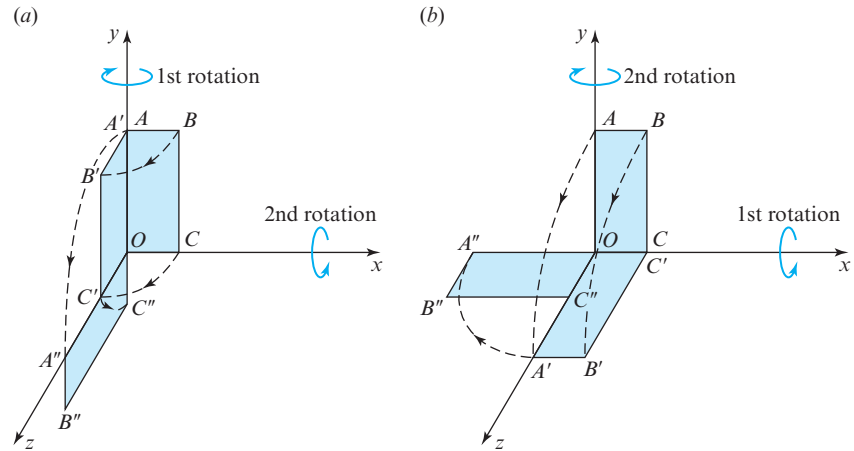


Figure 3.2 Three-dimensional angular displacements.

reach this conclusion by picturing the displacement as occurring in two steps. First, the body is assumed to translate through the displacement $\Delta \mathbf{R}_Q$ to the posture $x_2^* y_2^* z_2^*$. Next, the body is assumed to rotate about point Q^* to the posture $x_2' y_2' z_2'$.

Another way to interpret the displacement difference $\Delta \mathbf{R}_{PQ}$ is to imagine a moving coordinate system whose origin travels along with point Q but whose axes remain parallel

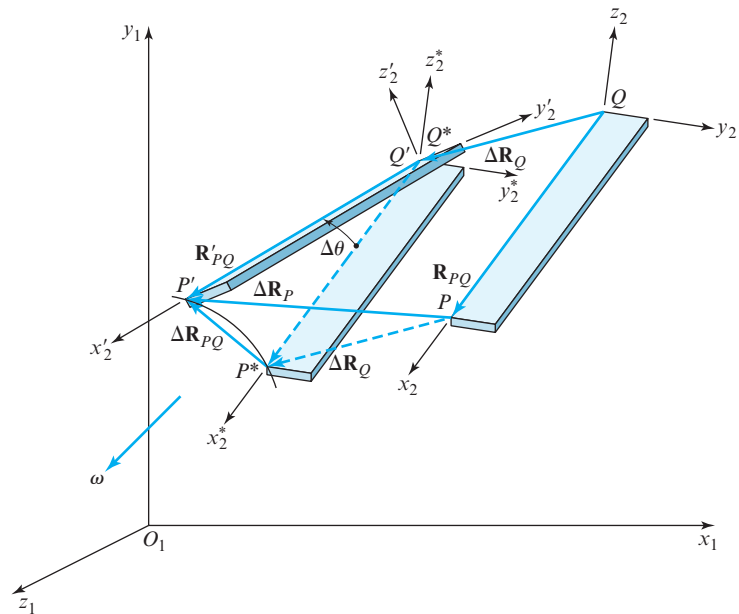


Figure 3.3 Displacement difference between points P and Q .

to the absolute axes $x_1y_1z_1$. Note that this coordinate system does not rotate. An observer in this moving coordinate system observes no motion for point Q , since it remains fixed in that observer's coordinate system. For the displacement of point P , such a moving observer will only observe the displacement-difference vector $\Delta \mathbf{R}_{PQ}$. It seems to such an observer that point Q remains fixed and that the body rotates about this seemingly fixed point, as shown in Fig. 3.4.

No matter whether the observer is in the fixed coordinate system $x_1y_1z_1$ or in the moving coordinate system $x_2y_2z_2$, the body appears to rotate through some total angle $\Delta\theta$ in its displacement from $x_2y_2z_2$ to $x'_2y'_2z'_2$. If we take the point of view of the fixed observer, the location of the axis of rotation is not obvious. As seen by the translating observer, the axis passes through the apparently stationary point Q ; all points in the body appear to travel in circular paths about this axis, and any line in the body whose direction is normal to this axis appears to undergo an identical angular displacement $\Delta\theta$. The rotating position-difference vector $\mathbf{R}_{PQ}$ generates a cone.

The *angular velocity* of a rotating body is now defined as a vector quantity $\boldsymbol{\omega}$ having a direction parallel to the instantaneous axis of rotation. The magnitude of the angular velocity is defined as the time rate of change of the angular orientation of any line in the body whose direction is normal to the axis of rotation. If we designate the angular displacement of any of these lines as $\Delta\theta$ and the time interval as Δt , then the magnitude of the angular velocity vector is

$$\omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt}. \quad (3.2)$$

Since we have agreed to treat counterclockwise rotations as positive, the sense of the angular velocity vector along the axis of rotation is in accordance with the right-hand rule.

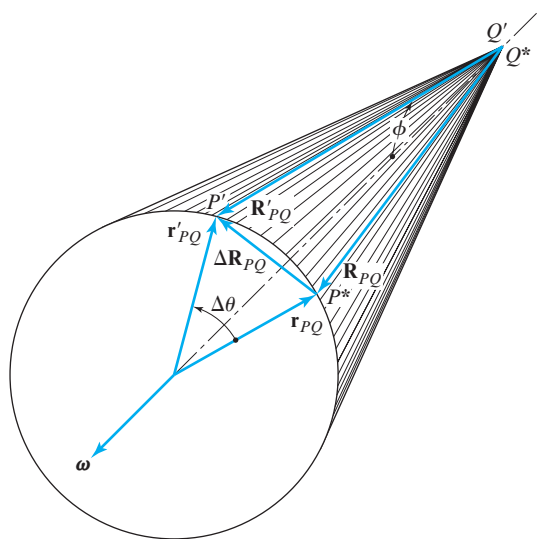


Figure 3.4 Displacement difference $\Delta \mathbf{R}_{PQ}$ as seen by a translating observer.

3.3 VELOCITY DIFFERENCE BETWEEN POINTS OF A RIGID BODY

Figure 3.5a shows another view of the same rigid-body displacement that was presented in Fig. 3.3. This is the view seen by an observer in the absolute coordinate system looking directly along the axis of rotation of the moving body, from the tip of the angular velocity vector. Therefore, the angular displacement $\Delta\theta$ is observed in true size, and the projections of *all* lines in the body rotate through this same angle during the displacement. The displacement vectors and the position-difference vectors, however, are not necessarily observed in true size; their projections may appear foreshortened under this viewing angle.

Figure 3.5b shows the same rigid-body displacement from the same viewing angle, but this time from the point of view of the translating observer. Thus, this figure corresponds to the base of the cone of Fig. 3.4. We note that the two vectors labeled $\mathbf{r}_{PQ}$ and $\mathbf{r}'_{PQ}$ are the foreshortened projections of $\mathbf{R}_{PQ}$ and $\mathbf{R}'_{PQ}$, and we observe that their magnitudes are

$$r_{PQ} = r'_{PQ} = R_{PQ} \sin \phi, \quad (a)$$

where ϕ is the constant angle from the angular velocity vector $\boldsymbol{\omega}$ to the rotating position-difference vector $\mathbf{R}_{PQ}$ as it traverses the cone. Figure 3.5b, can also be interpreted as a scale drawing corresponding to Eq. (2.60); that is,

$$\Delta\mathbf{R}_{PQ} = \mathbf{R}'_{PQ} - \mathbf{R}_{PQ}. \quad (b)$$

We can calculate the magnitude of the displacement-difference vector $\Delta\mathbf{R}_{PQ}$ by drawing the perpendicular bisector of this vector as shown in Fig. 3.5b. From this construction

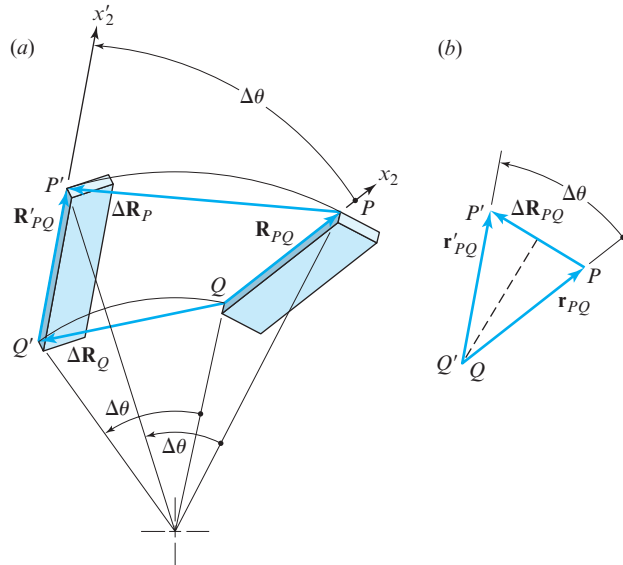


Figure 3.5 (a) True view of angular displacements; (b) vector subtraction to form displacement difference $\Delta\mathbf{R}_{PQ}$.

we have

$$\Delta R_{PQ} = 2r_{PQ} \sin \frac{\Delta\theta}{2}. \quad (c)$$

Substituting Eq. (a) into Eq. (c), we have

$$\Delta R_{PQ} = 2(R_{PQ} \sin \phi) \sin \frac{\Delta\theta}{2}. \quad (d)$$

If we limit ourselves to infinitesimal displacements, the sine of the angular displacement can be approximated by the angle itself; that is,

$$\Delta R_{PQ} = 2(R_{PQ} \sin \phi) \frac{\Delta\theta}{2} = \Delta\theta R_{PQ} \sin \phi. \quad (e)$$

Dividing by the time increment Δt , noting that the magnitude R_{PQ} and the angle ϕ are constant during the interval, and taking the limit gives

$$\lim_{\Delta t \rightarrow 0} \left(\frac{\Delta R_{PQ}}{\Delta t} \right) = \lim_{\Delta t \rightarrow 0} \left(\frac{\Delta\theta}{\Delta t} \right) R_{PQ} \sin \phi = \omega R_{PQ} \sin \phi. \quad (f)$$

Recalling the definition of ϕ as the constant angle from vector $\boldsymbol{\omega}$ to vector $\mathbf{R}_{PQ}$, we can restore the vector attributes of Eq. (f) by recognizing this result as the magnitude of a vector cross-product. Therefore,

$$\lim_{\Delta t \rightarrow 0} \left(\frac{\Delta \mathbf{R}_{PQ}}{\Delta t} \right) = \frac{d\mathbf{R}_{PQ}}{dt} = \boldsymbol{\omega} \times \mathbf{R}_{PQ}. \quad (g)$$

This form is so important and so useful that it is given its own name and notation; it is called the *velocity-difference* vector and is denoted $\mathbf{V}_{PQ}$; that is,

$$\mathbf{V}_{PQ} = \frac{d\mathbf{R}_{PQ}}{dt} = \boldsymbol{\omega} \times \mathbf{R}_{PQ}. \quad (3.3)$$

Let us now recall the displacement-difference equation, see Eq. (2.59)—namely,

$$\Delta \mathbf{R}_P = \Delta \mathbf{R}_Q + \Delta \mathbf{R}_{PQ}. \quad (h)$$

Dividing this equation by Δt , and taking the limit as the time increment goes to zero, gives

$$\lim_{\Delta t \rightarrow 0} \left(\frac{\Delta \mathbf{R}_P}{\Delta t} \right) = \lim_{\Delta t \rightarrow 0} \left(\frac{\Delta \mathbf{R}_Q}{\Delta t} \right) + \lim_{\Delta t \rightarrow 0} \left(\frac{\Delta \mathbf{R}_{PQ}}{\Delta t} \right). \quad (i)$$

Eq. (3.1), (g), and (3.3) then Eq. (i) can be written as

$$\mathbf{V}_P = \mathbf{V}_Q + \mathbf{V}_{PQ}. \quad (3.4)$$

This important equation is called the *velocity-difference equation* and, together with Eq. (3.3), forms one of the basic equations of all velocity analysis techniques.

Equation (3.4) can be written for any two points with no restriction. However, reviewing the previous derivation, we recognize that Eq. (3.3) cannot be applied to an arbitrary pair of points. This equation is *only valid if the two points are fixed in the same rigid body*.^{*} This restriction can, perhaps, be better remembered if all subscripts are written explicitly; that is,

$$\mathbf{V}_{P_2Q_2} = \boldsymbol{\omega}_2 \times \mathbf{R}_{P_2Q_2}. \quad (j)$$

Note that the link-number subscripts are the same throughout this equation, and so, in the interest of brevity, are often suppressed. If a mistaken attempt is made to apply Eq. (3.3) when points P and Q are not in the same rigid body, the error should be discovered, since it will not be clear which angular velocity vector should be used.

3.4 VELOCITY POLYGONS; VELOCITY IMAGES

One important approach to velocity analysis is graphic. As observed in graphic posture analysis, Sec. 2.8, it is primarily of use in two-dimensional mechanism problems when the solution of only a single posture is required. The major advantages are that a solution can be achieved quickly and that visualization of, and insight into, the problem are enhanced by the graphic approach [7].

Consider the planar motion of the unconstrained link ABC shown in Fig. 3.6*a*. Given the velocities of points A and B , the problem is to determine: (a) the angular velocity of the link, and (b) the velocity of point C . Let us agree that a scale diagram of the link has been drawn for the given posture and that any required position-difference vectors can be scaled from the diagram.

For the solution of Part (a), we first consider the velocity-difference equation, Eq. (3.4), relating points A and B ; that is,

$$\overset{\sqrt{\sqrt{\quad}}}{\mathbf{V}}_B = \overset{\sqrt{\sqrt{\quad}}}{\mathbf{V}}_A + \overset{??}{\mathbf{V}}_{BA}, \quad (a)$$

where the two unknowns are the magnitude and the direction of the velocity-difference vector, $\mathbf{V}_{BA}$. Fig. 3.6*b* shows the graphic solution to this equation. After an appropriate scale is chosen to represent velocity vectors, vectors $\mathbf{V}_A$ and $\mathbf{V}_B$ are both drawn to scale, starting from a common origin and in the two known directions. The vector spanning the termini of $\mathbf{V}_A$ and $\mathbf{V}_B$ is the velocity-difference vector $\mathbf{V}_{BA}$ and is correct, within graphic accuracy, in both magnitude and direction.

The velocity-difference vector can be written from Eq. (3.3) as

$$\mathbf{V}_{BA} = \boldsymbol{\omega} \times \mathbf{R}_{BA}. \quad (b)$$

Since the link is in planar motion, the angular velocity vector, $\boldsymbol{\omega}$, lies perpendicular to the plane of motion—that is, perpendicular to vectors $\mathbf{V}_{BA}$ and $\mathbf{R}_{BA}$. Considering only the

^{*} More precisely, the restriction is the requirement that the distance R_{PQ} remains constant. However, in applications, the above wording fits most real situations.

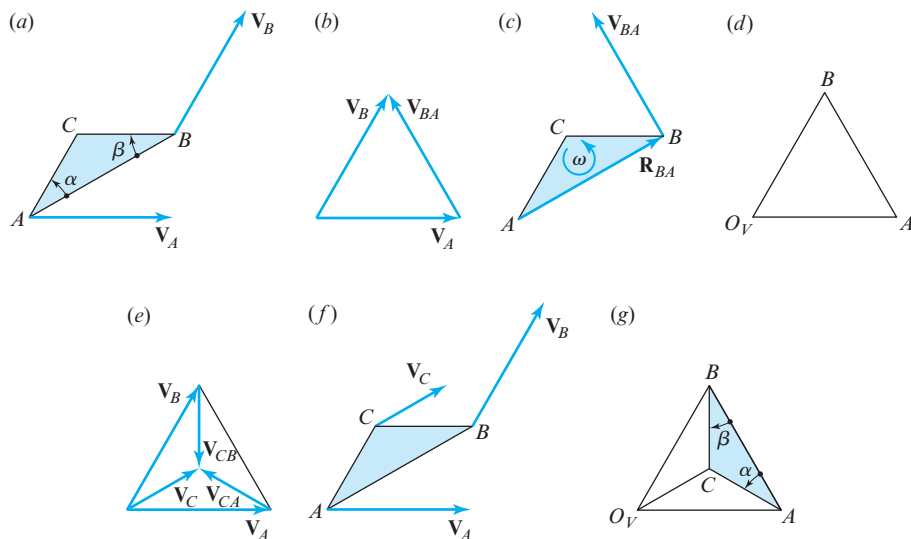


Figure 3.6 Graphic velocity analysis of link ABC.

magnitudes in Eq. (b) gives

$$V_{BA} = \omega R_{BA} \sin 90^\circ = \omega R_{BA}.$$

Rearranging this equation, the angular velocity of the link is

$$\omega = \frac{V_{BA}}{R_{BA}}. \quad (c)$$

Therefore, the magnitude of the angular velocity can be obtained by scaling V_{BA} from Fig. 3.6b and R_{BA} from Fig. 3.6a, being careful to properly apply the scale factors for units; it is common practice to evaluate ω in units of radians per second.

The magnitude, ω , is not a complete solution for the angular velocity vector; the direction must also be determined. As observed earlier, the ω vector is perpendicular to the plane of the link itself, since the motion is planar. However, this does not indicate whether ω is directed into or out of the plane of the motion. Taking the point of view of a translating observer, moving with point A but not rotating, we can visualize the link as rotating about point A; this is shown in Fig. 3.6c. The velocity difference V_{BA} is the only velocity seen by such an observer. Therefore, interpreting V_{BA} to indicate the direction of rotation of point B about point A, we discover the direction of ω , counterclockwise in this example. Although it is not strict vector notation, it is common practice in two-dimensional problems to indicate the final solution in the form $\omega = xxx \text{ rad/s ccw}$, where “xxx” is a numeric value indicating the magnitude, and “ccw” or “cw” indicates the direction.

The practice of constructing vector diagrams using thick dark lines, such as in Fig. 3.6b, makes them easy to read, but when the diagram is the graphic solution of an equation, it is not very accurate. For this reason, it is preferable to construct the graphic

solution with thin sharp lines, made with a hard drawing pencil, as shown in Fig. 3.6d. The solution begins by choosing a scale and a point labeled O_V to represent zero velocity. Absolute velocities such as $\mathbf{V}_A$ and $\mathbf{V}_B$ are constructed with their origins at O_V , and their termini are labeled as points A and B . The line *from A to B* then represents the velocity difference $\mathbf{V}_{BA}$. As we continue, we observe that these labels at the vertices are sufficient to determine the precise notation of all velocity differences represented by lines in the diagram. Note, for example, that $\mathbf{V}_{BA}$ is represented by the vector *from point A to point B*. With this labeling convention, no arrowheads or additional notation are necessary and do not clutter the diagram. Such a diagram is called a *velocity polygon* and adds considerable convenience to the graphic solution technique.

A danger of this convention, however, is that the reader may begin to think of the technique as a series of graphic “tricks” and lose sight of the fact that each line that is drawn can be, and should be, fully justified by a corresponding vector equation. The graphics are merely a convenient solution technique and not a substitute for a sound theoretical basis.

Returning to Fig. 3.6c, it may have appeared coincidental that the velocity-difference $\mathbf{V}_{BA}$ was perpendicular to $\mathbf{R}_{BA}$. Recalling Eq. (b), however, we see that it is a necessary outcome, resulting from the cross-product with the ω vector. We will take advantage of this relationship in the next step.

For the solution of Part (b), we can relate the absolute velocity of point C to the absolute velocities of both points A and B , by two velocity-difference equations; that is,

$$\mathbf{V}_C = \overset{\swarrow}{\mathbf{V}_A} + \overset{\nearrow}{\mathbf{V}_{CA}} = \overset{\swarrow}{\mathbf{V}_B} + \overset{\nearrow}{\mathbf{V}_{CB}}. \quad (d)$$

Since points A , B , and C are all fixed in the same link, each of the velocity-difference vectors, $\mathbf{V}_{CA}$ and $\mathbf{V}_{CB}$, is of the form $\omega \times \mathbf{R}$, using $\mathbf{R}_{CA}$ and $\mathbf{R}_{CB}$, respectively. As a result, $\mathbf{V}_{CA}$ is perpendicular to $\mathbf{R}_{CA}$, and $\mathbf{V}_{CB}$ is perpendicular to $\mathbf{R}_{CB}$. The directions of these two vectors are, therefore, indicated as known in Eq. (d).

Since ω has already been determined, it is easy to calculate the magnitudes of $\mathbf{V}_{CA}$ and $\mathbf{V}_{CB}$ using equations similar to Eq. (c); however, this is avoided here. Instead, we form the graphic solution to Eq. (d). Equation (d) states that a vector that is perpendicular to $\mathbf{R}_{CA}$ must be added to $\mathbf{V}_A$ and that the result is equal to the sum of $\mathbf{V}_B$ and a vector perpendicular to $\mathbf{R}_{CB}$. The solution is shown in Fig. 3.6e. In practice, the solution is commonly continued on the same diagram as Fig. 3.6d and results in Fig. 3.6g. A line perpendicular to $\mathbf{R}_{CA}$ (representing $\mathbf{V}_{CA}$) is drawn starting at point A (representing addition to $\mathbf{V}_A$); similarly, a line is drawn perpendicular to $\mathbf{R}_{CB}$ starting at point B . The point of intersection of these two lines is labeled C and it represents the solution to Eq. (d). The line from O_V to point C now represents the absolute velocity $\mathbf{V}_C$. This velocity can be transferred back to the link and interpreted as $\mathbf{V}_C$ in both magnitude and direction, as shown in Fig. 3.6f.

In seeing the shading and the labeled angles α and β in Figs. 3.6g and 3.6a, we are led to investigate whether the two triangles labeled ABC in each of these figures are similar in shape, as they appear to be. In reviewing the construction steps, we see that indeed they are similar, since the velocity-difference vectors $\mathbf{V}_{BA}$, $\mathbf{V}_{CA}$, and $\mathbf{V}_{CB}$ are each perpendicular to the respective position-difference vectors $\mathbf{R}_{BA}$, $\mathbf{R}_{CA}$, and $\mathbf{R}_{CB}$. This property is always true, regardless of the shape of the moving link; a similarly shaped figure appears in the velocity polygon. The sides of the polygon are always scaled up or down by a factor equal to the angular velocity of the link, and they are always rotated by 90° in the directions of their

angular velocities. These properties result from the fact that each velocity-difference vector between two points on the link results from a cross-product of the same ω vector with the corresponding position-difference vector. This similarly shaped figure in the velocity polygon is commonly referred to as the *velocity image* of the link, and every link has a corresponding velocity image in the velocity polygon.

The graphic procedures in this section were first published in 1883 by Mehmke as the following theorem [10]:

The end points of the velocity vectors of the points of a plane rigid body, when plotted from a common origin, produce a figure that is geometrically similar to the original figure (image diagram).

It is this theorem that allows clarity in the velocity polygon despite the minimal labeling that is required. This becomes evident in the upcoming two examples and continues throughout the text.

The concept of the velocity image allows the graphic solution to be obtained more efficiently. For example, there is no need to explicitly write Eq. (d). Once the solution has progressed to the state of Fig. 3.6d, the velocity-image points A and B are known. Therefore, one can use these two points as the base of a triangle similar to the shape of the link and construct the image point C directly. Care must be taken not to allow the triangle to be flipped over between the position diagram and the velocity image, but the solution can proceed quickly, accurately, and naturally, resulting in Fig. 3.6g. Again, we note that all steps in the solution are based on strictly derived vector equations and are not graphic tricks. It is wise to continue to write the corresponding vector equations until one is thoroughly familiar with the procedure and its vector basis.

The following are important properties of velocity images:

1. The velocity image of each link in the velocity polygon is a scale reproduction of the shape of the real link.
2. The velocity image of each link is rotated 90° from the real link in the direction of the angular velocity of that link.
3. The letters identifying the vertices of each link in the velocity polygon are the same as, and progress around the velocity image in the same order and in the same angular sense as, around the real link.
4. The ratio of the size of the velocity image of a link to the size of the link itself is equal to the magnitude of the angular velocity of the link. In general, this is not the same for different links in a mechanism.
5. The velocities of all points on a translating link are equal, and the angular velocity of the link is zero. Therefore, the velocity image of a link that is translating shrinks to a single point in the velocity polygon.
6. Point O_V in the velocity polygon is the image of all points with zero absolute velocity; it is the velocity image of the stationary link.
7. The absolute velocity of any point on any link is represented in the velocity polygon by the line from O_V to the image of the point. The velocity-difference vector between any two points, say P and Q , is represented by the line to image point P from image point Q .

To illustrate the graphic velocity analysis of mechanisms and the role of velocity images, we consider two typical linkage problems.

EXAMPLE 3.1

The four-bar linkage in the posture shown in Fig. 3.7a is driven by crank 2 at a constant angular velocity $\omega_2 = 900 \text{ rev/min}$ ccw. For this posture, determine the angular velocities of coupler link 3 and output link 4 and the velocities of point E in link 3 and point F in link 4.

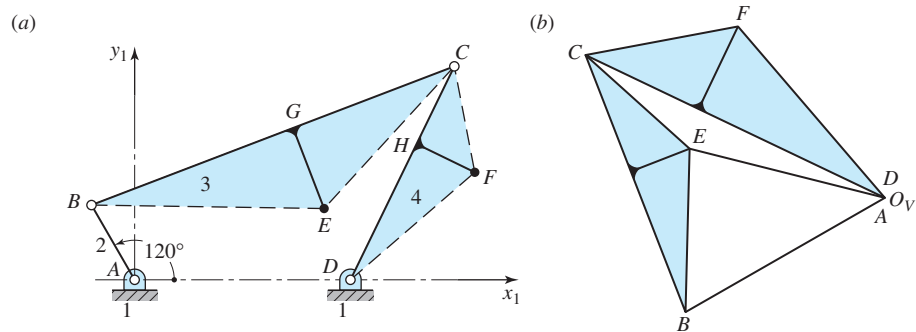


Figure 3.7 (a) $R_{BA} = 4 \text{ in}$, $R_{CB} = 18 \text{ in}$, $R_{CD} = 11 \text{ in}$, $R_{DA} = 10 \text{ in}$, $R_{GB} = 10 \text{ in}$, $R_{EG} = 4 \text{ in}$, $R_{HD} = 7 \text{ in}$, and $R_{FH} = 3 \text{ in}$; (b) velocity polygon.

GRAPHIC SOLUTION

First, we draw the linkage to a suitable scale. Then, we calculate the angular velocity of link 2 in radians per second; that is,

$$\omega_2 = \left(900 \frac{\text{rev}}{\text{min}}\right) \left(2\pi \frac{\text{rad}}{\text{rev}}\right) \left(\frac{1 \text{ min}}{60 \text{ s}}\right) = 94.25 \text{ rad/s ccw.} \quad (1)$$

Next we calculate the velocity of point B from Eqs. (3.3) and (3.4), noting that point A is fixed:

$$\begin{aligned} \mathbf{V}_B &= \mathbf{X}_A^0 + \mathbf{V}_{BA} = \omega_2 \times \mathbf{R}_{BA}, \\ V_B &= (94.25 \text{ rad/s}) \left(\frac{4}{12} \text{ ft}\right) = 31.42 \text{ ft/s.} \end{aligned} \quad (2)$$

It is important to note that the form “ $\omega \times \mathbf{R}$ ” was used for the velocity difference and not for the absolute velocity $\mathbf{V}_B$ directly. Next, we choose a suitable velocity scale and an arbitrary location for an origin, denoted as O_V , shown in Fig. 3.7b. Since $\mathbf{V}_A = \mathbf{0}$, the image point A is coincident with O_V . We construct the line AB perpendicular to $\mathbf{R}_{BA}$ and toward the lower left because of the counterclockwise sense of ω_2 ; this line represents $\mathbf{V}_{BA}$.

Now we write two equations for the velocity of point C . Since the velocities of points C_3 and C_4 must be equal (links 3 and 4 are pinned together at C), we have

$$\mathbf{V}_C = \overset{\text{✓✓}}{\mathbf{V}_B} + \overset{? \text{✓}}{\mathbf{V}_{CB}} = \overset{? \text{✓}}{\mathbf{X}_D^0} + \overset{? \text{✓}}{\mathbf{V}_{CD}}. \quad (3)$$

We construct two lines in the velocity polygon; line BC is drawn from image point B perpendicular to $\mathbf{R}_{CB}$, and line DC is drawn from image point D (coincident with O_V since $\mathbf{V}_D = \mathbf{0}$) perpendicular to $\mathbf{R}_{CD}$. We label the point of intersection of these two lines as image point C . When the lengths of these two lines are measured, we find that $V_{CB} = 38.4$ ft/s and $V_C = V_{CD} = 45.5$ ft/s. The angular velocities of links 3 and 4 can now be obtained:

$$\omega_3 = \frac{V_{CB}}{R_{CB}} = \frac{38.4 \text{ ft/s}}{(18/12) \text{ ft}} = 25.6 \text{ rad/s ccw}, \quad \text{Ans. (4)}$$

$$\omega_4 = \frac{V_{CD}}{R_{CD}} = \frac{45.5 \text{ ft/s}}{(11/12) \text{ ft}} = 49.64 \text{ rad/s ccw}, \quad \text{Ans. (5)}$$

where the directions of ω_3 and ω_4 are obtained using the technique illustrated in Fig. 3.6c.

There are several graphic methods for finding the velocity of point E . One method is to measure R_{EB} from the scale drawing of the linkage. Then, since points B and E are both attached to link 3, we calculate*

$$V_{EB} = \omega_3 R_{EB} = (25.6 \text{ rad/s}) \left(\frac{10.8}{12} \text{ ft} \right) = 23.04 \text{ ft/s}. \quad (6)$$

We next construct the line BE in the velocity polygon, drawn to the chosen scale and perpendicular to $\mathbf{R}_{EB}$. Then we solve† the velocity-difference equation

$$\overset{??}{\mathbf{V}}_E = \overset{\sqrt{\sqrt{\quad}}}{\mathbf{V}}_B + \overset{\sqrt{\sqrt{\quad}}}{\mathbf{V}}_{EB}. \quad (7)$$

From the velocity polygon, the velocity of point E is measured as

$$V_E = 27.6 \text{ ft/s } \angle 165.3^\circ. \quad \text{Ans.}$$

Alternatively, the velocity of point E can be obtained from the velocity-difference equation

$$\overset{??}{\mathbf{V}}_E = \overset{\sqrt{\sqrt{\quad}}}{\mathbf{V}}_C + \overset{\sqrt{\sqrt{\quad}}}{\mathbf{V}}_{EC} \quad (8)$$

by a procedure identical to that used for Eq. (7). This yields the triangle $O_V EC$ in the velocity polygon.

Suppose we wish to find $\mathbf{V}_E$ without the intermediate step of calculating ω_3 . In this case we write Eqs. (7) and (8) simultaneously; that is,

$$\mathbf{V}_E = \overset{\sqrt{\sqrt{\quad}}}{\mathbf{V}}_B + \overset{??}{\mathbf{V}}_{EB} = \overset{\sqrt{\sqrt{\quad}}}{\mathbf{V}}_C + \overset{??}{\mathbf{V}}_{EC}. \quad (9)$$

* There is no restriction in our derivation that requires that $\mathbf{R}_{EB}$ lie along the material portion of link 3 in order to use Eq. (6), only that points E and B remain a constant distance apart.

† Note that numeric values should not be substituted into Eq. (7) directly. This is a vector equation and requires vector addition, not scalar; performing these vector operations is precisely the purpose of constructing the velocity polygon.

Drawing lines EB (perpendicular to $\mathbf{R}_{EB}$) and EC (perpendicular to $\mathbf{R}_{EC}$) in the velocity polygon, we find their point of intersection and so solve Eq. (9).

Perhaps the easiest method of solving for $\mathbf{V}_E$, however, is to take advantage of the concept of the velocity image of link 3. Recognizing that the velocity-image points B and C have already been found, we construct triangle BEC in the velocity polygon, similar in shape to triangle BEC in Fig. 3.7a. This locates point E in the velocity polygon and, therefore, gives a solution for the velocity of point E .

The velocity of point F can also be found by any of the previous methods using points C , D , and F of link 4. The result is

$$V_F = 31.8 \text{ ft/s } \angle 130.9^\circ. \quad \text{Ans.}$$

For the purpose of comparison, an analytic solution is also presented here.

ANALYTIC SOLUTION

The first step is to perform a posture analysis of the linkage. Since this step was presented in Chap. 2, only the results are shown here. For the given link dimensions and the specified input variable $\theta_2 = 120^\circ$, the postures of links 3 and 4 are $\theta_3 = 20.92^\circ$ and $\theta_4 = 64.05^\circ$, respectively.

In vector form, the position-difference vectors corresponding to the links are:

$$\mathbf{R}_{BA} = 4 \text{ in } \angle 120^\circ = -2\hat{\mathbf{i}} + 3.464\hat{\mathbf{j}} \text{ in,}$$

$$\mathbf{R}_{CB} = 18 \text{ in } \angle 20.92^\circ = 16.813\hat{\mathbf{i}} + 6.427\hat{\mathbf{j}} \text{ in,}$$

$$\mathbf{R}_{CD} = 11 \text{ in } \angle 64.05^\circ = 4.813\hat{\mathbf{i}} + 9.891\hat{\mathbf{j}} \text{ in,}$$

$$\mathbf{R}_{DA} = 10 \text{ in } \angle 0^\circ = 10\hat{\mathbf{i}} \text{ in,}$$

$$\mathbf{R}_{EB} = 10.77 \text{ in } \angle -0.88^\circ = 10.769\hat{\mathbf{i}} - 0.165\hat{\mathbf{j}} \text{ in,}$$

$$\mathbf{R}_{FD} = 7.616 \text{ in } \angle 40.85^\circ = 5.761\hat{\mathbf{i}} + 4.981\hat{\mathbf{j}} \text{ in.}$$

We proceed as we did in the graphic solution—namely, the given input angular velocity is

$$\omega_2 = \left(900 \frac{\text{rev}}{\text{min}}\right) \left(2\pi \frac{\text{rad}}{\text{rev}}\right) \left(\frac{1 \text{ min}}{60 \text{ s}}\right) = 94.25 \text{ rad/s ccw.} \quad (10)$$

Then the velocity of point B is written as

$$\begin{aligned} \mathbf{V}_B &= \cancel{\mathbf{X}}_A^0 + \mathbf{V}_{BA} = \omega_2 \times \mathbf{R}_{BA} \\ &= -326.482\hat{\mathbf{i}} - 188.500\hat{\mathbf{j}} \text{ in/s} = 377 \text{ in/s } \angle -150^\circ. \end{aligned} \quad (11)$$

The velocity of point C can be written as

$$\begin{aligned} \mathbf{V}_C &= \cancel{\mathbf{V}}_B + \cancel{\mathbf{V}}_{CB} = \cancel{\mathbf{X}}_D^0 + \cancel{\mathbf{V}}_{CD} \\ &= \mathbf{V}_B + \omega_3 \times \mathbf{R}_{CB} = \omega_4 \times \mathbf{R}_{CD}. \end{aligned} \quad (12)$$

Substituting the position-difference vectors $\mathbf{R}_{CB}$ and $\mathbf{R}_{CD}$ and Eq. (11) into Eq. (12) and writing the result in matrix form gives

$$\begin{bmatrix} -326.482 \text{ in/s} \\ -188.500 \text{ in/s} \end{bmatrix} + \begin{bmatrix} -6.427 \text{ in} \\ 16.813 \text{ in} \end{bmatrix} \omega_3 = \begin{bmatrix} -9.891 \text{ in} \\ 4.813 \text{ in} \end{bmatrix} \omega_4. \quad (13)$$

Solving Eq. (13), the angular velocities of links 3 and 4, respectively, are

$$\omega_3 = 25.382 \text{ rad/s (ccw)} \quad \text{and} \quad \omega_4 = 49.501 \text{ rad/s (ccw)}. \quad \text{Ans. (14)}$$

The velocities of points E and F can be obtained from the velocity-difference equation using the velocities of points B and D ; that is,

$$\mathbf{V}_E = \mathbf{V}_B + \mathbf{V}_{EB} = \mathbf{V}_B + \omega_3 \times \mathbf{R}_{EB} \quad \text{and} \quad \mathbf{V}_F = \mathbf{V}_D + \mathbf{V}_{FD} = \mathbf{V}_D + \omega_4 \times \mathbf{R}_{FD}. \quad (15)$$

Substituting the known data into Eq. (15) gives

$$\begin{aligned} \mathbf{V}_E &= -322.294\hat{\mathbf{i}} + 84.839\hat{\mathbf{j}} \text{ in/s} & \text{and} & & \mathbf{V}_F &= -246.564\hat{\mathbf{i}} + 285.175\hat{\mathbf{j}} \text{ in/s} \\ &= 27.8 \text{ ft/s} \angle 165.25^\circ & & & &= 31.4 \text{ ft/s} \angle 130.85^\circ \end{aligned} \quad \text{Ans.}$$

Note that these answers are in good agreement with the answers obtained by the graphic method.

EXAMPLE 3.2

The offset slider-crank linkage in the posture shown in Fig. 3.8a is driven by slider 4 at a constant velocity $\mathbf{V}_C = -10\hat{\mathbf{i}}$ m/s. Determine the velocity of point D and the angular velocities of links 2 and 3.

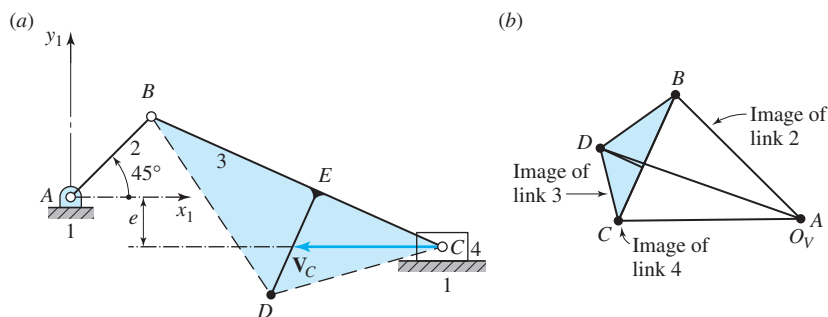


Figure 3.8 (a) $e = 20$ mm, $R_{BA} = 50$ mm, $R_{CB} = 140$ mm, $R_{EB} = 80$ mm, and $R_{DE} = 50$ mm; (b) velocity polygon.

SOLUTION

Choosing a suitable velocity scale and origin O_V , we draw the velocity vector $\mathbf{V}_C$. This locates the velocity image point C as shown in Fig. 3.8b. Solving the simultaneous velocity-difference equations for the velocity of point B , namely,

$$\mathbf{V}_B = \mathbf{V}_C + \mathbf{V}_{BC} = \mathbf{V}_A^0 + \mathbf{V}_{BA}$$

provides the location of the velocity image point B in the velocity polygon.

Having found the velocity image points B and C , we construct the image of link 3 as shown to locate point D . We then measure

$$V_D = 12.0 \text{ m/s} \quad \text{Ans.}$$

with the direction shown in the velocity polygon.

The angular velocities of links 2 and 3, respectively, are

$$\omega_2 = \frac{V_{BA}}{R_{BA}} = \frac{10.0 \text{ m/s}}{0.050 \text{ m}} = 200 \text{ rad/s ccw}, \quad \text{Ans.}$$

$$\omega_3 = \frac{V_{BC}}{R_{BC}} = \frac{7.5 \text{ m/s}}{0.140 \text{ m}} = 53.6 \text{ rad/s cw.} \quad \text{Ans.}$$

Note that the velocity image of each link is shown in Fig. 3.8b.

3.5 APPARENT VELOCITY OF A POINT IN A MOVING COORDINATE SYSTEM

In analyzing the velocities of various machine components, we frequently encounter situations in which it is convenient to describe how a point fixed in a moving link moves with respect to another point of a different moving link. However, it is not at all convenient to describe the absolute motion. This situation occurs when a rotating link contains a slot along which a point of another link is constrained to slide. With the motion of the link containing the slot and the relative sliding motion taking place within the slot as known quantities, we may wish to find the absolute motion of the sliding link. It is for this type of problem that the apparent-displacement vector is defined in Sec. 2.16 (see Fig. 2.29). We now wish to extend this concept to velocity analysis.

Consider Fig. 3.9 in which a rigid link with arbitrary motion has a coordinate system $x_2y_2z_2$ attached to it. After a time increment Δt , the coordinate system lies at $x'_2y'_2z'_2$. All points of link 2—for example, point P_2 —move with the coordinate system. During the same time interval, the initially coincident point P of link 3, denoted P_3 , is constrained to move along a known path with respect to link 2. This constraint is depicted in the figure as a slot carrying a pin from link 3; the center of the pin is point P_3 . Although it is pictured in this way, the constraint may occur in a variety of different forms. The only assumption here is that we know the path that point P_3 traces in coordinate system $x_2y_2z_2$ —that is, the locus of the tip of the apparent-position vector, $\mathbf{R}_{P_3/2}$.

Recall the apparent-displacement equation, Eq. (2.62); that is,

$$\Delta \mathbf{R}_{P_3} = \Delta \mathbf{R}_{P_2} + \Delta \mathbf{R}_{P_3/2}.$$

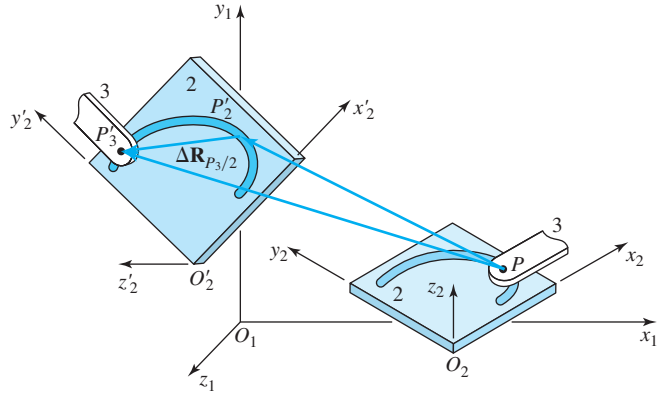


Figure 3.9 Apparent displacement.

Dividing this equation by the small time increment Δt and taking the limit gives

$$\lim_{\Delta t \rightarrow 0} \left(\frac{\Delta \mathbf{R}_{P_3}}{\Delta t} \right) = \lim_{\Delta t \rightarrow 0} \left(\frac{\Delta \mathbf{R}_{P_2}}{\Delta t} \right) + \lim_{\Delta t \rightarrow 0} \left(\frac{\Delta \mathbf{R}_{P_3/2}}{\Delta t} \right). \quad (a)$$

Defining the *apparent-velocity* vector as

$$\mathbf{V}_{P_3/2} = \lim_{\Delta t \rightarrow 0} \left(\frac{\Delta \mathbf{R}_{P_3/2}}{\Delta t} \right) = \frac{d\mathbf{R}_{P_3/2}}{dt}, \quad (3.5)$$

then, in the limit, Eq. (a) becomes

$$\mathbf{V}_{P_3} = \mathbf{V}_{P_2} + \mathbf{V}_{P_3/2}, \quad (3.6)$$

which is called the *apparent-velocity equation*.

We note from Eq. (3.5) that the apparent velocity resembles the absolute velocity except that it comes from the apparent displacement rather than the absolute displacement. Thus, $\mathbf{V}_{P_3/2}$ is the velocity of point P_3 as it would appear to an observer attached to moving link 2 and making observations in coordinate system $x_2y_2z_2$. This concept accounts for its name: the apparent-velocity vector. We also note that the absolute velocity, $\mathbf{V}_{P/1}$, is a special case of the apparent velocity where the observer happens to be fixed to the $x_1y_1z_1$ coordinate system.

Insight into the use of the apparent-velocity equation, Eq. (3.6), is obtained from the following examples.

EXAMPLE 3.3

An inversion of the slider-crank linkage is shown in Fig. 3.10a. For the given posture, with $\theta_2 = 30^\circ$, the angular velocity of link 2 is a constant 36 rad/s cw. Link 3 is pinned to the crank at A and slides on link 4 at point B . Find the angular velocity of link 4.

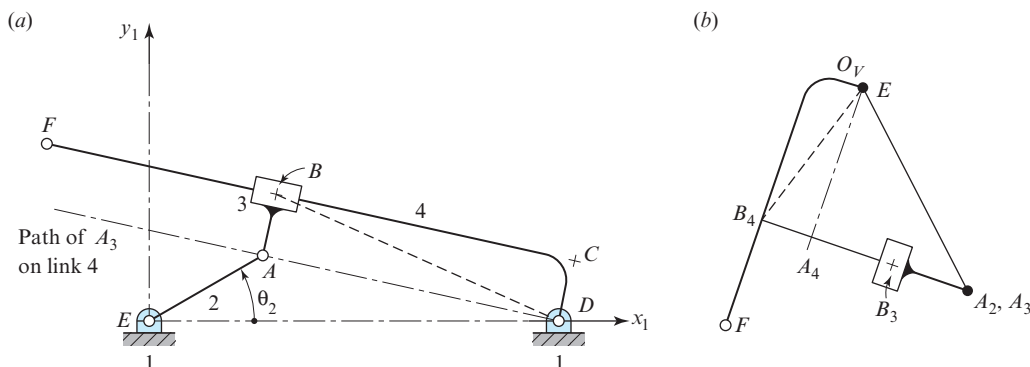


Figure 3.10 (a) $R_{AE} = 3$ in, $R_{BA} = R_{CD} = 2$ in, and $R_{DE} = 14$ in; (b) velocity polygon.

SOLUTION

From Eq. (3.4), the velocity of point A is

$$\begin{aligned}\mathbf{V}_A &= \mathbf{V}_E^0 + \mathbf{V}_{AE} = \boldsymbol{\omega}_2 \times \mathbf{R}_{AE}, \\ V_A &= (36 \text{ rad/s}) \left(\frac{3}{12} \text{ ft} \right) = 9 \text{ ft/s.}\end{aligned}\quad (1)$$

Choosing a suitable velocity scale and origin O_V , we draw velocity vector $\mathbf{V}_A$ to locate image point A in the velocity polygon (Fig. 3.10b).

Next, we distinguish two different points, B_3 and B_4 , at the location of sliding. Point B_3 is fixed in link 3, and point B_4 is fixed in link 4, but at the instant shown, the two points are coincident. Note that, as seen by an observer on link 4, point B_3 slides along link 4, thus defining a straight path along line CF . Thus, we can write the apparent-velocity equation, Eq. (3.6) as

$$\mathbf{V}_{B_3} = \mathbf{V}_{B_4} + \mathbf{V}_{B_3/4}. \quad (2)$$

When image point B_3 is related to image point A , and image point B_4 is related to image point D by velocity differences, expansion of Eq. (2) gives

$$\overset{\sqrt{\sqrt{}}}{\mathbf{V}}_A + \overset{?}{\sqrt{\sqrt{}}}{\mathbf{V}}_{B_3A} = \overset{\sqrt{\sqrt{}}}{\mathbf{V}}_D + \overset{?}{\sqrt{\sqrt{}}}{\mathbf{V}}_{B_4D} + \overset{?}{\sqrt{\sqrt{}}}{\mathbf{V}}_{B_3/4}, \quad (3)$$

where $\mathbf{V}_{B_3A}$ is perpendicular to $\mathbf{R}_{BA}$, $\mathbf{V}_{B_4D}$ (dashed line in Fig. 3.10b) is perpendicular to $\mathbf{R}_{BD}$, and $\mathbf{V}_{B_3/4}$ is directed along the tangent to the path of sliding at B .

Although Eq. (3) appears to have three unknowns, if we note that $\mathbf{V}_{B_3A}$ and $\mathbf{V}_{B_3/4}$ have identical orientations, the equation can be rearranged as

$$\overset{\swarrow}{\mathbf{V}}_A + \overset{\nearrow}{(\mathbf{V}_{B_3A} - \mathbf{V}_{B_3/4})} = \overset{\swarrow}{\mathbf{V}}_{B_4D}, \quad (4)$$

and the vector difference indicated in parentheses can be treated as a single vector with known orientation. The equation is thereby reduced to two unknowns and can be solved graphically to locate point B_4 in the velocity polygon.

The magnitude R_{BD} can be computed or measured from a scale drawing of the linkage, and the velocity-difference $\mathbf{V}_{B_4D}$ can be determined from the velocity polygon (the dashed line from O_V to B_4). Therefore, the angular velocity of link 4 is

$$\omega_4 = \frac{V_{B_4D}}{R_{BD}} = \frac{7.3 \text{ ft/s}}{(11.6/12) \text{ ft}} = 7.55 \text{ rad/s ccw.} \quad \text{Ans. (5)}$$

Another approach to this problem, which avoids the need to combine terms as in Eq. (4), is to extend the velocity polygon to include the images of links 2, 3, and 4. In so doing, it is necessary to note that, since links 3 and 4 are always perpendicular to each other ($\theta_3 = \theta_4 - 90^\circ$), they must rotate at the same rate. Therefore, the angular velocities of the two links are the same: $\omega_3 = \omega_4$. This allows the calculation of $\mathbf{V}_{BA} = \omega_3 \times \mathbf{R}_{BA}$ and the plotting of the velocity image point, B_3 . We also note that the velocity images of links 3 and 4 are of comparable size, since $\omega_3 = \omega_4$. However, they have quite different scales than the velocity image of link 2, line O_VA , since link 2 has a larger angular velocity.

Now, consider an observer riding on link 4 and ask what SHE would see for the path of point A in HER coordinate system. The answer to this is that this path is a straight line parallel to line CF , as indicated in Fig. 3.10a. Let us now define one point of this path as A_4 . Note that this point *does not move together with pins A_2 and A_3 ; it is a point in link 4 and it rotates along with the path around the fixed point D* . Since we can identify the path traced by A_2 and A_3 on the extension of link 4, we can write the apparent-velocity equation*

$$\mathbf{V}_{A_2} = \mathbf{V}_{A_4} + \mathbf{V}_{A_2/4}. \quad (6)$$

Since A_4 is a point in link 4, we can write

$$\mathbf{V}_{A_4} = \overset{0}{\mathbf{V}}_D + \mathbf{V}_{A_4D}. \quad (7)$$

* It is wrong to use the equation $\mathbf{V}_{A_4} = \mathbf{V}_{A_2} + \mathbf{V}_{A_4/2}$, since the *path* traced by point A_4 in a coordinate system attached to link 2 is *not known*. Although this equation shows a poor understanding, it still yields a correct solution. If the corresponding path (of point A_4 as observed from link 2) were found, it would be tangent to the path used for point A_2 as observed from link 4. Since the *tangents* to the two paths are the same, although the paths are not, the two solutions both yield the same numeric result. However, this is *not* true in the acceleration analysis in Chap. 4. Therefore, the concept should be studied and this “backward” use should be avoided.

Substituting Eq. (7) into Eq. (6) gives

$$\check{\mathbf{V}}_{A_2} = \check{\mathbf{V}}_{A_4D} + \check{\mathbf{V}}_{A_2/4}, \quad (8)$$

where $\mathbf{V}_{A_4D}$ is perpendicular to $\mathbf{R}_{AD}$, and $\mathbf{V}_{A_2/4}$ is tangent to the path that A_2 traces on extended link 4. Solving Eq. (8) locates velocity image point A_4 in the velocity polygon and gives $V_{A_4D} = 7.17$ ft/s and $V_{A_2/4} = 5.48$ ft/s. Also, measuring the diagram gives $R_{AD} = 11.4$ in. Substituting these values into the equation, we obtain

$$\omega_4 = \frac{V_{A_4D}}{R_{AD}} = \frac{7.17 \text{ ft/s}}{(11.4/12) \text{ ft}} = 7.55 \text{ rad/s ccw}. \quad (9)$$

Note that this result is in complete agreement with the result of Eq. (5).

The velocity of other points in link 4—for example, points C and F —can be obtained using the velocity polygon and are left as an exercise for the reader.

Further insight into the nature and use of the apparent-velocity equation is provided by the following example.

EXAMPLE 3.4

An airplane is traveling at a speed of 300 km/h on a circular path of radius 5 km with a center at C , as shown in Fig. 3.11. A rocket 30 km away from the airplane is traveling on a straight course at a speed of 2 000 km/h. Determine the velocity of the rocket as observed by the pilot of the airplane.

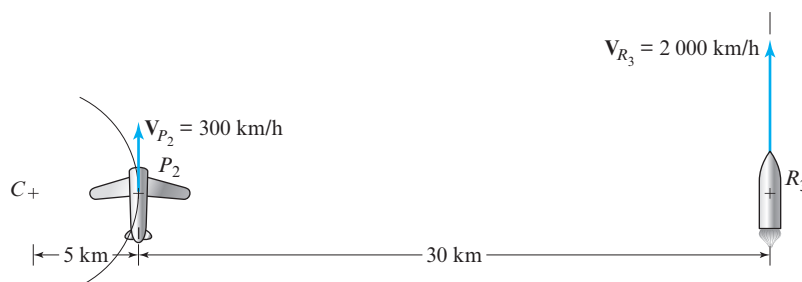


Figure 3.11 Airplane traveling on a circular path.

SOLUTION

Since the airplane is on a circular course, point C_2 , attached to the coordinate system of the airplane but coincident with C_1 , has no motion. Therefore, the angular velocity of the

airplane is

$$\omega_2 = \frac{V_{PC}}{R_{PC}} = \frac{(V_{P_2} - V_{C_2})}{R_{PC}} = \frac{(300 \text{ km/h} - 0)}{5 \text{ km}} = 60 \text{ rad/h ccw.}$$

The question posed explicitly requires the calculation of the apparent velocity $\mathbf{V}_{R_3/2}$, but the apparent-velocity equation applies *only between coincident points*. Therefore, we define another point, R_2 , attached to the rotating coordinate system of the airplane but located coincident with the rocket R_3 at the instant depicted. As part of the airplane, the velocity of this point can be written from Eq. (3.6) as

$$\mathbf{V}_{R_2} = \mathbf{V}_{P_2} + \omega_2 \times \mathbf{R}_{RP}.$$

Adding algebraically, since the vectors are parallel, gives

$$V_{R_2} = 300 \text{ km/h} + (60 \text{ rad/h})(30 \text{ km}) = 2\,100 \text{ km/h.}$$

The apparent velocity of the rocket as observed by the pilot of the airplane can be written as

$$\mathbf{V}_{R_3/2} = \mathbf{V}_{R_3} - \mathbf{V}_{R_2}.$$

Now, algebraically subtracting gives

$$V_{R_3/2} = 2\,000 \text{ km/h} - 2\,100 \text{ km/h} = -100 \text{ km/h.} \quad \text{Ans.}$$

Thus, as seen by the pilot of the airplane, the rocket appears to be *backing up* at a speed of 100 km/h. This result becomes better understood as we consider the motion of point R_2 . This imagined point is *attached to the airplane* and, therefore, seems stationary to the pilot. Yet, in the absolute coordinate system, this point is traveling faster than the rocket; the rocket is not keeping up with this point and, therefore, appears to the pilot to be backing up.

We can gain further insight into the nature of the apparent-velocity vector by studying Fig. 3.12. This figure shows the view of moving point P_3 as it would appear to the moving observer. To HER, the path traced on link 2 appears stationary, and the moving point moves along this path from P_3 to P'_3 . Working in this coordinate system, suppose we locate point C_2 as the center of curvature of the path of point P_3 . For small distances from P_2 , the path follows the circular arc $P_3P'_3$ with center C_2 and radius of curvature ρ . We now define the unit vector $\hat{\mathbf{u}}^t$ tangent to the path with positive sense in the direction of positive movement. The plane defined by this unit tangent vector $\hat{\mathbf{u}}^t$ and the center of curvature C_2 is called the *osculating plane*. If we choose a preferred side of this plane as the positive side and denote it by the positive unit vector $\hat{\mathbf{u}}^b$ (commonly referred to as the *binormal* vector), we can complete a right-handed Cartesian coordinate system by defining the unit vector normal to the path

$$\hat{\mathbf{u}}^n = \hat{\mathbf{u}}^b \times \hat{\mathbf{u}}^t. \quad (3.7)$$

Therefore, a rule to remember is that the unit normal vector $\hat{\mathbf{u}}^n$ is always 90° counterclockwise from the unit tangent vector $\hat{\mathbf{u}}^t$. This implies that the radius of curvature

speed with which the point P_3 progresses along the path and (b) the apparent-velocity vector is always tangent to the path traced by the point in the coordinate system of the observer.

The first of these two conclusions is seldom useful in the solution of problems, although it is an important concept. The second conclusion is extremely useful, since the apparent path traced by a moving point can often be visualized from the nature of the constraints, and, thus, the direction of the apparent-velocity vector becomes known. Note that only the tangent to the path $\hat{\mathbf{u}}'$ is used in this chapter; the radius of curvature ρ of the path is not needed yet but will become important when we analyze acceleration in Chap. 4.

3.6 APPARENT ANGULAR VELOCITY

Completeness suggests that we should define the term *apparent angular velocity*. When two rigid bodies rotate with different angular velocities, the vector difference between the two angular velocities is defined as the *apparent angular velocity*. For example, the apparent angular velocity of rotating links 2 and 3 can be written as

$$\omega_{3/2} = \omega_3 - \omega_2. \quad (3.10)$$

Therefore, the angular velocity of link 3 is

$$\omega_3 = \omega_2 + \omega_{3/2}. \quad (3.11)$$

Note that $\omega_{3/2}$ is the angular velocity of link 3 as it would appear to an observer attached to, and rotating with, link 2.

3.7 DIRECT CONTACT AND ROLLING CONTACT

Two links of a mechanism that are in direct contact have relative motion that may or may not involve sliding* at the point of contact. For example, consider the cam-and-follower mechanism shown in Fig. 3.13a; the cam, link 2, drives the follower, link 3, by direct contact. We see that if slip were not possible between link 2 and link 3 at the point of contact P , then triangle PAB would form a truss; that is, the mechanism would have a mobility of $m = 0$. Therefore, for the cam to drive the follower, sliding as well as rotation must take place at point P . We distinguish between two coincident points, both located at P at the instant shown: point P_2 , attached to link 2, and point P_3 , attached to link 3. Therefore, we can rearrange the apparent-velocity equation, Eq. (3.6), as

$$\mathbf{V}_{P_3/2} = \mathbf{V}_{P_3} - \mathbf{V}_{P_2}. \quad (3.12)$$

The graphic velocity solution to this equation is as follows. First, we note that the normal component of the apparent velocity must be zero; otherwise, the two links would either separate or interfere, both contrary to our basic assumption that contact continues. Therefore, the apparent velocity must act along the common tangent and is the velocity of the relative sliding motion along the direct-contact interface.

* The words *sliding* and *slipping* are used interchangeably in this text.

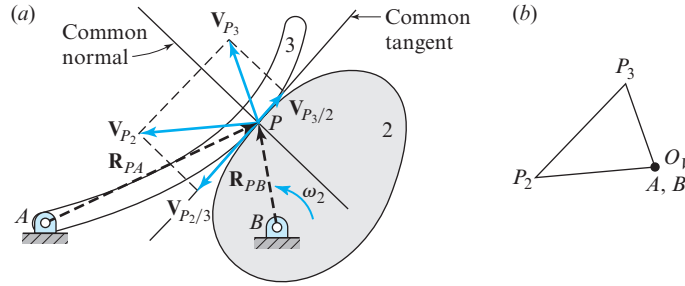


Figure 3.13 (a) Cam-and-follower mechanism; (b) velocity polygon.

For a given angular velocity of the driver (the cam) ω_2 , the velocity of point P_2 can be obtained from

$$\mathbf{V}_{P_2} = \omega_2 \times \mathbf{R}_{P_2B},$$

thus locating image point P_2 in the velocity polygon; see Fig. 3.13b. The component of $\mathbf{V}_{P_3}$ along the common normal must be equal to the component of $\mathbf{V}_{P_2}$ along the common normal, and this locates image point P_3 . Therefore, the apparent velocity, Eq. (3.12), can be measured from the velocity polygon.

It is also possible for there to be direct contact between two links of a mechanism without sliding between the links. For example, the cam and roller follower mechanism of Fig. 3.14a could have sufficient friction between the cam surface, link 2, and the roller, link 3, to restrain the roller to roll on the cam without slip. Henceforth, we will restrict our use of the term *rolling contact* to situations where *no slip* takes place. By “no slip” we imply that the apparent “slipping” velocity of Eq. (3.12) is zero:

$$\mathbf{V}_{P_3/2} = \mathbf{0}. \quad (3.13)$$

This equation is commonly referred to as the *rolling contact condition for velocity*. Substituting Eq. (3.13) into Eq. (3.12), the rolling contact condition can also be written as

$$\mathbf{V}_{P_3} = \mathbf{V}_{P_2}. \quad (3.14)$$

This says that *the absolute velocities of two points in rolling contact are equal*.

The graphic velocity analysis of this mechanism, assuming rolling contact at point P, is as follows. Given ω_2 , the velocity difference $\mathbf{V}_{P_2B}$ can be calculated and drawn to scale, thus locating point P_2 in the velocity polygon; see Fig. 3.14b. Using Eq. (3.13), the rolling contact condition, we also label this point P_3 . Next, writing simultaneous equations for the velocity of pin C, using $\mathbf{V}_{C_3P_3}$ and $\mathbf{V}_{C_4A_4}$, we locate the velocity-image points C_3 and C_4 . Finally, the angular velocities of links 3 and 4 are obtained from $\omega_3 = V_{CP}/R_{CP}$ and $\omega_4 = V_{CA}/R_{CA}$, respectively.

Another approach is to define a fictitious point C_2 , which is located instantaneously coincident with points C_3 and C_4 , but which is attached to and moving with link 2, as shown by the shaded triangle BPC_2 . When velocity-image concept is used for link 2,

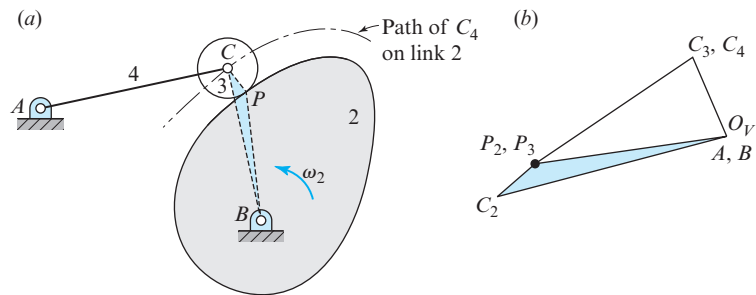


Figure 3.14 (a) Cam and roller follower mechanism; (b) velocity polygon.

velocity-image point C_2 can be located (see Fig. 3.14*b*). Noting that point C_4 (and C_3) traces a known path on link 2, we can write and solve the apparent-velocity equation involving $\mathbf{V}_{C_4/2}$. Then, the velocity $\mathbf{V}_{C_4}$ (and angular velocity ω_4 , if desired) is obtained without dealing with the point of direct contact. This second approach would be necessary if we had not assumed rolling contact (no slip) at P .

3.8 SYSTEMATIC STRATEGY FOR VELOCITY ANALYSIS

A careful review of the preceding sections and example problems should indicate that we have developed sufficient tools for dealing with situations that normally arise in the velocity analysis of planar mechanisms. It should also be noted that the word “relative” velocity has been carefully avoided. Instead, we note that whenever the desire for using “relative” velocity arises, there are always two points whose velocities are to be “related.” These two points are fixed either in the same rigid body or in two different rigid bodies. When the two points are in the same body, the velocity difference equation, Eq. (3.4), is appropriate. However, when it is desirable to switch to a point in a different body, then coincident points should be chosen and the apparent-velocity equation, Eq. (3.6), should be used. We can organize all possible situations into four cases as shown in Table 3.1.

Table 3.1 “Relative” Velocity Equations

Points are	Coincident	Separated
In same body	<i>Trivial case:</i> $\mathbf{V}_P = \mathbf{V}_Q.$	<i>Velocity difference:</i> $\mathbf{V}_P = \mathbf{V}_Q + \mathbf{V}_{PQ}$ $\mathbf{V}_{PQ} = \omega_j \times \mathbf{R}_{PQ}$
In different bodies	<i>Apparent velocity:</i> $\mathbf{V}_{P_i} = \mathbf{V}_{P_j} + \mathbf{V}_{P_i/j},$ where path $P_{i/j}$ is known. <i>Rolling contact velocity:</i> $\mathbf{V}_{P_i} = \mathbf{V}_{P_j}$ and $\mathbf{V}_{P_i/j} = 0.$	<i>Too general; use two steps.</i>

The notation has intentionally been made different to remind us that these are two totally different situations, and the formulae are not interchangeable between the two. We should not try to use an $\omega \times \mathbf{R}$ formula when using the apparent velocity; if we do try, then we will not find a single ω or a useful $\mathbf{R}$. Similarly, when using the velocity difference, there is no question of which ω to use since only one link pertains. Further advantages will become clear when we study accelerations in Chap. 4.

A detailed study of the graphic solutions of Example 3.1 through Example 3.4 clearly indicates how the suggested strategy in Table 3.1, and the labeling of the velocity polygons, is applied to the velocity analysis of planar mechanisms.

3.9 ALGEBRAIC VELOCITY ANALYSIS

For some types of mechanisms, a numeric solution by calculator or computer is often the most convenient. Also, when solutions for multiple postures are required, graphic methods become cumbersome. In this section we present an algebraic method for velocity analysis.

For purposes of illustration, consider the central (or in-line) slider-crank linkage shown in Fig. 3.15. This linkage is the mechanism found in most internal combustion engines, and, for this reason, we use the standard engine notation r and l to designate the length of the crank and the length of the connecting rod, respectively.

From the geometry of the linkage we can write

$$r \sin \theta = l \sin \phi, \quad (a)$$

$$x = r \cos \theta + l \cos \phi. \quad (b)$$

The angle ϕ can be eliminated as follows. From trigonometry,

$$l \cos \phi = l \sqrt{1 - \sin^2 \phi} = \sqrt{l^2 - l^2 \sin^2 \phi}. \quad (c)$$

Substituting Eq. (a) into Eq. (c) gives

$$l \cos \phi = \sqrt{l^2 - r^2 \sin^2 \theta}. \quad (d)$$

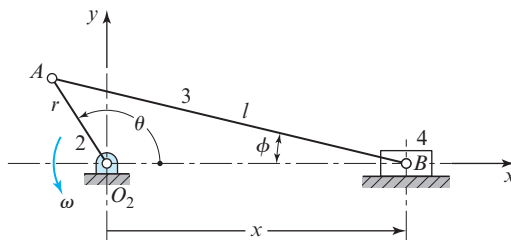


Figure 3.15 Central slider-crank linkage.

Then, substituting Eq. (d) into Eq. (b), the position of the slider (or piston) can be written as

$$x = r \cos \theta + \sqrt{l^2 - r^2 \sin^2 \theta}$$

or as

$$x = r \cos \theta + l \sqrt{1 - \left(\frac{r}{l} \sin \theta\right)^2}. \quad (3.15a)$$

Differentiating this equation with respect to time gives the exact equation for the velocity of the slider:

$$\dot{x} = -r\omega \left[\sin \theta + \frac{r \sin 2\theta}{2l \sqrt{1 - (r/l)^2 \sin^2 \theta}} \right]. \quad (3.15b)$$

We note that for most internal combustion engines, the ratio r/l is in the range $1/10 \leq r/l \leq 1/4$. This means that the maximum value of the second term under the radical is about $1/16$ or less. Therefore, it is common practice to use approximations for the position and the velocity of the slider. These approximations are obtained by using the binomial expansion to write the radical in Eq. (3.15a) as

$$\sqrt{1 - \left(\frac{r}{l} \sin \theta\right)^2} \approx 1 - \frac{r^2}{2l^2} \sin^2 \theta. \quad (e)$$

Note that sufficient accuracy is obtained for engine design by retaining only the first two terms of the binomial expansion. Substituting the identity

$$\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$$

into Eq. (e) and the result into Eq. (3.15a) gives

$$x = r \cos \theta + l \left[1 - \frac{r^2}{4l^2} (1 - \cos 2\theta) \right].$$

Rearranging this equation, the approximate position of the slider can be written as

$$x = l - \frac{r^2}{4l} + r \left(\cos \theta + \frac{r}{4l} \cos 2\theta \right). \quad (3.16a)$$

Differentiating Eq. (3.16a) with respect to time and rearranging, the approximate velocity of the slider can be written as

$$\dot{x} = -r\omega \left[\sin \theta + \frac{r}{2l} \sin 2\theta \right]. \quad (3.16b)$$

3.10 COMPLEX-ALGEBRAIC VELOCITY ANALYSIS

We recall from Sec. 2.9 that complex algebra provides an alternative algebraic formulation for problems in two-dimensional kinematics. As we noted there, the complex-algebraic formulation provides the advantage of increased accuracy over graphic methods, and, once a program is written, it is amenable to solution by digital computer at a large number of postures. On the other hand, the solution of the loop-closure equation for the unknown position variables is a nonlinear problem and can lead to tedious algebraic manipulations. Fortunately, the extension of the complex-algebraic approach to velocity analysis leads to a set of *linear* equations, and the solution is quite straightforward.

Recalling the complex polar form of a two-dimensional vector from Eq. (2.41); that is,

$$\mathbf{R} = R e^{j\theta},$$

we find the general form of its time derivative,

$$\dot{\mathbf{R}} = \frac{d\mathbf{R}}{dt} = \dot{R} e^{j\theta} + j\dot{\theta} R e^{j\theta}, \quad (3.17)$$

where $\dot{R}$ and $\dot{\theta}$ denote respectively the time rates of change of the magnitude and angle of vector $\mathbf{R}$.

We will see in the following two examples that the first term on the right-hand side of Eq. (3.17) commonly represents an apparent velocity, and the second term usually represents a velocity difference. The method shown in these examples was developed by Raven [12]. Although the original work provides methods applicable to both planar and spatial mechanisms, only the planar aspects are shown here.

EXAMPLE 3.5

For the inverted slider-crank linkage in the posture shown in Fig. 3.16a, link 2, the driver, has a known angular position θ_2 and a known angular velocity ω_2 . Derive expressions for the angular velocity of link 4 and the absolute velocity of point P in terms of the angular velocity of the driver.

SOLUTION

To simplify the notation, we will use the symbols* shown in Fig. 3.16b for the position-difference vectors; thus, $\mathbf{R}_{AB}$ is denoted $\mathbf{r}_1$, $\mathbf{R}_{C_2A}$ is denoted $\mathbf{r}_2$, and $\mathbf{R}_{C_2B}$ is denoted $\mathbf{r}_4$. In terms of these symbols, the loop-closure equation can be written as

$$\mathbf{r}_4 = \mathbf{r}_1 + \mathbf{r}_2, \quad (1)$$

* The symbols $\mathbf{R}$ and $\mathbf{r}$ are used interchangeably throughout this chapter.

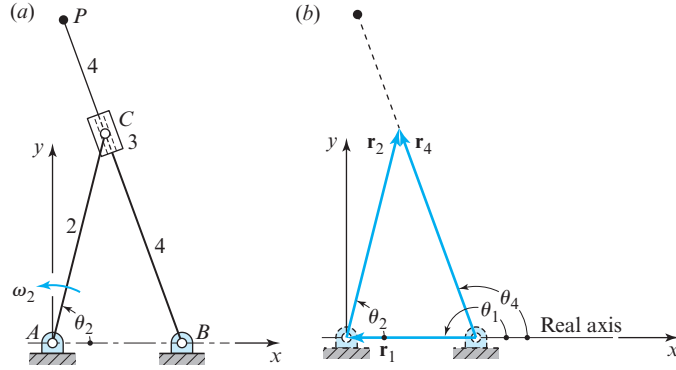


Figure 3.16 Inverted slider-crank linkage.

where vector $\mathbf{r}_4$ has unknown magnitude and direction, vector $\mathbf{r}_1$ has constant magnitude and direction,* and vector $\mathbf{r}_2$ has constant magnitude (its direction, θ_2 , varies). The variable angle θ_2 is the given input angle or, more precisely, the unknown variables will be solved as functions of θ_2 .

Recognizing this as case 1 (Sec. 2.8), we obtain the posture solution from Eqs. (2.43) and (2.44) as

$$r_4 = \sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos \theta_2} \quad (2)$$

and

$$\theta_4 = \tan^{-1} \left(\frac{r_2 \sin \theta_2}{r_2 \cos \theta_2 - r_1} \right). \quad (3)$$

The velocity solution is initiated by differentiating the loop-closure equation, Eq. (1), with respect to time. Applying the general formula, Eq. (3.17), to each term of Eq. (1) in turn, and keeping in mind that r_1 , θ_1 , and r_2 are constants, we obtain

$$\dot{r}_4 e^{j\theta_4} + j\dot{\theta}_4 r_4 e^{j\theta_4} = j\dot{\theta}_2 r_2 e^{j\theta_2}. \quad (4)$$

Since $\dot{\theta}_2$ and $\dot{\theta}_4$ are the same as ω_2 and ω_4 , respectively, and since we recognize that

$$\dot{\theta}_2 r_2 = V_{C_2}, \quad \dot{r}_4 = V_{C_2/4}, \quad \dot{\theta}_4 r_4 = V_{C_4},$$

we see that Eq. (4) is, in fact, the complex polar form of the apparent-velocity equation:

$$\mathbf{V}_{C_4} + \mathbf{V}_{C_2/4} = \mathbf{V}_{C_2}.$$

(This is pointed out for comparison only and is not a necessary step in the solution process.)

* Note particularly that the angle of $\mathbf{r}_1$ is $\theta_1 = 180^\circ$, not $\theta_1 = 0^\circ$.

The velocity solution is performed using Euler's formula, Eq. (2.40), to separate Eq. (4) into its real and imaginary components. This gives

$$\dot{r}_4 \cos \theta_4 - \omega_4 r_4 \sin \theta_4 = -\omega_2 r_2 \sin \theta_2, \quad (5)$$

$$\dot{r}_4 \sin \theta_4 + \omega_4 r_4 \cos \theta_4 = \omega_2 r_2 \cos \theta_2. \quad (6)$$

Using Cramer's rule, the expressions for the absolute velocity of point P and the angular velocity of link 4, in terms of the angular velocity of the driver, are

$$\dot{r}_4 = \omega_2 r_2 \sin(\theta_4 - \theta_2), \quad (7)$$

$$\omega_4 = \omega_2 \frac{r_2}{r_4} \cos(\theta_4 - \theta_2). \quad (8)$$

Although the variables r_4 and θ_4 given by Eqs. (2) and (3) could be substituted into Eqs. (2) and (3) to reduce the results to functions of θ_2 and ω_2 alone, the previous forms are considered sufficient. The reason is that, in writing a computer program, numeric values are normally found first for r_4 and θ_4 while performing the position analysis, and then these numeric results are used to obtain numeric values for $\dot{r}_4$ and ω_4 at each input angle θ_2 .

To find the velocity of point P , we first, write the position of point P as

$$\mathbf{R}_P = R_{PB} e^{j\theta_4}. \quad (9)$$

Then, differentiating this equation with respect to time, and remembering that R_{PB} is a constant length, gives

$$\mathbf{V}_P = j\omega_4 R_{PB} e^{j\theta_4}. \quad (10)$$

Next, substituting Eq. (8) into this equation gives

$$\mathbf{V}_P = j\omega_2 R_{PB} \frac{r_2}{r_4} \cos(\theta_4 - \theta_2) e^{j\theta_4}. \quad (11)$$

Therefore, the horizontal and vertical components of the velocity of point P in terms of the angular velocity of the driver are

$$V_P^x = -\omega_2 R_{PB} \frac{r_2}{r_4} \cos(\theta_4 - \theta_2) \sin \theta_4 \quad (12)$$

and

$$V_P^y = \omega_2 R_{PB} \frac{r_2}{r_4} \cos(\theta_4 - \theta_2) \cos \theta_4. \quad (13)$$

EXAMPLE 3.6

For the four-bar linkage in the arbitrary posture shown in Fig. 3.17a, derive expressions for the relationships between the angular velocity of the input link 2 and the angular velocities of the coupler link 3 and the output link 4.

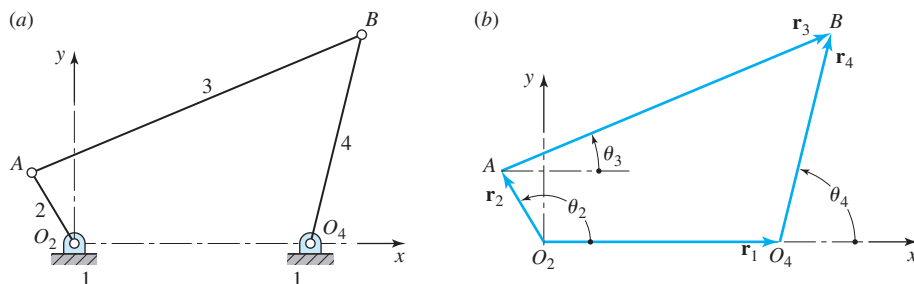


Figure 3.17 (a) Four-bar linkage; (b) the links are replaced by vectors.

SOLUTION

First, replace each link of Fig. 3.17a by a vector as shown in Fig. 3.17b. Then, the loop-closure equation can be written as

$$\mathbf{r}_2 + \mathbf{r}_3 - \mathbf{r}_4 - \mathbf{r}_1 = \mathbf{0}. \quad (1)$$

Next, write each vector in Eq. (1) in complex polar form; that is,

$$r_2 e^{j\theta_2} + r_3 e^{j\theta_3} - r_4 e^{j\theta_4} - r_1 e^{j\theta_1} = 0. \quad (2)$$

Taking the time derivative of this equation and noting that link 1 is the fixed link gives

$$jr_2 \dot{\theta}_2 e^{j\theta_2} + jr_3 \dot{\theta}_3 e^{j\theta_3} - jr_4 \dot{\theta}_4 e^{j\theta_4} = 0. \quad (3)$$

Now, transform Eq. (3) into rectangular form and separate the real and the imaginary terms. Then, noting that $\dot{\theta}_2 = \omega_2$, $\dot{\theta}_3 = \omega_3$, and $\dot{\theta}_4 = \omega_4$, gives

$$\begin{aligned} r_2 \omega_2 \cos \theta_2 + r_3 \omega_3 \cos \theta_3 - r_4 \omega_4 \cos \theta_4 &= 0, \\ r_2 \omega_2 \sin \theta_2 + r_3 \omega_3 \sin \theta_3 - r_4 \omega_4 \sin \theta_4 &= 0, \end{aligned} \quad (4)$$

where the input angular velocity ω_2 is known and the angular velocities ω_3 and ω_4 are the two unknown variables.

Finally, using Cramer's rule, the expressions for the angular velocities of the coupler link 3 and the output link 4 in terms of the input angular velocity, respectively, are

$$\omega_3 = \frac{r_2 \sin(\theta_2 - \theta_4)}{r_3 \sin(\theta_4 - \theta_3)} \omega_2 \quad \text{and} \quad \omega_4 = \frac{r_2 \sin(\theta_2 - \theta_3)}{r_4 \sin(\theta_4 - \theta_3)} \omega_2. \quad \text{Ans. (5)}$$

Note that the term $\sin(\theta_4 - \theta_3)$ appears in the denominators of both results. In general, the velocity analysis of a planar mechanism will have similar denominators in the solutions for each of the velocity unknowns; these denominators are the determinant of the matrix of coefficients of the unknowns, as is recognized by recalling Cramer's rule. The angle $(\theta_4 - \theta_3)$ is, in fact, the transmission angle γ , see Fig. 2.15 (Sec. 2.8). Recall that when the transmission angle becomes small, the ratio of the output speed, ω_4 , to the input speed, ω_2 , becomes very large and difficulties will result (Sec. 1.10).

From Examples 3.5 and 3.6, we note that the simultaneous equations to be solved are linear in the unknown variables. This is not a coincidence but is true for all velocity problems. It results from the fact that the general form of the velocity equation, Eq. (3.17), is linear in the velocity variables. When real and imaginary components are taken, the *coefficients* may become complicated, but the equations remain linear with respect to the velocity unknowns. Therefore, their solution is straightforward.

Recall that in Secs. 3.4 through 3.8, the graphic velocity methods, it was possible to choose an arbitrary scale factor for a velocity polygon. Therefore, if the input speed of a mechanism is doubled, the scale factor of the velocity polygon can also be doubled, and the same polygon is valid. This is another indication of the linearity of the velocity equations.

3.11 METHOD OF KINEMATIC COEFFICIENTS

A method that provides substantial geometric insight into the kinematic analysis and synthesis of a linkage is to differentiate the loop-closure equation with respect to the input variable(s), rather than differentiating with respect to time. This analytic approach is referred to as the *method of kinematic coefficients* [8]. The numeric values of the first-order kinematic coefficients can also be checked with the graphic approach of finding the locations of the instantaneous centers of velocity (Sec. 3.12).

This method is illustrated here by again solving the four-bar linkage problem that was presented in Example 3.1 and the offset slider-crank linkage problem that was presented in Example 3.2.

EXAMPLE 3.7

The four-bar linkage in the posture shown in Fig. 3.7a is driven by crank 2 at a constant angular velocity $\omega_2 = 900 \text{ rev/min}$ ccw. Determine the angular velocities of the coupler link and the output link, and the velocities of points *E* and *F*.

SOLUTION

The loop-closure equation is given by Eq. (1) in Example 3.6 (Fig. 3.17b) and is reproduced here:

$$\mathbf{r}_2 + \mathbf{r}_3 - \mathbf{r}_4 - \mathbf{r}_1 = \mathbf{0}. \quad (1)$$

The two scalar component equations are

$$\begin{aligned} r_2 \cos \theta_2 + r_3 \cos \theta_3 - r_4 \cos \theta_4 - r_1 \cos \theta_1 &= 0, \\ r_2 \sin \theta_2 + r_3 \sin \theta_3 - r_4 \sin \theta_4 - r_1 \sin \theta_1 &= 0. \end{aligned} \quad (2)$$

Since the input is the angular displacement of link 2, the method of kinematic coefficients necessitates differentiating Eqs. (2) with respect to the input variable θ_2 . Rearranging the result gives

$$-r_3 \sin \theta_3 \theta_3' + r_4 \sin \theta_4 \theta_4' = r_2 \sin \theta_2$$

and

$$r_3 \cos \theta_3 \theta'_3 - r_4 \cos \theta_4 \theta'_4 = -r_2 \cos \theta_2, \quad (3)$$

where

$$\theta'_3 = \frac{d\theta_3}{d\theta_2} \quad \text{and} \quad \theta'_4 = \frac{d\theta_4}{d\theta_2} \quad (4)$$

are referred to as the *first-order kinematic coefficients* of links 3 and 4, respectively.

Algebraic forms for the kinematic coefficients can be obtained using Cramer's rule. Writing Eqs. (3) in matrix form gives

$$\begin{bmatrix} -r_3 \sin \theta_3 & r_4 \sin \theta_4 \\ r_3 \cos \theta_3 & -r_4 \cos \theta_4 \end{bmatrix} \begin{bmatrix} \theta'_3 \\ \theta'_4 \end{bmatrix} = \begin{bmatrix} r_2 \sin \theta_2 \\ -r_2 \cos \theta_2 \end{bmatrix}. \quad (5)$$

The determinant of the (2×2) coefficient matrix can be written as

$$\Delta = -r_3 r_4 \sin(\theta_4 - \theta_3) \quad (6)$$

and provides geometric insight into special postures of the linkage (Example 3.6). For example, when this determinant tends toward zero, the kinematic coefficients tend toward infinity. Note that the determinant is zero (that is, the transmission angle $\theta_4 - \theta_3$ is zero) either when (a) $\theta_3 = \theta_4$ or when (b) $\theta_3 = \theta_4 \pm 180^\circ$ —that is, when links 3 and 4 are fully extended or aligned on top of each other.

From Eqs. (5), the first-order kinematic coefficients of links 3 and 4 are

$$\theta'_3 = \frac{r_2 \sin(\theta_2 - \theta_4)}{r_3 \sin(\theta_4 - \theta_3)} \quad \text{and} \quad \theta'_4 = \frac{r_2 \sin(\theta_2 - \theta_3)}{r_4 \sin(\theta_4 - \theta_3)}. \quad (7)$$

Recall from Example 3.1 that the position values are $\theta_3 = 20.92^\circ$ and $\theta_4 = 64.05^\circ$. Therefore, the first-order kinematic coefficients of links 3 and 4 are

$$\theta'_3 = +0.2718 \text{ rad/rad} \quad \text{and} \quad \theta'_4 = +0.5265 \text{ rad/rad}, \quad (8)$$

where the positive signs indicate that links 3 and 4 are rotating in the same direction as the input link, 2.

The angular velocities of links 3 and 4, obtained from the chain rule, are

$$\omega_3 = \theta'_3 \omega_2 \quad \text{and} \quad \omega_4 = \theta'_4 \omega_2. \quad (9)$$

Substituting Eqs. (7) into Eqs. (9) gives the same symbolic results as Eqs. (5) in Example 3.6. Also, substituting Eqs. (8) and the input angular velocity into Eqs. (9), the angular velocities of links 3 and 4 are

$$\omega_3 = 25.62 \text{ rad/s ccw} \quad \text{and} \quad \omega_4 = 49.62 \text{ rad/s ccw}. \quad \text{Ans. (10)}$$

These answers are in good agreement with the results obtained in Example 3.1 from the graphic method [Eqs. (4) and (5)] and the analytic method [Eqs. (14)].

The position of coupler point E with respect to ground pivot A (Fig. 3.7a) can be written as

$$\mathbf{r}_E = \mathbf{r}_2 + \mathbf{r}_{EB}. \quad (11)$$

The x and y components of this vector equation are

$$\begin{aligned} x_E &= r_2 \cos \theta_2 + r_{EB} \cos (\theta_3 - \phi), \\ y_E &= r_2 \sin \theta_2 + r_{EB} \sin (\theta_3 - \phi), \end{aligned} \quad (12)$$

where $\phi = \tan^{-1} (R_{EG}/R_{GB}) = \tan^{-1} (4 \text{ in}/10 \text{ in}) = 21.80^\circ$.

Differentiating Eqs. (12) with respect to the input variable θ_2 , the first-order kinematic coefficients for point E are

$$\begin{aligned} x'_E &= \frac{dx_E}{d\theta_2} = -r_2 \sin \theta_2 - r_{EB} \sin (\theta_3 - \phi) \theta'_3, \\ y'_E &= \frac{dy_E}{d\theta_2} = r_2 \cos \theta_2 + r_{EB} \cos (\theta_3 - \phi) \theta'_3. \end{aligned} \quad (13)$$

Substituting the known values into these equations, the first-order kinematic coefficients for point E are

$$x'_E = -3.419 \text{ in/rad} \quad \text{and} \quad y'_E = +0.926 \text{ in/rad}. \quad (14)$$

The velocity of point E can be written from the chain rule as

$$\mathbf{V}_E = (x'_E \hat{\mathbf{i}} + y'_E \hat{\mathbf{j}}) \omega_2. \quad (15)$$

Substituting Eqs. (14) and the input angular velocity into this equation, the velocity of point E is

$$\mathbf{V}_E = -26.85 \hat{\mathbf{i}} + 7.28 \hat{\mathbf{j}} \text{ ft/s}. \quad \text{Ans.}$$

Therefore, the speed of point E is $V_E = 27.8 \text{ ft/s}$, which agrees closely with the result obtained from the graphic method in Example 3.1 ($V_E = 27.6 \text{ ft/s}$).

The velocity of point F can be obtained in a similar manner, and the result is

$$\mathbf{V}_F = -20.60 \hat{\mathbf{i}} + 23.82 \hat{\mathbf{j}} \text{ ft/s}. \quad \text{Ans.}$$

Therefore, the speed of point F is $V_F = 31.5 \text{ ft/s}$, which agrees closely with the result obtained from the graphic method in Example 3.1 ($V_F = 31.8 \text{ ft/s}$).

EXAMPLE 3.8

The offset slider-crank linkage in the posture shown in Fig. 3.8a is driven by slider 4 at a constant speed of $V_C = 10 \text{ m/s}$ to the left. Determine the angular velocities of links 2 and 3 and the velocity of point D .

SOLUTION

The loop-closure equation for the offset slider-crank linkage, in complex polar form, is

$$jr_1 + r_2 e^{j\theta_2} + r_3 e^{j\theta_3} - r_4 = 0. \quad (1)$$

The two scalar equations are

$$\begin{aligned} r_2 \cos \theta_2 + r_3 \cos \theta_3 - r_4 &= 0, \\ r_1 + r_2 \sin \theta_2 + r_3 \sin \theta_3 &= 0. \end{aligned} \quad (2)$$

Since the input is the linear displacement of link 4, then differentiating Eqs. (2) with respect to r_4 gives

$$\begin{aligned} -r_2 \sin \theta_2 \theta'_2 - r_3 \sin \theta_3 \theta'_3 &= 1, \\ r_2 \cos \theta_2 \theta'_2 + r_3 \cos \theta_3 \theta'_3 &= 0, \end{aligned} \quad (3)$$

where

$$\theta'_2 = \frac{d\theta_2}{dr_4} \quad \text{and} \quad \theta'_3 = \frac{d\theta_3}{dr_4} \quad (4)$$

are the first-order kinematic coefficients of links 2 and 3. Symbolic equations for the first-order kinematic coefficients are obtained here using Cramer's rule. The first step is to write Eqs. (3) in matrix form; that is,

$$\begin{bmatrix} -r_2 \sin \theta_2 & -r_3 \sin \theta_3 \\ r_2 \cos \theta_2 & r_3 \cos \theta_3 \end{bmatrix} \begin{bmatrix} \theta'_2 \\ \theta'_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}. \quad (5)$$

The determinant of the (2×2) coefficient matrix in Eq. (5) is written as

$$\Delta = r_2 r_3 \sin(\theta_3 - \theta_2). \quad (6)$$

Note that the determinant is zero when (a) $\theta_2 = \theta_3$ or when (b) $\theta_2 = \theta_3 \pm 180^\circ$ —that is, when links 2 and 3 are fully extended or aligned on top of each other.

From Eq. (5), the first-order kinematic coefficients of links 2 and 3 can be written as

$$\theta'_2 = \frac{\cos \theta_3}{r_2 \sin(\theta_3 - \theta_2)} \quad \text{and} \quad \theta'_3 = \frac{-\cos \theta_2}{r_3 \sin(\theta_3 - \theta_2)}. \quad (7)$$

For the given link dimensions and the given input position $r_4 = 164$ mm, the angular positions of links 2 and 3 are $\theta_2 = 45^\circ$ and $\theta_3 = 337^\circ$, respectively. Substituting these values into Eqs. (7), the first-order kinematic coefficients of links 2 and 3 are

$$\theta'_2 = -19.856 \text{ rad/m} \quad \text{and} \quad \theta'_3 = +5.447 \text{ rad/m}, \quad (8)$$

where the negative sign shows that link 2 rotates clockwise while the positive sign shows that link 3 rotates counterclockwise for positive input motion of link 4 (to the right).

The angular velocities of links 2 and 3 can be written from the chain rule as

$$\omega_2 = \theta'_2 \dot{r}_4 \quad \text{and} \quad \omega_3 = \theta'_3 \dot{r}_4, \quad (9)$$

where the velocity of the slider is $\dot{r}_4 = V_C = -10$ m/s. Substituting Eqs. (8) and the velocity of the slider into Eqs. (9), the angular velocities of the two links are

$$\omega_2 = 198.56 \text{ rad/s ccw and } \omega_3 = -54.47 \text{ rad/s (cw)} \quad \text{Ans.}$$

These answers agree closely with the results obtained from the velocity polygon method in Example 3.2 ($\omega_2 = 200$ rad/s ccw and $\omega_3 = 53.6$ rad/s cw) and are more accurate.

The position of point D with respect to ground pivot A (Fig. 3.8a) can be written as

$$\mathbf{r}_D = \sqrt{r_2^2 + r_{DB}^2} \angle \theta_D. \quad (10)$$

The x and y components of this vector equation are

$$x_D = r_2 \cos \theta_2 + r_{DB} \cos (\theta_3 - \beta), \quad (11a)$$

$$y_D = r_2 \sin \theta_2 + r_{DB} \sin (\theta_3 - \beta), \quad (11b)$$

where $\beta = (R_{DE}/R_{EB}) = \tan^{-1} (50 \text{ mm}/80 \text{ mm}) = 32.01^\circ$.

Differentiating Eqs. (11) with respect to input variable r_4 , the first-order kinematic coefficients for point D are

$$x'_D = -r_2 \sin \theta_2 \theta'_2 - r_{DB} \sin (\theta_3 - \beta) \theta'_3, \quad (12a)$$

$$y'_D = r_2 \cos \theta_2 \theta'_2 + r_{DB} \cos (\theta_3 - \beta) \theta'_3. \quad (12b)$$

Substituting the known values into these equations, the first-order kinematic coefficients are

$$x'_D = 1.123 \text{ m/m} \quad \text{and} \quad y'_D = -0.407 \text{ m/m}. \quad (13)$$

The velocity of point D can be written as

$$\mathbf{V}_D = (x'_D \hat{\mathbf{i}} + y'_D \hat{\mathbf{j}}) \dot{r}_4. \quad (14)$$

Substituting Eqs. (13) and the input velocity $\dot{r}_4 = -10$ m/s into this equation, the velocity of point D is

$$\mathbf{V}_D = -11.23 \hat{\mathbf{i}} + 4.07 \hat{\mathbf{j}} \text{ m/s} = 11.94 \text{ m/s} \angle 160.08^\circ. \quad \text{Ans. (15)}$$

Therefore, the instantaneous speed of point D agrees closely with the result obtained from the velocity polygon method in Example 3.2 ($V_D = 12$ m/s).

The method of kinematic coefficients also provides geometric insight into the motion of mechanisms that have links that are in rolling contact. The following two examples will help to illustrate this important concept.

EXAMPLE 3.9

For the mechanism in the posture shown in Fig. 3.18, the wheel is rolling without slipping on the ground. Determine the first-order kinematic coefficients of links 3 and 4. If input link 2 is rotating with a constant angular velocity of 10 rad/s ccw, then determine: (a) the angular velocities of links 3 and 4 and (b) the velocity of the center of the wheel, point G .

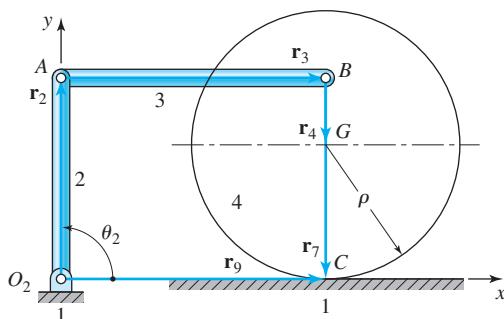


Figure 3.18 $R_{AO_2} = r_2 = 150$ mm, $R_{BA} = r_3 = 200$ mm, $R_{GB} = r_4 = 50$ mm, and $\rho = 100$ mm.

SOLUTION

The loop-closure equation for the mechanism is

$$\sqrt{I} + \sqrt{?} + \sqrt{?} + \sqrt{C1} - \frac{C2}{r_9} = 0, \quad (1)$$

where the first constraint, $C1$, is $\theta_7 = \theta_9 + 270^\circ$ and the second constraint, $C2$, designates the rolling contact constraint defined later by Eq. (5).

The horizontal and vertical components of Eq. (1) give the scalar equations

$$r_2 \cos \theta_2 + r_3 \cos \theta_3 + r_4 \cos \theta_4 + r_7 \cos \theta_7 - r_9 \cos \theta_9 = 0, \quad (2a)$$

$$r_2 \sin \theta_2 + r_3 \sin \theta_3 + r_4 \sin \theta_4 + r_7 \sin \theta_7 - r_9 \sin \theta_9 = 0. \quad (2b)$$

Since the input is the angular displacement of link 2, we differentiate Eqs. (2) with respect to θ_2 and, by setting $\theta_9 = 0^\circ$, we obtain

$$-r_2 \sin \theta_2 - r_3 \sin \theta_3 \theta'_3 - r_4 \sin \theta_4 \theta'_4 - r'_9 = 0 \quad (3a)$$

and

$$r_2 \cos \theta_2 + r_3 \cos \theta_3 \theta'_3 + r_4 \cos \theta_4 \theta'_4 = 0, \quad (3b)$$

where the first-order kinematic coefficients of links 3 and 4 and the change in position of contact point C , respectively, are

$$\theta'_3 = \frac{d\theta_3}{d\theta_2}, \quad \theta'_4 = \frac{d\theta_4}{d\theta_2}, \quad \text{and} \quad r'_9 = \frac{dr_9}{d\theta_2}. \quad (4)$$

Also, note that the rotation of the wheel, θ_4 , and the change in the distance, r_9 , are not independent, since the wheel is rolling on the ground without slipping. This rolling contact (no-slip) condition is constraint $C2$ and can be written as shown in Sec. 2.18:

$$-\Delta r_9 = \rho (\Delta \theta_4 - \Delta \theta_7),$$

where ρ is the radius of the wheel and, from constraint $C1$, we note that $\Delta \theta_7 = 0$. Therefore, the rolling contact condition is

$$-\Delta r_9 = \rho \Delta \theta_4. \quad (5)$$

The negative sign on the left-hand side of this constraint equation is the result of the decreasing magnitude of r_9 for a positive change in the input variable—that is, $\Delta \theta_2$.

In the limit, for infinitesimal displacements, constraint Eq. (5) can be written in terms of first-order kinematic coefficients as

$$-r'_9 = \rho \theta'_4. \quad (6)$$

Substituting Eq. (6) into Eqs. (3), and writing the resulting equations in matrix form, gives

$$\begin{bmatrix} -r_3 \sin \theta_3 & \rho - r_4 \sin \theta_4 \\ r_3 \cos \theta_3 & r_4 \cos \theta_4 \end{bmatrix} \begin{bmatrix} \theta'_3 \\ \theta'_4 \end{bmatrix} = \begin{bmatrix} +r_2 \sin \theta_2 \\ -r_2 \cos \theta_2 \end{bmatrix}. \quad (7)$$

The determinant of the (2×2) coefficient matrix in Eq. (7) can be written as

$$\Delta = r_3 [r_4 \sin (\theta_4 - \theta_3) - \rho \cos \theta_3]. \quad (8)$$

From Eq. (7), using Cramer's rule, the symbolic equations for the first-order kinematic coefficients of links 3 and 4 are

$$\theta'_3 = \frac{r_2 [\rho \cos \theta_2 - r_4 \sin (\theta_4 - \theta_2)]}{\Delta}, \quad (9a)$$

$$\theta'_4 = \frac{r_2 r_3 \sin (\theta_3 - \theta_2)}{\Delta}. \quad (9b)$$

For the posture shown in Fig. 3.18, we have $\theta_2 = 90^\circ$, $\theta_3 = 0^\circ$, and $\theta_4 = -90^\circ$. Therefore, the determinant in Eq. (8) can be written as

$$\Delta = -r_3 (r_4 + \rho). \quad (10)$$

Substituting the known data and Eq. (10) into Eqs. (9), and simplifying, gives

$$\theta'_3 = 0 \quad \text{and} \quad \theta'_4 = \frac{r_2}{r_4 + \rho} = 1 \text{ rad/rad.} \quad \text{Ans. (11)}$$

Therefore, the angular velocity of link 3 is

$$\omega_3 = \theta'_3 \omega_2 = 0, \quad \text{Ans.}$$

and the angular velocity of the wheel, link 4, is

$$\omega_4 = \theta'_4 \omega_2 = 10 \text{ rad/s ccw.} \quad \text{Ans.}$$

Note that these results are in agreement with our intuition; in this posture, link 3 is not rotating, and link 4 is rotating at the same angular velocity as the input.

The velocity of the center of the wheel, point G , can be written as

$$\mathbf{V}_G = r'_9 \omega_2 \angle \theta_9. \quad (12)$$

Substituting Eq. (11) into Eq. (6) gives $r'_9 = -\rho = -100 \text{ mm}$, and substituting this into Eq. (12) gives

$$\mathbf{V}_G = r'_9 \omega_2 \angle \theta_9 = (-100 \text{ mm/rad}) (10 \text{ rad/s}) \angle 0^\circ = -1.0 \text{ m/s} \angle 0^\circ. \quad \text{Ans.}$$

This result can be verified by direct calculation by noting that

$$\mathbf{V}_G = \mathcal{N}_{C_4}^0 + \omega_4 \times (\rho \hat{\mathbf{j}}) = (10 \text{ rad/s}) \hat{\mathbf{k}} \times (100 \text{ mm}) \hat{\mathbf{j}} = -1.0 \hat{\mathbf{i}} \text{ m/s.} \quad \text{Ans.}$$

EXAMPLE 3.10

For the mechanism in the posture shown in Fig. 3.19, gear 3 is rolling without slip on input gear 2. Determine: (a) the first-order kinematic coefficients of links 3, 4, and 5; and (b) the angular velocities of links 3, 4, and 5 if the input gear is rotating with a constant angular velocity of 10 rad/s ccw.

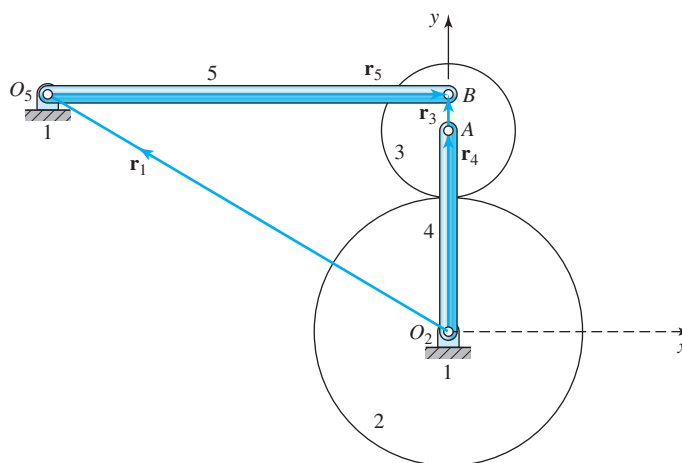


Figure 3.19 $\rho_2 = 20 \text{ mm}$, $\rho_3 = 10 \text{ mm}$, $R_{BA} = r_3 = 5 \text{ mm}$, $R_{AO_2} = r_4 = 30 \text{ mm}$, and $R_{BO_5} = r_5 = 60 \text{ mm}$.

SOLUTION

The loop-closure equation for the mechanism is

$$\mathbf{r}_4 + \mathbf{r}_3 - \mathbf{r}_5 - \mathbf{r}_1 = \mathbf{0}, \quad (1)$$

where the C refers to the rolling contact constraint that is defined later by Eq. (5a).

The two scalar component equations are

$$r_4 \cos \theta_4 + r_3 \cos \theta_3 - r_5 \cos \theta_5 - r_1 \cos \theta_1 = 0, \quad (2a)$$

$$r_4 \sin \theta_4 + r_3 \sin \theta_3 - r_5 \sin \theta_5 - r_1 \sin \theta_1 = 0. \quad (2b)$$

Since the input is the angular displacement of gear 2, differentiating Eqs. (2) with respect to θ_2 gives

$$-r_4 \sin \theta_4 \theta'_4 - r_3 \sin \theta_3 \theta'_3 + r_5 \sin \theta_5 \theta'_5 = 0, \quad (3a)$$

$$r_4 \cos \theta_4 \theta'_4 + r_3 \cos \theta_3 \theta'_3 - r_5 \cos \theta_5 \theta'_5 = 0, \quad (3b)$$

where

$$\theta'_3 = \frac{d\theta_3}{d\theta_2}, \quad \theta'_4 = \frac{d\theta_4}{d\theta_2}, \quad \text{and} \quad \theta'_5 = \frac{d\theta_5}{d\theta_2} \quad (4)$$

are the first-order kinematic coefficients of links 3, 4, and 5.

Note that the variables θ_2 , θ_3 , and θ_4 are not independent, since gear 3 is constrained to roll on gear 2. The rolling contact condition can be written to ensure that the arc lengths passed during a small movement are equal on the two surfaces. This rolling contact (no-slip) condition can be written as shown in Sec. 2.18:

$$\rho_2 (\Delta\theta_2 - \Delta\theta_4) = -\rho_3 (\Delta\theta_3 - \Delta\theta_4), \quad (5a)$$

where $\Delta\theta_2$, $\Delta\theta_3$, and $\Delta\theta_4$ are the angular displacements of links 2, 3, and 4 from the posture shown in Fig. 3.19. The negative sign for the right-hand term arises from the reversal in the direction of the angular displacement differences.

Equation (5a) can be divided by a change in the input angular displacement, $\Delta\theta_2$, and the limit taken for small increments. This yields the equivalent constraint equation written in terms of first-order kinematic coefficients; that is,

$$\rho_2 (\theta'_2 - \theta'_4) = -\rho_3 (\theta'_3 - \theta'_4). \quad (5b)$$

Since the input is the angular displacement of gear 2, then, by definition, $\theta'_2 = 1$, and by substituting $\rho_2 = 2\rho_3$, Eq. (5b) can be written as

$$\theta'_3 = 3\theta'_4 - 2. \quad (6)$$

Substituting Eq. (6) into Eqs. (3) and writing the resulting equations in matrix form gives

$$\begin{bmatrix} -r_4 \sin \theta_4 - 3r_3 \sin \theta_3 & r_5 \sin \theta_5 \\ r_4 \cos \theta_4 + 3r_3 \cos \theta_3 & -r_5 \cos \theta_5 \end{bmatrix} \begin{bmatrix} \theta'_4 \\ \theta'_5 \end{bmatrix} = \begin{bmatrix} -2r_3 \sin \theta_3 \\ 2r_3 \cos \theta_3 \end{bmatrix}. \quad (7)$$

The determinant of the (2×2) coefficient matrix in Eqs. (7) can be written as

$$\Delta = r_5 [r_4 \sin(\theta_4 - \theta_5) + 3r_3 \sin(\theta_3 - \theta_5)]. \quad (8)$$

From Eqs. (7), using Cramer's rule, the symbolic equations for the first-order kinematic coefficients of links 4 and 5 are

$$\theta'_4 = \frac{2r_3 r_5 \sin(\theta_3 - \theta_5)}{\Delta}, \quad (9a)$$

$$\theta'_5 = \frac{2r_3 r_4 \sin(\theta_3 - \theta_4)}{\Delta}. \quad (9b)$$

For the posture shown in Fig. 3.19, we have $\theta_3 = 90^\circ$, $\theta_4 = 90^\circ$, and $\theta_5 = 0^\circ$; therefore, Eq. (8) can be written as

$$\Delta = r_5 (r_4 + 3r_3). \quad (10)$$

Also, Eqs. (9) reduce to

$$\theta'_4 = \frac{2r_3}{r_4 + 3r_3} = \frac{2}{9} \text{ rad/rad} \quad \text{and} \quad \theta'_5 = 0. \quad (11)$$

It is interesting to note that link 5 is not rotating in this posture; that is, the angular velocity $\omega_5 = 0$ at this instant. Substituting Eq. (11) into Eq. (6), the first-order kinematic coefficient of gear 3 is

$$\theta'_3 = -\frac{4}{3} \text{ rad/rad}. \quad (12)$$

The angular velocity of link j can now be written as

$$\omega_j = \theta'_j \omega_2. \quad (13)$$

Therefore, substituting Eqs. (11) and (12) into Eq. (13), and setting $\omega_2 = 10 \text{ rad/s}$ ccw, the angular velocities of gear 3 and links 4 and 5 are

$$\omega_3 = -13.33 \text{ rad/s}, \quad \omega_4 = +2.22 \text{ rad/s}, \quad \text{and} \quad \omega_5 = 0, \quad \text{Ans. (14)}$$

where the negative sign indicates clockwise and the positive sign indicates counterclockwise rotation.

Note that kinematic coefficients are functions of *posture only*; that is, they are not directly functions of time. Also, note that the units of the first-order kinematic coefficients depend on the specified input variable and the unknown variable under consideration. The units may be nondimensional (rad/rad or length/length), length (length/rad), or the reciprocal of length (rad/length). Table 3.2 summarizes the first-order kinematic coefficients that are related to link j of a mechanism (that is, vector $\mathbf{r}_j$), having (a) unknown angle θ_j and/or (b) unknown magnitude r_j .

Table 3.2 Summary of First-order Kinematic Coefficients

	Variable of Interest Angle θ_j (Use symbol θ'_j for kinematic coefficient regardless of input)	Variable of Interest Magnitude r_j (Use symbol r'_j for kinematic coefficient regardless of input)
Input	$\omega_j = \theta'_j \dot{\psi}$	$\dot{r}_j = r'_j \dot{\psi}$
ψ = angle θ_i	$\theta'_j = \frac{d\theta_j}{d\psi}$ (dimensionless, rad/rad)	$r'_j = \frac{dr_j}{d\psi}$ (length, length/rad)
Input	$\omega_j = \theta'_j \dot{\psi}$	$\dot{r}_j = r'_j \dot{\psi}$
ψ = distance r_i	$\theta'_j = \frac{d\theta_j}{dr_i}$ (1/length, rad/length)	$r'_j = \frac{dr_j}{dr_i}$ (dimensionless, length/length)

3.12 INSTANTANEOUS CENTERS OF VELOCITY

One of the more interesting concepts in planar kinematics is that of an instantaneous center of velocity for a pair of rigid bodies that move with respect to one another. For spatial motion, an axis exists that is common to both bodies and about which each body is rotating and translating with respect to the other.* The study here is restricted to planar motion; therefore, each axis is perpendicular to the plane of the motion and reduces to a point in the plane. We shall refer to these points as *instantaneous centers of velocity*, henceforth referred to as *instant centers*. (Some texts prefer the name *velocity poles*; see the footnote in Sec. 3.20.)

The instantaneous center of velocity is defined as *the instantaneous location of a pair of coincident points of two different rigid bodies in planar motion for which the absolute velocities of the two points are equal*. The instantaneous center of velocity may also be defined as the location of a pair of coincident points of two different rigid bodies in planar motion for which the apparent velocity of one of the points is zero, as seen by an observer on the other body.

This chapter only considers instant centers for mechanisms with mobility $m = 1$. However, instant centers can also be used for velocity analysis of mechanisms with mobility greater than unity (Chap. 5).

For planar motion, an instant center between two bodies can be regarded as a pair of coincident points, one attached to each body, about which one body may have a rotational velocity, but no translational velocity, with respect to the other. This property is true only instantaneously, and a new pair of coincident points becomes the instant center at the next instant. It is not correct, therefore, to speak of an instant center as the center of rotation, since it is generally not located at the center of curvature of the apparent locus that a point of one body generates with respect to the coordinate system of the other. Even with this restriction, however, we will find that instant centers contribute substantially to our understanding of the kinematics of planar motion.

* For three-dimensional motion, this axis is referred to as the *instantaneous screw axis*. The classic work covering the theory of screws is [1].

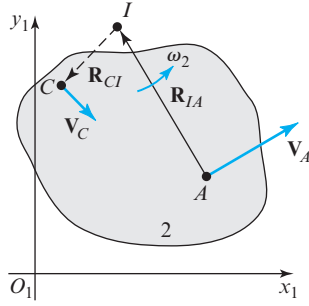


Figure 3.20 Rigid body 2 in planar motion.

Let us consider rigid body 2, shown in Fig. 3.20, having an arbitrary planar motion with respect to the x_1y_1 plane (body 1); the motion might be translation, rotation, or a combination of both. Suppose that a chosen point A of the body has a known velocity $\mathbf{V}_A$ and that the body has a known angular velocity ω_2 . With these two quantities known, the velocity of any other point of the body, such as point C , can be found from the velocity-difference equation, Eq. (3.4):

$$\mathbf{V}_C = \mathbf{V}_A + \mathbf{V}_{CA} = \mathbf{V}_A + \omega_2 \times \mathbf{R}_{CA}. \quad (a)$$

Suppose we now choose a point I of this same body 2 whose position difference from point A is

$$\mathbf{R}_{IA} = \frac{\omega_2 \times \mathbf{V}_A}{\omega_2^2}. \quad (3.18)$$

As shown by the vector cross-product, point I is located on the perpendicular to $\mathbf{V}_A$, and vector $\mathbf{R}_{IA}$ is rotated from the direction of $\mathbf{V}_A$ in the direction of ω_2 , as illustrated in Fig. 3.20. The length of $\mathbf{R}_{IA}$ can be calculated from Eq. (3.18), and point I can be located. The velocity of point I can be written from Eq. (a) as

$$\mathbf{V}_I = \mathbf{V}_A + \mathbf{V}_{IA} = \mathbf{V}_A + \omega_2 \times \mathbf{R}_{IA} = \mathbf{V}_A + \omega_2 \times \frac{\omega_2 \times \mathbf{V}_A}{\omega_2^2}. \quad (b)$$

Recalling the vector triple product identity $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{c} \cdot \mathbf{a}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$ and substituting this identity into Eq. (b) gives

$$\mathbf{V}_I = \mathbf{V}_A + \frac{\overbrace{\omega_2 (\mathbf{V}_A \cdot \omega_2)}^0 - \underbrace{\mathbf{V}_A (\omega_2 \cdot \omega_2)}_{\omega_2^2}}{\omega_2^2} = \mathbf{V}_A - \mathbf{V}_A = \mathbf{0}. \quad (c)$$

Since the absolute velocity of point I is zero, the same as the velocity of the coincident point of the fixed link, then point I is the instant center between body 2 and body 1.

The velocity of any other point of the moving body can now be determined using the instant center, point I . Note that the choice of the instant center simplifies the

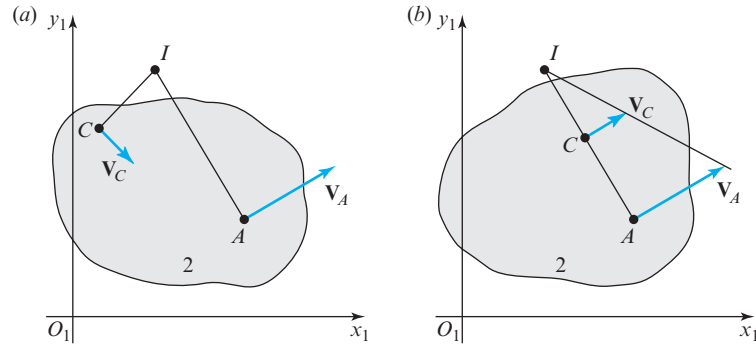


Figure 3.21 Locating an instant center from two known velocities.

velocity-difference equation. For example, the velocity of point C in Eq. (a) can now be written as

$$\mathbf{V}_C = \mathbf{V}_I^0 + \mathbf{V}_{CI} = \boldsymbol{\omega}_2 \times \mathbf{R}_{CI}. \quad (3.19)$$

The direction of the velocity of this point is as shown in Fig. 3.20.

An instant center can be located more easily when the absolute velocities of two points are known. In Fig. 3.21a, suppose that points A and C have known velocities $\mathbf{V}_A$ and $\mathbf{V}_C$, respectively. The perpendiculars to $\mathbf{V}_A$ and $\mathbf{V}_C$ intersect at the instant center, point I. Figure 3.21b shows how to locate instant center I when points A, C, and I happen to fall on the same straight line.

The instant center between two bodies, in general, is not a stationary point. Its location changes with respect to both bodies as the motion progresses and describes a path or locus on each. These paths of the instant centers, called *centrodes*, are discussed in Sec. 3.20.

Since we have adopted the convention of numbering the links of a mechanism, it is convenient to designate an instant center by using the numbers of the two links associated with it. Thus, I_{32} , for example, identifies the instant center between links 3 and 2. This same instant center can also be identified as I_{23} , since the order of the numbers has no significance. A mechanism has as many instant centers as there are ways of pairing the link numbers. Thus, the number of instant centers in an n -link mechanism is

$$N = \frac{n(n-1)}{2}. \quad (3.20)$$

3.13 ARONHOLD-KENNEDY THEOREM OF THREE CENTERS

Consider the four-bar linkage shown in Fig. 3.22a. According to Eq. (3.20), the number of instant centers for this linkage is six. We can identify four of the instant centers by inspection (henceforth called *primary instant centers*). The four pins are primary instant centers I_{12} , I_{23} , I_{34} , and I_{14} , since each satisfies the definition. For example, instant center I_{23} , is a point of link 2 about which link 3 appears to rotate; it is a point of link 3 that has

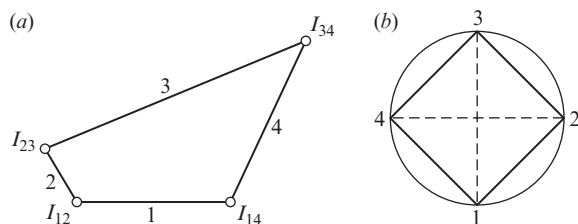


Figure 3.22 Locations of the primary instant centers.

no apparent velocity as seen from link 2. This instant center is a pair of coincident points of links 2 and 3 that have the same absolute velocities.

A good method of keeping track of which instant centers have been identified is to arrange the link numbers around the perimeter of a circle (called the *Kennedy circle*), as shown in Fig. 3.22b. Then, as each instant center is identified, a solid line is drawn connecting the corresponding pair of link numbers. Figure 3.22b shows that I_{12} , I_{23} , I_{34} , and I_{14} (primary instant centers) have been identified, as indicated by the solid lines. The figure also includes two dashed lines indicating that I_{13} and I_{24} have not yet been identified (henceforth called *secondary instant centers*). The locations of the secondary instant centers cannot be determined simply by applying the definition visually. After locating the primary instant centers by inspection, the secondary instant centers are located by applying the Aronhold-Kennedy theorem of three centers (often just called *Kennedy's theorem*).^{*} This theorem states: *Three instant centers shared by three rigid bodies in motion with respect to one another (whether or not they are connected) all lie on the same straight line.*

The theorem can be proven by contradiction, as shown in Fig. 3.23. Link 1 is a stationary frame, and instant center I_{12} is located where link 2 is pin-connected to link 1. Similarly, I_{13} is located at the pin connecting links 1 and 3. The shapes of links 2 and 3 are arbitrary (they can be regarded as infinite planes). The Aronhold-Kennedy theorem states that the three instant centers I_{12} , I_{13} , and I_{23} must all lie on the same straight line (the line connecting the two pins), commonly referred to as the *line of centers*. Let us suppose that this were not true; in fact, let us suppose that I_{23} were located at the point labeled P in Fig. 3.23. Then the velocity of P as a point of link 2 would have the orientation $\mathbf{V}_{P_2}$, perpendicular to $\mathbf{R}_{PI_{12}}$. However, the velocity of P as a point of link 3 would have the orientation $\mathbf{V}_{P_3}$, perpendicular to $\mathbf{R}_{PI_{13}}$. The orientations of these two velocities are inconsistent with the definition that an instant center must have equal absolute velocities as a part of either link. The point P chosen, therefore, cannot be the instant center I_{23} . This same contradiction in the orientations of $\mathbf{V}_{P_2}$ and $\mathbf{V}_{P_3}$ occurs for any location chosen for point P unless the point is chosen on the straight line through the instant centers I_{12} and I_{13} .

^{*} The Aronhold-Kennedy theorem is named after its two independent discoverers, S. H. Aronhold, 1872, and A. B. W. Kennedy, 1886. It is usually known as the *Aronhold theorem* in German-speaking countries, and is usually called *Kennedy's theorem* in English-speaking countries.

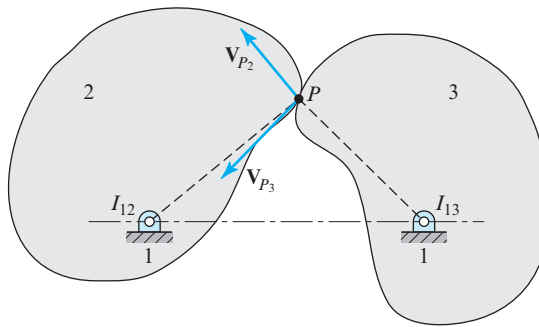


Figure 3.23

Aronhold-Kennedy theorem.

3.14 LOCATING INSTANTANEOUS CENTERS OF VELOCITY

In the previous two sections, we have considered several methods of locating instant centers. They can often be located by inspecting the figure of a mechanism and visually seeking out coincident point pairs that fit the definition, such as pin-joint centers. Also, after these primary instant centers are identified, the secondary instant centers can be determined using the theorem of three centers. Section 3.13 demonstrated that an instant center between a moving body and the fixed link can be identified if the directions of the absolute velocities of two points of the body are known or if the absolute velocity of one point and the angular velocity of the body are known. The purpose of this section is to expand the list of techniques to include instant centers of mechanisms involving direct contact and instant centers at infinity.

For example, consider the cam-and-follower system of Fig. 3.24. The instant centers I_{12} and I_{13} are primary instant centers, since, by inspection, they are located at the two pin centers. However, the location of the secondary instant center I_{23} (indicated by the dashed line in the Kennedy circle) is not immediately obvious. According to the Aronhold-Kennedy theorem, it must lie on the straight line connecting I_{12} and I_{13} , but where on this line? After some reflection, we see that the orientation of the apparent velocity, $\mathbf{V}_{A2/3}$, must be along the common tangent to the two moving links at their point of contact. As seen by an observer on link 3, this velocity must result from the apparent relative rotation of body 2 about the instant center I_{23} . Therefore, I_{23} must lie on the line that is perpendicular to $\mathbf{V}_{A2/3}$. This line now locates I_{23} , as shown in Fig. 3.24. The concept

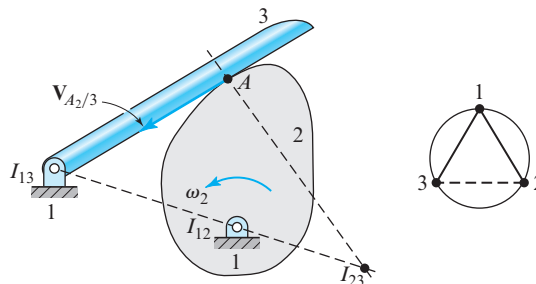


Figure 3.24 Instant centers of a disk cam with a flat-faced follower.

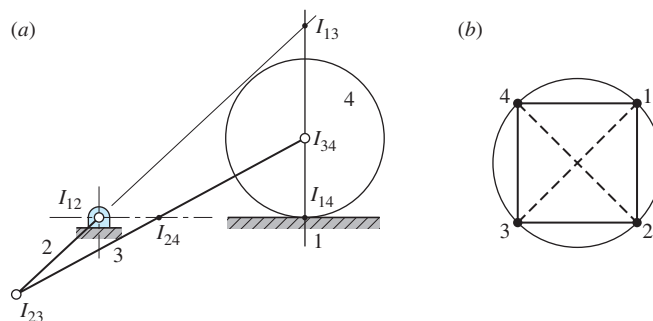


Figure 3.25 Instant center at a point of rolling contact.

illustrated in this example should be remembered, since it is often useful in locating the instant centers of mechanisms involving direct contact.

A special case of direct contact, as we have previously noted, is rolling contact with no slip. Considering the mechanism of Fig. 3.25, we can immediately locate the primary instant centers I_{12} , I_{23} , and I_{34} . As demonstrated by the previous example, if the contact between links 1 and 4 involves any slippage, we can only say that instant center I_{14} is located on the vertical line through the point of contact. However, if we also know that there is no slip—that is, *if there is rolling contact—then the instant center is located at the point of contact*. This is also a general principle, as can be seen by comparing the definition of rolling contact, Eq. (3.14) and the definition of an instant center; they are equivalent. Therefore, I_{14} is regarded as a primary instant center, as shown in Fig. 3.25b.

Another special case of direct contact is evident between links 3 and 4 of the inverted slider-crank linkage in Fig. 3.26. In this case, there *is* an apparent (slip) velocity, $\mathbf{V}_{A3/4}$, between points A of links 3 and 4, but there is *no apparent rotation between the links*. Here, as in Fig. 3.24, the instant center I_{34} lies along a common perpendicular to the known line of slipping, but now it is located infinitely far away in the direction defined by this perpendicular line (since there is no apparent rotation). This infinite distance can be demonstrated by considering the kinematic inversion of the mechanism in which link 4

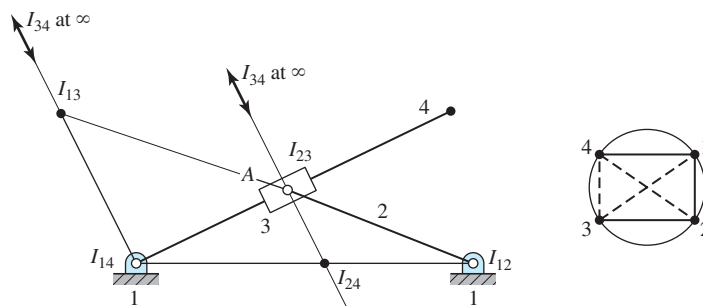


Figure 3.26 Inverted slider-crank linkage and locations of instant centers.

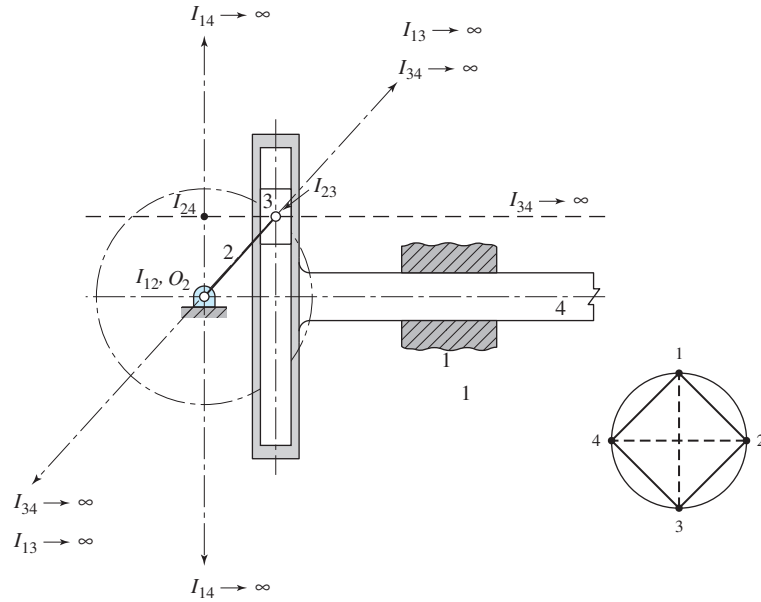


Figure 3.27 Scotch-yoke linkage and locations of instant centers.

becomes stationary. In this case, Eq. (3.18) for the inverted mechanism can be written as

$$|\mathbf{R}_{I_{34}A}| = \left| \frac{\omega_{3/4} \times \mathbf{V}_{A_{3/4}}}{\omega_{3/4}^2} \right| = \left| \hat{\omega}_{3/4} \times \left(\frac{\mathbf{V}_{A_{3/4}}}{\omega_{3/4}} \right) \right| = \infty. \quad (3.21)$$

Since there is no relative rotation between links 3 and 4, the denominator in this equation is zero, and the distance from I_{23} to I_{34} is infinite. The direction of I_{34} , stated earlier, is confirmed by the cross-product in Eq. (3.21).

Note that I_{34} can be regarded as either a primary instant center or a secondary instant center. The remaining secondary instant centers, I_{13} and I_{24} , can be obtained from the Aronhold-Kennedy theorem. To locate I_{13} , a line must be drawn through I_{14} parallel to the line through instant centers I_{23} and I_{34} . The point of intersection of this line with the line through I_{12} and I_{23} is I_{13} . The point of intersection of the line through I_{23} and I_{34} with the line through I_{14} and I_{12} is I_{24} .

An interesting example that has instant centers at infinity is the *Scotch-yoke* linkage, shown in Fig. 3.27, which is a variation of the slider-crank linkage. The four primary instant centers are I_{12} , I_{23} , I_{34} , and I_{14} . The secondary instant center I_{24} can be found directly from the Aronhold-Kennedy theorem. Also, from the same theorem, the secondary instant center I_{13} must lie on line $I_{12}I_{23}$. However, since link 3 is in curvilinear translation (that is, link 3 does not change angle with respect to the fixed link), instant center I_{13} is located at infinity.

One more example is presented here to reinforce these concepts.

EXAMPLE 3.11

Locate all instant centers of the five-bar mechanism shown in Fig. 3.28. There is rolling contact between links 1 and 2, and slipping between links 2 and 4 and between links 2 and 5.

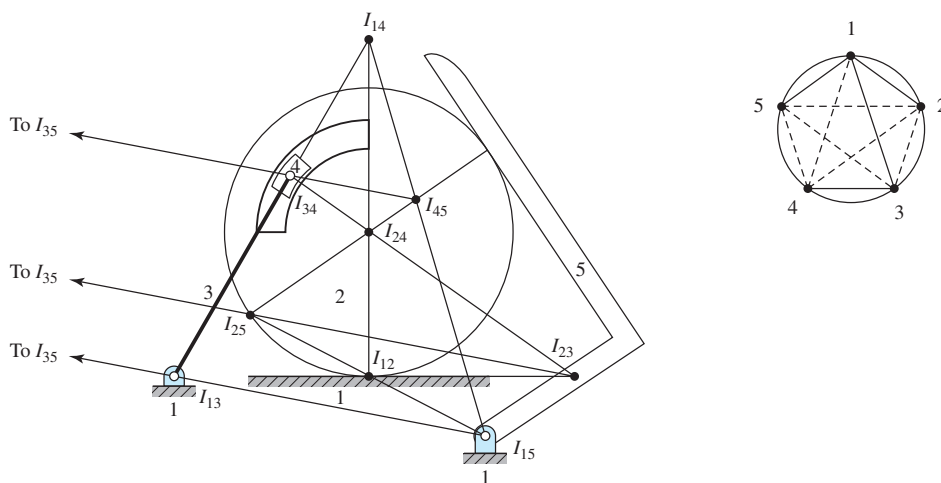


Figure 3.28 Five-bar mechanism and locations of instant centers.

SOLUTION

According to Eq. (3.20), the number of instant centers for this mechanism is 10. As indicated by the Kennedy circle, there are four primary instant centers and six secondary instant centers. The three pin joints are the primary instant centers I_{13} , I_{34} , and I_{15} , and the point of rolling contact between links 1 and 2 is the primary instant center I_{12} . The instant center I_{24} is regarded as a secondary instant center, since it can be located by drawing perpendicular lines to the directions of the apparent velocities at two of the corners of link 4. However, this instant center could also be regarded as a primary instant center, since it can be obtained from observation as the center of the apparent rotation between links 2 and 4. One line for the secondary instant center I_{25} is the perpendicular to the direction of slipping between links 2 and 5, and the other line is the line through the instant centers $I_{12}I_{15}$. The locations of the remaining four secondary instant centers, I_{23} , I_{35} , I_{45} , and I_{14} , can be found by repeated applications of the theorem of three centers. This is left as an exercise for the reader.

It should be noted before closing this section that in the previous examples the locations of instant centers were all found without requiring knowledge of the actual operating speed of the mechanism. This is another indication of the linearity of the equations relating velocities, as indicated in Sec. 3.9. For a single-degree-of-freedom mechanism, the locations of all instant centers are uniquely determined by geometry alone and do not depend on the operating speed [5].

3.15 VELOCITY ANALYSIS USING INSTANT CENTERS

In this section, we demonstrate how the properties of instant centers can provide a simple graphic approach to the velocity analysis of planar mechanisms. For illustration, consider the four-bar linkage in the posture indicated in Fig. 3.29a. For a known angular velocity of the input link, 2, the focus is to determine the velocities of pin B , coupler point D , and point E of link 4.

The procedure is to denote the four pins as the four primary instant centers and then locate the two secondary instant centers from Kennedy's theorem (Sec. 3.13). Now consider the instant center I_{24} . This instant center, by definition, is a pair of coincident points common to both links 2 and 4 and has the same absolute velocities in both links. Therefore, the velocity of this instant center can be written from Eq. (3.19) as

$$\mathbf{V}_{I_{24}} = \boldsymbol{\omega}_2 \times \mathbf{R}_{I_{24}I_{12}} = \boldsymbol{\omega}_4 \times \mathbf{R}_{I_{24}I_{14}}$$

and is shown on Fig. 3.29b.

Now we can obtain the velocity of any other point of link 4. For example, consider points B' and E' , which are on the line of centers $I_{24}I_{12}I_{14}$ and are chosen to have the same radii from I_{14} as pin B and point E . Therefore, their velocities have magnitudes equal to those of $\mathbf{V}_B$ and $\mathbf{V}_E$, respectively, and these can be laid out in their proper directions, as shown in Fig. 3.29b.

To obtain the velocity of the coupler point D , consider the line of centers $I_{23}I_{12}I_{13}$, which is shown in Fig. 3.30. Using the given input angular velocity, ω_2 , and the primary instant center I_{12} , we find the absolute velocity of the instant center I_{23} . Here, this step is straightforward, since $\mathbf{V}_{I_{23}} = \mathbf{V}_A$. Locating point D' on this line of centers (in the same manner as B' and E' , above), we find $\mathbf{V}_{D'}$ as shown and use its magnitude to plot the desired

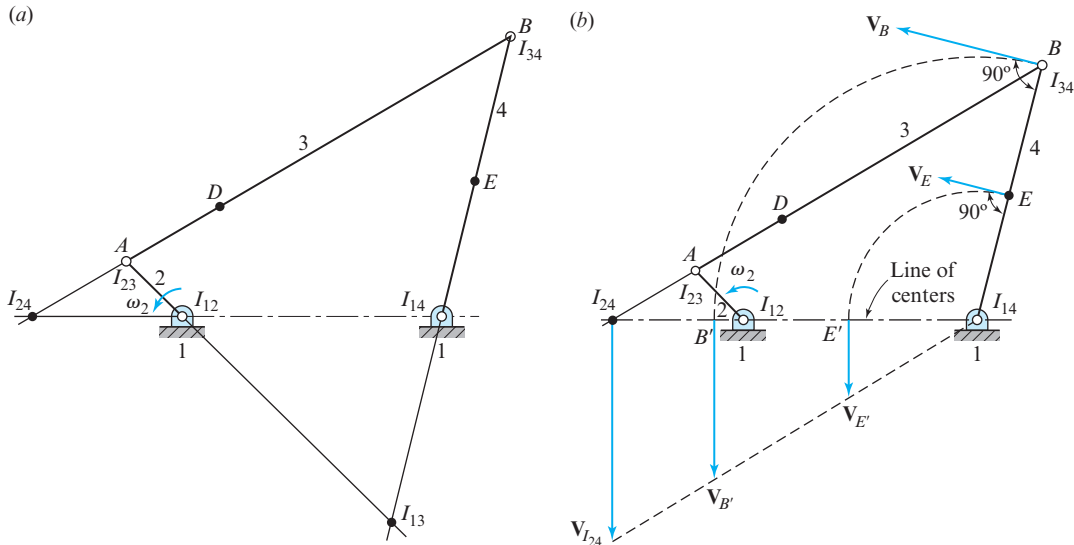


Figure 3.29 Graphic velocity analysis using instant centers.

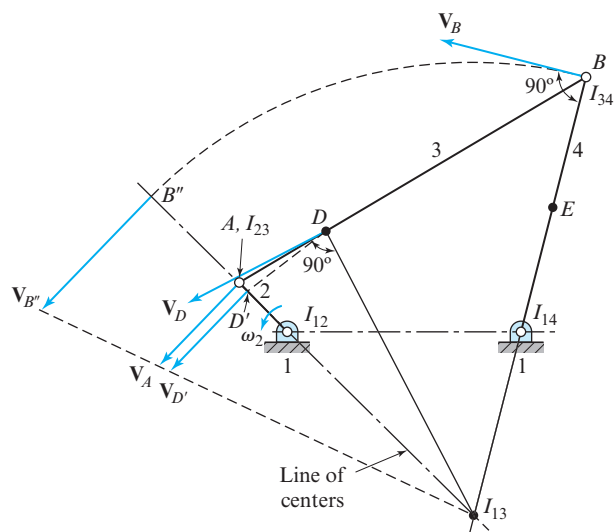


Figure 3.30 Velocities of pin B and point D .

velocity $\mathbf{V}_D$. Note that, according to its definition, the instant center I_{13} , as a point of link 3, has zero velocity at this instant. Also, note that pin B can be considered a point of link 3; its velocity can be determined in a similar manner by determining $\mathbf{V}_{B''}$, as shown in Fig. 3.30.

The *line-of-centers* method of velocity analysis using instant centers is summarized as follows:

1. Identify the three link numbers associated with the given velocity, the velocity to be determined, and the reference link. The reference link, in general, is link 1, since typically absolute velocity information is given and requested.
2. Locate the three instant centers defined by the links of step 1 and draw the line of centers.
3. Find the velocity of the common instant center by treating it as a point of the link whose velocity is given.
4. With the velocity of the common instant center known, consider it as a point of the link whose velocity is to be determined. The velocity of any other point in that link can now be determined.

The following example will help to illustrate the line-of-centers method and also demonstrate how to treat instant centers at infinity.

EXAMPLE 3.12

For the device shown in Fig. 3.31, some of the links are visible, whereas other links are enclosed in a housing. However, the location of instant center I_{25} is known, as shown in the figure. If the velocity of point C of link 6 is $\mathbf{V}_C = 10$ m/s to the right, determine the angular velocity of link 2.

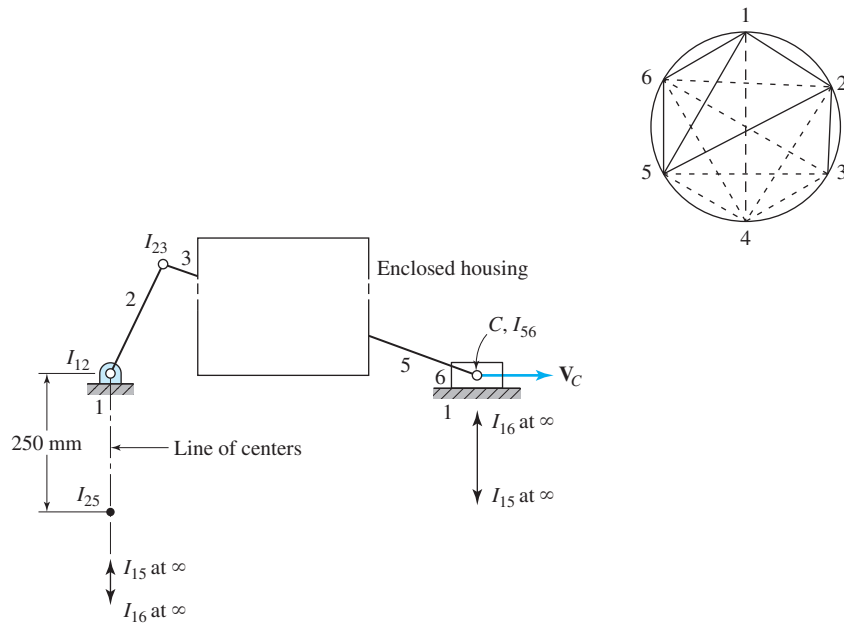


Figure 3.31 Links enclosed in a housing.

SOLUTION

Since $\mathbf{V}_{C5/1} = \mathbf{V}_C$, and we are required to find $\omega_{2/1}$, then we need to use the instant centers I_{15} , I_{12} , and I_{25} . Note that $\omega_{2/1}$ is the same as ω_2 , but the additional subscript has been written to emphasize the presence of the third link (the frame). The primary instant center I_{16} is at infinity, as shown. The instant center I_{15} lies on line $I_{12}I_{25}$ and on line $I_{16}I_{56}$, and since these two lines are parallel, then I_{15} is also at infinity.

Considering I_{25} a point of link 5, we wish to find the velocity of this point from the given $\mathbf{V}_C$. We have difficulty in locating point C' on the line of centers at the same radius from I_{15} as point C , since I_{15} is at infinity. How can we proceed?

Recalling the discussion of Sec. 3.14 and Eq. (3.21), since I_{15} is at infinity, the relative motion between links 5 and 1 is translation, and $\omega_{5/1} = 0$. Therefore, *every* point of link 5 has the same absolute velocity, including $\mathbf{V}_{I_{25}} = \mathbf{V}_C$. Thus, we lay out $\mathbf{V}_{I_{25}}$ on Fig. 3.31.

Next, we turn our attention to link 2. We treat I_{25} as a point of link 2 rotating about I_{12} , and solve for the angular velocity of link 2:

$$\omega_2 = \frac{V_{I_{25}}}{R_{I_{25}I_{12}}} = \frac{10 \text{ m/s}}{0.25 \text{ m}} = 40 \text{ rad/s ccw.} \quad \text{Ans.}$$

Noticing the apparent paradox between the directions of $\mathbf{V}_C$ and ω_2 , we may speculate on the validity of our solution. This is resolved, however, by opening the enclosed housing and discovering that the linkage is as shown in Fig. 3.32.

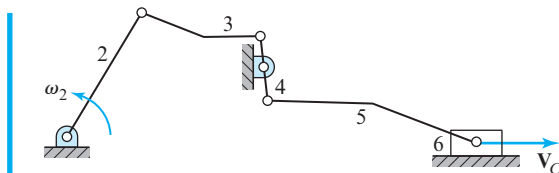


Figure 3.32 Explanation of the velocity paradox.

3.16 ANGULAR-VELOCITY-RATIO THEOREM

The instant center I_{24} of the four-bar linkage shown in Fig. 3.33 is a pair of coincident points of links 2 and 4. The absolute velocity $\mathbf{V}_{I_{24}}$ is the same whether I_{24} is considered as a point of link 2 or a point of link 4. Considering it each way, we can write

$$\mathbf{V}_{I_{24}} = \mathbf{V}_{I_{12}}^0 + \omega_{2/1} \times \mathbf{R}_{I_{24}I_{12}} = \mathbf{V}_{I_{14}}^0 + \omega_{4/1} \times \mathbf{R}_{I_{24}I_{14}}. \quad (a)$$

Again, note that $\omega_{2/1}$ and $\omega_{4/1}$ are the same as ω_2 and ω_4 , respectively, but the additional subscript is used to emphasize the presence of the third link (the frame).

Considering only the magnitudes of Eq. (a), the equation can be rewritten as

$$\frac{\omega_{4/1}}{\omega_{2/1}} = \frac{R_{I_{24}I_{12}}}{R_{I_{24}I_{14}}}. \quad (b)$$

This illustrates the *angular-velocity-ratio theorem*, which states that *the angular-velocity ratio of any two bodies in planar motion with respect to a third body is inversely proportional to the segments into which the common instant center cuts the line of centers*. Written in general notation for the motion of two bodies j and k with respect to a third body i , the equation is

$$\frac{\omega_{k/i}}{\omega_{j/i}} = \frac{R_{I_{jk}I_{ji}}}{R_{I_{jk}I_{ki}}}. \quad (3.22)$$

Choosing an arbitrary positive direction along the line of centers, you should prove for yourself that *the angular-velocity ratio is positive when the common instant center falls outside the other two instant centers and negative when it falls between them*.

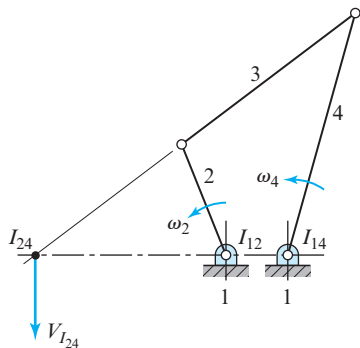


Figure 3.33
Angular-velocity-ratio theorem.

3.17 RELATIONSHIPS BETWEEN FIRST-ORDER KINEMATIC COEFFICIENTS AND INSTANT CENTERS

First-order kinematic coefficients (Sec. 3.11) can be expressed in terms of the locations of instant centers. For a planar linkage, the angular velocity of link j can be written from the chain rule as

$$\omega_j = \theta'_j \omega_i, \quad (3.23)$$

where ω_i is the input angular velocity. Therefore, the first-order kinematic coefficient of link j can be written as

$$\theta'_j = \frac{\omega_j}{\omega_i}.$$

Consistent with the angular-velocity-ratio theorem, Eq. (3.22), the first-order kinematic coefficient of link j can be written as

$$\theta'_j = \frac{R_{I_{ij}I_{i1}}}{R_{I_{ij}I_{j1}}}, \quad (3.24)$$

where I_{i1} and I_{j1} are the absolute instant centers of links i and j , respectively, and I_{ij} is the common instant center to links i and j .

EXAMPLE 3.13

For the four-bar linkage in Example 3.1, the lengths of the input link and the frame can be written as $R_{I_{12}I_{23}} = 4$ in and $R_{I_{12}I_{14}} = 10$ in, respectively. Determine the angular velocities of the coupler link and the output link using instant centers and first-order kinematic coefficients.

SOLUTION

The first-order kinematic coefficients of links 3 and 4 can be written from Eq. (3.24) as

$$\theta'_3 = \frac{R_{I_{23}I_{12}}}{R_{I_{23}I_{13}}} \quad \text{and} \quad \theta'_4 = \frac{R_{I_{24}I_{12}}}{R_{I_{24}I_{14}}}.$$

From Fig. 3.29—that is, the Aronhold-Kennedy theorem—we find $R_{I_{23}I_{13}} = 15$ in and $R_{I_{24}I_{12}} = 11.2$ in. Therefore, the first-order kinematic coefficients of the two links are

$$\theta'_3 = \frac{4 \text{ in}}{15 \text{ in}} = 0.267 \text{ rad/rad} \quad \text{and} \quad \theta'_4 = \frac{11.2 \text{ in}}{21.2 \text{ in}} = 0.528 \text{ rad/rad}.$$

Substituting these values and the input angular velocity $\omega_2 = 94.25$ rad/s ccw into Eqs. (3.23), the angular velocity of the coupler link and the output link, respectively, are

$$\omega_3 = 25.16 \text{ rad/s ccw} \quad \text{and} \quad \omega_4 = 49.77 \text{ rad/s ccw}.$$

These answers agree well with the results obtained from the velocity polygon method of Example 3.1 ($\omega_3 = 25.6$ rad/s ccw and $\omega_4 = 49.64$ rad/s ccw).

Note that Eqs. (3.23) and (3.24), as now written, are not valid for a linkage where link i is a slider. For example, consider a slider-crank linkage with the slider (denoted as link i) regarded as the input. The first-order kinematic coefficient of link j can be written from the chain rule as

$$\theta'_j = \frac{\omega_j}{\dot{r}_i}, \quad (3.25)$$

where $\dot{r}_i$ is the velocity of the slider. When $R_{I_{ij}I_{1i}}$ becomes infinite, Eq. (3.24) for the first-order kinematic coefficient of link j becomes

$$\theta'_j = \frac{\pm 1}{R_{I_{ij}I_{1j}}}, \quad (3.26)$$

where the plus or minus sign depends on the positive direction chosen along the line of centers. Consistent with the statement after Eq. (3.22), the angular-velocity ratio is positive when the relative instant center falls outside the two absolute centers and negative when it falls between them. The following example is used to illustrate Eqs. (3.25) and (3.26).

EXAMPLE 3.14

For the offset slider-crank linkage in the posture shown in Fig. 3.34, the input velocity of slider 4 is 10 m/s to the right. Determine the angular velocities of the crank and coupler link using instant centers and first-order kinematic coefficients.

SOLUTION

The first-order kinematic coefficients of links 2 and 3 can be written from Eq. (3.26) as

$$\theta'_2 = \frac{\pm 1}{R_{I_{24}I_{12}}} \quad \text{and} \quad \theta'_3 = \frac{\pm 1}{R_{I_{34}I_{13}}}.$$

We choose the velocity of slider 4 as positive to the right, therefore we have $\hat{\mathbf{u}}^t = \hat{\mathbf{i}}$, and then, by Eq. (3.7), we have $\hat{\mathbf{u}}^n = \hat{\mathbf{u}}^b \times \hat{\mathbf{u}}^t = \hat{\mathbf{k}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}}$ (upward) as the positive direction along the line of centers.* From a scale drawing of the linkage, we measure $R_{I_{24}I_{12}} = +51$ mm

* Note that this sign convention is the same as saying that positive input motion of slider 4 is the same as a positive (counterclockwise) rotation of link 4 about instant center I_{14} now located at infinity in the positive $\hat{\mathbf{u}}^n$ direction.

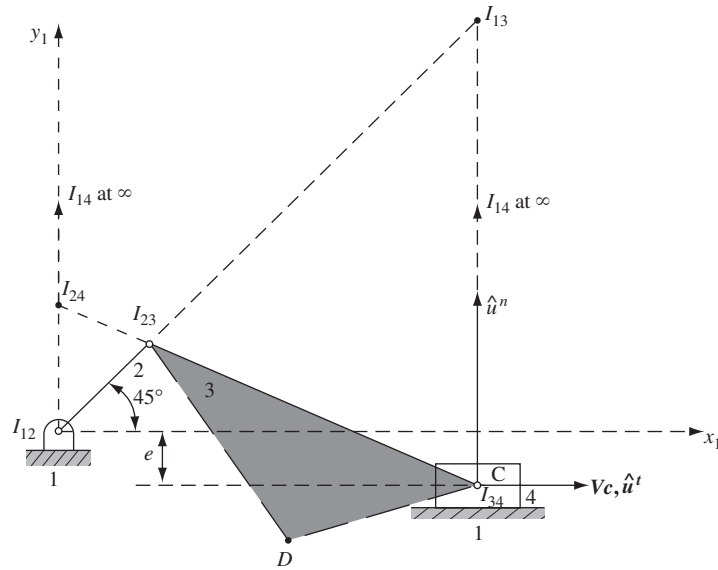


Figure 3.34 $e = 20$ mm, $R_{BA} = 50$ mm, $R_{CB} = 140$ mm, $R_{EB} = 80$ mm, and $R_{DE} = 50$ mm.

and $R_{I_{34}I_{13}} = -185$ mm and find that the kinematic coefficients are

$$\theta'_2 = \frac{\pm 1}{0.051 \text{ m}} = -19.6 \text{ rad/m} \quad \text{and} \quad \theta'_3 = \frac{\pm 1}{-0.185 \text{ m}} = +5.41 \text{ rad/m},$$

where θ'_2 is negative since I_{24} lies between I_{12} and I_{14} (which is at infinity upward, in the positive $\hat{\mathbf{u}}^n$ direction along the line of centers). Similarly, θ'_3 is positive since I_{34} is outside of I_{13} and I_{14} .

Rearranging Eq. (3.25), and substituting the kinematic coefficients and the input velocity, $\dot{r}_4 = 10$ m/s, the angular velocities of links 2 and 3, respectively, are

$$\omega_2 = 196 \text{ rad/s cw} \quad \text{and} \quad \omega_3 = 54.1 \text{ rad/s ccw}.$$

Allowing for the reversal of the input velocity, these answers agree closely with the results obtained from the graphic method in Example 3.2 ($\omega_2 = 200$ rad/s ccw and $\omega_3 = 53.6$ rad/s cw). Also, since measurements are used for finding the locations of the instant centers, these results are probably still not as accurate as those of Example 3.8 ($\omega_2 = 198.56$ rad/s ccw and $\omega_3 = 54.47$ rad/s cw).

The velocity of a point, say P , fixed in a link of a mechanism can be written as

$$\mathbf{V}_P = V_P \hat{\mathbf{u}}_P^t \quad (3.27)$$

or as

$$\mathbf{V}_P = (x'_P \hat{\mathbf{i}} + y'_P \hat{\mathbf{j}}) \dot{\psi}, \quad (3.28)$$

where $\dot{\psi}$ is the generalized input velocity to the mechanism. The magnitude of the velocity, commonly referred to as the speed, can be written from the chain rule as

$$V_P = r'_P \dot{\psi}, \quad (3.29)$$

where the first-order kinematic coefficient is defined as

$$r'_P = \pm \sqrt{x_P'^2 + y_P'^2}. \quad (3.30)$$

Here, the sign convention is as follows: We use the positive sign if the instantaneous change in the input position is positive, and we use the negative sign if the instantaneous change in the input position is negative.

Rearranging Eq. (3.27), the unit tangent vector to the point trajectory can be written as

$$\hat{\mathbf{u}}_P^t = \frac{\mathbf{V}_P}{V_P}. \quad (3.31)$$

Then, substituting Eqs. (3.28) and (3.29) into this relation gives

$$\hat{\mathbf{u}}_P^t = \left(\frac{x'_P}{r'_P} \right) \hat{\mathbf{i}} + \left(\frac{y'_P}{r'_P} \right) \hat{\mathbf{j}}. \quad (3.32)$$

Consistent with Eq. (3.7), the unit normal vector to the point trajectory can now be written as $\hat{\mathbf{u}}_P^n = \hat{\mathbf{u}}^b \times \hat{\mathbf{u}}_P^t$. Substituting Eq. (3.32) into this relation, the unit normal vector can be written as

$$\hat{\mathbf{u}}_P^n = \left(\frac{-y'_P}{r'_P} \right) \hat{\mathbf{i}} + \left(\frac{x'_P}{r'_P} \right) \hat{\mathbf{j}}. \quad (3.33)$$

3.18 FREUDENSTEIN'S THEOREM

In the analysis or design of a linkage, it is sometimes important to know the postures of the linkage at which the extreme values of the output velocity occur, or, more precisely, the input position values at which the ratio of the output and input velocities reaches its extremes.

The earliest work in determining extreme values is apparently that of Krause [9], who stated that the velocity ratio ω_2/ω_4 of the drag-link four-bar linkage (Fig. 3.35) reaches an extreme value when the connecting rod and follower, links 3 and 4, become perpendicular to each other. Rosenauer, however, demonstrated that this is not strictly true [13]. Following

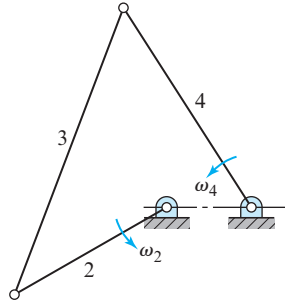


Figure 3.35 Drag-link four-bar linkage.

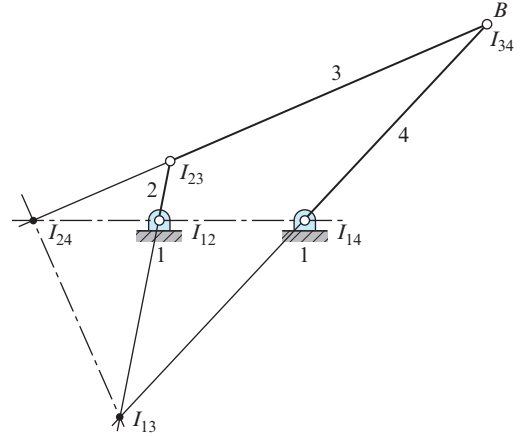


Figure 3.36 Collineation axis.

Krause, Freudenstein developed a simple graphic method for determining the postures of the four-bar linkage at which the extreme values of the output velocity do occur [6].

Freudenstein's theorem makes use of the line connecting instant centers I_{13} and I_{24} (Fig. 3.36), called the *collineation axis*. The theorem states: *At an extreme of the output to input angular-velocity ratio of a four-bar linkage, the collineation axis is perpendicular to the coupler link.**

Using the angular-velocity-ratio theorem, Eq. (3.22), we write

$$\frac{\omega_4}{\omega_2} = \frac{R_{I_{24}I_{12}}}{R_{I_{24}I_{12}} + R_{I_{12}I_{14}}}.$$

Since $R_{I_{12}I_{14}}$ is the fixed length of the frame, the extremes of the velocity ratio occur when $R_{I_{24}I_{12}}$ is either a maximum or a minimum. Such positions may occur on either or both sides of the instant center I_{12} . Thus the problem reduces to finding the geometry of the linkage for which $R_{I_{24}I_{12}}$ is an extremum.

During motion of the linkage, I_{24} travels along the line $I_{12}I_{14}$, as shown by the theorem of three centers, but, at an extreme value of the velocity ratio, I_{24} must instantaneously be at rest (its direction of travel on this line must be reversing). This occurs when the velocity of I_{24} , considered a point of link 3, is directed along the coupler link. This is true only when the coupler link is perpendicular to the collineation axis, since I_{13} is the instant center of link 3.

An inversion of the theorem (treating link 2 as fixed) states that an *extreme value of the velocity ratio ω_3/ω_2 of a four-bar linkage occurs when the collineation axis is perpendicular to the follower (link 4).*

* A. S. Hall, Jr., contributed a rigorous proof of this theorem in an appendix to Freudenstein's paper [6].

3.19 INDICES OF MERIT; MECHANICAL ADVANTAGE

In this section we will study various ratios, angles, and other parameters of mechanisms that tell whether a mechanism is acceptable for a particular application (Sec. 1.10). Many such parameters have been defined by various authors over the years, and there is no common agreement on a single “index of merit” for all mechanisms. Yet the many used have a number of features in common, including the fact that most can be related to velocity ratios of the mechanism, and, therefore, can be determined solely from its geometry. In addition, most depend on some knowledge of the application of the mechanism, especially the input and output variables. It is often desirable in the analysis or synthesis of a mechanism to plot an index of merit for a revolution of the input crank and to note in particular the minimum and maximum values when evaluating a design or its suitability for a given application.

In Sec. 3.16, we learned that the ratio of the angular velocity of the output link to the input velocity of a mechanism is inversely proportional to the segments into which the common instant center cuts the line of centers. Thus, in the four-bar linkage shown in Fig. 3.37, if links 2 and 4 are the input and output cranks, respectively, then

$$\frac{\omega_4}{\omega_2} = \frac{R_{IA}}{R_{ID}}$$

is an equation for the output-to-input-velocity ratio. We also learned in Sec. 3.18 that the extremes of this ratio occur when the collineation axis is perpendicular to the coupler, link 3.

If we now assume that the linkage of Fig. 3.37 has no friction or inertia forces during its operation or that these are negligible compared with the input torque, T_2 , applied to link 2, and the output torque, T_4 , the resisting load torque on link 4, then we can derive a relation between T_2 and T_4 . Since friction and inertia forces are negligible, the input power applied to link 2 is equal to the power applied to link 4 by the load; hence,

$$T_2\omega_2 = T_4\omega_4 \quad (a)$$

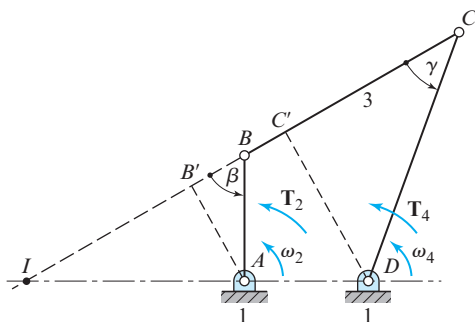


Figure 3.37 Four-bar linkage.

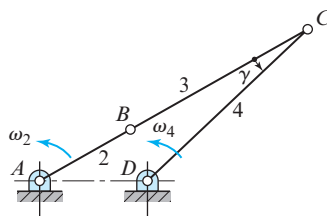


Figure 3.38 Four-bar linkage in toggle.

or

$$\frac{T_4}{T_2} = \frac{\omega_2}{\omega_4} = \frac{R_{ID}}{R_{IA}}. \quad (3.34)$$

As stated in Sec. 1.10, the *mechanical advantage* of a mechanism is the instantaneous ratio of the output force (torque) to the input force (torque). Here we see that the mechanical advantage is the reciprocal of the velocity ratio. Either can be used as an index of merit in judging the ability of a mechanism to transmit force or power.

The mechanism is redrawn in Fig. 3.38 at a posture where links 2 and 3 are aligned. At such a posture, R_{IA} and ω_4 are passing through zero; hence, an extreme value (infinity) of the mechanical advantage is obtained. A mechanism in such a posture is said to be *in toggle*. Such toggle postures are often used to produce a high mechanical advantage; an example is the clamping mechanism of Fig. 2.8.

Proceeding further, we construct $B'A$ and $C'D$ perpendicular to line IBC in Fig. 3.37. Also, we assign labels β and γ to the acute angles made by the coupler or its extension and the input and output links, respectively. Then, by similar triangles,

$$\frac{R_{ID}}{R_{IA}} = \frac{R_{C'D}}{R_{B'A}} = \frac{R_{CD} \sin \gamma}{R_{BA} \sin \beta}. \quad (b)$$

Then, using Eq. (3.34), another expression for mechanical advantage is

$$\frac{T_4}{T_2} = \frac{\omega_2}{\omega_4} = \frac{R_{CD} \sin \gamma}{R_{BA} \sin \beta}. \quad (3.35)$$

This equation demonstrates that the mechanical advantage is infinite whenever the angle β is 0° or 180° —that is, whenever the mechanism is in a toggle posture.

In Secs. 1.10 and 2.9, we defined the angle γ between the coupler link and the follower link as the *transmission angle*. This angle is also often used as an index of merit for a four-bar linkage. Equation (3.35) indicates that the mechanical advantage diminishes when the transmission angle is much less than a right angle. If the transmission angle becomes too small, the mechanical advantage becomes small, and even a small amount of friction may cause a mechanism to lock or jam. To avoid this, a common rule of thumb is that a four-bar linkage should not be used in a region where the transmission angle is less than, say, 45° or 50° . The better four-bar linkage, based on the quality of its force transmission, has a transmission angle that deviates from 90° by the smaller amount.

In other mechanisms—for example, meshing gear teeth or a cam-and-follower system—*pressure angle* is used as an index of merit. The pressure angle is defined as the acute angle between the direction of the output force and the direction of the velocity of the point where the output force is applied. Pressure angles are discussed more thoroughly in Chaps. 6, 7, and 8. In the four-bar linkage, the pressure angle is the complement of the transmission angle.

Another index of merit that has been proposed [4] is the determinant of the matrix of coefficients of the simultaneous equations relating the dependent velocities of a mechanism. In Example 3.6, for example, we saw that the dependent velocities of a

four-bar linkage are related by

$$(r_3 \sin \theta_3) \omega_3 - (r_4 \sin \theta_4) \omega_4 = -(r_2 \sin \theta_2) \omega_2,$$

$$(r_3 \cos \theta_3) \omega_3 - (r_4 \cos \theta_4) \omega_4 = -(r_2 \cos \theta_2) \omega_2.$$

The matrix of coefficients is called the *Jacobian* of the system and its determinant is

$$\Delta = \begin{vmatrix} r_3 \sin \theta_3 & -r_4 \sin \theta_4 \\ r_3 \cos \theta_3 & -r_4 \cos \theta_4 \end{vmatrix} = r_3 r_4 \sin(\theta_4 - \theta_3).$$

As is clear from Cramer's rule, the solutions for the dependent velocities, in this case ω_3 and ω_4 , must include this determinant in the denominator. This is borne out in the solution of the four-bar linkage, Eqs. (5) in Example 3.6. Although the form of this determinant changes for different mechanisms, such a determinant can always be defined and always appears in the denominators of all dependent velocity solutions.

If this determinant becomes small, the mechanical advantage also becomes small and the usefulness of the mechanism is reduced in such regions. This same determinant also appears in the denominator of the dependent accelerations (Sec. 4.12) and all other quantities that require derivatives of the loop-closure equation. If this determinant is small, the mechanism will function poorly in all respects—force transmission, motion transformation, sensitivity to manufacturing errors, and so on.

3.20 CENTRODES

We noted in Sec. 3.12 that the location of an instant center is defined only instantaneously and changes as the mechanism moves. When the changing locations of an instant center are found for all possible postures of a single-degree-of-freedom mechanism, a curve or

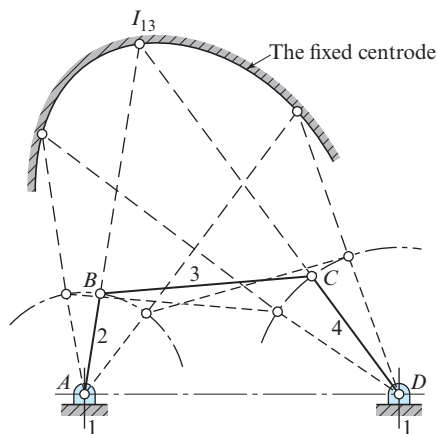


Figure 3.39 Fixed centrode.

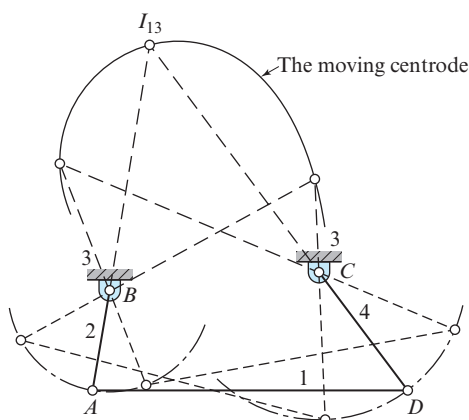


Figure 3.40 Moving centrode.

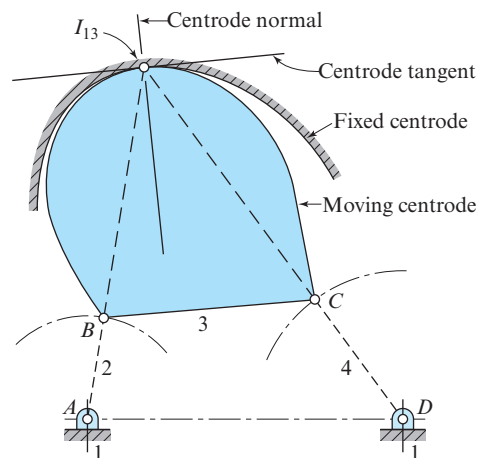


Figure 3.41 Rolling contact between centrodes.

locus, called a *centrode*,* is defined. For example, consider the four-bar linkage shown in Fig. 3.39. The instant center I_{13} (located at the intersection of the extensions of links 2 and 4) traces out a curve, called the *fixed centrode* on link 1, as the linkage moves through all possible postures. The literature [11] indicates that the fixed centrode of a four-bar linkage is a planar algebraic curve of degree eight.

Figure 3.40 shows the inversion of the same four-bar linkage in which link 3 is fixed and link 1 is movable. When this inversion moves through all possible input positions, I_{13} traces a *different* curve on link 3. For the original linkage, with link 1 fixed, this is the curve traced by I_{13} on the coordinate system of the moving link 3; it is called the *moving centrode*.

Figure 3.41 shows the moving centrode, attached to link 3, and the fixed centrode, attached to link 1. It is imagined here that links 1 and 3 have been machined to the actual shapes of the respective centrodes and that links 2 and 4 have been removed entirely. If the moving centrode is now permitted to roll on the fixed centrode without slip, link 3 has exactly the same motion as it had in the original linkage. This remarkable property, which stems from the fact that a point of rolling contact is an instant center, turns out to be quite useful in the synthesis of linkages.

* Opinion seems divided on whether these loci should be termed *centrodes* or *polodes*. Generally, those preferring the name *instant center* call them *centrodes*, and those preferring the name *velocity pole* call them *polodes*. The French name *roulettes* has also been applied. The three-dimensional equivalents are ruled surfaces and are referred to as *axodes*. Note that there is a subtle distinction between the words *instant center* and *velocity pole*, and this is explained in Sec. 4.12.

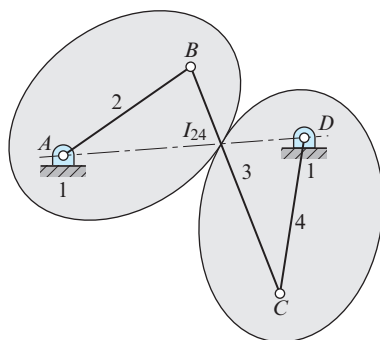


Figure 3.42 Elliptic gears.

We can restate this property as follows: *The plane motion of one rigid body with respect to another is completely equivalent to the rolling motion of one centrode on the other.* Beyer [2] provides a detailed discussion of centrodes and their fundamental significance in the synthesis of planar mechanisms. The instantaneous point of rolling contact of the centrodes is the instant center, as shown in Fig. 3.41. Also shown are the common tangent to the two centrodes and the common normal, called the *centrode tangent* and the *centrode normal*. These coordinates are sometimes used as the axes of a coordinate system, commonly referred to as the *canonical* coordinate system. This system is used for developing equations of a coupler curve and other geometric properties of the motion, called the *instantaneous invariants* [3].

The centrodes of Fig. 3.41 were generated by the instant center I_{13} on links 1 and 3. Another pair of centrodes, both moving, is generated on links 2 and 4 when instant center I_{24} is considered. These two centrodes roll upon each other and describe the identical motion between links 2 and 4 that would result from the operation of the original four-bar linkage. Figure 3.42 shows these centrodes as two ellipses for the case of a crossed double-crank linkage with equal cranks. This construction can be used as the basis for the development of a pair of elliptic gears.

3.21 REFERENCES

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PROBLEMS*

- †3.1** The position vector of a point is given by the equation $\mathbf{R} = 100e^{j\pi t}$, where R is in inches. Find the velocity of the point at $t = 0.40$ s.
- †3.2** The path of a point is defined by the equation $\mathbf{R} = (t^2 + 4)e^{-j\pi t/10}$, where R is in meters. Find the velocity of the point at $t = 20$ s.
- †3.3** Automobile A is traveling south at 55 mi/h and automobile B is traveling north 60° east at 40 mi/h. Find the velocity difference between B and A and the apparent velocity of B to the driver of A .
- †3.4** Wheel 2 rotates at 600 rev/min cw and drives wheel 3 without slipping. Find the velocity difference between points B and A .

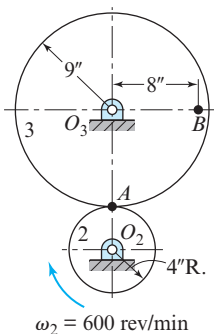


Figure P3.4

- †3.5** The distance between points A and B , located along the radius of a wheel, is $R_{BA} = 300$ mm. The speeds of points A and B are $V_A = 80$ m/s and $V_B = 140$ m/s, respectively. Find the diameter of the wheel, the velocities $\mathbf{V}_{AB}$ and $\mathbf{V}_{BA}$, and the angular velocity of the wheel.

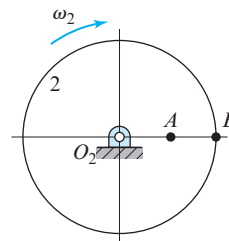


Figure P3.5

- †3.6** An airplane takes off from point B and flies east at 350 mi/h. Simultaneously, another airplane at point A , 200 miles southeast, takes off and flies northeast at 390 mi/h. (a) How close will the airplanes come to each other if they fly at the same altitude? (b) If both airplanes leave at 6:00 p.m., at what time will this occur?

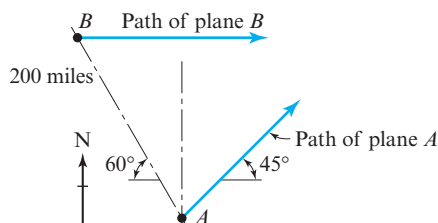


Figure P3.6 $R_{AB} = 200$ mi.

- †3.7 Include a wind of 30 mi/h from the west with the data of Prob. 3.6. (a) If airplane A flies at the same heading, what is its new path? (b) What change does the wind make in the results of Prob. 3.6?

- †3.8 For the double-slider linkage in the posture shown, the velocity of point B is 40 m/s. Find the velocity of point A and the angular velocity of link 3.

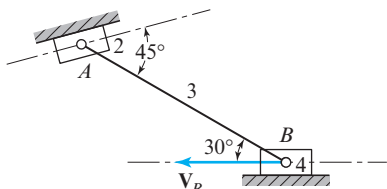


Figure P3.8 $R_{AB} = 400$ mm.

- †3.9 The four-bar linkage in the posture shown is driven by crank 2 at $\omega_2 = 45$ rad/s ccw. Find the angular velocities of links 3 and 4.

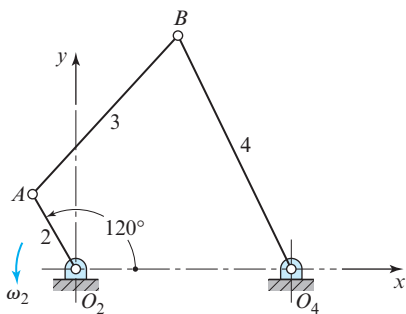


Figure P3.9 $R_{AO_2} = 4$ in, $R_{BA} = 10$ in, $R_{O_4O_2} = 10$ in, and $R_{BO_4} = 12$ in.

* When assigning problems, the instructor may wish to specify the method of solution to be used, because a variety of approaches are presented in the text.

- †3.10 The four-bar linkage in the posture shown is driven by crank 2 at $\omega_2 = 60$ rad/s cw. Find the angular velocities of links 3 and 4, and the velocity of pin B and point C on link 3.

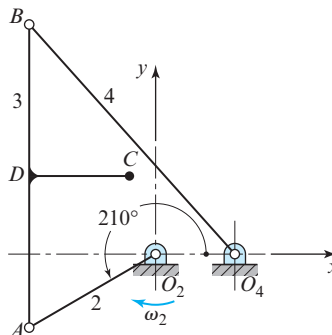


Figure P3.10 $R_{AO_2} = 150$ mm, $R_{BA} = 300$ mm, $R_{O_4O_2} = 75$ mm, $R_{BO_4} = 300$ mm, $R_{DA} = 150$ mm, and $R_{CD} = 100$ mm.

- †3.11 The four-bar linkage in the posture shown is driven by crank 2 at $\omega_2 = 48$ rad/s ccw. Find the angular velocity of link 3 and the velocity of point C on link 4.

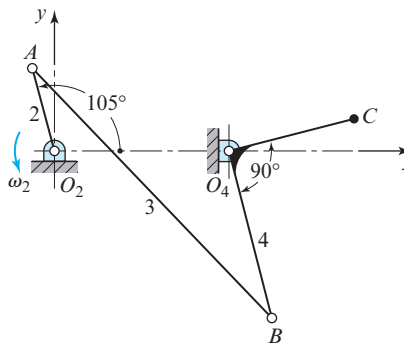


Figure P3.11 $R_{AO_2} = 8$ in, $R_{BA} = 32$ in, $R_{O_4O_2} = 16$ in, $R_{BO_4} = 16$ in, and $R_{CO_4} = 12$ in.

- †3.12 For the parallelogram four-bar linkage, demonstrate that ω_3 is always zero and that $\omega_4 = \omega_2$. How would you describe the motion of link 4 with respect to link 2?

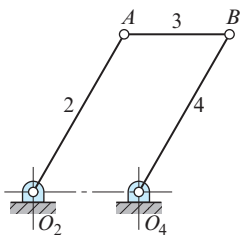


Figure P3.12

- †3.13** The antiparallel, or crossed, four-bar linkage in the posture shown is driven by link 2 at $\omega_2 = 1$ rad/s ccw. Find the velocities of points C and D.

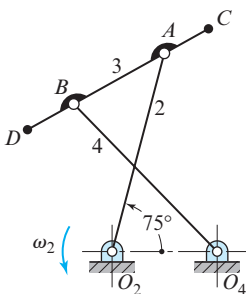


Figure P3.13 $R_{AO_2} = R_{BO_4} = 300$ mm, $R_{BA} = R_{O_4O_2} = 150$ mm, and $R_{CA} = R_{DB} = 75$ mm.

- †3.14** For the four-bar linkage in the posture shown, link 2 has an angular velocity of 60 rad/s ccw. Find the angular velocities of links 3 and 4, and the velocity of point C.

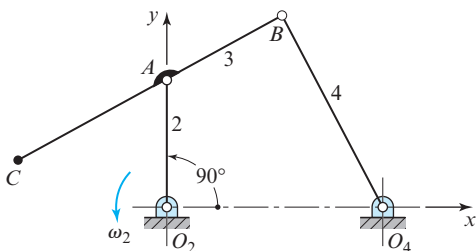


Figure P3.14 $R_{AO_2} = R_{BA} = 6$ in, $R_{O_4O_2} = R_{BO_4} = 10$ in, and $R_{CA} = 8$ in.

- †3.15** Crank 2 of the inverted slider-crank linkage, in the posture shown, is driven at $\omega_2 = 60$ rad/s ccw. Find the angular velocities of links 3 and 4, and the velocity of point B.

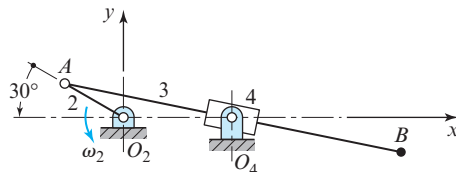


Figure P3.15 $R_{AO_2} = 75$ mm, $R_{BA} = 400$ mm, and $R_{O_4O_2} = 125$ mm.

- †3.16** For the four-bar linkage in the posture shown, crank 2 has an angular velocity of 30 rad/s cw. Find the velocity of coupler point C and the angular velocities of links 3 and 4.

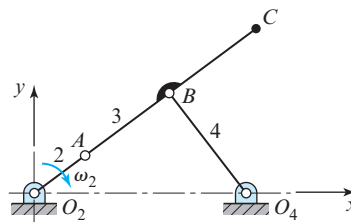


Figure P3.16 $R_{AO_2} = 3$ in, $R_{BA} = R_{CB} = 5$ in, $R_{O_4O_2} = 10$ in, and $R_{BO_4} = 6$ in.

- †3.17** For the modified slider-crank linkage in the posture shown, crank 2 has an angular velocity of 10 rad/s ccw. Find the angular velocity of link 6 and the velocities of points B, C, and D.

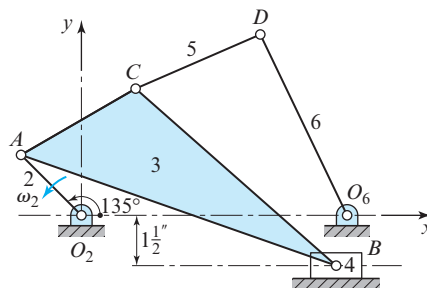


Figure P3.17 $R_{AO_2} = 2.5$ in, $R_{BA} = 10$ in, $R_{CB} = 8$ in, $R_{DC} = 4$ in, $R_{O_6O_2} = 8$ in, and $R_{DO_6} = 6$ in.

- †3.18 For the four-bar linkage shown, the angular velocity of crank 2 is a constant 16 rad/s cw. Plot a polar velocity diagram for the velocity of point B for all crank positions. Check the positions of maximum and minimum velocities by using Freudenstein's theorem.

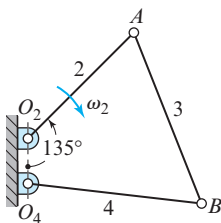


Figure P3.18 $R_{AO_2} = 350$ mm, $R_{BA} = 425$ mm, $R_{O_4O_2} = 100$ mm, and $R_{BO_4} = 400$ mm.

- †3.19 For the four-bar linkage in the posture shown, link 2 is driven at $\omega_2 = 36$ rad/s cw. Find the angular velocity of link 3 and the velocity of point B .

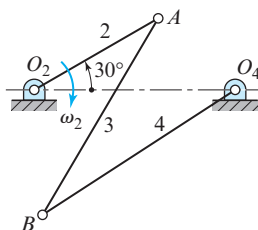


Figure P3.19 $R_{AO_2} = 5$ in, $R_{BA} = R_{BO_4} = 8$ in, and $R_{O_4O_2} = 7$ in.

- †3.20 For the four-bar linkage in the posture shown, the angular velocity of the input link 2 is 8 rad/s ccw. Find the velocity of point C and the angular velocity of link 3.

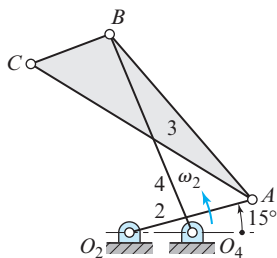


Figure P3.20 $R_{AO_2} = 150$ mm, $R_{BA} = R_{BO_4} = 250$ mm, $R_{O_4O_2} = 75$ mm, $R_{CA} = 300$ mm, and $R_{CB} = 100$ mm.

- †3.21 For the four-bar linkage in the posture shown, link 2 has an angular velocity of 56 rad/s ccw. Find the velocity of point C .

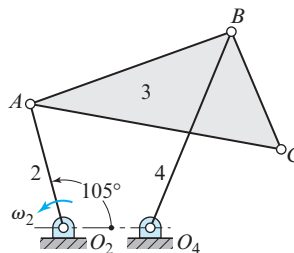


Figure P3.21 $R_{AO_2} = 150$ mm, $R_{BA} = R_{BO_4} = 250$ mm, $R_{O_4O_2} = 100$ mm, and $R_{CA} = 300$ mm.

- †3.22 For the double-slider linkage in the posture shown, the angular velocity of the input crank 2 is 42 rad/s cw. Find the velocities of points B , C , and D .

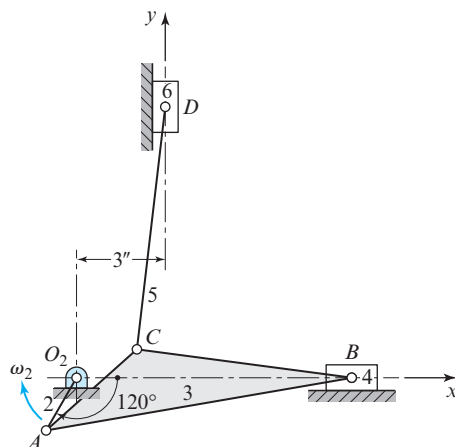


Figure P3.22 $R_{AO_2} = 2$ in, $R_{BA} = 10$ in, $R_{CA} = 4$ in, $R_{CB} = 7$ in, and $R_{DC} = 8$ in.

- †3.23 For the linkage used in a two-cylinder 60° V-engine consisting, in part, of an articulated connecting rod crank 2 rotates at 2 000 rev/min cw. Find the velocities of points B , C , and D .

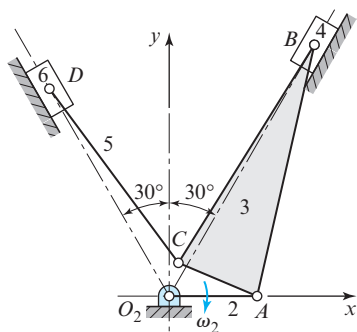


Figure P3.23 $R_{AO_2} = 2$ in, $R_{BA} = R_{CB} = 6$ in, $R_{CA} = 2$ in, and $R_{DC} = 5$ in.

- 3.24** For the inverted slider-crank linkage in the posture shown, the angular velocity of the crank is $\omega_2 = 24$ rad/s cw. Make a complete velocity analysis of the linkage. What is the absolute velocity of point B ? What is its apparent velocity to an observer moving with link 4?

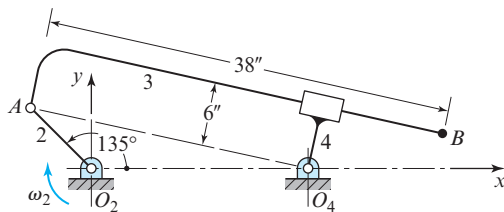


Figure P3.24 $R_{AO_2} = 8$ in and $R_{O_4O_2} = 20$ in.

- 3.25** For the linkage in the posture shown, the velocity of point A is $1\hat{i}$ ft/s. Find the velocity of coupler point B .

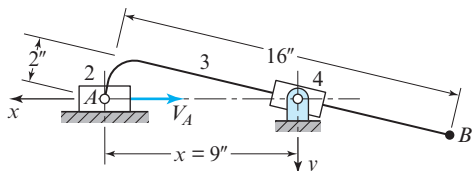


Figure P3.25

- †3.26** A variation of the Scotch-yoke linkage in the posture shown is driven by crank 2 at $\omega_2 = 36$ rad/s ccw. Find the velocity of the crosshead, link 4.

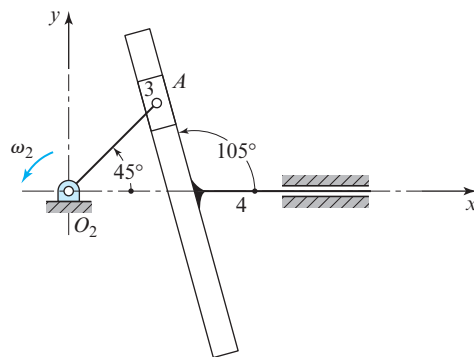


Figure P3.26 $R_{AO_2} = 250$ mm.

- 3.27** Perform a complete velocity analysis of the modified four-bar linkage for $\omega_2 = 72$ rad/s ccw.

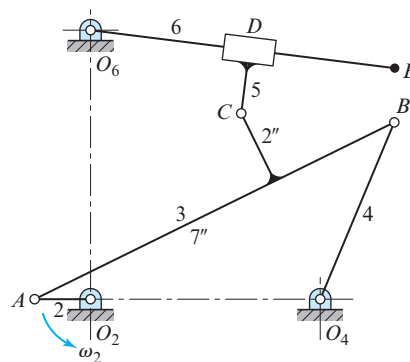


Figure P3.27 $R_{AO_2} = R_{DC} = 1.5$ in, $R_{BA} = 10.5$ in, $R_{O_4O_2} = 6$ in, $R_{BO_4} = 5$ in, $R_{O_6O_2} = 7$ in, and $R_{EO_6} = 8$ in.

- 3.28** For the mechanism in the posture shown, the velocity of point C is $V_C = 10$ in/s to the left. There is rolling contact between links 1 and 2,

- 3.33** The diagram shows a planar schematic approximation of an automotive front suspension. The *roll center* is the term used by the industry to describe the point about which the auto body seems to rotate (roll) with respect to the ground. The assumption is made that there is pivoting but no slip between the tires and the road. After making a sketch, use the concepts of instant centers to find a technique to locate the roll center.

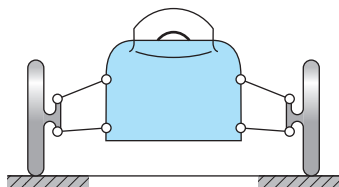


Figure P3.33

- 3.34** Locate all instant centers for the linkage of Prob. 3.22.
- 3.35** Locate all instant centers for the mechanism of Prob. 3.25.
- 3.36** Locate all instant centers for the mechanism of Prob. 3.26.
- 3.37** Locate all instant centers for the mechanism of Prob. 3.27.
- 3.38** Locate all instant centers for the mechanism of Prob. 3.28.
- 3.39** Locate all instant centers for the mechanism of Prob. 3.29.
- 3.40** The posture of the input link 2 is $\mathbf{R}_{AO_4} = -120\hat{\mathbf{i}}$ mm, and the velocity of point A is $\mathbf{V}_A = 15\hat{\mathbf{i}}$ m/s. Determine the first-order kinematic coefficients for the mechanism. Find the angular velocities of links 3 and 4.

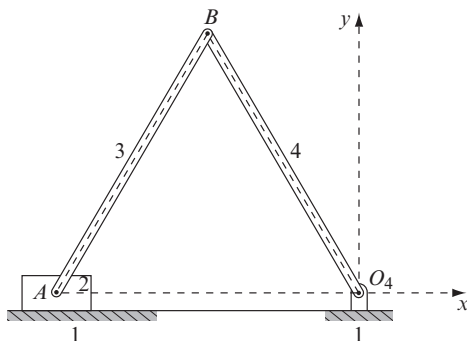


Figure P3.40 $R_{BO_4} = R_{BA} = 120$ mm.

- 3.41** For the rack-and-pinion mechanism in the posture shown, link 2 is the input, and pinion 3 is rolling without slipping on rack 4 at point D. Determine the first-order kinematic coefficients of links 3 and 4. If the constant input velocity is $\mathbf{V}_G = 3\hat{\mathbf{i}}$ in/s, determine the angular velocities of rack 4 and pinion 3.

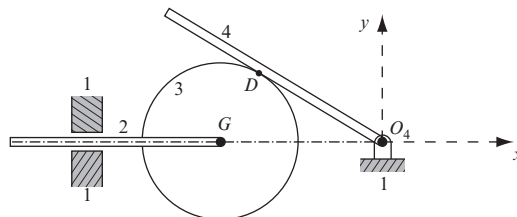


Figure P3.41 $R_{GO_4} = 10$ in, and $R_{DG} = r_3 = 5$ in.

- 3.42** For the rack-and-pinion mechanism of Example 2.8 (Figs. 2.33 and 2.34), the dimensions are $R_1 = 800$ mm, $R_9 = 550$ mm, $\theta_{34} = 60^\circ$, and $r_3 = 500$ mm. In the posture where $R_2 = 750$ mm, input link 2 has a velocity of $\mathbf{V}_A = 0.150\hat{\mathbf{j}}$ m/s. Determine the first-order kinematic coefficients to obtain the velocity of rack 4 and the angular velocity of pinion 3.
- 3.43** For the mechanism in the posture shown, $R_{AO_4} = 10$ in, and the input velocity is $\mathbf{V}_A = -5\hat{\mathbf{i}}$ in/s. Determine the first-order kinematic coefficients to obtain the angular velocity of link 3 and the slipping velocity between links 3 and 4.

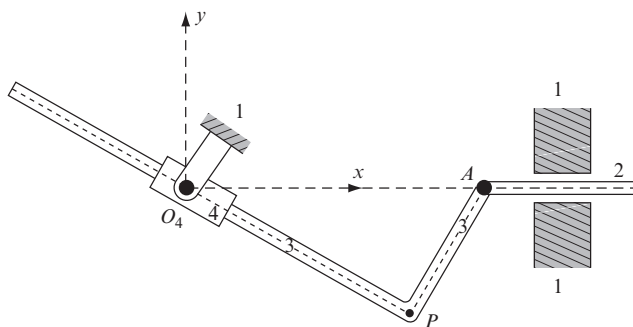


Figure P3.43 $R_{PA} = 5$ in, and $\angle APO_4 = 90^\circ$.

3.44 For the mechanism in the posture shown in Fig. P3.30, input crank 2 has an angular velocity $\omega_2 = 10 \text{ rad/s}$ ccw, and there is rolling contact between links 5 and 6 at point F . Determine the first-order kinematic coefficients of links 3, 4, 5, and 6. Find the angular velocities of links 3, 4, 5, and 6, and the velocities of points E and G .

3.45 For the mechanism in the posture shown, where $R_{AO_4} = 40 \text{ mm}$, pinion 4, is rolling without slipping on rack 3 at point B . Determine the first-order kinematic coefficients of rack 3 and pinion 4. If $\mathbf{V}_A = 150\hat{i} \text{ mm/s}$, determine the angular velocities of rack 3 and pinion 4, and the velocity of point E . Also, determine the velocity along rack 3 of the point of contact between links 3 and 4 (that is, point B).

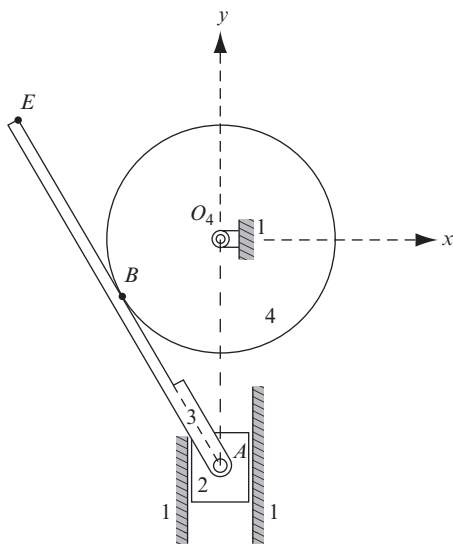


Figure P3.45 $\rho_4 = 20 \text{ mm}$, and $R_{EB} = R_{BA}$.

3.46 For the mechanism in the posture shown, where $\theta_2 = 150^\circ$, $R_{PA} = R_{AO_4}$, and $R_{PB} = R_{BA}$, determine the first-order kinematic coefficients of links 3, 4, and 5. If the angular velocity of the input link 2

is $\omega_2 = 5 \text{ rad/s}$ cw, determine: (a) the angular velocities of links 3 and 4, (b) the velocity of link 5, and (c) the velocity of point P fixed in link 4.

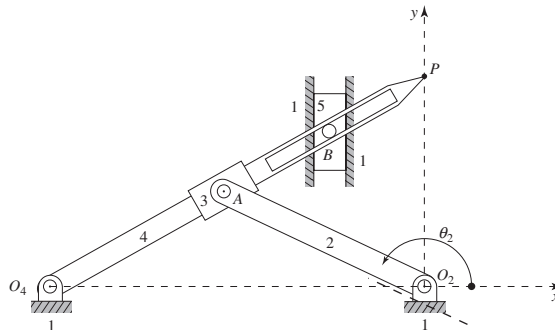


Figure P3.46 $R_{AO_2} = 10 \text{ in}$ and $R_{PO_4} = 20 \text{ in}$.

3.47 For the inverted slider-crank linkage in the posture shown, where $\theta_4 = 60^\circ$, the input link, 2, is moving parallel to the x axis. Determine the first-order kinematic coefficients of links 3 and 4. Also, determine the conditions for the determinant of the coefficient matrix to become zero. If $V_B = 15 \text{ in/s}$ constant to the right, determine the angular velocities of links 3 and 4.

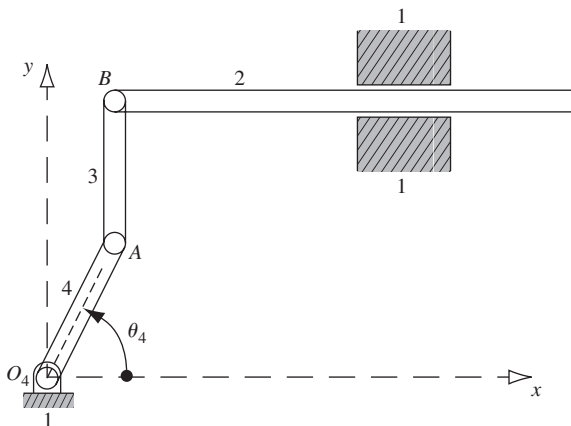


Figure P3.47 $R_{AO_4} = R_{BA} = 4 \text{ in}$.

3.48 For the rack-and-pinion mechanism in the posture shown in Fig. P2.15, the input link, 2, is vertical, and $\angle BAO_2$ is 150° . The dimensions are $\rho_5 = 2.5$ in, $\mathbf{R}_{O_5O_2} = -8\hat{i} + 4\hat{j}$ in, $R_{AO_2} = 2$ in, and $R_{BA} = 6$ in. (a) Show the locations of all instant centers. (b) Using instant centers, determine the first-order kinematic coefficients of link 3, rack 4, and pinion 5. (c) If $\omega_2 = 10$ rad/s cw, determine the angular velocity of link 3, the velocity of rack 4, and the angular velocity of pinion 5.

3.49 For the mechanism in the posture shown in Fig. P2.16, $\rho_2 = 1$ in, $\rho_5 = 2$ in, $R_{BA} = 7.071$ in, and $R_{BC} = 6$ in. Determine the first-order kinematic coefficients of links 3, 4, and 5. If link 2 is driven at $\omega_2 = 5$ rad/s ccw, determine the angular velocities of links 3, 4, and 5.

3.50 For the mechanism in the posture shown in Fig. P2.17, link 4 is parallel to the x axis and link 5 is coincident with the y axis. The radius of wheel 3 is $\rho_3 = 0.75$ in, $R_{O_2O_5} = 7.0$ in, $R_{BA} = 5.5$ in, and $R_{AO_5} = 2.6$ in. Determine the first-order kinematic coefficients of links 3, 4, and 5. If the input link, 2, has an angular velocity of $\omega_2 = 15$ rad/s cw, determine the angular velocities of links 3, 4, and 5.

3.51 For the mechanism in the posture shown, wheel 3 is rolling without slipping on the ground link at point C while sliding in the slot in link 2. Write the vector loop equation and determine the first-order kinematic coefficients of the mechanism. If the angular velocity of the input is $\omega_2 = 30$ rad/s ccw, determine: the angular velocity of the wheel; and the apparent velocity of the center of the wheel, point A , with respect to the slot in link 2.

3.52 For the linkage in the posture shown, link 2 is the input, link 3 is horizontal, and link 4 is vertical. Write the vector loop equation and determine the kinematic coefficients of the mechanism. If the angular velocity of link 2 is $\omega_2 = 30$ rad/s ccw, determine the angular velocity of links 3 and 4.

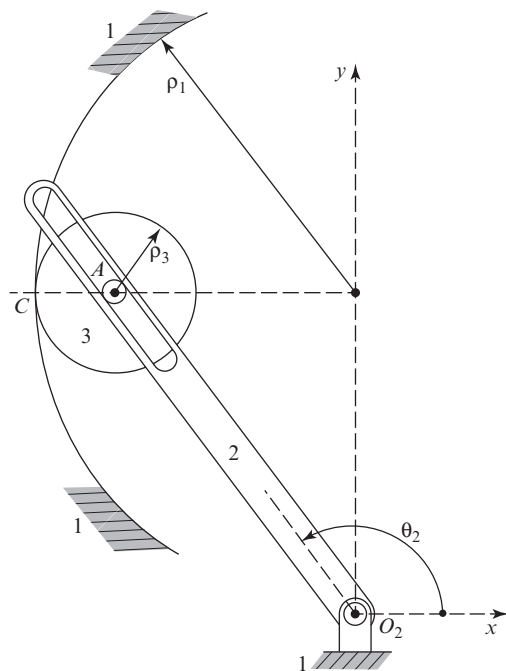


Figure P3.51 $\rho_1 = 60$ mm, $\rho_3 = 15$ mm, $\mathbf{R}_{CO_2} = -60\hat{i} + 60\hat{j}$ mm, and $R_{AO_2} = 75$ mm.

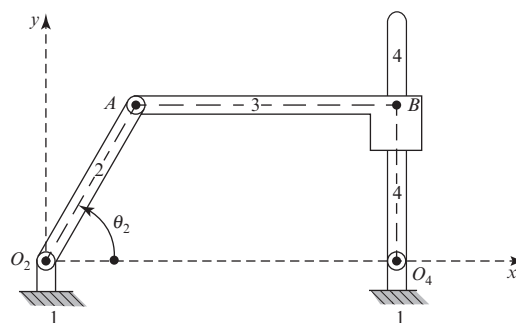


Figure P3.52 $R_{O_4O_2} = 12$ in, $R_{AO_2} = 6$ in, and $R_{BA} = 9$ in.

- 3.53** For the mechanism in the posture shown, determine the first-order kinematic coefficients of the mechanism. If the velocity of link 2 is $V_{A_2} = 0.30$ m/s in the direction shown, determine: (a) the angular velocity of link 3, (b) the apparent velocity of pin A_2 with respect to the slot in link 3, and (c) the velocity of point B .

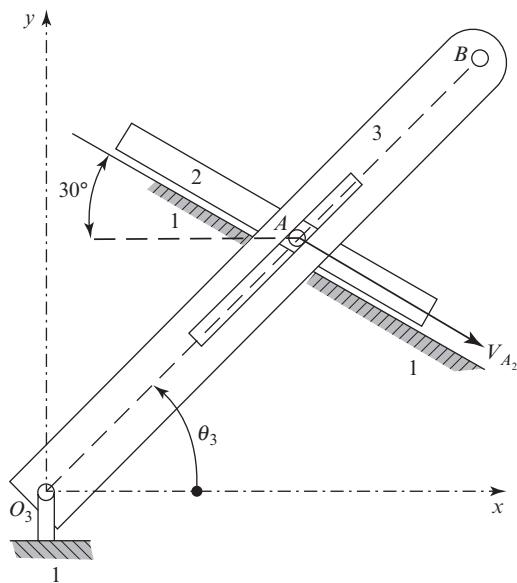


Figure P3.53 $R_{A_2O_3} = 750$ mm, and $R_{BO_3} = 1250$ mm.

- 3.54** For the mechanism in the posture shown, the internal track of input gear 2 is in rolling contact with gear 3 at point C , and the external track is in rolling contact with rack 5 at point F . Gear 3 is also in rolling contact with the fixed gear (link 1) at point E . Determine the first-order kinematic coefficients of gear 3 and the rack. Also, determine the angular velocities of gear 3 and link 4, and the velocity of the rack if gear 2 has an angular velocity $\omega_2 = 77$ rad/s ccw.

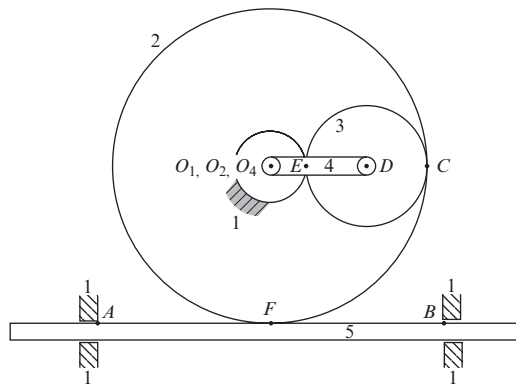


Figure P3.54 $\rho_1 = R_{EO_1} = 4$ in, $\rho_2 = R_{CO_2} = 18$ in, $\rho_3 = R_{CD} = 7$ in, and $R_{FA} = R_{BF} = 20$ in.

- 3.55** For the mechanism in the posture shown, the input arm, link 2, is pinned to the ground at O_1 and is pinned to the center of gear 3 at A . The center of gear 4 is also pinned to the ground at O_1 , and gear 5 is pinned to the ground at O_5 . Gears 3, 4, and 5 are all in rolling contact at point B . Determine the first-order kinematic coefficients of gears 3, 4, and 5. If the angular velocity is $\omega_2 = 15$ rad/s cw, use the kinematic coefficients to determine the angular velocities of gears 3, 4, and 5.

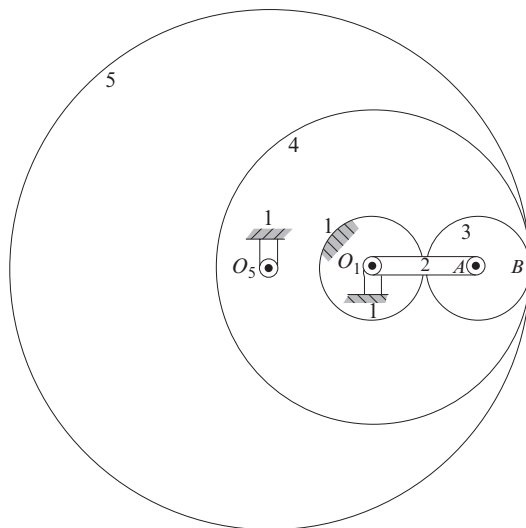


Figure P3.55 $\rho_1 = 100$ mm, $\rho_3 = 100$ mm, $\rho_4 = 300$ mm, and $\rho_5 = 500$ mm.

- 3.56** For the mechanism in the posture shown, the radius of the wheel (link 5) is rolling on the circular ground link. Determine the first-order kinematic coefficients of the mechanism. If the input link has a constant velocity of $\mathbf{V}_{A_2} = 5\mathbf{i}$ m/s, determine the angular velocities of links 3, 4, and 5.

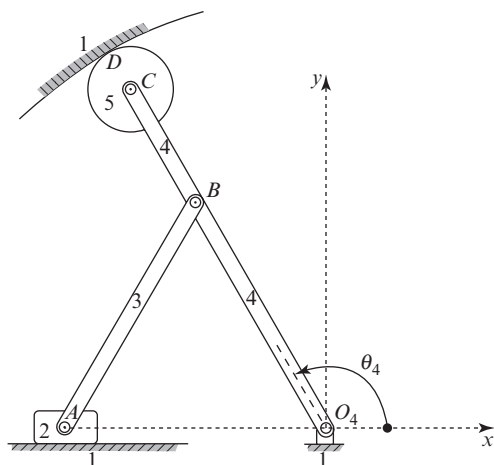


Figure P3.56 $R_{BO_4} = 120$ mm, $R_{CO_4} = 180$ mm, $R_{BA} = 120$ mm, and $\rho_5 = 20$ mm.

- 3.57** For the mechanism in the posture shown, the pinion (link 3) rolls without slip on rack 4 at point B. Determine the first-order kinematic coefficients

of the mechanism. If the velocity of input link 2 is $V_{A_2} = 4.8$ in/s constant upward, determine the angular velocity of the pinion and the velocity of the rack.

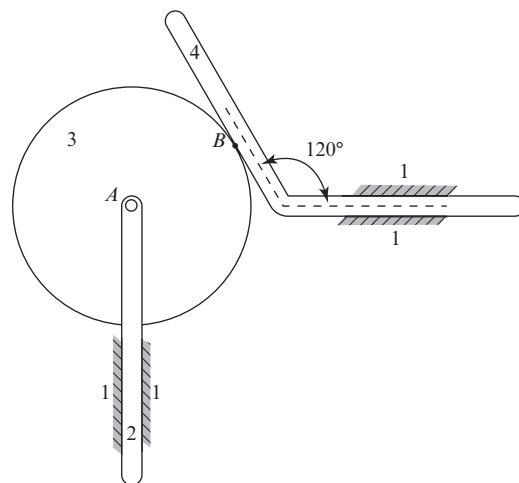


Figure P3.57 $\rho_3 = 2$ in.

- 3.58** For the mechanism in the posture shown, determine the first-order kinematic coefficients of links 3, 4, 5, and 6. If the constant input velocity $\mathbf{V}_{A_2} = 0.090\mathbf{j}$ m/s, determine the angular velocities of links 3 and 5, and the velocity of point P.

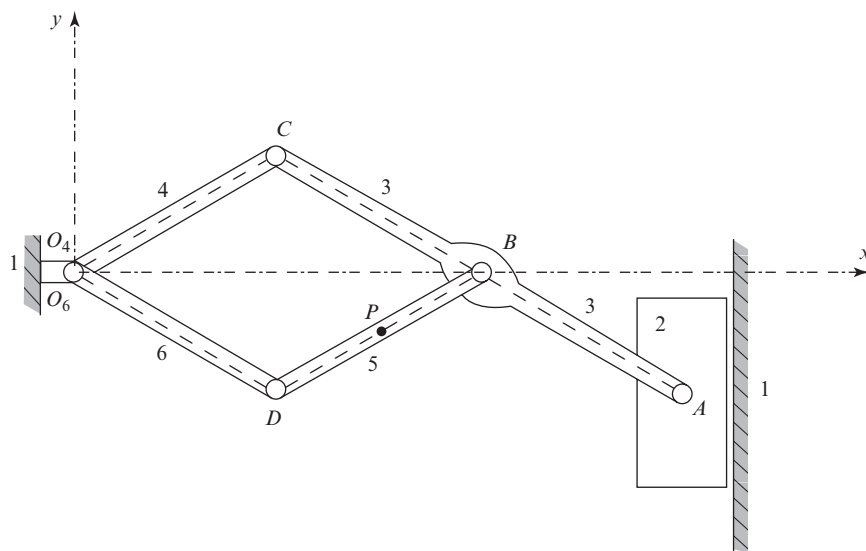


Figure P3.58 $R_{CO_4} = R_{DO_6} = R_{BC} = R_{BD} = R_{AB} = 40$ mm, $R_{PD} = 20$ mm, and $\mathbf{R}_A = (104 \text{ mm})\mathbf{i} + y_A\mathbf{j}$.

- 3.59** For the mechanism in the posture shown, line AB is vertical and line CD is horizontal. Determine the first-order kinematic coefficients of the mechanism. If the angular velocity of the input link 2 is $\omega_2 = 10 \text{ rad/s}$ cw, determine the angular velocities of links 3 and 4, and the velocity of point D fixed in link 4 with respect to point C fixed in link 3.

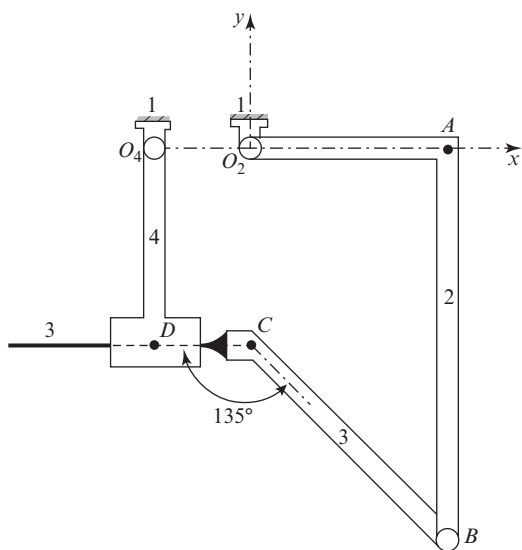


Figure P3.59 $R_{AO_2} = 4 \text{ in}$, $R_{BA} = 8 \text{ in}$, $R_{DO_4} = 4 \text{ in}$, and $R_{O_2O_4} = 2 \text{ in}$.

- 3.60** For the mechanism in the posture shown, gear 3 rolls without slipping on link 4 at point C . Determine the first-order kinematic coefficients of the mechanism. If the velocity of input link 2 is a constant, $\mathbf{V}_B = -0.5 \hat{\mathbf{j}} \text{ m/s}$, determine the angular velocity of gear 3 and the velocity of link 4.

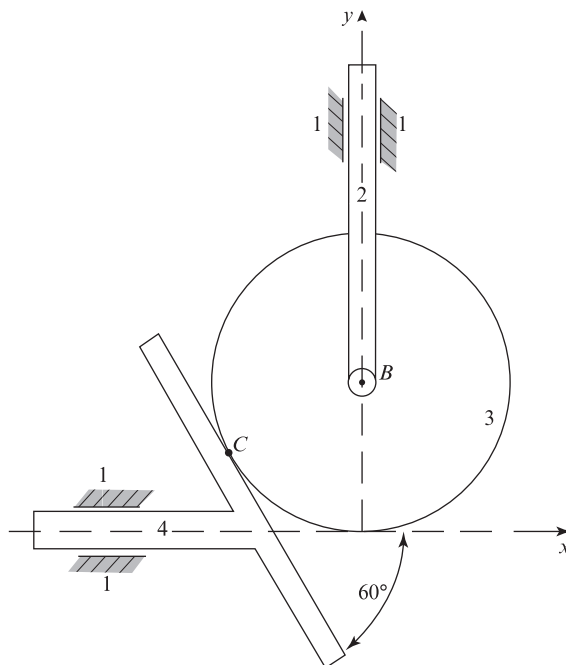


Figure P3.60 $\rho_3 = 25 \text{ mm}$.

- 3.61** For the mechanism in the posture shown, link 3 is vertical and link 4 is horizontal. Determine the first-order kinematic coefficients of the mechanism and coupler point C . If input link 2 has a constant angular velocity $\omega_2 = 15 \text{ rad/s}$ ccw, then determine the velocity of point C .

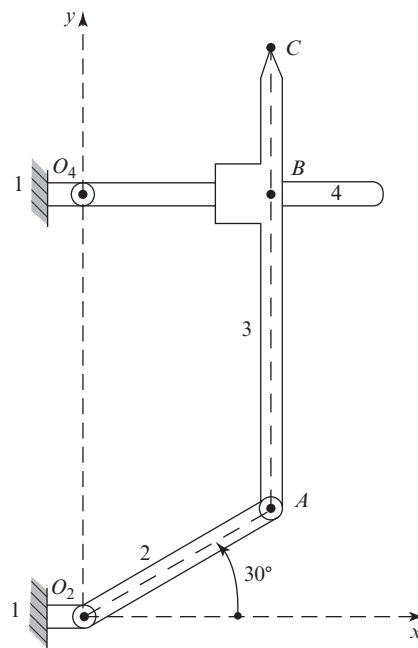
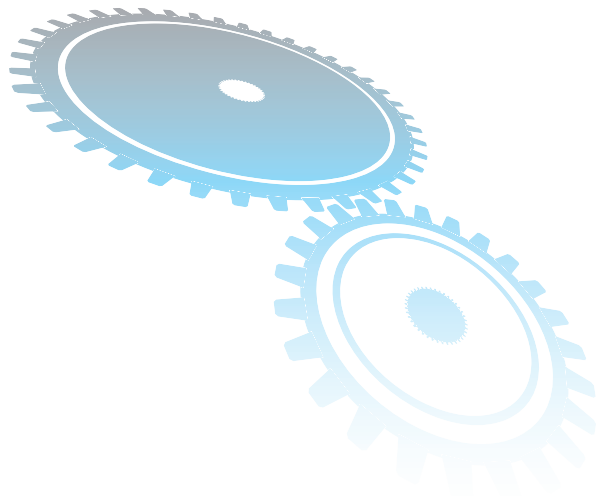


Figure P3.61 $R_{O_4O_2} = 12 \text{ in}$, $R_{AO_2} = 6 \text{ in}$, and $R_{CA} = 13 \text{ in}$.

4

Acceleration



4.1 DEFINITION OF ACCELERATION

Consider a moving point, first observed at location P , where it has a velocity, $\mathbf{V}_P$, as shown in Fig. 4.1a. After a short time interval, Δt , the point is observed to have moved along a path to a new location, P' , where it has velocity $\mathbf{V}'_P$, which may be different from $\mathbf{V}_P$ in both magnitude and direction. The change in the velocity of the point, between the two positions, can be evaluated as shown in the velocity polygon, Fig. 4.1b; namely,

$$\Delta \mathbf{V}_P = \mathbf{V}'_P - \mathbf{V}_P.$$

The *average acceleration* of point P during the given time interval is defined as $\Delta \mathbf{V}_P / \Delta t$. The *instantaneous acceleration* (henceforth, referred to simply as the *acceleration*) of point P is defined as the time rate of change of its velocity—that is, the limit of the average acceleration for an infinitesimally small time interval:

$$\mathbf{A}_P = \lim_{\Delta t \rightarrow 0} \left(\frac{\Delta \mathbf{V}_P}{\Delta t} \right) = \frac{d\mathbf{V}_P}{dt} = \frac{d^2 \mathbf{R}_P}{dt^2}. \quad (4.1)$$

Since velocity is a vector quantity, the change in velocity, $\Delta \mathbf{V}_P$, and the acceleration, $\mathbf{A}_P$, are also vector quantities—that is, they have both magnitude and direction. Also, like velocity, the acceleration vector is properly defined only for a point; the term should not be applied to a line, a coordinate system, a volume, or any other collection of points, since the accelerations of different points may be different.

As in the case of velocity (Chap. 3), the acceleration of a moving point may appear differently to different observers. Acceleration does not depend on the actual location of the observer but does depend on the motion of the observer or, more precisely, on the

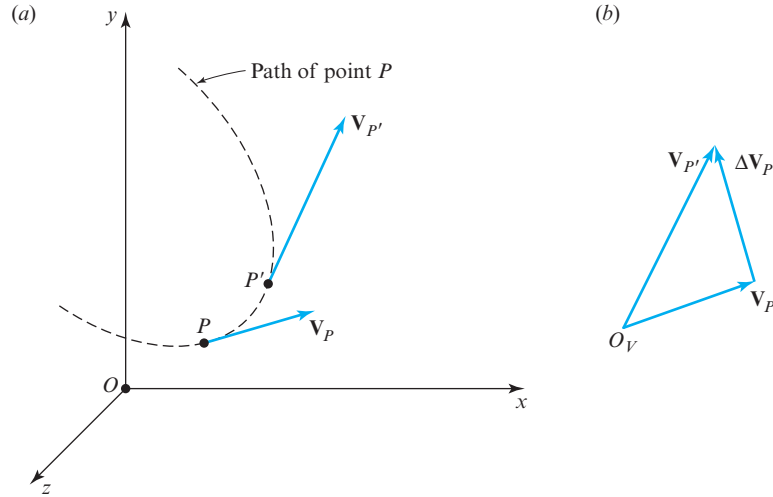


Figure 4.1 Change in velocity of a moving point; (b) velocity.

motion of the observer's coordinate system. If acceleration is sensed by an observer using the absolute coordinate system, it is referred to as *absolute acceleration* and is denoted by the symbol $\mathbf{A}_{P/1}$ or simply $\mathbf{A}_P$, which is consistent with the notation used for position, displacement, and velocity.

Recall that the velocity of point P , moving along path AB (Fig. 4.2a) can be written from Eq. (3.9) as

$$\mathbf{V}_P = \dot{s} \hat{\mathbf{u}}^t, \quad (a)$$

where $\dot{s}$ is the instantaneous speed of P along the path, and $\hat{\mathbf{u}}^t$ is the unit vector tangent to the path and with positive sense in the direction of positive displacement Δs (Sec. 3.5) of point P . We identify the *osculating plane*, defined by $\hat{\mathbf{u}}^t$, and the instantaneous *center of curvature* of the path of P (denoted as C). Designating the preferred positive side of the osculating plane by the *binormal* unit vector, $\hat{\mathbf{u}}^b$, we complete the right-hand vector triad $\hat{\mathbf{u}}^b \hat{\mathbf{u}}^t \hat{\mathbf{u}}^n$ by defining the *unit normal* vector, $\hat{\mathbf{u}}^n = \hat{\mathbf{u}}^b \times \hat{\mathbf{u}}^t$, consistent with Eq. (3.7). Thus, $\hat{\mathbf{u}}^t$ and $\hat{\mathbf{u}}^n$ are tangent and normal, respectively, to the path at the instantaneous position of point P .

The acceleration of point P is obtained by differentiating Eq. (a) with respect to time; that is

$$\mathbf{A}_P = \dot{s} \dot{\hat{\mathbf{u}}}^t + \ddot{s} \hat{\mathbf{u}}^t, \quad (b)$$

where $\ddot{s}$ is the instantaneous time rate of change in the speed of P along the path. The first term on the right-hand side of this equation can be written in the form

$$\dot{s} \dot{\hat{\mathbf{u}}}^t = \frac{ds}{dt} \left(\frac{d\hat{\mathbf{u}}^t}{d\phi} \frac{d\phi}{ds} \frac{ds}{dt} \right), \quad (c)$$

where ϕ represents the inclination angle of the unit tangent vector, $\hat{\mathbf{u}}^t$, with respect to an arbitrary axis selected in the osculating plane (Fig. 4.2a uses an axis parallel to the x axis).

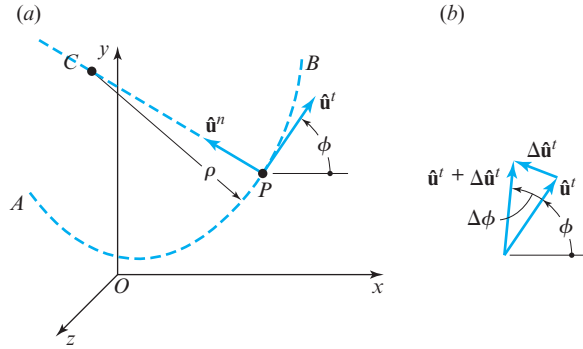


Figure 4.2 Motion of point P generates path AB .

As point P moves along the path AB , $\hat{u}^t$ is a function of ϕ , which, in turn, is a function of the distance, s , along the path.

In Fig. 4.2b, the unit tangent vector $\hat{u}^t$ changes, the vectors $\hat{u}^t$ and $\hat{u}^t + \Delta \hat{u}^t$ have been transferred to a common origin, and we observe that

$$\lim_{\Delta \phi \rightarrow 0} \left(\frac{\Delta \hat{u}^t}{\Delta \phi} \right) = \frac{d\hat{u}^t}{d\phi}. \quad (d)$$

From trigonometry, the left-hand side of this equation can be written as

$$\lim_{\Delta \phi \rightarrow 0} \left(\frac{2 \sin(\Delta \phi / 2) \hat{u}^n}{\Delta \phi} \right) = \hat{u}^n. \quad (e)$$

Therefore, equating Eq. (d) with Eq. (e), we have the relation

$$\frac{d\hat{u}^t}{d\phi} = \hat{u}^n. \quad (f)$$

The term $d\phi/ds$ in Eq. (c) is the rate of change of the inclination angle with respect to the change in distance, s , along the path and can be written as

$$\frac{d\phi}{ds} = \kappa = \frac{1}{\rho}, \quad (g)$$

where κ is called the *curvature* of the path and, as indicated in Eq. (g), it is the reciprocal of the *radius of curvature* of the path.

Substituting Eqs. (f) and (g) into Eq. (c), and rearranging, gives

$$\dot{s} \hat{u}^t = \frac{\dot{s}^2}{\rho} \hat{u}^n. \quad (h)$$

Finally, substituting Eq. (h) into Eq. (b), the acceleration of point P can now be written as

$$\mathbf{A}_P = \frac{\dot{s}^2}{\rho} \hat{u}^n + \ddot{s} \hat{u}^t. \quad (4.2)$$

The important observation from Eq. (4.2) is that, in general, the acceleration vector of a point has two perpendicular components: a normal component of magnitude $\dot{s}^2/\rho$ (called

normal since it is oriented along the $\hat{\mathbf{u}}^n$ axis),* and a tangential component of magnitude $\ddot{s}$ along the $\hat{\mathbf{u}}^t$ axis. Hence, the acceleration of point P can be written as

$$\mathbf{A}_P = \mathbf{A}_P^n + \mathbf{A}_P^t, \quad (4.3)$$

where the superscript n is for the normal direction and the superscript t is for the tangential direction. Note that the normal component of acceleration is commonly referred to as *centripetal acceleration*.

4.2 ANGULAR ACCELERATION

The previous section demonstrated that the acceleration of a point is a vector quantity having a magnitude and a direction. But a point has no dimensions (Sec. 2.1), so we cannot speak of the angular acceleration of a point. Rather, *rectilinear acceleration*, or simply “acceleration,” deals with the motion of a point, whereas *angular acceleration* deals with the motion of a rigid body.

Suppose a rigid body has an angular velocity $\boldsymbol{\omega}$ at one instant in time, and an instant later it has an angular velocity $\boldsymbol{\omega}'$. The difference,

$$\Delta\boldsymbol{\omega} = \boldsymbol{\omega}' - \boldsymbol{\omega}, \quad (a)$$

is also a vector quantity. The angular velocities $\boldsymbol{\omega}$ and $\boldsymbol{\omega}'$ may have different magnitudes as well as different directions. Thus, we define *angular acceleration* as the time rate of change of the angular velocity of a rigid body and designate it by the symbol $\boldsymbol{\alpha}$, that is,

$$\boldsymbol{\alpha} = \lim_{\Delta t \rightarrow 0} \left(\frac{\Delta\boldsymbol{\omega}}{\Delta t} \right) = \frac{d\boldsymbol{\omega}}{dt} = \dot{\boldsymbol{\omega}}. \quad (4.4)$$

As is the case with $\Delta\boldsymbol{\omega}$, there is no reason to believe that $\boldsymbol{\alpha}$ has the same direction as either $\boldsymbol{\omega}$ or $\boldsymbol{\omega}'$; it may have an entirely new direction.

We further note that the angular acceleration vector, $\boldsymbol{\alpha}$, applies to the absolute rotation of the entire rigid body and hence may be subscripted by the number of the coordinate system of the rigid body (for example, $\boldsymbol{\alpha}_2$ or $\boldsymbol{\alpha}_{2/1}$).

4.3 ACCELERATION DIFFERENCE BETWEEN POINTS OF A RIGID BODY

In Sec. 3.3, we determined the velocity difference between two points of a rigid body moving with both translation and rotation. Also, we showed that the velocity of a point in a rigid body can be obtained as the sum of the velocity of any reference point of the body and a term, called the *velocity difference*, caused by the angular velocity of the body. Thus, the velocity of any point, P , in a rigid body can be obtained from the velocity-difference equation, Eq. (3.4); that is,

$$\mathbf{V}_P = \mathbf{V}_Q + \mathbf{V}_{PQ}, \quad (a)$$

* The reader should verify that the normal component of acceleration is always directed toward the center of curvature of the path, no matter which orientations are positive for $\hat{\mathbf{u}}^t$ or $\hat{\mathbf{u}}^n$.

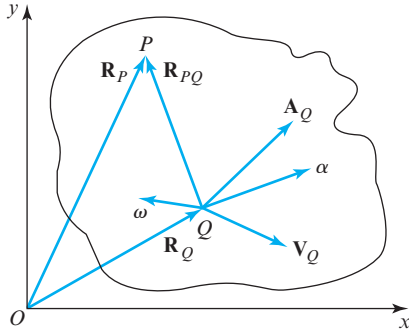


Figure 4.3

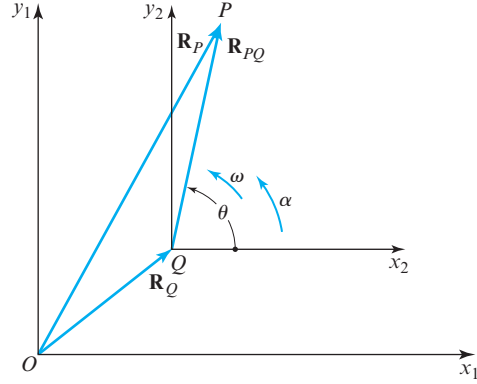


Figure 4.4

where $\mathbf{V}_Q$ is the velocity of the reference point, Q , and $\mathbf{V}_{PQ}$ is the velocity difference and is given by

$$\mathbf{V}_{PQ} = \boldsymbol{\omega} \times \mathbf{R}_{PQ}. \quad (b)$$

Here, $\boldsymbol{\omega}$ is the angular velocity of the body and $\mathbf{R}_{PQ}$ is the position-difference vector that defines the position of point P with respect to point Q . Substituting Eq. (b) into Eq. (a) gives

$$\mathbf{V}_P = \mathbf{V}_Q + \boldsymbol{\omega} \times \mathbf{R}_{PQ}. \quad (c)$$

We employ a similar nomenclature in Fig. 4.3 and specify that a reference point, Q , of the rigid body has an acceleration $\mathbf{A}_Q$ and that the body has an angular acceleration $\boldsymbol{\alpha}$ in addition to the angular velocity $\boldsymbol{\omega}$. Note that, in general, $\boldsymbol{\alpha}$ need not have the same orientation as $\boldsymbol{\omega}$.

The acceleration of point P is obtained by differentiating Eq. (c) with respect to time; that is

$$\dot{\mathbf{V}}_P = \dot{\mathbf{V}}_Q + \boldsymbol{\omega} \times \dot{\mathbf{R}}_{PQ} + \dot{\boldsymbol{\omega}} \times \mathbf{R}_{PQ}. \quad (d)$$

Since we know that $\dot{\mathbf{V}}_P = \mathbf{A}_P$, $\dot{\mathbf{V}}_Q = \mathbf{A}_Q$, $\dot{\mathbf{R}}_{PQ} = \boldsymbol{\omega} \times \mathbf{R}_{PQ}$, and $\dot{\boldsymbol{\omega}} = \boldsymbol{\alpha}$, then the acceleration of point P can be written from Eq. (d) as

$$\mathbf{A}_P = \mathbf{A}_Q + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{R}_{PQ}) + \boldsymbol{\alpha} \times \mathbf{R}_{PQ}, \quad (4.5)$$

where the first term, $\mathbf{A}_Q$, is the acceleration of the reference point, Q , and the remaining two terms are caused by the rotation of the body. To visualize the directions of these two terms, let us first study them for a two-dimensional application.

In Fig. 4.4, let P and Q be two points of a rigid body that has a combination of translation and rotation with respect to the ground reference plane, x_1y_1 . We also define a moving system, x_2y_2 , with origin at Q , but restrict this system to only translation; thus,

x_2 must remain parallel to x_1 . As given quantities, we specify the velocity and acceleration of reference point Q ; we also specify the angular velocity, ω , and the angular acceleration, α , of the rigid body. These angular rates can be treated as scalars, since the corresponding vectors always have axes perpendicular to the plane of the motion. They may have different senses, however, and the scalar quantities may be either positive or negative.

The location of point P can be specified by the position-difference equation,

$$\mathbf{R}_P = \mathbf{R}_Q + \mathbf{R}_{PQ}. \quad (e)$$

Equation (a) can also be written in two-dimensional form as

$$\mathbf{R}_P = \mathbf{R}_Q + R_{PQ} \angle \theta \quad (f)$$

or in complex polar form as

$$\mathbf{R}_P = \mathbf{R}_Q + R_{PQ} e^{j\theta}. \quad (g)$$

Differentiating Eq. (g) with respect to time gives the velocity of point P ; that is,

$$\mathbf{V}_P = \mathbf{V}_Q + j\omega R_{PQ} e^{j\theta}. \quad (h)$$

Note that this is the complex polar form of Eq. (c) for planar motion, where the second term on the right-hand side corresponds to the velocity-difference vector, $\mathbf{V}_{PQ}$. The magnitude of this term is ωR_{PQ} and the direction is $je^{j\theta}$, which is perpendicular to $\mathbf{R}_{PQ}$, rotated in the sense of ω , as shown in Fig. 4.5.

Differentiating Eq. (h) with respect to time gives the acceleration of point P ; that is,

$$\mathbf{A}_P = \mathbf{A}_Q - \omega^2 R_{PQ} e^{j\theta} + j\alpha R_{PQ} e^{j\theta}. \quad (i)$$

The second and third terms on the right-hand side of this equation correspond exactly with the second and third terms of Eq. (4.5). The second term is called the *normal* or *centripetal* component of acceleration. For planar motion, the magnitude of this

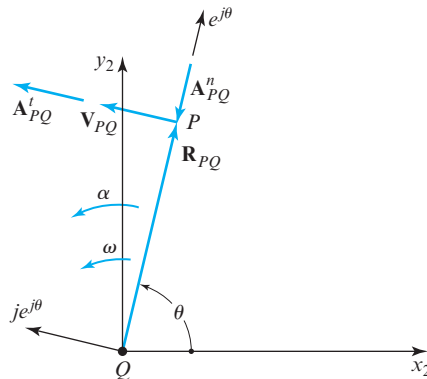


Figure 4.5

$V_{PQ} = \omega R_{PQ}$, $A_{PQ}^n = \omega^2 R_{PQ}$, and $A_{PQ}^t = \alpha R_{PQ}$.

component is

$$\omega^2 R_{PQ} = \frac{V_{PQ}^2}{R_{PQ}},$$

and the direction $-e^{j\theta}$ is opposite to the position-difference vector, $\mathbf{R}_{PQ}$. This is the normal component of the acceleration difference and we designate its magnitude

$$A_{PQ}^n = \omega^2 R_{PQ} = \frac{V_{PQ}^2}{R_{PQ}}. \quad (4.6)$$

The third term of Eq. (i) is associated with the angular acceleration of the body. The magnitude is αR_{PQ} and the direction is $je^{j\theta}$, which is along the same line as the velocity difference, $\mathbf{V}_{PQ}$. Note that P traces out a circular arc in its motion about translating reference point Q . Since the third term is perpendicular to the position-difference vector $\mathbf{R}_{PQ}$ and hence tangent to the circular arc, it is convenient to call it the *tangential* component of the acceleration difference and designate its magnitude

$$A_{PQ}^t = \alpha R_{PQ}. \quad (4.7)$$

The normal and tangential components of the acceleration difference—that is, Eqs. (4.6) and (4.7)—are shown in Fig. 4.5 for two-dimensional motion.

Let us now examine the final two terms of Eq. (4.5) again, but this time for three-dimensional applications. The direction of the normal component of acceleration; that is,

$$\mathbf{A}_{PQ}^n = \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{R}_{PQ}), \quad (4.8)$$

is shown in Fig. 4.6. This component is in the plane containing $\boldsymbol{\omega}$ and $\mathbf{R}_{PQ}$ and is perpendicular to $\boldsymbol{\omega}$. The magnitude is

$$|\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{R}_{PQ})| = \omega^2 R_{PQ} \sin \phi = \frac{V_{PQ}^2}{R_{PQ} \sin \phi},$$

where $R_{PQ} \sin \phi$ is the radius of the circle in Fig. 4.6.

According to the definition of the vector cross-product, the tangential component of acceleration,

$$\mathbf{A}_{PQ}^t = \boldsymbol{\alpha} \times \mathbf{R}_{PQ}, \quad (4.9)$$

is perpendicular to the plane containing $\boldsymbol{\alpha}$ and $\mathbf{R}_{PQ}$ with a sense indicated by the right-hand rule. Because of the angular acceleration, $\boldsymbol{\alpha}$, we visualize point P accelerating around a circle, as shown in Fig. 4.7; the plane of this circle is normal to the plane containing $\boldsymbol{\alpha}$ and $\mathbf{R}_{PQ}$. Using the definition of the vector cross-product, the magnitude of this tangential component of acceleration can be written as

$$|\boldsymbol{\alpha} \times \mathbf{R}_{PQ}| = \alpha R_{PQ} \sin \theta,$$

where $R_{PQ} \sin \theta$ is the radius of the circle in Fig. 4.7.

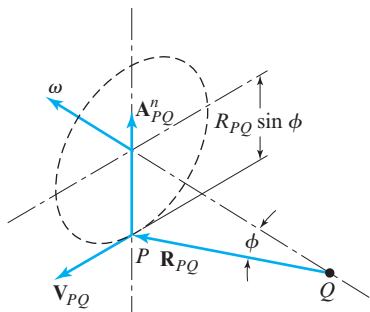


Figure 4.6 $\mathbf{V}_{PQ} = \boldsymbol{\omega} \times \mathbf{R}_{PQ}$
and $\mathbf{A}_{PQ}^n = \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{R}_{PQ})$.

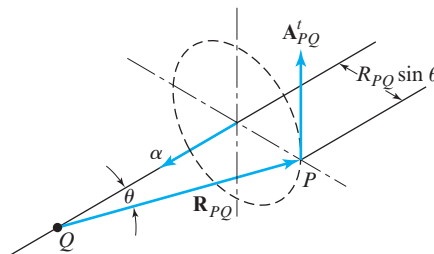


Figure 4.7 $\mathbf{A}_{PQ}^t = \boldsymbol{\alpha} \times \mathbf{R}_{PQ}$.

Again, we emphasize that, in general, $\boldsymbol{\alpha}$ and $\boldsymbol{\omega}$ may have different directions in three-dimensional applications.

Let us now summarize the results of this section. The acceleration of a point fixed in a rigid body can be obtained from the sum of three terms. The first term is the acceleration of a reference point (Q in Fig. 4.3), which, of course, depends on the motion of the particular point selected. The remaining two terms are due to the rotation of the body. One of these is the normal component and depends on the angular velocity of the body; the other is the tangential component and depends on the time rate of change of the angular velocity.

Equation (4.5) can also be written as

$$\mathbf{A}_P = \mathbf{A}_Q + \mathbf{A}_{PQ}, \quad (4.10)$$

which is called the *acceleration-difference equation*. It may be convenient to designate the components of the acceleration-difference term as

$$\mathbf{A}_{PQ} = \mathbf{A}_{PQ}^n + \mathbf{A}_{PQ}^t.$$

Then, the acceleration-difference equation can be written as

$$\mathbf{A}_P = \mathbf{A}_Q + \mathbf{A}_{PQ}^n + \mathbf{A}_{PQ}^t. \quad (4.11)$$

Acceleration problems can be solved using the acceleration-difference equation in a manner similar to our use of the velocity-difference equation, Eq. (3.4).

The following example illustrates both the graphic and analytic methods of acceleration analysis.

EXAMPLE 4.1

For the four-bar linkage in the posture shown in Fig. 4.8a, the constant input angular velocity is $\omega_2 = 200$ rad/s ccw. Determine the accelerations of points A and B , and the angular accelerations of links 3 and 4.

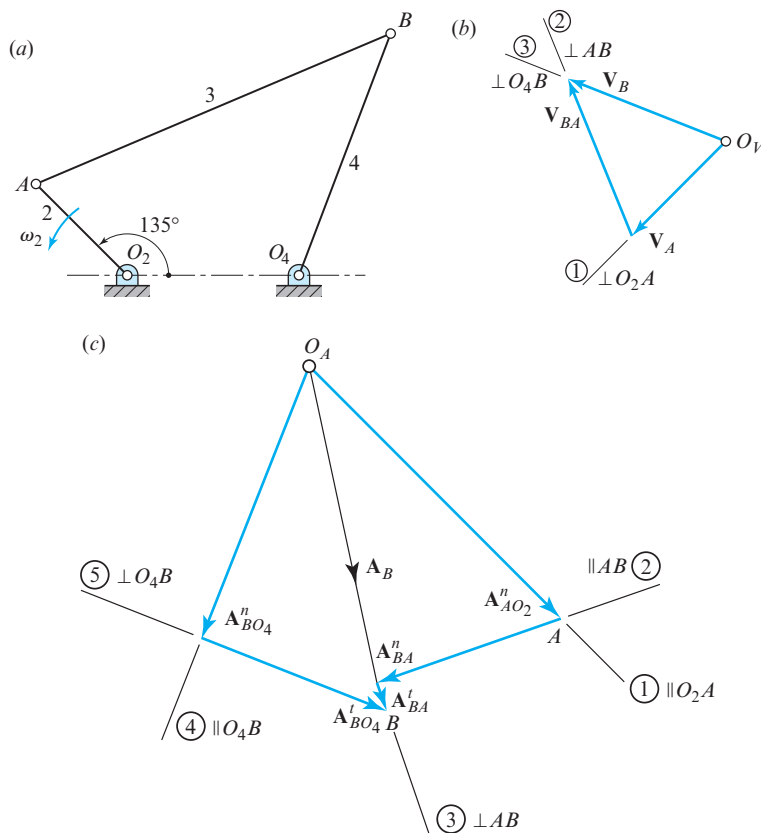


Figure 4.8 (a) $R_{O_4O_2} = 8$ in, $R_{AO_2} = 6$ in, $R_{BA} = 18$ in, and $R_{BO_4} = 12$ in; (b) velocity polygon; (c) acceleration polygon.

GRAPHIC SOLUTION

The velocity polygon must be drawn first, since the velocities of points A and B , and the angular velocities of links 3 and 4 are required for the acceleration analysis.

The magnitude of the velocity of point A is

$$V_A = \omega_2 R_{AO_2} = (200 \text{ rad/s})(6/12 \text{ ft}) = 100 \text{ ft/s.}$$

Using this value, the velocity polygon can now be drawn as shown in Fig. 4.8*b*. Then, from the polygon, we measure the magnitudes of the velocity difference of points B and A , and the velocity of point B , respectively, as

$$V_{BA} = 128 \text{ ft/s and } V_B = 129 \text{ ft/s.}$$

From the acceleration polygon, the two unknown magnitudes are measured as

$$A_{BO_4}^t = 11\,900 \text{ ft/s}^2 \text{ and } A_{BA}^t = 2\,000 \text{ ft/s}^2.$$

The line from point O_A to point B in the polygon is the magnitude of the acceleration of point B and is measured as

$$A_B = 20\,500 \text{ ft/s}^2. \quad \text{Ans.}$$

The angular accelerations of links 3 and 4 are then computed as follows:

$$\alpha_3 = \frac{A_{BA}^t}{R_{BA}} = \frac{2\,000 \text{ ft/s}^2}{(18/12) \text{ ft}} = 1\,333 \text{ rad/s}^2 \text{ cw}, \quad \text{Ans.}$$

$$\alpha_4 = \frac{A_{BO_4}^t}{R_{BO_4}} = \frac{11\,900 \text{ ft/s}^2}{(12/12) \text{ ft}} = 11\,900 \text{ rad/s}^2 \text{ cw}, \quad \text{Ans.}$$

where the directions are obtained from an examination of the tangential components of the acceleration polygon.

ANALYTIC SOLUTION

The first two steps are to perform a posture analysis and a velocity analysis of the linkage. Since these steps were presented in Chaps. 2 and 3, only the results are presented here.

The results of the posture analysis are shown in Fig. 4.9. In vector form, the position-difference vectors corresponding to the links are

$$\mathbf{R}_{AO_2} = \left(\frac{6}{12}\right) \text{ ft } \angle 135^\circ = -0.353\,55\hat{\mathbf{i}} + 0.353\,55\hat{\mathbf{j}} \text{ ft},$$

$$\mathbf{R}_{BA} = \left(\frac{18}{12}\right) \text{ ft } \angle 22.4^\circ = 1.386\,82\hat{\mathbf{i}} + 0.571\,61\hat{\mathbf{j}} \text{ ft},$$

$$\mathbf{R}_{BO_4} = \left(\frac{12}{12}\right) \text{ ft } \angle 68.4^\circ = 0.368\,12\hat{\mathbf{i}} + 0.929\,78\hat{\mathbf{j}} \text{ ft}.$$

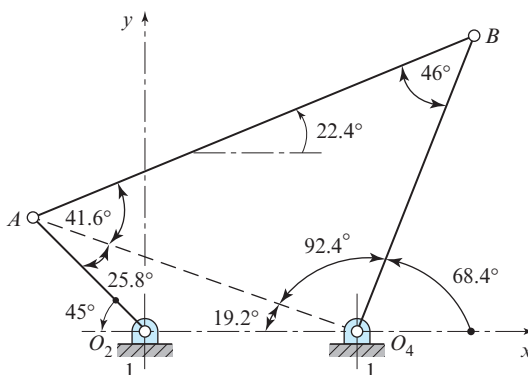


Figure 4.9

For the given input angular velocity $\omega_2 = 200\hat{\mathbf{k}}$ rad/s, the results of the velocity analysis are

$$\omega_3 = 84.253\hat{\mathbf{k}} \text{ rad/s and } \omega_4 = 129.39\hat{\mathbf{k}} \text{ rad/s.}$$

Next, the known acceleration components are

$$\mathbf{A}_{AO_2}^n = \omega_2 \times (\omega_2 \times \mathbf{R}_{AO_2}) = 14\,142\hat{\mathbf{i}} - 14\,142\hat{\mathbf{j}} \text{ ft/s}^2 = \mathbf{A}_A, \quad \text{Ans. (2)}$$

$$\mathbf{A}_{BA}^n = \omega_3 \times (\omega_3 \times \mathbf{R}_{BA}) = -9\,844\hat{\mathbf{i}} - 4\,058\hat{\mathbf{j}} \text{ ft/s}^2, \quad (3)$$

$$\mathbf{A}_{BO_4}^n = \omega_4 \times (\omega_4 \times \mathbf{R}_{BO_4}) = -6\,163\hat{\mathbf{i}} - 15\,567\hat{\mathbf{j}} \text{ ft/s}^2. \quad (4)$$

Although the angular accelerations α_3 and α_4 have unknown magnitudes, we can incorporate them into the solution in the following manner:

$$\mathbf{A}_{BA}^t = \alpha_3 \times \mathbf{R}_{BA} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 0 & 0 & \alpha_3 \\ 1.386\,82 \text{ ft} & 0.571\,61 \text{ ft} & 0 \end{vmatrix} = -0.571\,61\alpha_3\hat{\mathbf{i}} + 1.386\,82\alpha_3\hat{\mathbf{j}} \text{ ft}, \quad (5)$$

$$\mathbf{A}_{BO_4}^t = \alpha_4 \times \mathbf{R}_{BO_4} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 0 & 0 & \alpha_4 \\ 0.368\,12 \text{ ft} & 0.929\,78 \text{ ft} & 0 \end{vmatrix} = -0.929\,78\alpha_4\hat{\mathbf{i}} + 0.368\,12\alpha_4\hat{\mathbf{j}} \text{ ft}. \quad (6)$$

Writing the acceleration-difference equation for point B and noting that $\mathbf{A}_{AO_2}^t = \mathbf{0}$ gives

$$\mathbf{A}_{BO_4}^n + \mathbf{A}_{BO_4}^t = \mathbf{A}_{AO_2}^n + \mathbf{A}_{BA}^n + \mathbf{A}_{BA}^t. \quad (7)$$

Then, substituting Eqs. (2) through (6) into Eq. (7) and separating the $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ components, we obtain the following pair of simultaneous equations:

$$(0.571\,61 \text{ ft})\alpha_3 - (0.929\,78 \text{ ft})\alpha_4 = 10\,461 \text{ ft/s}^2, \quad (8)$$

$$-(1.386\,82 \text{ ft})\alpha_3 + (0.368\,12 \text{ ft})\alpha_4 = -2\,633 \text{ ft/s}^2. \quad (9)$$

Solving these equations, the angular accelerations of links 3 and 4, respectively, are

$$\alpha_3 = -1\,300\hat{\mathbf{k}} \text{ rad/s}^2 \quad \text{and} \quad \alpha_4 = -12\,050\hat{\mathbf{k}} \text{ rad/s}^2, \quad \text{Ans. (10)}$$

where the two negative signs indicate that both angular accelerations are clockwise.

The acceleration of point B can now be written from Eq. (4.11) as

$$\mathbf{A}_B = \mathbf{A}_{BO_4}^n + \mathbf{A}_{BO_4}^t. \quad (11)$$

The procedure is to substitute Eq. (10) into Eq. (6), which gives

$$\mathbf{A}_{BO_4}^t = 11\,204\hat{\mathbf{i}} - 4\,436\hat{\mathbf{j}} \text{ ft/s}^2. \quad (12)$$

Then, substituting Eqs. (4) and (12) into Eq. (11), the acceleration of point B is

$$\mathbf{A}_B = 5\,041\hat{\mathbf{i}} - 20\,003\hat{\mathbf{j}} \text{ ft/s}^2 = 20\,628 \text{ ft/s}^2 \angle -75.86^\circ. \quad \text{Ans.}$$

Note that the four answers from the analytic method are in good agreement with, but more accurate than, the results of the graphic method.

4.4 ACCELERATION POLYGONS; ACCELERATION IMAGES

The acceleration image of a link in an acceleration polygon can be obtained in much the same manner as the velocity image of a link in a velocity polygon (Sec. 3.4).

According to Rosenauer and Willis [7], the theorem of Mehmke can be written as:

The end points of the acceleration vectors of the points of a plane rigid body, when plotted from a common origin, produce a figure which is geometrically similar to the original figure (image diagram).

It is this theorem that results in the acceleration image presented in this section. Also, it is a result of this theorem that allows clarity in the acceleration polygon despite the very minimal labeling that is required. This becomes evident in the following examples and continues throughout the text.

EXAMPLE 4.2

For the four-bar linkage of Example 3.1, in the posture shown in Fig. 4.10, the angular velocity of crank 2 is a constant $900 \text{ rev/min} = 94.25 \text{ rad/s}$ ccw. Determine the angular accelerations of links 3 and 4, and the accelerations of points E and F .

SOLUTION

First, we consider the accelerations of points B and C . Since the angular acceleration of link 2 is zero, point B has only a normal component of acceleration; namely,

$$A_B = A_{BA}^n = \omega_2^2 R_{BA} = (94.25 \text{ rad/s})^2 (4/12 \text{ ft}) = 2\,961 \text{ ft/s}^2.$$

Now, we choose a convenient scale and an acceleration origin, O_A . Note that points A and D have zero acceleration, and therefore their images are coincident with the acceleration

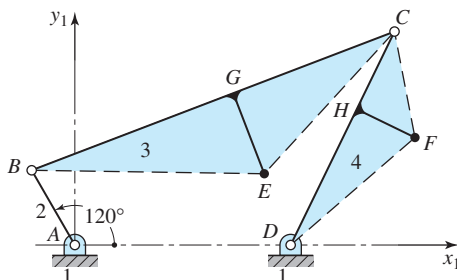


Figure 4.10

$R_{BA} = 4 \text{ in.}$, $R_{CB} = 18 \text{ in.}$,
 $R_{CD} = 11 \text{ in.}$, $R_{DA} = 10 \text{ in.}$,
 $R_{GB} = 10 \text{ in.}$, $R_{EG} = 4 \text{ in.}$,
 $R_{HD} = 7 \text{ in.}$, and $R_{FH} = 3 \text{ in.}$

origin, O_A . Then, we draw the acceleration vector, $\mathbf{A}_B$ (opposite in direction to the vector $\mathbf{R}_{BA}$), to locate the acceleration image point B , as shown in Fig. 4.11. Next, we write Eq. (4.11) to relate the acceleration of point C to those of points B and D in the acceleration polygon; that is,

$$\mathbf{A}_C = \mathbf{A}_B + \mathbf{A}_{CB}^n + \mathbf{A}_{CB}^t = \mathbf{A}_{CD}^n + \mathbf{A}_{CD}^t. \quad (1)$$

Using the posture and velocity results obtained from Example 3.1 (see Fig. 3.7b), we calculate the magnitudes of the two normal components of Eqs. (1); that is,

$$A_{CB}^n = \frac{V_{CB}^2}{R_{CB}} = \frac{(38.4 \text{ ft/s})^2}{(18/12) \text{ ft}} = 983 \text{ ft/s}^2,$$

$$A_{CD}^n = \frac{V_{CD}^2}{R_{CD}} = \frac{(45.5 \text{ ft/s})^2}{(18/12) \text{ ft}} = 2\,258 \text{ ft/s}^2.$$

These two normal components are constructed with directions opposite to the vectors $\mathbf{R}_{CB}$ and $\mathbf{R}_{CD}$, respectively. As required by Eqs. (1), they are added to the acceleration polygon originating from acceleration image points B and D , respectively, as indicated by two dashed lines in Fig. 4.11. Perpendicular dashed lines are then drawn through the termini of these two normal components, representing the addition of the two tangential components, $\mathbf{A}_{CB}^t$ and $\mathbf{A}_{CD}^t$. The point of intersection of the two tangential components is labeled the acceleration image point, C .

The angular accelerations of links 3 and 4 are now obtained from measurements of the two tangential components:

$$\alpha_3 = \frac{A_{CB}^t}{R_{CB}} = \frac{170 \text{ ft/s}^2}{(18/12) \text{ ft}} = 113 \text{ rad/s}^2 \text{ ccw}, \quad \text{Ans.}$$

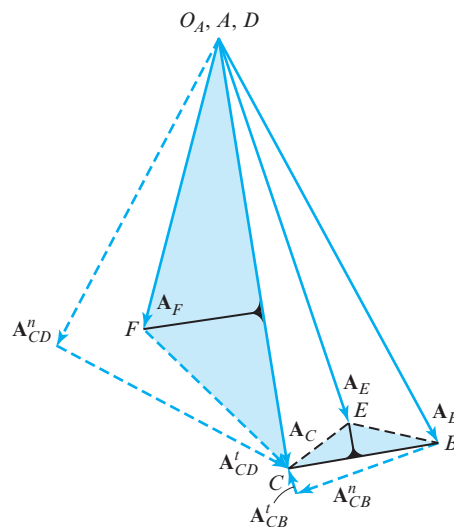


Figure 4.11 Acceleration polygon.

$$\alpha_4 = \frac{A_{CD}^t}{R_{CD}} = \frac{1\,670 \text{ ft/s}^2}{(11/12) \text{ ft}} = 1\,822 \text{ rad/s}^2 \text{ cw.} \quad \text{Ans.}$$

There are several methods to determine the acceleration of point E of link 3. One method is to write two acceleration-difference equations, starting from points B and C , which are also in link 3; that is,

$$\mathbf{A}_E = \mathbf{A}_B + \mathbf{A}_{EB}^n + \mathbf{A}_{EB}^t = \mathbf{A}_C + \mathbf{A}_{EC}^n + \mathbf{A}_{EC}^t. \quad (2)$$

The solution of these two equations can be obtained from the same methods used for Eq. (1), if desired. A second method to determine the acceleration of point E is to use the now known angular acceleration, α_3 , to calculate one or both of the tangential components in Eq. (2).

A third method and probably the easiest (and the one used in Fig. 4.11) is to form an acceleration image triangle, BCE , for link 3. This image triangle is formed by using the line of the acceleration-difference vector, $\mathbf{A}_{CB}$, as a base line, then set a scale for triangle BCE of link 3.* Any of the three methods provides the location of acceleration image point E . The magnitude of the acceleration of point E is measured as

$$A_E = 2\,580 \text{ ft/s}^2. \quad \text{Ans.}$$

Similarly, the magnitude of the acceleration of point F is found by using acceleration image triangle DCF and is measured as

$$A_F = 1\,960 \text{ ft/s}^2. \quad \text{Ans.}$$

EXAMPLE 4.3

Find the acceleration images corresponding to the links of the slider-crank linkage in the posture shown in Fig. 4.12a. The crank (link 2) is rotating counterclockwise with constant angular velocity $\omega_2 = 1 \text{ rad/s}$.

SOLUTION

In Fig. 4.12a, links 2 and 3 are represented by triangles O_2DA and ABC , respectively. Note that the velocity and acceleration polygons are found in exactly the same manner as in the solution of the previous example, and the results are shown in Figs. 4.12b and 4.12c, respectively. Note also that each acceleration image is formed from the *total acceleration-difference* vectors, not from the component vectors.

Since the angular acceleration of the crank is zero ($\alpha_2 = 0$), the corresponding acceleration image is rotated 180° from the orientation of the crank. Note that link 3 has

* We must be careful that the shape of the acceleration image is not “flipped over” with respect to the original shape. A convenient test is to notice that, for link 3 of this example, because the labels BCE appear in clockwise order for the original link shape, they still appear in clockwise order in the velocity and the acceleration images.

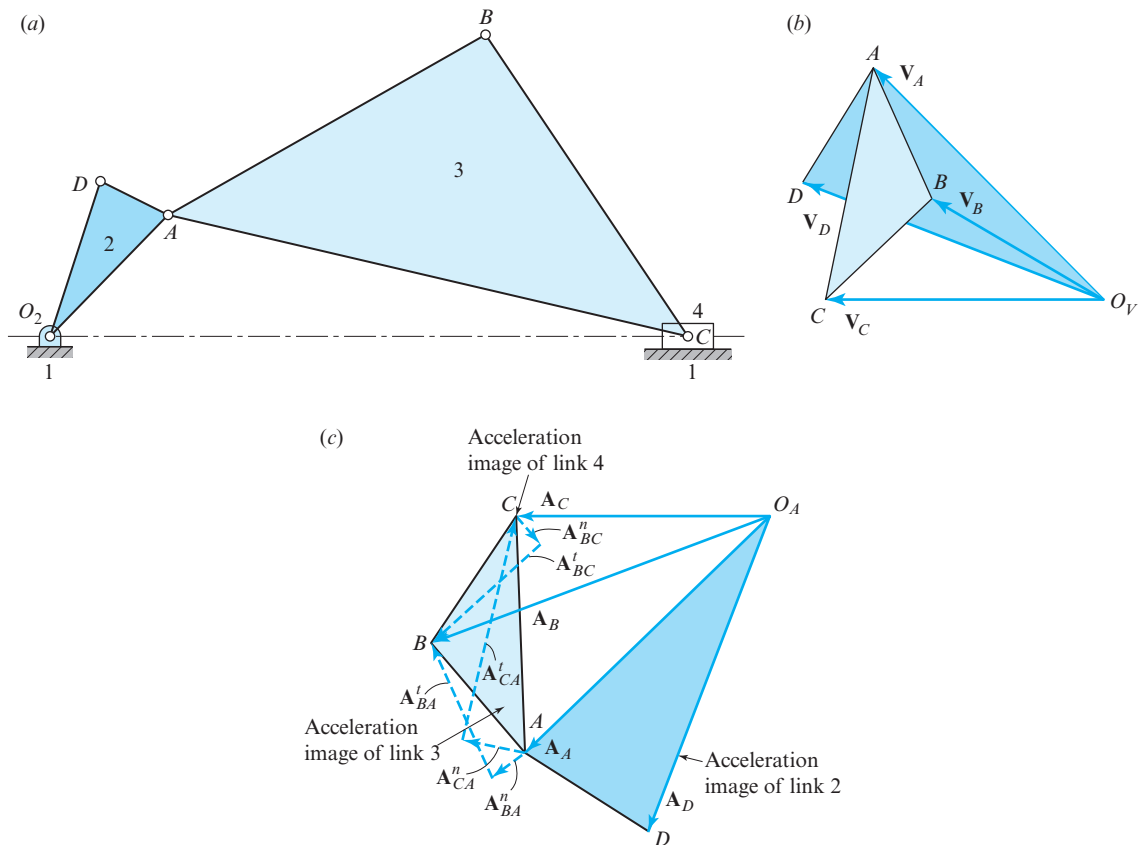


Figure 4.12 (a) Slider-crank linkage; (b) velocity polygon; (c) acceleration polygon.

counterclockwise angular acceleration and that its acceleration image is oriented much less than 180° from the orientation of the link. Thus, the orientation of each acceleration image depends on both the angular velocity and the angular acceleration of the link under consideration.

Summary The following are important properties of acceleration images:

1. The acceleration image of a link in the acceleration polygon is a scale reproduction of the shape of the link itself.
2. The letters identifying the vertices in the acceleration polygon are the same as those of the corresponding link, and they progress around the acceleration image in the same order and in the same angular sense as around the link itself.
3. The acceleration origin, O_A , in the acceleration polygon is the image of all points with zero absolute acceleration. It is the acceleration image of the fixed link.

4. The absolute acceleration of any point on any link is represented by the line from O_A to the image of the point in the acceleration polygon. The acceleration difference between two points in the same link, say B and C , is represented by the line to acceleration image point B from acceleration image point C .
5. For a nonrotating (translating) link, the accelerations of all points on the link are equal, and the angular velocity and angular acceleration of the link are both zero. Therefore, the acceleration image of a link that is translating shrinks to a single point in the acceleration polygon.
6. The orientation of the acceleration image of link j is given by

$$\delta_j = 180^\circ - \tan^{-1} \frac{\alpha_j}{\omega_j^2}, \quad (4.12)$$

where δ_j is the angle in degrees, measured in the (positive) counterclockwise direction from the orientation of link j to its acceleration image.

4.5 APPARENT ACCELERATION OF A POINT IN A MOVING COORDINATE SYSTEM

In Sec. 3.5, we found it helpful to develop the apparent-velocity equation for situations where it is convenient to describe the path along which a point moves relative to another moving link, but where it is not convenient to describe the absolute motion of the same point. Let us now investigate the acceleration of such a point.

Figure 4.13 illustrates the point, P , of link 3 (denoted as P_3) that moves along a known path (sometimes referred to as the “slot”) relative to the moving reference frame $x_2y_2z_2$. Recall from Chaps. 2 and 3 that point P_2 is point P fixed in moving link 2 and is instantaneously coincident with point P_3 . The problem now is to find an equation relating the accelerations of points P_3 and P_2 in terms of parameters that can either be calculated or measured in a typical mechanism.

In Fig. 4.14, we recall how the same situation would be perceived by a moving observer attached to link 2. To HER, the path of P_3 (the “slot”) would appear stationary, and point P_3 would appear to move along the tangent to the path with the apparent velocity

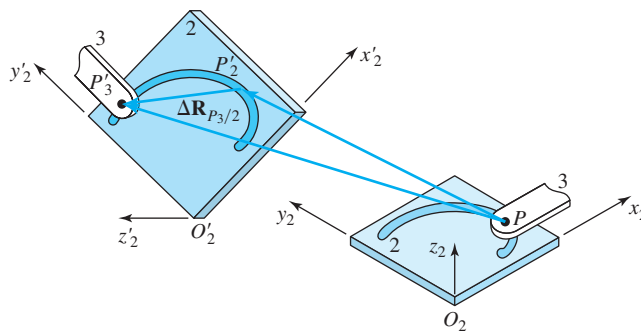


Figure 4.13 Apparent displacement.

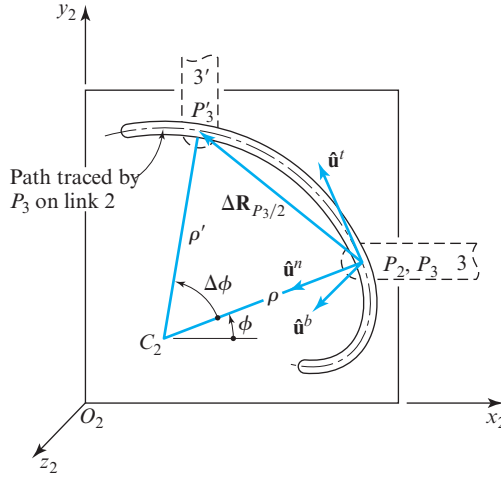


Figure 4.14 Apparent displacement of point P_3 as seen by an observer on link 2.

$\mathbf{V}_{P_3/2}$. In Sec. 3.5 (Fig. 3.12), we define the moving coordinate system $\hat{\mathbf{u}}^t \hat{\mathbf{u}}^n \hat{\mathbf{u}}^b$, where $\hat{\mathbf{u}}^t$ is the unit tangent vector to the path of point P in the direction of positive movement, and $\hat{\mathbf{u}}^b$ is normal to the plane containing $\hat{\mathbf{u}}^t$ and the center of curvature C , and is directed positive to the preferred side of the plane. A third unit vector, $\hat{\mathbf{u}}^n$, is obtained from Eq. (3.7), $\hat{\mathbf{u}}^n = \hat{\mathbf{u}}^b \times \hat{\mathbf{u}}^t$, thus completing a right-handed Cartesian coordinate system. Also, we derive the apparent velocity equation, Eq. (3.9), as

$$\mathbf{V}_{P_3/2} = \frac{ds}{dt} \hat{\mathbf{u}}^t, \quad (a)$$

where the scalar, s , is the arc distance along the path measuring the travel of P_3 .

Now, the radius of curvature, ρ , sweeps through some small angle, $\Delta\phi$, as P_3 travels the small arc distance Δs , as seen by the moving observer ($P_3 P_3'$, Fig. 4.14) during a short time interval, Δt . The changes of the angle and the arc distance are related by

$$\Delta\phi = \frac{\Delta s}{\rho}. \quad (b)$$

Note here that the center of curvature C can lie along either the positive or the negative extension of $\hat{\mathbf{u}}^n$. Therefore, the radius of curvature ρ , measured from P to C , can have either a positive value or a negative value according to the sense of $\hat{\mathbf{u}}^n$. Also, this implies that the angle $\Delta\phi$ is positive when counterclockwise, as seen from the positive $\hat{\mathbf{u}}^b$.

Dividing Eq. (b) by Δt and taking the limit for infinitesimally small Δt , we find with the aid of Eq. (a),

$$\frac{d\phi}{dt} = \frac{1}{\rho} \frac{ds}{dt} = \frac{V_{P_3/2}}{\rho}. \quad (c)$$

This is the angular rate at which the radius of curvature ρ (and also the unit vectors $\hat{\mathbf{u}}^t$ and $\hat{\mathbf{u}}^n$) appears to rotate as seen by an observer in coordinate system 2 as point P_3

moves along its path. We can give this rotation rate its proper vector properties as an apparent angular velocity by noting that the axis of this rotation is parallel to $\hat{\mathbf{u}}^b$. Thus, we define the apparent angular velocity vector as

$$\dot{\boldsymbol{\phi}} = \frac{d\phi}{dt} \hat{\mathbf{u}}^b = \frac{V_{P3/2}}{\rho} \hat{\mathbf{u}}^b. \quad (d)$$

Next, we seek the time derivative of unit tangent vector $\hat{\mathbf{u}}^t$ so that we can differentiate Eq. (a) to obtain the apparent acceleration. Since $\hat{\mathbf{u}}^t$ is a unit vector, its length does not change; however, it does have a derivative because of its change in direction: its rotation. In the absolute coordinate system, $\hat{\mathbf{u}}^t$ is subject to the apparent angular velocity, $\dot{\boldsymbol{\phi}}$, and also to the angular velocity, $\boldsymbol{\omega}_2$, with which the moving coordinate system 2 is rotating. Therefore, the time derivative of $\hat{\mathbf{u}}^t$ can be written as

$$\frac{d\hat{\mathbf{u}}^t}{dt} = (\boldsymbol{\omega}_2 + \dot{\boldsymbol{\phi}}) \times \hat{\mathbf{u}}^t = \boldsymbol{\omega}_2 \times \hat{\mathbf{u}}^t + \dot{\boldsymbol{\phi}} \times \hat{\mathbf{u}}^t. \quad (e)$$

Substituting Eq. (d) into this equation gives

$$\frac{d\hat{\mathbf{u}}^t}{dt} = \boldsymbol{\omega}_2 \times \hat{\mathbf{u}}^t + \frac{V_{P3/2}}{\rho} \hat{\mathbf{u}}^b \times \hat{\mathbf{u}}^t = \boldsymbol{\omega}_2 \times \hat{\mathbf{u}}^t + \frac{V_{P3/2}}{\rho} \hat{\mathbf{u}}^n. \quad (f)$$

Similarly, the time derivative of $\hat{\mathbf{u}}^n$ can be written as

$$\begin{aligned} \frac{d\hat{\mathbf{u}}^n}{dt} &= (\boldsymbol{\omega}_2 + \dot{\boldsymbol{\phi}}) \times \hat{\mathbf{u}}^n = \boldsymbol{\omega}_2 \times \hat{\mathbf{u}}^n + \dot{\boldsymbol{\phi}} \times \hat{\mathbf{u}}^n \\ &= \boldsymbol{\omega}_2 \times \hat{\mathbf{u}}^n + \frac{V_{P3/2}}{\rho} \hat{\mathbf{u}}^b \times \hat{\mathbf{u}}^n = \boldsymbol{\omega}_2 \times \hat{\mathbf{u}}^n - \frac{V_{P3/2}}{\rho} \hat{\mathbf{u}}^t. \end{aligned} \quad (g)$$

Now, taking the time derivative of Eq. (a) and using Eq. (f), we find that

$$\frac{d\mathbf{V}_{P3/2}}{dt} = \frac{ds}{dt} \frac{d\hat{\mathbf{u}}^t}{dt} + \frac{d^2s}{dt^2} \hat{\mathbf{u}}^t = \frac{ds}{dt} \boldsymbol{\omega}_2 \times \hat{\mathbf{u}}^t + \frac{ds}{dt} \frac{V_{P3/2}}{\rho} \hat{\mathbf{u}}^n + \frac{d^2s}{dt^2} \hat{\mathbf{u}}^t.$$

Finally, using Eqs. (a) and (c), this equation reduces to

$$\frac{d\mathbf{V}_{P3/2}}{dt} = \boldsymbol{\omega}_2 \times \mathbf{V}_{P3/2} + \frac{V_{P3/2}^2}{\rho} \hat{\mathbf{u}}^n + \frac{d^2s}{dt^2} \hat{\mathbf{u}}^t. \quad (h)$$

Note that the three components on the right-hand side of Eq. (h) are *not* all defined as *apparent-acceleration* components. To be consistent, the apparent acceleration includes only those components *that are seen by an observer attached to the moving coordinate system*. Equation (h) is derived in the absolute coordinate system and includes the rotation effect of $\boldsymbol{\omega}_2$ that is not sensed by the moving observer. The apparent acceleration can easily be determined, however, by setting $\boldsymbol{\omega}_2$ to be zero in Eq. (h). The two remaining components define the apparent acceleration and can be written as

$$\mathbf{A}_{P3/2} = \mathbf{A}_{P3/2}^n + \mathbf{A}_{P3/2}^t, \quad (4.13)$$

where

$$\mathbf{A}_{P_3/2}^n = \frac{V_{P_3/2}^2}{\rho} \hat{\mathbf{u}}^n \quad (4.14)$$

is called the *normal component* of the apparent acceleration, indicating that it is always normal to the path and is always directed from point P toward the center of curvature (that is, in the $\hat{\mathbf{u}}^n$ direction when ρ is positive or in the $-\hat{\mathbf{u}}^n$ direction when ρ is negative), and

$$\mathbf{A}_{P_3/2}^t = \frac{d^2 s}{dt^2} \hat{\mathbf{u}}^t \quad (4.15)$$

is called the *tangential component*, indicating that it is always tangent to the path (along the $\hat{\mathbf{u}}^t$ direction, but it may be either positive or negative).

Next, from Fig. 4.14, we write the position of point P_3 as

$$\mathbf{R}_{P_3} = \mathbf{R}_{C_2} - \rho \hat{\mathbf{u}}^n.$$

With the help of Eq. (g), the time derivative of this equation can be written as*

$$\mathbf{V}_{P_3} = \mathbf{V}_{C_2} - \boldsymbol{\omega}_2 \times (\rho \hat{\mathbf{u}}^n) + \rho \frac{V_{P_3/2}}{\rho} \hat{\mathbf{u}}^t = \mathbf{V}_{C_2} - \boldsymbol{\omega}_2 \times (\rho \hat{\mathbf{u}}^n) + \mathbf{V}_{P_3/2}. \quad (i)$$

Then, differentiating this equation, again with respect to time, the acceleration of point P_3 can be written as

$$\mathbf{A}_{P_3} = \mathbf{A}_{C_2} - \boldsymbol{\alpha}_2 \times (\rho \hat{\mathbf{u}}^n) - \boldsymbol{\omega}_2 \times \frac{d(\rho \hat{\mathbf{u}}^n)}{dt} + \frac{d\mathbf{V}_{P_3/2}}{dt},$$

and, with the help of Eqs. (g) and (h), this becomes

$$\mathbf{A}_{P_3} = \mathbf{A}_{C_2} + \boldsymbol{\alpha}_2 \times (-\rho \hat{\mathbf{u}}^n) + \boldsymbol{\omega}_2 \times [\boldsymbol{\omega}_2 \times (-\rho \hat{\mathbf{u}}^n)] + 2\boldsymbol{\omega}_2 \times \mathbf{V}_{P_3/2} + \frac{V_{P_3/2}^2}{\rho} \hat{\mathbf{u}}^n + \frac{d^2 s}{dt^2} \hat{\mathbf{u}}^t. \quad (j)$$

The first three terms on the right-hand side of Eq. (j) are the components of the acceleration, $\mathbf{A}_{P_2}$ [Eq. (4.5)], and the final two terms are the normal and tangential components of the apparent acceleration, $\mathbf{A}_{P_3/2}$ [Eqs. (4.13)–(4.15)]. For the fourth term, we define the following new symbol:

$$\mathbf{A}_{P_3P_2}^c = 2\boldsymbol{\omega}_2 \times \mathbf{V}_{P_3/2}. \quad (4.16)$$

This vector is called the *Coriolis component of acceleration*. Unlike the components of the apparent acceleration, it is not sensed by an observer attached to the moving coordinate

* The first two terms on the right-hand side of Eq. (i) are equal to $\mathbf{V}_{P_2}$; thus, Eq. (i) is equivalent to the apparent-velocity equation. Note, however, that although $\rho \hat{\mathbf{u}}^n = \mathbf{R}_{C_2P_2}$ is true instantaneously, the derivatives of the two terms are not equal—that is, the vectors do not rotate at the same rate. Thus, several terms could be missed if the apparent-velocity equation were differentiated instead.

system, 2. Still, it is a necessary term in Eq. (j), and it is a part of the difference between $\mathbf{A}_{P_3}$ and $\mathbf{A}_{P_2}$ sensed by an absolute observer.

Finally, Eq. (j) can now be written in the following form, called the *apparent-acceleration equation*:

$$\mathbf{A}_{P_3} = \mathbf{A}_{P_2} + \mathbf{A}_{P_3P_2}^c + \mathbf{A}_{P_3/2}^n + \mathbf{A}_{P_3/2}^t, \quad (4.17)$$

where the definitions of the individual components are given by Eqs. (4.16), (4.14), and (4.15), respectively.

It is extremely important to recognize the following features of the apparent-acceleration equation:

1. It serves the objectives of this section, since it relates the accelerations of two *coincident points on different links* in a meaningful way.
2. There is only *one unknown* among the three new components defined. The normal component and the Coriolis component can be calculated from Eqs. (4.14) and (4.16) from the velocity information; they do not contribute any new unknowns. The tangential component given by Eq. (4.15), however, almost always has an unknown magnitude in application, since d^2s/dt^2 is usually not known.
3. It is important in each application to note the dependence of Eq. (4.17) on the ability to recognize the point path that P_3 traces on coordinate system 2. This path is the basis for the axes of the normal and tangential components and is also necessary for determining the radius of curvature ρ for Eq. (4.14).

A word of warning: *The path described by P_3 on link 2 is not necessarily the same as the path described by P_2 on link 3.* In Fig. 4.14, the path of P_3 on link 2 is clear; it is the curved slot. However, the path of P_2 on link 3 is not at all clear. As a result, there is a natural “right” and “wrong” way to write the apparent-acceleration equation for that situation. The equation

$$\mathbf{A}_{P_2} = \mathbf{A}_{P_3} + \mathbf{A}_{P_2P_3}^c + \mathbf{A}_{P_2/3}^n + \mathbf{A}_{P_2/3}^t$$

is a perfectly valid equation, but it is *useless*, since that path, and hence the radius of curvature of that path, are not known for the normal component. Note, also, that $\mathbf{A}_{P_3P_2}^c$ makes use of ω_2 , while $\mathbf{A}_{P_2P_3}^c$ would make use of ω_3 . *We must be extremely careful to write the appropriate equation for each application, recognizing which path is known.*

The following three examples demonstrate the importance of the apparent-acceleration equation, Eq. (4.17), in the graphic approach to the acceleration analysis of mechanisms.

EXAMPLE 4.4

For the sliding-block linkage in the posture shown in Fig. 4.15a, the block (link 3) is sliding outward on link 2 at a uniform rate of 30 m/s, while link 2 is rotating at a constant angular velocity of 50 rad/s ccw. Determine the acceleration of point A of the block.

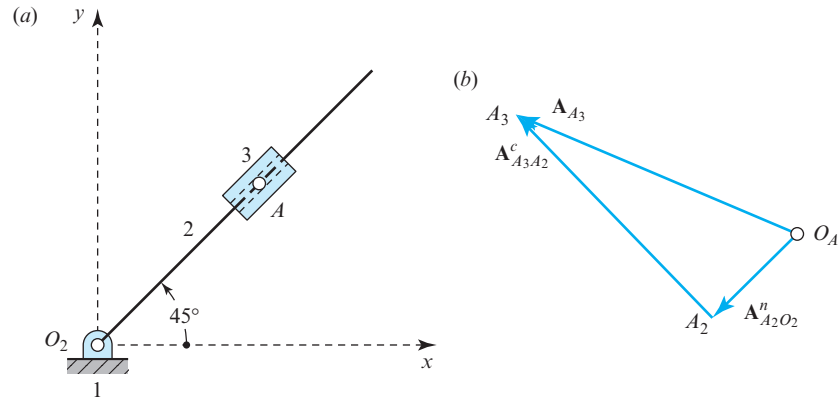


Figure 4.15 (a) $R_{AO_2} = 500$ mm; (b) acceleration polygon.

SOLUTION

The acceleration of A_2 —that is, the point immediately under the block (point A_3) but attached to link 2—can be written from Eq. (4.11) as

$$\mathbf{A}_{A_2} = \mathbf{A}_{O_2}^0 + \mathbf{A}_{A_2O_2}^n + \mathbf{A}_{A_2O_2}^t,$$

where

$$A_{A_2O_2}^n = \omega_2^2 R_{A_2O_2} = (50 \text{ rad/s})^2 (0.500 \text{ m}) = 1\,250 \text{ m/s}^2.$$

Therefore, we can draw this normal component of acceleration to scale, locating the acceleration image point A_2 in Fig. 4.15b. Next, we recognize that point A_3 is constrained to travel along link 2. This provides a path for which we can write the apparent-acceleration equation:

$$\mathbf{A}_{A_3} = \mathbf{A}_{A_2} + \mathbf{A}_{A_3A_2}^c + \mathbf{A}_{A_3/2}^n + \mathbf{A}_{A_3/2}^t.$$

The final three components for this equation can be computed; that is,

$$A_{A_3A_2}^c = 2\omega_2 V_{A_3/2} = 2(50 \text{ rad/s})(30 \text{ m/s}) = 3\,000 \text{ m/s}^2, \quad (1)$$

$$A_{A_3/2}^n = \frac{V_{A_3/2}^2}{\rho} = \frac{(30 \text{ m/s})^2}{\infty} = 0, \quad (2)$$

and

$$A_{A_3/2}^t = \frac{d^2s}{dt^2} = 0 \text{ (because of uniform rate along path)}. \quad (3)$$

The component given by Eq. (1) can now be drawn on the acceleration polygon. Noting that the sense of this component comes from the sense of the vector cross-product of

Eq. (4.16), this locates the acceleration image point A_3 . Therefore, the acceleration of point A_3 of the block is measured from the polygon as

$$A_{A_3} = 3\,250 \text{ m/s}^2. \quad \text{Ans.}$$

EXAMPLE 4.5

Perform an acceleration analysis of the inverted slider-crank linkage in the posture shown in Fig. 4.16a. The input link 2 is rotating at a constant angular velocity $\omega_2 = 18 \text{ rad/s}$ cw.

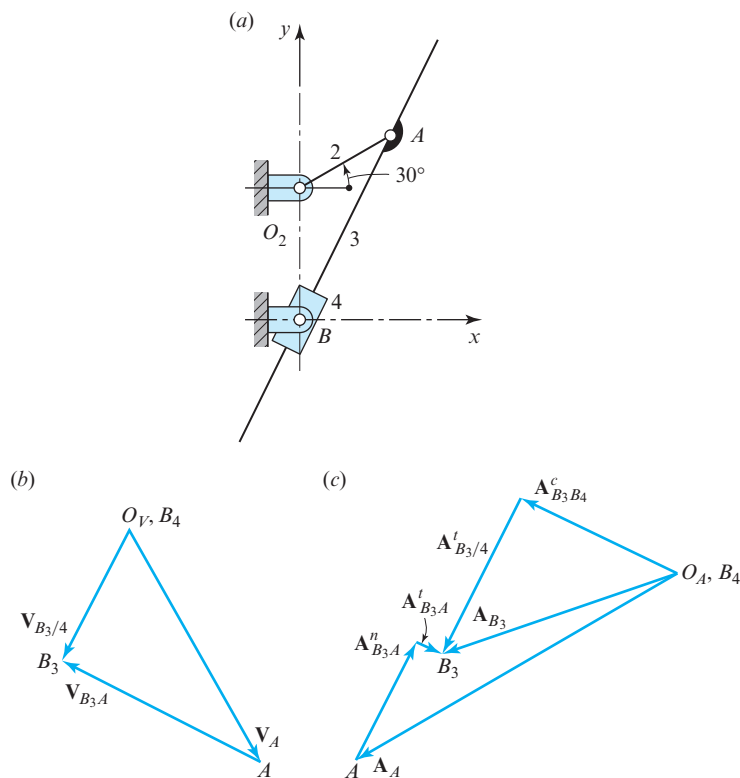


Figure 4.16 (a) $R_{AO_2} = 8 \text{ in}$ and $R_{BO_2} = 10 \text{ in}$; (b) velocity polygon; and (c) acceleration polygon.

SOLUTION

First, a complete velocity analysis is performed, as shown by the velocity polygon in Fig. 4.16b, and the results are

$$V_A = 12.0 \text{ ft/s}, \quad V_{B_3A} = 10.0 \text{ ft/s}, \quad \text{and} \quad V_{B_3/4} = 6.7 \text{ ft/s},$$

$$\omega_3 = \omega_4 = 7.67 \text{ rad/s cw.}$$

To solve for accelerations, we first calculate the acceleration of pin A:

$$\mathbf{A}_A = \cancel{\mathbf{A}_{O_2}^0} + \mathbf{A}_{AO_2}^n + \cancel{\mathbf{A}_{AO_2}^t},$$

where

$$A_{AO_2}^n = \omega_2^2 R_{AO_2} = (18 \text{ rad/s})^2 (8/12) \text{ ft} = 216.0 \text{ ft/s}^2. \quad \text{Ans.}$$

Then, we draw this acceleration to scale, locating the acceleration image point A in Fig. 4.16c. Next, we write the acceleration-difference equation

$$\mathbf{A}_{B_3} = \mathbf{A}_A + \mathbf{A}_{B_3A}^n + \mathbf{A}_{B_3A}^t, \quad (1)$$

where

$$A_{B_3A}^n = \frac{V_{B_3A}^2}{R_{BA}} = \frac{(10.0 \text{ ft/s})^2}{(15.6/12) \text{ ft}} = 76.9 \text{ ft/s}^2,$$

and is directed from B toward A. This component is added to the acceleration polygon in Fig. 4.16c.

The final component, $\mathbf{A}_{B_3A}^t$, has an unknown magnitude but is perpendicular to $\mathbf{R}_{BA}$. Since Eq. (1) has three unknowns, it cannot be solved by itself. Therefore, a second equation is required to solve for $\mathbf{A}_{B_3}$. Consider the view of an observer located on link 4; SHE would see point B_3 moving on a straight-line path along the centerline of block 4. Using this path, we write the apparent-acceleration equation, Eq. (4.17), as

$$\mathbf{A}_{B_3} = \cancel{\mathbf{A}_{B_4}^0} + \mathbf{A}_{B_3B_4}^c + \mathbf{A}_{B_3/4}^n + \mathbf{A}_{B_3/4}^t. \quad (2)$$

Note that point B_4 has zero acceleration, since it is pinned to the ground link. The two known component magnitudes of Eq. (2) are

$$A_{B_3B_4}^c = 2\omega_4 V_{B_3/4} = 2(7.67 \text{ rad/s})(6.7 \text{ ft/s}) = 103 \text{ ft/s}^2$$

and

$$A_{B_3/4}^n = \frac{V_{B_3/4}^2}{\rho} = \frac{(6.5 \text{ ft/s})^2}{\infty} = 0.$$

The Coriolis component is added to the acceleration polygon, originating at point B_4 , which is coincident with O_A . Finally, $\mathbf{A}_{B_3/4}^t$, with unknown magnitude and sense, is graphically added along a line defined by the path tangent. It crosses the unknown-length line of $\mathbf{A}_{B_3A}^t$, Eq. (1), locating the acceleration image point B_3 . From the acceleration polygon, the results are measured as

$$A_{B_3/4}^t = 103 \text{ ft/s}^2, A_{B_3A}^t = 17 \text{ ft/s}^2, \quad \text{and} \quad A_{B_3} = 145 \text{ ft/s}^2. \quad \text{Ans.}$$

The angular accelerations of links 3 and 4 are

$$\alpha_3 = \alpha_4 = \frac{A_{B_3A}^t}{R_{BA}} = \frac{17 \text{ ft/s}^2}{(15.6/12) \text{ ft}} = 13.1 \text{ rad/s}^2 \text{ ccw.} \quad \text{Ans.}$$

We note that, in this example, the path of B_3 on link 4 and the path of B_4 on link 3 can both be visualized, and either can be used in deciding the approach. However, even though B_4 is pinned to ground (link 1), the path of point B_3 on link 1 is not known. Therefore, the term $\mathbf{A}_{B_3/1}^n$ cannot be calculated directly.

EXAMPLE 4.6

For the inverted slider-crank linkage of Example 3.3 in the posture shown in Fig. 4.17a, the constant angular velocity of link 2 is $\omega_2 = 36 \text{ rad/s}$ cw. Determine the angular acceleration of link 4.

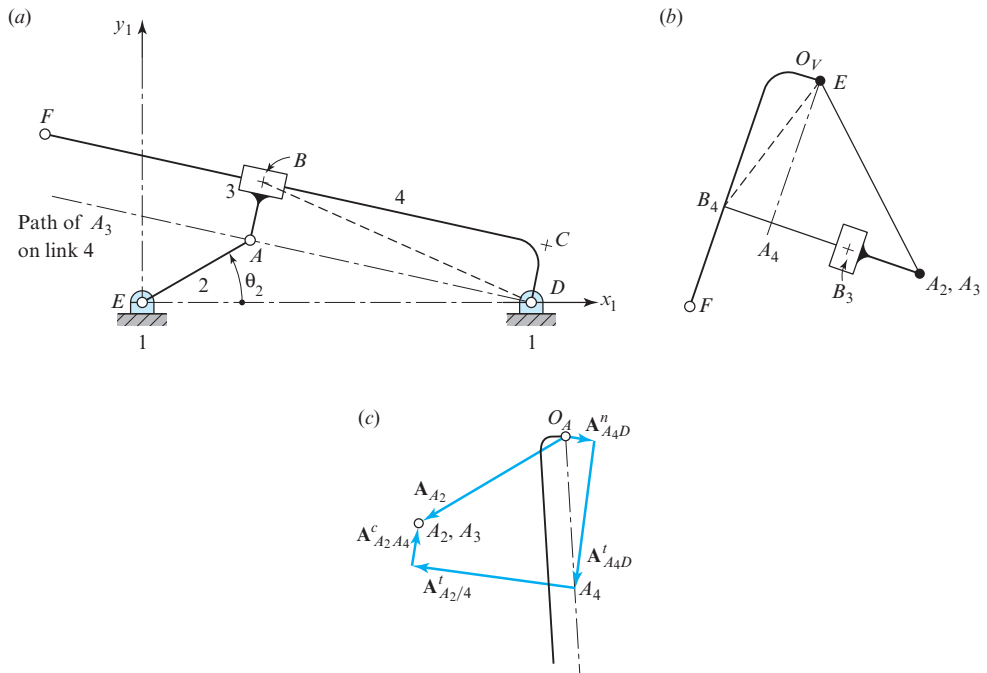


Figure 4.17 (a) Inverted slider-crank linkage; (b) velocity polygon; and (c) acceleration polygon.

SOLUTION

The velocity analysis of this linkage in the given posture is performed in Example 3.3. The results are

$$V_{A_2} = 9 \text{ ft/s}, \quad V_{A_4D} = 7.17 \text{ ft/s}, \quad V_{A_2/4} = 5.48 \text{ ft/s}, \quad \text{and} \quad \omega_3 = \omega_4 = 7.55 \text{ rad/s ccw}.$$

The velocity polygon is shown in Fig. 3.10b and is repeated here as Fig. 4.17b

The acceleration of point A on link 2 can be written as

$$\mathbf{A}_{A_2} = \cancel{\mathbf{A}_E^0} + \mathbf{A}_{A_2E}^n + \mathbf{A}_{A_2E}^t, \quad (1)$$

where

$$A_{A_2E}^n = \omega_2^2 R_{A_2E} = (36 \text{ rad/s})^2 \left(\frac{3}{12} \text{ ft} \right) = 324 \text{ ft/s}^2. \quad (2)$$

We plot this to scale, locating the acceleration image point, A_2 , as shown in Fig. 4.17c.

Note that point A_2 travels along the straight-line path indicated, as seen by an observer on link 4. Knowing this path, we write

$$\overset{\swarrow\swarrow}{\mathbf{A}}_{A_2} = \overset{? \swarrow}{\mathbf{A}}_{A_4} + \overset{\swarrow\swarrow}{\mathbf{A}}_{A_2A_4}^c + \overset{\mathbf{0}}{\mathbf{A}}_{A_2/4}^n + \overset{? \swarrow}{\mathbf{A}}_{A_2/4}^t, \quad (3)$$

where

$$A_{A_2A_4}^c = 2\omega_4 V_{A_2/4} = 2(7.55 \text{ rad/s})(5.48 \text{ ft/s}) = 82.7 \text{ ft/s}^2, \quad (4)$$

and $A_{A_2/4}^n = 0$, since $\rho = \infty$. The acceleration, $\mathbf{A}_{A_4}$, in Eq. (3), was marked as having only one unknown, since

$$\mathbf{A}_{A_4} = \overset{\mathbf{0}}{\cancel{\mathbf{A}}_D} + \overset{\swarrow\swarrow}{\mathbf{A}}_{A_4D}^n + \overset{? \swarrow}{\mathbf{A}}_{A_4D}^t, \quad (5)$$

where

$$A_{A_4D}^n = \frac{V_{A_4D}^2}{R_{A_4D}} = \frac{(7.17 \text{ ft/s})^2}{(11.5/12) \text{ ft}} = 53.6 \text{ ft/s}^2. \quad (6)$$

This component is now added from O_A , followed by a line of unknown length for the $\mathbf{A}_{A_4D}^t$ component. Since image point A_4 is not yet known, components $\mathbf{A}_{A_2A_4}^c$ and $\mathbf{A}_{A_2/4}^t$ cannot be added as indicated by Eq. (3). However, these two components can be transferred to the other side of the equation and graphically subtracted from image point A_2 , thus completing the acceleration polygon. The angular acceleration of link 4 can then be found:

$$\alpha_4 = \frac{A_{A_4D}^t}{R_{AD}} = \frac{279.9 \text{ ft/s}^2}{(11.5/12) \text{ ft}} = 292 \text{ rad/s}^2 \text{ ccw.} \quad \text{Ans.}$$

This need to subtract vectors is common in acceleration problems involving the Coriolis component and should be studied carefully. Note that the equation involving $\mathbf{A}_{A_4/2}$ cannot be used, since ρ and, therefore, $A_{A_4/2}^n$ would become an additional (third) unknown.

Since the angular acceleration of link 3 must be equal to the angular acceleration of link 4, then the acceleration of point B_3 can also be obtained.

4.6 APPARENT ANGULAR ACCELERATION

Completeness suggests that we should define the term *apparent angular acceleration*. When two rigid bodies rotate with different angular accelerations, the vector difference

between them is defined as the apparent angular acceleration,

$$\alpha_{3/2} = \alpha_3 - \alpha_2.$$

The apparent-angular-acceleration equation can also be written as

$$\alpha_3 = \alpha_2 + \alpha_{3/2}. \quad (4.18)$$

We recognize that $\alpha_{3/2}$ is the angular acceleration of body 3 as it would appear to an observer attached to, and rotating with, body 2.

4.7 DIRECT CONTACT AND ROLLING CONTACT

We recall from Sec. 3.7 that the relative motion between two bodies in direct contact at a point can occur in two different ways: there may be an apparent slipping velocity between the bodies or there may be no such slip. The purpose of this section is to extend these concepts to include acceleration. The following two examples illustrate these two cases; the first example is for direct contact with slipping, and the second example is for rolling contact.

EXAMPLE 4.7

Consider the circular cam, link 2, in direct contact with the oscillating flat-faced follower, link 3, in the posture shown in Fig. 4.18a. The angular velocity and angular acceleration of link 2 are $\omega_2 = 10$ rad/s cw and $\alpha_2 = 25$ rad/s² cw, respectively. Determine the angular acceleration of link 3.

SOLUTION

The velocity polygon must be drawn first, since the velocities of points *B* and *C*, and the angular velocity of link 3 are required for the acceleration analysis.

The magnitude of the velocity of point *B* is

$$V_B = \omega_2 R_{BA} = (10 \text{ rad/s}) (3/12 \text{ ft}) = 2.5 \text{ ft/s}.$$

Using this value, the velocity polygon is now drawn as shown in Fig. 4.18b. Then, from the polygon, we measure the magnitude of the velocity difference of points *C*₃ and *D* as

$$V_{C_3D} = 2.13 \text{ ft/s}.$$

Therefore, the angular velocity of link 3 is

$$\omega_3 = \frac{V_{C_3D}}{R_{CD}} = \frac{2.13 \text{ ft/s}}{(2.56/12) \text{ ft}} = 9.98 \text{ rad/s cw},$$

where the direction is obtained from an examination of the velocity polygon.

Now, to write an acceleration equation relating points of link 2 and link 3, we look for a pair of coincident points where the curvature of the path is known. Considering

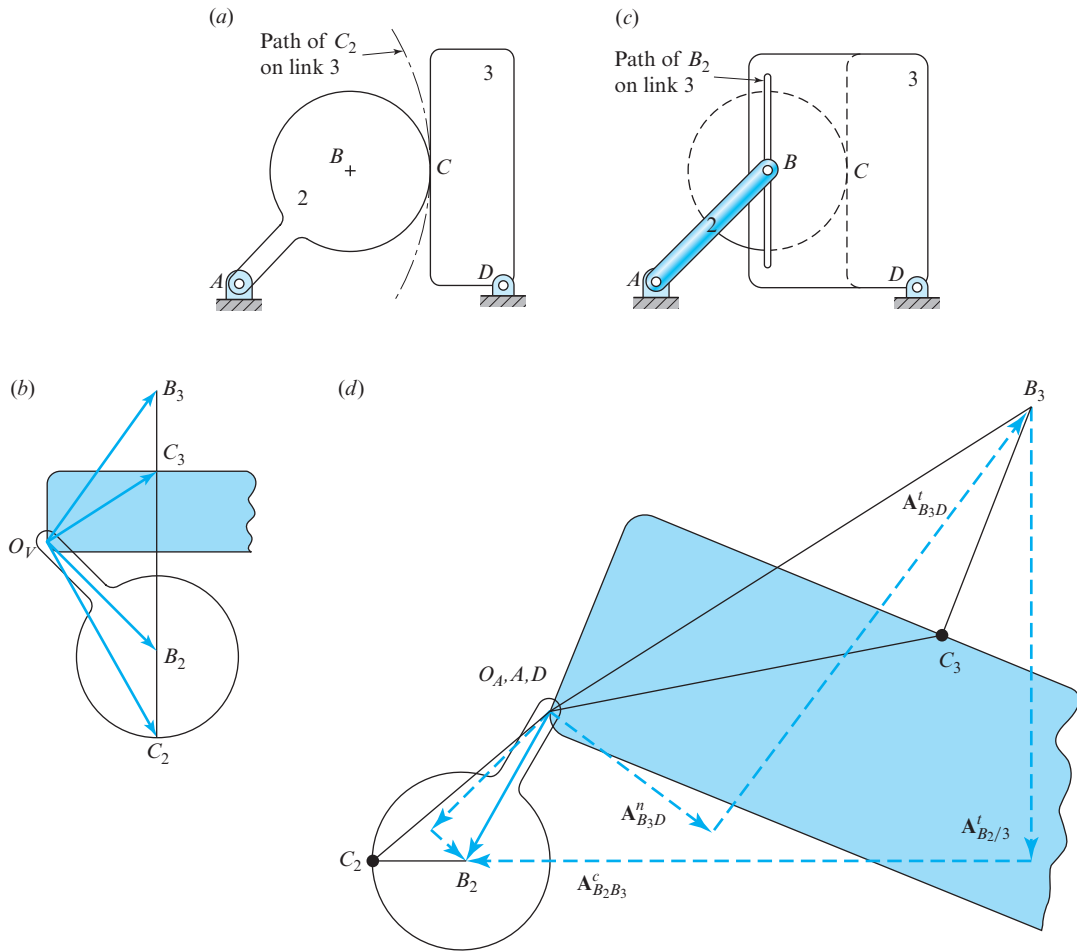


Figure 4.18 (a) $R_{BA} = 3$ in, $R_{CB} = 1.5$ in, $R_{CD}^x = -1.4$ in, $R_{CD}^y = 2.14$ in, and $R_{DA} = 5$ in; (b) velocity polygon; (c) equivalent mechanism; (d) acceleration polygon.

the path traced by point B_2 on (extended) link 3, we note that it remains a constant distance from the surface of link 3. This path is a straight line, as shown by the “slot” in the equivalent mechanism of Fig. 4.18c. Note that link 3 of this equivalent mechanism has motion equivalent to the original mechanism.

Since this “slot” can be used for a path, it becomes clear how to proceed; the appropriate equation is

$$\sqrt{\sqrt{\mathbf{A}_{B_2} = \mathbf{A}_{B_3} + \mathbf{A}_{B_2B_3}^c + \mathbf{A}_{B_2/3}^n + \mathbf{A}_{B_2/3}^t}} \quad (1)$$

where B_3 is a point coincident with B_2 but attached to link 3.

Since the path (the slot) of Eq. (1) is a straight line, we can extend the velocity polygon (Fig. 4.18*b*) to find the velocities

$$V_{B_2/3} = 4.17 \text{ ft/s} \quad \text{and} \quad V_{B_3} = 2.99 \text{ ft/s}.$$

Next, the term on the left-hand side of Eq. (1) can be written as

$$\mathbf{A}_{B_2} = \overset{0}{\cancel{\mathbf{A}_A}} + \mathbf{A}_{B_2A}^n + \mathbf{A}_{B_2A}^t,$$

where

$$A_{B_2A}^n = \omega_2^2 R_{B_2A} = (10 \text{ rad/s})^2 (3/12 \text{ ft}) = 25 \text{ ft/s}^2,$$

$$A_{B_2A}^t = \alpha_2 R_{B_2A} = (25 \text{ rad/s}^2)(3/12 \text{ ft}) = 6.25 \text{ ft/s}^2.$$

These are plotted on the acceleration polygon as shown in Fig. 4.18*d*.

Then, we calculate the known magnitudes of terms on the right-hand side of Eq. (1):

$$A_{B_2/3}^n = \frac{V_{B_2/3}^2}{\rho} = \frac{(4.17 \text{ ft/s})^2}{\infty} = 0 \quad (2)$$

and

$$A_{B_2B_3}^c = 2\omega_3 V_{B_2/3} = 2(10 \text{ rad/s})(4.17 \text{ ft/s}) = 83.4 \text{ ft/s}^2. \quad (3)$$

The acceleration of B_3 is determined from the acceleration-difference equation,

$$\overset{??}{\mathbf{A}_{B_3}} = \overset{0}{\cancel{\mathbf{A}_D}} + \overset{\sqrt{\sqrt{}}}{\mathbf{A}_{B_3D}^n} + \overset{? \sqrt{}}{\mathbf{A}_{B_3D}^t}, \quad (4)$$

where

$$A_{B_3D}^n = \frac{V_{B_3D}^2}{R_{B_3D}} = \frac{(2.99 \text{ ft/s})^2}{(3.6/12 \text{ ft})} = 29.8 \text{ ft/s}^2. \quad (5)$$

Substituting Eqs. (2) through (5) into Eq. (1) and rearranging terms, we arrive at an equation with only two unknowns:

$$\overset{\sqrt{\sqrt{}}}{\mathbf{A}_{B_2}} - \overset{\sqrt{\sqrt{}}}{\mathbf{A}_{B_2B_3}^c} - \overset{? \sqrt{}}{\mathbf{A}_{B_2/3}^t} = \overset{\sqrt{\sqrt{}}}{\mathbf{A}_{B_3D}^n} + \overset{? \sqrt{}}{\mathbf{A}_{B_3D}^t}. \quad (6)$$

This equation is solved graphically as shown in Fig. 4.18*d*, and the results are

$$A_{B_2/3}^t = 67.2 \text{ ft/s}^2 \quad \text{and} \quad A_{B_3D}^t = 78.7 \text{ ft/s}^2.$$

Once image point B_3 has been determined, image point C_3 is determined by constructing the acceleration image of triangle DB_3C_3 , all on link 3. Figure 4.18*d* has been extended to illustrate the acceleration images of links 2 and 3 to aid in visualization and to illustrate

once again that there is no obvious relation between the final locations of image points C_2 and C_3 .

Finally, the angular acceleration of link 3 is determined:

$$\alpha_3 = \frac{A_{B_3D}^t}{R_{B_3D}} = \frac{78.7 \text{ ft/s}^2}{(3.6/12\text{ft})} = 262 \text{ rad/s}^2 \text{ cw.} \quad \text{Ans.}$$

Recall in Sec. 3.7 that we defined the term *rolling contact* to imply that no slip is in progress and developed the rolling contact condition, Eq. (3.13), to indicate that the apparent velocity at such a point is zero. Here we will investigate the apparent acceleration at a point of rolling contact.

Consider the case of a circular wheel, 3, in rolling contact with a fixed straight link, 2, as shown in Fig. 4.19. Although this is admittedly a very simplified case, the arguments made and the conclusions reached are completely general and apply to any rolling contact situation, no matter what the shapes of the two bodies or whether either is the ground link. To keep this clear in our mind, the ground link has been numbered here as 2.

Once the acceleration, $\mathbf{A}_C$, of the center point of the wheel is known, the origin, O_A , can be chosen, and the acceleration polygon can be started by plotting $\mathbf{A}_C$. In relating the accelerations of points P_3 and P_2 at the point of rolling contact, however, we are dealing with two coincident points of different bodies. Therefore, it is appropriate to write the apparent-acceleration equation, Eq. (4.17) [and not the acceleration-difference equation, Eq. (4.10)]. To do this, we must identify a path that one of these points traces on the other body. The path that point P_3 traces on link 2 is sketched in the figure.* Although the precise shape of this path depends on the shapes of the two contacting links, provided that there is no slip, there is always a cusp at the point of rolling contact, and the tangent to this cusp-shaped path is always normal to the surfaces that are in contact.

Since this path is known, we are free to write the apparent-acceleration equation:

$$\mathbf{A}_{P_3} = \mathbf{A}_{P_2} + \mathbf{A}_{P_3P_2}^c + \mathbf{A}_{P_3/2}^n + \mathbf{A}_{P_3/2}^t.$$

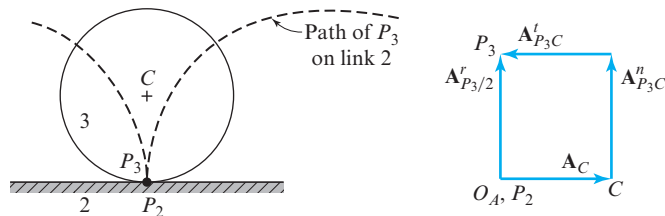


Figure 4.19 Apparent acceleration at a point of rolling contact.

* This particular curve is called a *cycloid*.

In evaluating the components, we keep in mind the rolling contact velocity condition, Eq. (3.13); that is, $\mathbf{V}_{P_3/2} = \mathbf{0}$. Therefore,

$$\mathbf{A}_{P_3P_2}^c = 2\omega_2 \times \mathbf{V}_{P_3/2} = \mathbf{0} \text{ and } A_{P_3/2}^n = \frac{V_{P_3/2}^2}{\rho} = 0.$$

Therefore, only one component of the apparent-acceleration equation—namely, $\mathbf{A}_{P_3/2}^t$ —can be nonzero. Because of possible confusion in calling this term a tangential component (tangent to the cusp-shaped path) while its direction is *normal* to the rolling surfaces, we adopt a new superscript and refer to it as the *rolling contact acceleration*, $\mathbf{A}_{P_3/2}^r$.

For rolling contact with no slip, the apparent-acceleration equation (henceforth referred to as the *rolling contact condition for acceleration*) becomes

$$\mathbf{A}_{P_3} = \mathbf{A}_{P_2} + \mathbf{A}_{P_3/2}^r, \quad (4.19)$$

where the component $\mathbf{A}_{P_3/2}^r$ is always normal to the surfaces at the point of rolling contact.

EXAMPLE 4.8

Consider the circular roller, 4, rolling without slip on the oscillating flat-faced follower, 3, in the posture shown in Fig. 4.20. The angular velocity and angular acceleration of input link 2 are $\omega_2 = 10 \text{ rad/s}$ cw and $\alpha_2 = 25 \text{ rad/s}^2$ cw, respectively. Determine the angular accelerations of the roller and the follower.

SOLUTION

Since this problem is an extension of Example 4.7, in which a roller (link 4) has been included, then the velocity polygon can be easily completed as in Fig. 4.20b. The rolling contact condition for velocity implies that $\mathbf{V}_{C_4} = \mathbf{V}_{C_3}$, which allows drawing the velocity image of link 4 as indicated.

Next, we might be tempted to write the apparent-acceleration equation for point C_4 :

$$\mathbf{A}_{C_4} = \overset{\checkmark\checkmark}{\mathbf{A}_{B_2A_2}^n} + \overset{\checkmark\checkmark}{\mathbf{A}_{B_2A_2}^t} + \overset{\checkmark\checkmark}{\mathbf{A}_{C_4B_4}^n} + \overset{?\checkmark}{\mathbf{A}_{C_4B_4}^t} = \overset{\checkmark\checkmark}{\mathbf{A}_{C_3D_3}^n} + \overset{?\checkmark}{\mathbf{A}_{C_3D_3}^t} + \overset{?\checkmark}{\mathbf{A}_{C_4/3}^r}. \quad (1)$$

Unfortunately, this equation cannot be solved, since it contains three unknowns—namely, the two angular accelerations, α_3 and α_4 , and the rolling contact acceleration, $\mathbf{A}_{C_4/3}^r$. A different solution strategy must be sought.

We proceed as in Example 4.7, Fig. 4.18d, all the way to the solution for α_3 and to finding $\mathbf{A}_{C_3}$. From this, we will have constructed a good portion of the acceleration polygon of Fig. 4.20c.

Now we can relate the acceleration of point C_4 to that of point B_4 ; that is,

$$\overset{??}{\mathbf{A}_{C_4}} = \overset{\checkmark\checkmark}{\mathbf{A}_{B_4}} + \overset{\checkmark\checkmark}{\mathbf{A}_{C_4B_4}^n} + \overset{?\checkmark}{\mathbf{A}_{C_4B_4}^t}, \quad (2)$$

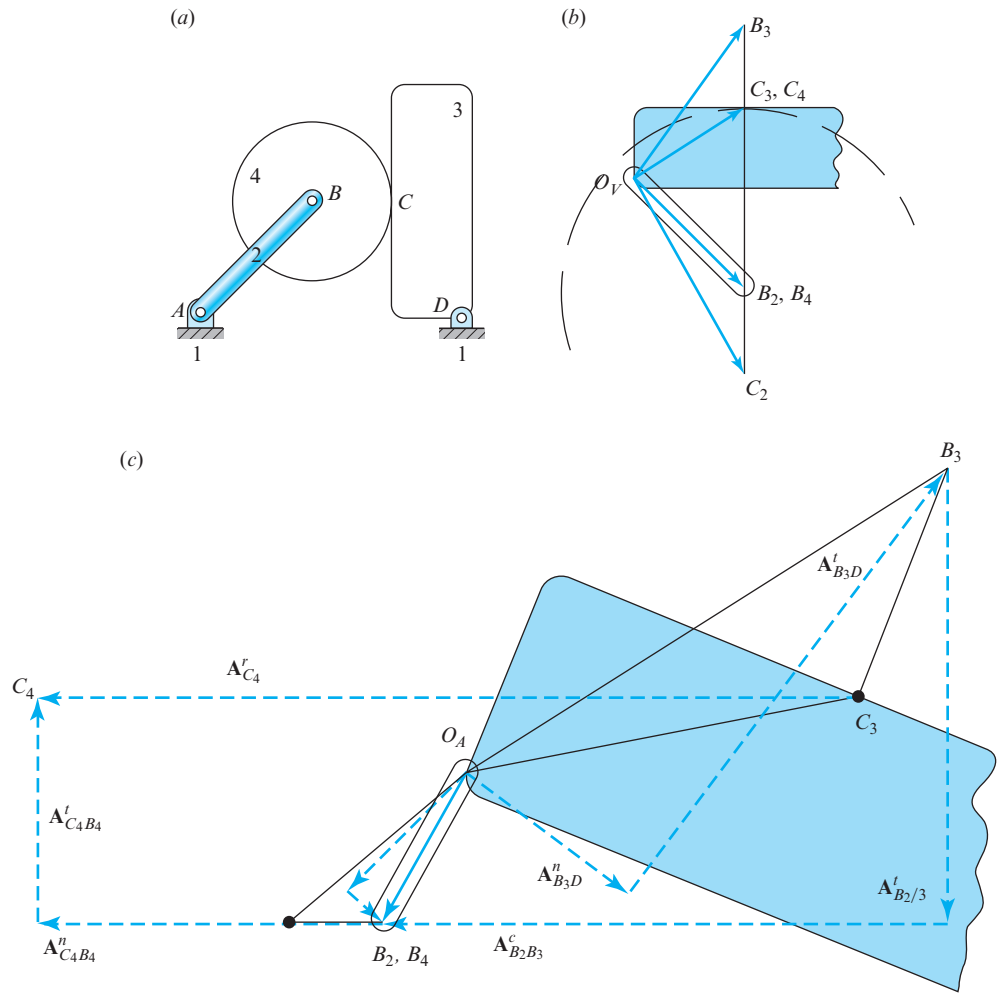


Figure 4.20 (a) Rolling contact mechanism; (b) velocity polygon; and (c) acceleration polygon.

where

$$A_{C_4B_4}^n = \frac{V_{C_4B_4}^2}{R_{CB}} = \frac{(2.87 \text{ ft/s})^2}{(1.5/12 \text{ ft})} = 65.9 \text{ ft/s}^2. \quad (3)$$

We can also write the rolling contact acceleration condition [Eq. (4.19)] as

$$\mathbf{A}_{C_4}^{??} = \mathbf{A}_{C_3}^{\checkmark\checkmark} + \mathbf{A}_{C_4/3}^{r\checkmark}. \quad (4)$$

Remembering that $\mathbf{A}_{C_4/3}^r$ is perpendicular to the surfaces at C , we graphically construct the simultaneous solution to Eqs. (2) and (4) as shown in Fig. 4.20c. Finally, the angular acceleration of roller 4 is obtained as follows:

$$\alpha_4 = \frac{A_{C_4B_4}^t}{R_{CB}} = \frac{65.9 \text{ ft/s}^2}{(1.5/12 \text{ ft})} = 527 \text{ rad/s}^2 \text{ ccw.} \quad \text{Ans.}$$

The angular acceleration of the follower is identical to that found in Example 4.7; namely,

$$\alpha_3 = 262 \text{ rad/s}^2 \text{ cw.} \quad \text{Ans.}$$

In general, it is almost always necessary to determine the motions of the two links adjacent on either side of a point of rolling contact (links 3 and 4 in Example 4.8) and then to perform the rolling contact computation working from both sides. It is almost never possible to work straight through a point of direct contact, as tried in Eq. (1), above. This almost always requires the visualization of an equivalent mechanism and the solution of a Coriolis equation. However, with a little practice, it is usually quite straightforward.

4.8 SYSTEMATIC STRATEGY FOR ACCELERATION ANALYSIS

Review of the preceding sections and example problems will demonstrate that we have now developed sufficient tools for dealing with those situations that arise repeatedly in the acceleration analysis of planar rigid-body mechanical systems. It will also be noted that the word “relative” acceleration has been carefully avoided. Instead, as with velocity analysis, whenever the desire for using “relative” acceleration arises, there are always two points whose accelerations are to be “related”; also, these two points are fixed either to the same rigid body or to two different rigid bodies. Therefore, as with velocity analysis (Table 3.1), we organize all situations into the four cases shown in Table 4.1.

In Table 4.1 we see that, when the two points are separated by a distance, only the acceleration-difference equation is appropriate for use, and two points on the *same link* should be used. When it is desirable to switch to another link, then *coincident points* should be chosen and the apparent-acceleration equation should be used. The path of one of these points in a coordinate system on the other link is then required.

Even the notation has been made different to continually remind us that these are two totally different situations and the formulae are not interchangeable between the two. We should not try to use an $\alpha \times \mathbf{R}$ formula when the apparent acceleration is required; however, if we do try, then we will not find an appropriate α or $\mathbf{R}$. Similarly, when using the acceleration difference, there is no question of which α to use, since only one link pertains. The two questions that always arise with respect to apparent acceleration are: (a) when should we include the Coriolis term, and when should we only use normal and tangential components? and (b) which ω should we use in the Coriolis term? The answer to the first question is straightforward. Whenever we use the apparent-acceleration equation, the Coriolis term should always be included; if it should not be there, such as when the “path” is not rotating, it should be included anyway, and the calculation will give it a magnitude of zero. The answer to the second question is also straightforward: Whenever we use the

Table 4.1 “Relative” Acceleration Equations

Points are	Coincident	Separated
In same body	<i>Trivial Case:</i> $\mathbf{A}_P = \mathbf{A}_Q$	<i>Acceleration difference:</i> $\mathbf{A}_P = \mathbf{A}_Q + \mathbf{A}_{PQ}^n + \mathbf{A}_{PQ}^t$ $\mathbf{A}_{PQ}^n = \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{R}_{PQ})$ $\mathbf{A}_{PQ}^t = \boldsymbol{\alpha} \times \mathbf{R}_{PQ}$
In different bodies	<i>Apparent Acceleration:</i> $\mathbf{A}_{P_i} = \mathbf{A}_{P_j} + \mathbf{A}_{P_i P_j}^c + \mathbf{A}_{P_i/j}^n + \mathbf{A}_{P_i/j}^t$ <i>where path $P_{i/j}$ is known, and</i> $\mathbf{A}_{P_i P_j}^c = 2\boldsymbol{\omega}_j \times \mathbf{V}_{P_i/j}$ $\mathbf{A}_{P_i/j}^n = \frac{V_{P_i/j}^2}{\rho} \hat{\mathbf{u}}^n$ $\mathbf{A}_{P_i/j}^t = \frac{d^2 s}{dt^2} \hat{\mathbf{u}}^t$	<i>Too general; use two steps.</i>
	<i>Rolling Contact Acceleration:</i> $\mathbf{A}_{P_i} = \mathbf{A}_{P_j} + \mathbf{A}_{P_i/j}^r$ <i>where path $\mathbf{A}_{P_i/j}^r$ is normal to surfaces at point of contact.</i>	

apparent-acceleration equation, we must always visualize a path that a point P_i makes on another link j ; then $\boldsymbol{\omega}_j$ for the link that contains the path is used. As indicated in Sec. 4.7 and Table 4.1, rolling contact acceleration is a special case of apparent acceleration.

Careful review of Examples 4.2 through 4.8 demonstrates how the strategy suggested in Table 4.1 and the labeling strategy are applied in a variety of problems of acceleration analysis.

4.9 ALGEBRAIC ACCELERATION ANALYSIS

In this section we will continue the analytic approach that we began in Sec. 3.9. Again, we focus on the central (in-line) slider-crank linkage of Fig. 3.15, reproduced here as Fig. 4.21.

The equation for the exact acceleration of the slider might be obtained by differentiating Eq. (3.15b) with respect to time. However, this is a very difficult differentiation, and the resulting expression is very unwieldy. Alternatively, differentiating Eq. (a) in Sec. 3.9

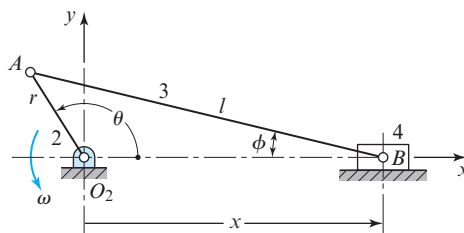


Figure 4.21 Central slider-crank linkage.

twice with respect to time, and after considerable manipulation, the exact acceleration of the slider can be written as

$$\ddot{x} = -r\omega^2 \left(\cos\theta + \frac{r\cos 2\theta}{l\cos\phi} + \frac{r^3\sin^2 2\theta}{4l^3\cos^3\phi} \right) - r\alpha \left(\sin\theta + \frac{r\sin 2\theta}{2l\cos\phi} \right). \quad (4.20)$$

This is still a rather complex expression, and it becomes even more complex when ϕ is eliminated by using Eq. (d) of Sec. 3.9; that is,

$$\cos\phi = \sqrt{1 - \left(\frac{r}{l} \sin\theta \right)^2}. \quad (4.21)$$

In some mechanism applications, approximate expressions are used for velocity and acceleration analyses. In Sec. 3.9, the binomial expansion was used to approximate the velocity of the slider [Eq. (3.16b)]. Here we will obtain an approximate expression for the acceleration of the slider. For small values of (r/l) , the last term in the first bracket of Eq. (4.20) can be neglected. Also, in this case, $\cos\phi$ is near unity; therefore, the acceleration of the slider can be written as

$$\ddot{x} = -r\omega^2 \left(\cos\theta + \frac{r}{l} \cos 2\theta \right) - r\alpha \left(\sin\theta + \frac{r}{2l} \sin 2\theta \right). \quad (4.22a)$$

Note that this result could have been obtained directly by differentiating Eq. (3.16b) with respect to time. Also, in the special case that the input angular velocity is constant ($\alpha = 0$), then the acceleration of the slider is

$$\ddot{x} = -r\omega^2 \left(\cos\theta + \frac{r}{l} \cos 2\theta \right). \quad (4.22b)$$

4.10 COMPLEX-ALGEBRAIC ACCELERATION ANALYSIS

Let us now see how Raven's method (Sec. 3.10) is extended to the analysis of accelerations. The general approach is outlined here for the offset slider-crank linkage shown in Fig. 4.22.

In complex polar form, the loop-closure equation is

$$r_2 e^{j\theta_2} + r_3 e^{j\theta_3} - r_1 e^{-j(\pi/2)} - r_4 e^{j0} = 0. \quad (a)$$

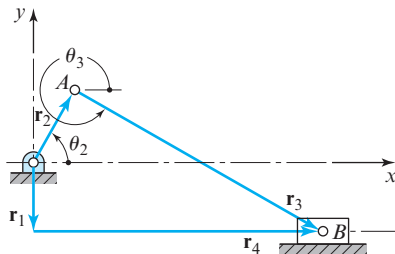


Figure 4.22 Offset slider-crank linkage.

If we separate this equation into its real and imaginary components, we obtain the two posture equations:

$$r_2 \cos \theta_2 + r_3 \cos \theta_3 - r_4 = 0, \quad (b)$$

$$r_2 \sin \theta_2 + r_3 \sin \theta_3 + r_1 = 0. \quad (c)$$

With r_1, r_2 , and r_3 given, and for a known input posture, θ_2 , these two equations can be solved for the two variables θ_3 and r_4 . In this example, Eq. (c) can be solved for the variable θ_3 ; that is,

$$\theta_3 = \sin^{-1} \left(\frac{-r_1 - r_2 \sin \theta_2}{r_3} \right). \quad (4.23)$$

Then Eq. (b) can be solved for the variable r_4 ; that is,

$$r_4 = r_2 \cos \theta_2 + r_3 \cos \theta_3. \quad (4.24)$$

Differentiating Eq. (a) with respect to time gives the velocity equation, which is equivalent to the equation of the velocity polygon in exponential form. After replacing $\dot{\theta}$ with ω , the result is

$$jr_2\omega_2 e^{j\theta_2} + jr_3\omega_3 e^{j\theta_3} - \dot{r}_4 e^{j0} = 0. \quad (d)$$

Raven's method, as we have seen, consists of making the trigonometric transformation and separating the result into real and imaginary terms, as was done in obtaining Eqs. (b) and (c) from Eq. (a). When this procedure is carried out for the velocity equation in this example, Eq. (d), we find

$$\omega_3 = -\frac{r_2 \cos \theta_2}{r_3 \cos \theta_3} \omega_2 \quad (4.25)$$

and

$$\dot{r}_4 = -r_2 \sin \theta_2 \omega_2 - r_3 \sin \theta_3 \omega_3. \quad (4.26)$$

We use the same approach to obtain the acceleration equations by first differentiating Eq. (d). Then, separating the result into its real and imaginary components gives two equations that can be solved for the two acceleration unknowns; namely,

$$\alpha_3 = \frac{r_2 \sin \theta_2 \omega_2^2 - r_2 \cos \theta_2 \alpha_2 + r_3 \sin \theta_3 \omega_3^2}{r_3 \cos \theta_3}, \quad (4.27)$$

$$\ddot{r}_4 = -r_2 \cos \theta_2 \omega_2^2 - r_2 \sin \theta_2 \alpha_2 - r_3 \cos \theta_3 \omega_3^2 - r_3 \sin \theta_3 \alpha_3. \quad (4.28)$$

If the input angular velocity is constant—that is, if $\alpha_2 = 0$ —then the acceleration results are

$$\alpha_3 = \frac{r_2 \sin \theta_2 \omega_2^2 + r_3 \sin \theta_3 \omega_3^2}{r_3 \cos \theta_3}, \quad (4.29)$$

$$\ddot{r}_4 = -r_2 \cos \theta_2 \omega_2^2 - r_3 \cos \theta_3 \omega_3^2 - r_3 \sin \theta_3 \alpha_3. \quad (4.30)$$

The same procedure, when carried out for the four-bar linkage, gives results that can be used for computer solutions of both the crank-rocker and the drag-link. Unfortunately, they cannot be used for other four-bar linkages unless an arrangement is included in the program to cause the solution to stop when an extreme posture is reached.

If the loop-closure equation for the four-bar linkage (Fig. 3.18) is written as

$$\mathbf{r}_1 + \mathbf{r}_2 + \mathbf{r}_3 - \mathbf{r}_4 = \mathbf{0}, \quad (e)$$

where the subscripts are the link numbers and where link 2 is the driver having a constant input angular velocity, then the acceleration relations that are obtained by Raven's method are

$$\alpha_3 = \frac{r_2 \cos(\theta_2 - \theta_4) \omega_2^2 + r_3 \cos(\theta_3 - \theta_4) \omega_3^2 - r_4 \omega_4^2}{r_3 \sin(\theta_4 - \theta_3)}, \quad (4.31)$$

$$\alpha_4 = \frac{r_2 \cos(\theta_2 - \theta_3) \omega_2^2 - r_4 \cos(\theta_3 - \theta_4) \omega_4^2 + r_3 \omega_3^2}{r_4 \sin(\theta_4 - \theta_3)}. \quad (4.32)$$

4.11 METHOD OF KINEMATIC COEFFICIENTS

Again we employ the four-bar linkage as an example of this method of solution. The angular accelerations of links 3 and 4 can be obtained by differentiating Eqs. (3) of Example 3.7 with respect to the input angle θ_2 ; that is,

$$-r_3 \cos \theta_3 \theta_3'^2 - r_3 \sin \theta_3 \theta_3'' + r_4 \cos \theta_4 \theta_4'^2 + r_4 \sin \theta_4 \theta_4'' = r_2 \cos \theta_2, \quad (4.33)$$

$$-r_3 \sin \theta_3 \theta_3'^2 + r_3 \cos \theta_3 \theta_3'' + r_4 \sin \theta_4 \theta_4'^2 - r_4 \cos \theta_4 \theta_4'' = r_2 \sin \theta_2, \quad (4.34)$$

where, by definition,

$$\theta_3'' = \frac{d^2 \theta_3}{d\theta_2^2} \quad \text{and} \quad \theta_4'' = \frac{d^2 \theta_4}{d\theta_2^2},$$

and they are referred to as the *second-order kinematic coefficients* of links 3 and 4. Writing Eqs. (4.33) and (4.34) in matrix form gives

$$\begin{bmatrix} -r_3 \sin \theta_3 & r_4 \sin \theta_4 \\ r_3 \cos \theta_3 & -r_4 \cos \theta_4 \end{bmatrix} \begin{bmatrix} \theta_3'' \\ \theta_4'' \end{bmatrix} = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}, \quad (4.35)$$

where

$$\begin{aligned} B_1 &= r_2 \cos \theta_2 + r_3 \cos \theta_3 \theta_3'^2 - r_4 \cos \theta_4 \theta_4'^2, \\ B_2 &= r_2 \sin \theta_2 + r_3 \sin \theta_3 \theta_3'^2 - r_4 \sin \theta_4 \theta_4'^2, \end{aligned} \quad (4.36)$$

and they are known from position and velocity analyses. Also, note that the (2×2) coefficient matrix on the left-hand side of Eq. (4.35) is the same as the (2×2) coefficient matrix in the velocity analysis* [Eq. (5) in Example 3.7]. This is not a coincidence. The coefficient matrix is the same for all kinematic derivatives—that is, first-order, second-order, third-order kinematic coefficients, and so on. Therefore, this matrix provides a built-in check of all higher-order differentiation. The determinant of the coefficient matrix is as given by Eq. (6) in Example 3.7; that is,

$$\Delta = -r_3 r_4 \sin(\theta_4 - \theta_3). \quad (4.37)$$

The second-order kinematic coefficients for links 3 and 4 can be obtained from Eq. (4.35) by using Cramer's rule, and the results are

$$\theta_3'' = \frac{B_1 \cos \theta_4 + B_2 \sin \theta_4}{r_3 \sin(\theta_4 - \theta_3)} \quad \text{and} \quad \theta_4'' = \frac{B_1 \cos \theta_3 + B_2 \sin \theta_3}{r_4 \sin(\theta_4 - \theta_3)}. \quad (4.38)$$

Note that the second-order kinematic coefficients for the four-bar linkage are nondimensional (rad/rad^2). The angular accelerations of links 3 and 4, obtained from the chain rule, are

$$\alpha_3 = \theta_3'' \omega_2^2 + \theta_3' \alpha_2 \quad \text{and} \quad \alpha_4 = \theta_4'' \omega_2^2 + \theta_4' \alpha_2. \quad (4.39)$$

Table 4.2 summarizes the second-order kinematic coefficients that are related to link j of a planar mechanism (that is, vector $\mathbf{r}_j$) having (a) variable angle θ_j and/or (b) variable magnitude r_j .

To illustrate the method of kinematic coefficients for determining the angular accelerations of links and the accelerations of moving points, consider the following three examples.

Table 4.2 Summary of Second-Order Kinematic Coefficients

	Variable of Interest Angle θ_j (Use symbol θ_j'' for kinematic coefficient regardless of input)	Variable of Interest Magnitude r_j (Use symbol r_j'' for kinematic coefficient regardless of input)
Input	$\alpha_j = \theta_j'' \dot{\psi}^2 + \theta_j' \ddot{\psi}$	$\ddot{r}_j = r_j'' \dot{\psi}^2 + r_j' \ddot{\psi}$
$\psi = \text{angle } \theta_i$	$\theta_j' = \frac{d\theta_j}{d\psi}$ (dimensionless, rad/rad)	$r_j' = \frac{dr_j}{d\psi}$ (length, length/rad)
	$\theta_j'' = \frac{d^2\theta_j}{d\psi^2}$ (dimensionless, rad/rad ²)	$r_j'' = \frac{d^2r_j}{d\psi^2}$ (length, length/rad ²)
Input	$\alpha_j = \theta_j'' \dot{\psi}^2 + \theta_j' \ddot{\psi}$	$\ddot{r}_j = r_j'' \dot{\psi}^2 + r_j' \ddot{\psi}$
$\psi = \text{magnitude } r_i$	$\theta_j' = \frac{d\theta_j}{dr_i}$ (1/length, rad/length)	$r_j' = \frac{dr_j}{dr_i}$ (dimensionless, length/length)
	$\theta_j'' = \frac{d^2\theta_j}{dr_i^2}$ (1/length ² , rad/length ²) ²	$r_j'' = \frac{d^2r_j}{dr_i^2}$ (1/length, length/length ²)

* This coefficient matrix is called the *Jacobian* of the system.

EXAMPLE 4.9

For the four-bar linkage of Example 4.2, determine the angular accelerations of links 3 and 4, and the absolute accelerations of points E and F .

SOLUTION

Recall that the first-order kinematic coefficients from Eq. (8) in Example 3.7 are

$$\theta'_3 = 0.2718 \text{ rad/rad} \quad \text{and} \quad \theta'_4 = 0.5265 \text{ rad/rad}. \quad (1)$$

Substituting Eqs. (1) and the known data into Eqs. (4.36) and substituting the results into Eq. (4.38), the second-order kinematic coefficients of links 3 and 4 are

$$\theta''_3 = 0.0131 \text{ rad/rad}^2 \quad \text{and} \quad \theta''_4 = -0.2030 \text{ rad/rad}^2. \quad (2)$$

Substituting the constant angular velocity, $\omega_2 = 94.25 \text{ rad/s}$ ccw, and Eqs. (1) and (2) into Eqs. (4.39), the angular accelerations of links 3 and 4 are

$$\alpha_3 = 116 \text{ rad/s}^2 \text{ ccw} \quad \text{and} \quad \alpha_4 = 1801 \text{ rad/s}^2 \text{ cw}. \quad \text{Ans.}$$

These answers are in good agreement with the results obtained from the graphic method of Example 4.2—that is, $\alpha_3 = 113 \text{ rad/s}^2$ ccw and $\alpha_4 = 1822 \text{ rad/s}^2$ cw.

To obtain the acceleration of point E , we differentiate Eqs. (13) in Example 3.7 with respect to the input angle θ_2 . Therefore, the second-order kinematic coefficients for point E are

$$\begin{aligned} x''_E &= -r_2 \cos \theta_2 - r_{EB} \cos(\theta_3 - \phi) \theta_3'^2 - r_{EB} \sin(\theta_3 - \phi) \theta_3'', \\ y''_E &= -r_2 \sin \theta_2 - r_{EB} \sin(\theta_3 - \phi) \theta_3'^2 + r_{EB} \cos(\theta_3 - \phi) \theta_3''. \end{aligned} \quad (3)$$

Substituting Eqs. (1) and (2) and the known data into Eqs. (3) gives

$$x''_E = 1.2067 \text{ in/rad}^2 \quad \text{and} \quad y''_E = -3.3108 \text{ in/rad}^2. \quad (4)$$

Differentiating the velocity of point E , from Eq. (15) in Example 3.7, with respect to time, the acceleration of point E can be written as

$$\mathbf{A}_E = (x''_E \omega_2^2 + x'_E \alpha_2) \hat{\mathbf{i}} + (y''_E \omega_2^2 + y'_E \alpha_2) \hat{\mathbf{j}}. \quad (5)$$

Recall that the first-order kinematic coefficients for point E , from Eq. (14) in Example 3.7, are

$$x'_E = -3.4191 \text{ in/rad} \quad \text{and} \quad y'_E = 0.9269 \text{ in/rad}. \quad (6)$$

Substituting Eqs. (4) and (6), and the angular velocity and acceleration of the input link into Eq. (5) gives

$$\mathbf{A}_E = 892.32 \hat{\mathbf{i}} - 2448.24 \hat{\mathbf{j}} \text{ ft/s}^2. \quad \text{Ans.}$$

Therefore, the magnitude of the acceleration of point E is $A_E = 2\,606\text{ ft/s}^2$. This result is in good agreement with the answer in Example 4.2—that is, $A_E = 2\,580\text{ ft/s}^2$.

The acceleration of point F can be obtained in a similar manner. The first-order and second-order kinematic coefficients for point F are

$$\begin{aligned}x'_F &= -2.626\,2\text{ in/rad}, \quad y'_F = 3.068\,6\text{ in/rad}, \\x''_F &= -0.586\,0\text{ in/rad}^2, \quad y''_F = -2.551\,7\text{ in/rad}^2.\end{aligned}\tag{7}$$

Similar to Eq. (5), the acceleration of point F can be written as

$$\mathbf{A}_F = (x''_F\omega_2^2 + x'_F\alpha_2)\hat{\mathbf{i}} + (y''_F\omega_2^2 + y'_F\alpha_2)\hat{\mathbf{j}}.\tag{8}$$

Substituting Eqs. (7) and the angular velocity and acceleration of the input into Eq. (8), the acceleration of point F is

$$\mathbf{A}_F = -433\hat{\mathbf{i}} - 1\,886\hat{\mathbf{j}}\text{ ft/s}^2.\tag{Ans.}$$

Therefore, the magnitude of the acceleration of point F is $A_F = 1\,935\text{ ft/s}^2$. This result is in good agreement with the answer of Example 4.2—that is, $A_F = 1\,960\text{ ft/s}^2$.

EXAMPLE 4.10

For the offset slider-crank linkage of Example 3.2 in the posture shown in Fig. 3.8a, link 4 is driven at a constant speed of 10 m/s to the left. Determine the angular accelerations of links 2 and 3, and the acceleration of coupler point D .

SOLUTION

The angular accelerations of links 2 and 3 are obtained by differentiating Eqs. (3) in Example 3.8; with respect to the input displacement, r_4 ; that is,

$$\begin{aligned}-r_2 \cos \theta_2 \theta_2'' - r_2 \sin \theta_2 \theta_2'' - r_3 \cos \theta_3 \theta_3'' - r_3 \sin \theta_3 \theta_3'' &= 0, \\-r_2 \sin \theta_2 \theta_2'' + r_2 \cos \theta_2 \theta_2'' - r_3 \sin \theta_3 \theta_3'' + r_3 \cos \theta_3 \theta_3'' &= 0,\end{aligned}\tag{1}$$

where the second-order kinematic coefficients of links 2 and 3 are defined as

$$\theta_2'' = \frac{d^2\theta_2}{dr_4^2} \quad \text{and} \quad \theta_3'' = \frac{d^2\theta_3}{dr_4^2}.$$

We recall the first-order kinematic coefficients from Eq. (8) in Example 3.8; they are

$$\theta_2' = -19.856\text{ rad/m} \quad \text{and} \quad \theta_3' = 5.447\text{ rad/m}.\tag{2}$$

Substituting Eqs. (2) and the known data into Eqs. (1), the second-order kinematic coefficients of links 2 and 3 are

$$\theta_2'' = -246.25\text{ rad/m}^2 \quad \text{and} \quad \theta_3'' = 162.73\text{ rad/m}^2.\tag{3}$$

The angular accelerations of links 2 and 3 can be written from Table 4.2 as

$$\alpha_2 = \theta_2'' \dot{r}_4^2 + \theta_2' \ddot{r}_4 \quad \text{and} \quad \alpha_3 = \theta_3'' \dot{r}_4^2 + \theta_3' \ddot{r}_4. \quad (4)$$

Substituting Eqs. (2) and (3), and the constant input speed $\dot{r}_4 = -10$ m/s into Eq. (4), the angular accelerations of links 2 and 3 are

$$\alpha_2 = -24\,625 \text{ rad/s}^2 \text{ (cw)} \quad \text{and} \quad \alpha_3 = 16\,273 \text{ rad/s}^2 \text{ ccw}. \quad \text{Ans.}$$

To obtain the acceleration of point D , we differentiate Eqs. (12) in Example 3.8 with respect to the input displacement, r_4 . The second-order kinematic coefficients for point D are

$$\begin{aligned} x_D'' &= -r_2 \cos \theta_2 \theta_2'' - r_2 \sin \theta_2 \theta_2'' - r_{DB} \cos(\theta_3 - \beta) \theta_3'' - r_{DB} \sin(\theta_3 - \beta) \theta_3'', \\ y_D'' &= -r_2 \sin \theta_2 \theta_2'' + r_2 \cos \theta_2 \theta_2'' - r_{DB} \sin(\theta_3 - \beta) \theta_3'' + r_{DB} \cos(\theta_3 - \beta) \theta_3''. \end{aligned} \quad (5)$$

Substituting Eqs. (2) and (3) and the known data into Eqs. (5), we find

$$x_D'' = 6.52 \text{ m/m}^2 \quad \text{and} \quad y_D'' = -12.31 \text{ m/m}^2. \quad (6)$$

Differentiating the velocity of point D , Eq. (14) in Example 3.8, with respect to time, the acceleration of point D can be written as

$$\mathbf{A}_D = (x_D'' \dot{r}_4^2 + x_D' \ddot{r}_4) \hat{\mathbf{i}} + (y_D'' \dot{r}_4^2 + y_D' \ddot{r}_4) \hat{\mathbf{j}}. \quad (7)$$

Next, we recall the first-order kinematic coefficients for point D from Eq. (13) in Example 3.8 as

$$x_D' = 1.123 \text{ m/m} \quad \text{and} \quad y_D' = -0.407 \text{ m/m}. \quad (8)$$

Substituting Eqs. (6) and (8) and the constant input speed into Eq. (7), we get

$$\mathbf{A}_D = 652 \hat{\mathbf{i}} - 1\,231 \hat{\mathbf{j}} \text{ m/s}^2. \quad \text{Ans.}$$

Therefore, the magnitude of the acceleration of point D is $A_D = 1\,393 \text{ m/s}^2$.

EXAMPLE 4.11

Consider the marine steering gear, called *Rapson's slide*, in the posture shown in Fig. 4.23. Link 4, O_4B , is the tiller, and link 2, AC , is the actuating rod. If the velocity of link 2 is a constant 15 ft/s to the left, determine the angular acceleration of the tiller and the acceleration of point B .

SOLUTION

The vectors that are chosen for the kinematic analysis of the mechanism are shown in Fig. 4.23. Note that point D is a point of the ground link, 1, such that it lies vertically below